# SUPPLEMENTARY MATERIAL TO "EXPLICIT HAMILTONIAN BALL PACKINGS: GENERAL CAPACITIES, HIGHER DIMENSIONS AND THE SIEGEL–YAO CRITERION"

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Abstract. This supplement contains the complete proofs of the constructions of the main text that are not in Paper I: the finite comparator, the explicit planar rearrangement of convex functions and its lift to the fibres, the scalar comparison datum for the four-dimensional cubic model, the bordered handle construction with its layer maps, finite coordinate primitive and Moser fields, the diagonal affine quadric model and its transfer to the ball, the two models for one large ball, the input-name programs, the finite compilation, the trajectory certificates and the termination of the algorithm. Appendix A reproduces the statements of Paper I used in the proofs.

*Authorship and the role of AI.* See the corresponding section at the end of the main text.

Contents

1. A checking route 1
2. The finite comparator 2
3. Rearranging convex functions by explicit isotopies 8
4. The scalar comparison datum for the cubic model 19
5. Dimension four: general capacities 26
6. The bordered handle construction 27
7. The layer maps 35
8. The finite-atlas toric construction 39
9. Dimension at least six: capacities below one half 42
10. The affine quadric transfer 46
11. One large ball: the models 52
12. One large ball in dimension at least eight: the shell return and the packing 56
13. Dimension six: the polynomial model 62
14. Dimension six: compact transfers and the large ball 66
15. Input names and elementary programs 73
16. Finite scalar compilation 74
17. Finite trajectory certificates 78
18. The algorithm and its termination 80
Appendix A. Statements quoted from Paper I 82
References 88

## 1. A checking route

This section lists, in the order of the construction, the statements to check for Theorem M1 and its corollaries. Numbers with the prefix M refer to the main text, as do the numbered results.

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Paper I [Son26] gives the four-dimensional chain for the linear datum as steps F1–F12 (Paper I, § "The four-dimensional chain"). Steps F5–F11 (cubic model, handle, transfer, embeddings, common isotopy, Hamiltonian, statements) hold for all valid cubic data, and steps F9–F11 in every dimension; they are recalled in §M2.4. The case distinction of Construction M3.4 selects one of four branches. The statements of Paper I used in the proofs are recalled in §A, and the Lean declarations for each step are listed in Table M4. Steps F2–F4 of Paper I are replaced, or extended, as follows.

**H1** *Comparison datum on a polytope.* In: a rational downward polytope $\Delta$ and a concave minorant $P$. Out: $K$ and a Hamiltonian isotopy of $\mathbb{C}^m$ preserving $V_\Delta$, Poisson-commuting with the face moments on a collar, with gap $\gamma$. Proved: Proposition M2.9 and Theorem M2.11 (§2); the lift is Theorem 3.13 (§3, with the planar machine of Paper I, § "Area-preserving disc maps from Morse charts"). New: a finite rational iterate with explicit gap replaces the fixed point of the rearrangement contraction of [Ope26].

**H2** *Dimension four, general capacities.* In: $\rho_i < \frac{1}{3}$ with $\sum_i \rho_i^2 < 1$. Out: valid cubic data for Paper I, Theorem (Packing in dimension four). The scalar iteration, the planar wells and the Alexander path, §4; the theorem, §5. Proved: Propositions 4.2, 4.7, 4.8, Theorem 5.2.

**H3** *Packing in a model.* In: a model input with handle data and collar bounds, a comparison datum, and $\rho_i < a_i$ with the profile minorant. Out: embeddings $\psi_i$ of $B(\rho_i)$ into $\operatorname{int} V_\Delta(E)$. Proved: Theorem 6.18 (§§6–8); the composition with the transfer is Proposition M2.7.

**H4** *All $\rho_i < \frac{1}{2}$, $n \geq 3$.* In: $\rho$ with $\sum_i \rho_i^n < 1$. Out: embeddings into $B^\circ(1)$ with disjoint images. The diagonal affine quadric curve, §§9 and 10. Proved: Theorems 10.12 and 9.7.

**H5** *One $\rho_j \geq \frac{1}{2}$, $n \geq 4$.* In: $A = \rho_j$ and valid auxiliary data. Out: the identity on the large ball and embeddings of the others into the shell $S_a$. The mixed model and the slit-shell return, §§11.1 and 12. Proved: Lemma 12.7, Theorem 12.9.

**H6** *One $\rho_j \geq \frac{1}{2}$, $n = 3$.* In: $A = \rho_j$ and valid rational auxiliary data. Out: embeddings into $B^\circ(1)$ with disjoint images. The polynomial cubic model and the bubble, §§11.2, 13 and 14. Proved: Proposition 14.3, Theorem 14.7.

**H7** *Finite form and algorithm.* Any $H$ satisfying Pack $\mapsto (B, P)$ with $\beta_B P$ satisfying Pack, found by a machine from input names (Theorem M5.2, Theorem 16.15; Theorem M5.3, Theorems 18.4 and 17.5). New: a sound rational Euler certificate, and termination proved from the explicit $H$.

Steps F9–F11 of Paper I then give Theorem M1(i), and with H7 part (iii) and the corollaries.

## 2. The finite comparator

This section proves the statements on comparison data of §M2.3: the polyhedral form of the coordinate rearrangement, the detailed form of Proposition M2.9, the smoothing and composition lemmas, and Theorem M2.11. The setting and the basic properties of $R_j$ are those of Paper I, § "Coordinate rearrangement and the equal-ball identities"; for $m = 1$ the equal-ball identities of Paper I give a linear comparison function, and Remark 2.8 shows that no linear one exists for some unequal inputs.

2.1. **Rational polyhedral functions.** A function on $\Delta$ is *rational convex polyhedral* (resp. *rational concave piecewise affine*) if it is the restriction of $\max_{\iota \in I} \phi_\iota$ (resp. $\min_{u \in U} \psi_u$) for a finite nonempty family of rational affine functions.

- Sums, maxima and nonnegative rational multiples of rational convex polyhedral functions are again rational convex polyhedral, and so is $-P$ for $P$ rational concave piecewise affine.

- Sums of rational concave piecewise affine functions are rational concave piecewise affine, and so is $-(a-|p|)_+ = \min(0,|p|-a)$ for rational $a$.

Hence $P_{\mathrm{b}} = H_{\mathrm{b}} - \sum_i (a_i - |p|)_+$ with affine $H_{\mathrm{b}}$ is concave piecewise affine, and rational when $H_{\mathrm{b}}$ and the $a_i$ are rational. This covers the cubic model (Lemma 5.1) and the higher-dimensional models (Section M3).

Rationality is used only in Lemma 2.1 and Proposition 2.3(a),(d), to make the iterates and $\mu$, $\lambda$, $N$ rational; every other statement, and every inequality of Proposition 2.3, holds for real data. The four-dimensional equal case ($a = t_0/\sqrt{k}$) uses the closed-form datum of Paper I, Proposition (Linear comparison datum for equal balls) (Paper I, § "Equal balls in dimension four").

## 2.2. **Polyhedral form.**

**Lemma 2.1** (Polyhedral form)**.** *Let $F = \max_{\iota\in I}\phi_\iota$ be rational convex polyhedral on $\Delta$. Let $\Pi \subset \mathbb{R}^{m-1}\times\mathbb{R}\times\mathbb{R}\times\mathbb{R}$ be the rational polyhedron of points $(y,x,t,a)$ satisfying*

$$(y,x)\in\Delta, \qquad a\ge 0, \qquad b_{\nu j}(a+x) + \textstyle\sum_{i\neq j} b_{\nu i}y_i \le \ell_\nu \quad (b_{\nu j}>0),$$
$$t \ge \phi_\iota(y,a), \qquad t\ge \phi_\iota(y,a+x) \qquad (\iota\in I).$$

*Then the epigraph $\{(p,t) : p\in\Delta,\ t\ge (R_jF)(p)\}$ of $R_jF$ over $\Delta$ is the image of $\Pi$ under $(y,x,t,a)\mapsto(y,x,t)$. Eliminating $a$ from the defining inequalities of $\Pi$ by one step of Fourier–Motzkin elimination gives finitely many rational inequalities $\langle\alpha_l,p\rangle+\zeta_l t\le\delta_l$ ($l\in\mathcal{S}$) that define this epigraph, every $\zeta_l$ is $\le 0$, the set $\mathcal{S}_- = \{l:\zeta_l<0\}$ is nonempty, and*

$$R_jF = \max_{l\in\mathcal{S}_-}\frac{\langle\alpha_l,p\rangle-\delta_l}{-\zeta_l} \qquad \text{on } \Delta.$$

*In particular $R_jF$ is rational convex polyhedral, and so is $\mathcal{R}F$.*

*Proof.* On the first line of constraints, the second line reads $t\ge\Phi_F(y,x,a)$. Since the minimum is attained, the projection of $\Pi$ is the epigraph.

One Fourier–Motzkin step [Sch86, §12.2] eliminates $a$: compare every lower bound with every upper bound and keep the $a$-free inequalities. This yields finitely many rational inequalities.

For $p\in\Delta$ the admissible $t$ form $[(R_jF)(p),\infty)$. Hence $\zeta_l\le 0$, the inequalities with $\zeta_l=0$ hold on $\Delta$, and $\mathcal{S}_-\neq\emptyset$; the least admissible $t$ is the stated maximum. Iterating $m$ times gives the claim for $\mathcal{R}F$. □

**Lemma 2.2** (Properties of $R_j$)**.** *Let $F,G$ be continuous convex functions on $\Delta$ and $j\in\{1,\dots,m\}$.*

*(i) $R_jF$ is continuous and convex on $\Delta$, and $\int_\Delta R_jF = \int_\Delta F$.*

*(ii) If $F\le G$ then $R_jF\le R_jG$; $R_j(F+c)=R_jF+c$ and $R_j(\alpha F)=\alpha R_jF$ for constants $c\in\mathbb{R}$, $\alpha\ge 0$; $\|R_jF-R_jG\|\le\|F-G\|$.*

*(iii) For each $y$, $x\mapsto(R_jF)(y,x)$ is nondecreasing, $(R_jF)(y,0)=\min_{[0,L(y)]}F(y,\cdot)$ and $(R_jF)(y,L(y))=\max_{[0,L(y)]}F(y,\cdot)$; in particular $\max_\Delta R_jF=\max_\Delta F$.*

*(iv) If $F$ is nondecreasing in $p_i$ for some $i\neq j$, so is $R_jF$. Consequently $\mathcal{R}F$ is nondecreasing in every coordinate, and $\mathcal{R}F(0)=\min_\Delta\mathcal{R}F$.*

*(v) Let $\Delta$ be rational. If $F$ is rational convex polyhedral, so is $R_jF$. If $F=\max_{\iota\in I}\phi_\iota$, the epigraph of $R_jF$ over $\Delta$ is the projection to $(y,x,t)$ of the rational polyhedron in the variables $(y,x,t,a)$ defined by $p=(y,x)\in\Delta$, $a\ge 0$, $b_{\nu j}(a+x)+\sum_{i\neq j}b_{\nu i}y_i\le\ell_\nu$ for every $\nu$ with $b_{\nu j}>0$, and $t\ge\phi_\iota(y,a)$, $t\ge\phi_\iota(y,a+x)$ for every $\iota\in I$; eliminating $a$ by Fourier–Motzkin elimination gives finitely many rational affine functions whose maximum is $R_jF$.*

*Proof.* Its items (i)–(v) are, respectively, Paper I, Lemma (Properties of $R_j$)(ii),(iii); (iv); (i) with the last assertion of (iv); (v); and Lemma 2.1. □

2.3. **The finite fixed point.** For $0 < \lambda < 1$ and concave continuous $P$ on $\Delta$, let $\mathcal{C}$ be the set of continuous convex functions on $\Delta$, and put $T_\lambda(F) = \mathcal{R}(\lambda F - P)$. Its fixed point $K_\lambda$ is used as in [Ope26, §2.2].

**Proposition 2.3** (Finite comparator, detailed form)**.** *Let $\tau_\Delta \in (0,1)$ be rational and let $P$ be rational concave piecewise affine on $\Delta$ with mean $\mu = \mathrm{vol}(\Delta)^{-1}\int_\Delta P > 0$ and $P \geq \beta_P$ on $\Delta \setminus \tau_\Delta\Delta$ for some $\beta_P > 0$. Let $d \in \mathbb{Q}$ with $0 < d < \min(\mu, \beta_P)$, and put*

$$\eta_\Delta = (1-\tau_\Delta)(\mu - d), \qquad M_P = \max\bigl(1,\ d - \min_\Delta P + 1\bigr), \qquad \lambda = 1 - \frac{\eta_\Delta}{2(M_P + \eta_\Delta)}.$$

*(a) $\mu$, $\min_\Delta P$, $\max_\Delta P$ and $\|P\|$ are rational and are computed from $(\Delta, P)$ by finitely many rational operations; hence so are $\eta_\Delta$, $M_P$ and $\lambda$, and $\frac{1}{2} < \lambda < 1$, $\lambda\eta_\Delta/(1-\lambda) = 2M_P + \eta_\Delta > d - \min_\Delta P$.*

*(b) $T_\lambda$ maps $\mathcal{C}$ to itself and is a contraction with constant $\lambda$ for the uniform norm. Its unique fixed point $K_\lambda$ is convex, nondecreasing in every coordinate, has mean $-\mu/(1-\lambda)$, and $\|K_\lambda\| \leq \|P\|/(1-\lambda)$.*

*(c) $(1-\lambda)\max_\Delta K_\lambda < -d$, and $\mathcal{R}(K_\lambda - P) \leq K_\lambda - d$ on $\Delta$.*

*(d) The iterates $K_0 = 0$, $K_{r+1} = T_\lambda(K_r)$ are rational convex polyhedral and satisfy $\|K_r - K_\lambda\| \leq \lambda^r\|P\|/(1-\lambda)$. Each $K_{r+1}$ is obtained from $K_r$ by $m$ Fourier–Motzkin eliminations (Lemma 2.1).*

*(e) Let $N_0$ be the least integer $N_0 \geq 0$ with $\lambda^{N_0}\|P\|/(1-\lambda) < d/16$. For every $N \geq N_0$,*

$$N_0 \leq 1 + \frac{1}{1-\lambda}\log^+\frac{16\|P\|}{d(1-\lambda)} \qquad \text{and} \qquad \mathcal{R}(K_N - P) \leq K_N - \tfrac{7}{8}d \quad \text{on } \Delta.$$

*Proof.* (a) Write $G = -P = \max_{u\in U}(-\psi_u)$.

*Extrema.* For rational convex polyhedral $H$, Paper I, Lemma (Properties of $R_j$)(iv),(v) gives $\min_\Delta H = (\mathcal{R}H)(0)$, a maximum of finitely many rational affine values by Lemma 2.1. With $\max_\Delta \phi = -\min_\Delta(-\phi)$ for affine $\phi$, this yields rational $\min_\Delta P$, $\max_\Delta P$ and $\|P\|$.

*Integrals.* Let $D \subset \mathbb{R}^r$ be cut out by rational halfspaces, including those of a cube $[-M, M]^r$. For $\Delta$ take $M = 1 + \sum_j \sum_{\nu:\, b_{\nu j} > 0} \ell_\nu / b_{\nu j}$. We integrate a rational polynomial $f$ over $D$ by induction on $r$.

- Write $p = (x, t)$. The inequalities give upper bounds $t \leq u_\alpha(x)$ and lower bounds $t \geq l_\beta(x)$ (at least one of each, from the cube), plus constraints on $x$. Remove repetitions, so that the affine functions $u_\alpha$ are pairwise distinct, and so are the $l_\beta$; index both lists by integers.
- Let $D_{\alpha\beta} \subset \mathbb{R}^{r-1}$ be defined by the $x$-constraints, the projected cube, $l_\beta \leq u_\alpha$, and the tie-broken conditions

  $$u_\alpha \leq u_{\alpha'}\ (\alpha' > \alpha), \quad u_\alpha < u_{\alpha'}\ (\alpha' < \alpha), \qquad l_{\beta'} \leq l_\beta\ (\beta' > \beta), \quad l_{\beta'} < l_\beta\ (\beta' < \beta).$$

- For each $x$ in the projection of $D$ there is exactly one $\alpha$ such that $u_\alpha(x)$ is the minimum of the upper bounds and no $u_{\alpha'}$ with $\alpha' < \alpha$ attains it, namely the least index attaining the minimum; likewise for $\beta$. Hence the $D_{\alpha\beta}$ partition the projection of $D$, and over $D_{\alpha\beta}$ the fibre of $D$ is $[l_\beta, u_\alpha]$. Each $D_{\alpha\beta}$ differs from the polyhedron obtained by replacing every strict inequality by the non-strict one only within finitely many sets $\{u_\alpha = u_{\alpha'}\}$, $\{l_\beta = l_{\beta'}\}$ with distinct affine functions, which are contained in hyperplanes or empty (two distinct affine functions that differ by a nonzero constant never agree), hence null. So the integral over $D_{\alpha\beta}$ equals that over the closed polyhedron, to which the induction applies.

With $f = \sum_k f_k(x)t^k$, this gives

$$\int_D f = \sum_{\alpha,\beta} \int_{D_{\alpha\beta}} g_{\alpha\beta}, \qquad g_{\alpha\beta}(x) = \sum_{k=0}^{n} f_k(x)\, \frac{u_\alpha(x)^{k+1} - l_\beta(x)^{k+1}}{k+1}.$$

Induction on $r$ computes $\int_D f$ exactly, also for empty or degenerate $D$.

*Mean.* Take rational $R = 1 + \max_u\big(|\psi_u(0)| + M\sum_j |\partial_j \psi_u|\big) > |G|$ and $E = \{(p,t) : p \in \Delta,\ G(p) \le t \le R\}$. Then $\int_\Delta P = \mathrm{vol}(E) - R\,\mathrm{vol}(\Delta)$, so $\mu \in \mathbb{Q}$.

Since $d < \mu$, $\eta_\Delta > 0$; since $M_P \ge 1$, $1 - \lambda \in (0, \frac{1}{2})$. A direct computation gives $\lambda\eta_\Delta/(1-\lambda) = 2M_P + \eta_\Delta > d - \min_\Delta P$.

(b) $\mathcal{C}$ is complete, and $T_\lambda(\mathcal{C}) \subset \mathcal{C}$ is a $\lambda$-contraction by Paper I, Lemma (Properties of $R_j$)(ii),(iv). Monotonicity of $K_\lambda$ follows from (v). Integrating $K_\lambda = \mathcal{R}(\lambda K_\lambda - P)$ with (iii) gives the mean. The bound $\|K_\lambda\| \le \lambda\|K_\lambda\| + \|P\|$ uses $\mathcal{R}0 = 0$.

(c) Let $M_\lambda = \max_\Delta K_\lambda$ and suppose $\lambda M_\lambda \le M_\lambda + d$. We show $\lambda K_\lambda - P < M_\lambda$ on $\Delta$. On $\Delta \setminus \tau_\Delta\Delta$ this follows from $P \ge \beta_P > d$. If $p = \tau_\Delta v$ with $v \in \Delta$: by (b), $K_\lambda(0) = \min K_\lambda \le -\mu/(1-\lambda)$, and convexity on $[0, v]$ gives

$$K_\lambda(p) \le M_\lambda - (1-\tau_\Delta)\big(M_\lambda - K_\lambda(0)\big) \le M_\lambda - \frac{\eta_\Delta}{1-\lambda}.$$

By (a), $\lambda K_\lambda(p) - P(p) \le M_\lambda + d - \min_\Delta P - (2M_P + \eta_\Delta) < M_\lambda$.

But $\max \mathcal{R}G = \max G$ forces $M_\lambda = \max(\lambda K_\lambda - P)$, a contradiction. So $(1-\lambda)K_\lambda < -d$, hence $K_\lambda - P \le \lambda K_\lambda - P - d$. Then Paper I, Lemma (Properties of $R_j$)(iv) gives $\mathcal{R}(K_\lambda - P) \le K_\lambda - d$.

(d) Induction with $m$ applications of Lemma 2.1; the bound follows from the contraction and (b).

(e) Put $n_0 = (1-\lambda)^{-1}\log^+\big(16\|P\|/(d(1-\lambda))\big)$. For integers $n > n_0$, $\lambda^n \le e^{-(1-\lambda)n} < d(1-\lambda)/(16\|P\|)$, so $N_0 \le \lfloor n_0 \rfloor + 1$. With $\delta_N = \|K_N - K_\lambda\| < d/16$, (c) and Paper I, Lemma (Properties of $R_j$)(iv) give

$$\mathcal{R}(K_N - P) \le K_\lambda - d + \delta_N \le K_N - d + 2\delta_N < K_N - \tfrac{7}{8}d. \qquad \square$$

The construction uses

$$N = \left\lceil \frac{16\|P\|}{d(1-\lambda)(\lambda^{-1}-1)} \right\rceil + 1.$$

By Bernoulli, $\lambda^{-N} \ge 1 + N(\lambda^{-1}-1) > 16\|P\|/[d(1-\lambda)]$, so $N \ge N_0$; any rational $\bar{P} \ge \|P\|$ may replace $\|P\|$. The number $\#I$ of affine pieces of $K_N = \max_{\iota\in I}\phi_\iota$ is read off from the eliminations and enters only through $c_K$ in Lemma 2.4.

### 2.4. **Smoothing.**

**Lemma 2.4** (Log-sum-exp smoothing)**.** *Let $\phi_\iota$ ($\iota \in I$, $1 \le \#I < \infty$) be affine functions on $\mathbb{R}^m$, $c_K > 0$, and*

$$K = \frac{1}{c_K}\log\sum_{\iota\in I} e^{c_K\phi_\iota}.$$

*(a)* $\max_\iota \phi_\iota \le K \le \max_\iota \phi_\iota + \log(\#I)/c_K$ *on $\mathbb{R}^m$; if $\#I = 1$, $K$ is the single $\phi_\iota$.*
*(b) $K$ is real-analytic and convex on $\mathbb{R}^m$, and $K \circ p$ is real-analytic on $\mathbb{C}^m$.*
*(c)* $\nabla K = \sum_\iota \varpi_\iota \nabla\phi_\iota$ *with weights $\varpi_\iota = e^{c_K\phi_\iota}/\sum_{\iota'} e^{c_K\phi_{\iota'}} \in (0,1]$, $\sum_\iota \varpi_\iota = 1$; in particular $|\partial_i K| \le \max_\iota |\partial_i\phi_\iota|$, and $K$ is nondecreasing in $p_i$ if every $\phi_\iota$ is.*

*If $d > 0$ and $c_K$ is an integer with $c_K > 32\,\#I/d$, then* $\log(\#I)/c_K < d/32$.

*Proof.* (a) With $\phi_{\max} = \max_\iota \phi_\iota$, $e^{c_K\phi_{\max}} \le \sum_\iota e^{c_K\phi_\iota} \le \#I\, e^{c_K\phi_{\max}}$. (c) is direct differentiation. (b) $K$ is real-analytic as the log of a positive sum of exponentials of affine functions, and $K\circ p$ is real-analytic since $p_j = \pi|w_j|^2$ is polynomial. Differentiating (c), the Hessian $\partial_a\partial_b K$ is $c_K$ times the covariance matrix of $(\nabla\phi_\iota)$ under the weights $\varpi_\iota$, hence positive semidefinite. Finally $\log(\#I)/c_K < \#I/c_K < d/32$. □

2.5. **Composition of coordinate realisations.**

**Lemma 2.5** (Composition of coordinate realisations)**.** *Let $F_0$ be continuous and convex on $\Delta$, $F_j = R_jF_{j-1}$ for $1\le j\le m$, and $e_1,\dots,e_m \ge 0$. For each $j$ let $Q^{(j)}\colon [0,1]\times\mathbb{C}^m\to\mathbb{R}$ be smooth (with one-sided derivatives in $s$ at $s=0,1$), and let $(q_s^{(j)})_{s\in[0,1]}$ be a smooth isotopy of $\mathbb{C}^m$ (smooth in $(s,w)$ in the same sense) generated by $Q_s^{(j)}$, that is, $\partial_s q_s^{(j)} = X_s^{(j)}\circ q_s^{(j)}$ with $\iota_{X_s^{(j)}}\omega_V = -dQ_s^{(j)}$, with $q_0^{(j)} = \mathrm{id}$ and $q_s^{(j)}(V_\Delta) = V_\Delta$. Assume:*

*(1) there is $\varsigma_j > 0$ such that $\{Q_s^{(j)}, \langle b_\nu, p\rangle\} = 0$ on $p^{-1}\{\langle b_\nu,p\rangle > \ell_\nu - \varsigma_j\}$ for every $\nu$ and $s$;*
*(2) $F_{j-1}\big(p(q_1^{(j)}(w))\big) \le F_j\big(p(w)\big) + e_j$ for every $w\in V_\Delta$.*

*For $t\in[\frac{r-1}{m},\frac{r}{m}]$, $r=1,\dots,m$, and $j = m+1-r$, put, with $S$ of Definition A.1,*

$$q_t = q^{(j)}_{S(mt-r+1)}\circ q_1^{(j+1)}\circ\cdots\circ q_1^{(m)}.$$

*Then:*

*(a) $q_t$ is well defined, $q_0 = \mathrm{id}$, $q_1 = q_1^{(1)}\circ\cdots\circ q_1^{(m)}$, $q_t(V_\Delta) = V_\Delta$, and $(q_t)$ is a smooth isotopy of $\mathbb{C}^m$ generated by the smooth function $Q_t = mS'(mt-r+1)\,Q^{(j)}_{S(mt-r+1)}$ on $[\frac{r-1}{m},\frac{r}{m}]$; $Q_t = 0$ at $t = r/m$;*
*(b) with $\varsigma = \min_j\varsigma_j$, $\{Q_t, \langle b_\nu,p\rangle\} = 0$ on $p^{-1}\{\langle b_\nu,p\rangle > \ell_\nu - \varsigma\}$ for every $\nu$ and $t$;*
*(c) $F_0\big(p(q_1(w))\big) \le (\mathcal{R}F_0)\big(p(w)\big) + \sum_j e_j$ for every $w\in V_\Delta$.*

*Proof.* (a) Paper I, Lemma (Flat concatenation)(b) with $X=\mathbb{C}^m$, $N=m$, $t_r = r/m$, $\psi^{(r)} = q^{(m+1-r)}$ and clocks $\sigma_r(t) = S(mt-r+1)$ realises $q_t$ as the concatenation and gives the claims on $q_0$, $q_1$, smoothness and the generator $\sigma_r'(t)Q^{(j)}_{\sigma_r(t)}$, which vanishes at the seams. Each factor preserves $V_\Delta$, hence so does $q_t$.

(b) $Q_t$ is a multiple of $Q^{(j)}_{S(mt-r+1)}$, and $\varsigma\le\varsigma_j$.

(c) For $w\in V_\Delta$ put $u_m = w$ and $u_{j-1} = q_1^{(j)}(u_j)\in V_\Delta$, so $u_0 = q_1(w)$. Chaining (2) over $j$ gives the claim, since $F_m = \mathcal{R}F_0$. □

In the proof of Theorem M2.11, Lemma 2.5 applies with $F_0 = K_N - P$, the isotopies $q^{(j)}$ of Theorem M2.10 (for $F_{j-1}$, coordinate $j$) and $e_j = d/(8(m+1))$: these are smooth Hamiltonian isotopies with smooth compactly supported generators, (2) is Theorem M2.10(c), and (1) holds with the collar $\varsigma_j$ of Theorem M2.10(b) (a function vanishing on an open set has zero bracket there; for $b_{\nu j}=0$ the bracket vanishes identically). Parts (a),(b) give smoothness across the seams and condition (1) of Definition M2.4; part (c) with Proposition 2.3(e) gives $(K_N-P)(p(q_1(w))) \le K_N(p(w)) - \frac34 d$.

2.6. **The general comparator.** Fibre Hamiltonians follow $\iota_{X_Q}\omega_V = -dQ$; $\|\cdot\|$ is the uniform norm on $\Delta$. Cutoffs and clocks use the flat step $S$ of Definition A.1, all of whose derivatives vanish at 0 and 1.

For $0<\lambda<1$ let $T_\lambda(F) = \mathcal{R}(\lambda F - P)$; for concave $P$ it is a $\lambda$-contraction on continuous convex functions by Lemma 2.2(ii).

*Proof of Proposition M2.9.* The data are those of Proposition 2.3, and $K_r$ are its iterates, so they are rational convex polyhedral by part (d). Since $\lambda^{-N} = (1+(\lambda^{-1}-1))^N > N(\lambda^{-1}-1) \geq 16\|P\|/(d(1-\lambda))$, we have $\lambda^N\|P\|/(1-\lambda) < d/16$; hence $N \geq N_0$, and part (e) gives the inequality. □

Everything holds verbatim with real coefficients ("convex polyhedral"), including the statements below; this version is used for real capacities.

Theorem M2.10 is proved in §3 (Theorem 3.13). The construction is planar: sublevel sets of a smoothed convex potential in $(v,\theta)$ are carried, via the chart $(v,\theta) \mapsto -\sqrt{v/\pi}\, e^{2\pi i\theta}$, onto centred discs (Theorems A.12 and A.13), and the planar generator is lifted fibrewise in $w_j$ with a cutoff in $y$; in (b) $\varsigma_j = 2^{-N_j}$ for the least $N_j$ below the collar width. The equal-ball instance of the planar construction is Paper I, § "The explicit inner map for equal balls".

*Proof of Theorem M2.11.* Lemma 2.5 applies, as explained after its proof. Its parts (a) and (b) give a smooth Hamiltonian isotopy with $q_0 = \mathrm{id}$, $q_t(V_\Delta) = V_\Delta$ and condition (1) of Definition M2.4. Each $F_j$ is rational convex polyhedral, and $F_m = \mathcal{R}(K_N - P) \leq K_N - \frac{7}{8}d$ (Proposition M2.9), so part (c) with $\sum_j e_j < \frac{1}{8}d$ gives $(K_N - P)(p(q_1(w))) \leq K_N(p(w)) - \frac{3}{4}d$ for $w \in V_\Delta$. As $K_N \leq K < K_N + d/32$ (Lemma 2.4), $(K-P)(p(q_1(w))) < K(p(w)) - \frac{11}{16}d$, which is condition (2). Finally $K$ is real-analytic. □

2.7. **One fibre coordinate.** For $m = 1$ (dimension four), $\Delta = [0,h]$ with rational $h > 0$, $L \equiv h$, and $R_1$ is the increasing rearrangement of Lemma A.6. $R_1$ does not increase slice Lipschitz constants (used in the planar realisation of §3 to convert a level error into a value error), and the elimination of Lemma 2.1 can be replaced by inverting a distribution function.

**Lemma 2.6** (Lipschitz bounds)**.** *Let $L \geq 0$, let $f$ be continuous and convex on $[0,L]$ with $|f(v) - f(v')| \leq \Lambda_f |v - v'|$, and let $f^\sharp$ be as in Lemma A.6. Then $0 \leq f^\sharp(x+b) - f^\sharp(x) \leq \Lambda_f b$ whenever $0 \leq x \leq x+b \leq L$. Consequently, for every $m$, if $F$ is continuous and convex on $\Delta$ and $F(y,\cdot)$ is $\Lambda_f$-Lipschitz on every slice, then so is $(R_jF)(y,\cdot)$; if $F = \max_\iota \phi_\iota$, one may take $\Lambda_f = \max_\iota |\partial_j \phi_\iota|$.*

*Proof.* The lower bound is Lemma A.6(c). Let $[a, a+x]$ be minimising for $x$; then $a' = \min(a, L - x - b)$ is admissible for $x+b$. If $a' < a$, then $a - a' = a + x + b - L \leq b$ and $a' + x + b = L \geq a + x$; in either case $|a' - a| \leq b$ and $0 \leq (a' + x + b) - (a + x) \leq b$, so $f^\sharp(x+b) \leq \max(f(a'), f(a'+x+b)) \leq f^\sharp(x) + \Lambda_f b$. The statement for $R_j$ follows from Paper I, Lemma (Properties of $R_j$)(i); for $F = \max_\iota \phi_\iota$ each slice is a maximum of affine functions of slopes $\partial_j \phi_\iota$. □

**Lemma 2.7** (Rearrangement of a piecewise affine function)**.** *Let $f$ be continuous, convex and piecewise affine on $[0,h]$, with knots $0 = \xi_0 < \xi_1 < \cdots < \xi_n = h$ and rational knots and values. Put $f_{\min} = \min f$. Then $\mathcal{L}_f$ is continuous, piecewise affine and strictly increasing on $[f_{\min}, \max f]$, with $\mathcal{L}_f(\max f) = h$ and knots among the values $f(\xi_i)$, and*

$$R_1 f(x) = f_{\min} \quad (0 \leq x \leq \mathcal{L}_f(f_{\min})), \qquad R_1 f(x) = \mathcal{L}_f^{-1}(x) \quad (\mathcal{L}_f(f_{\min}) \leq x \leq h).$$

*$R_1 f$ is piecewise affine with rational values and with knots among $0$, $h$ and the rational numbers $\mathcal{L}_f(f(\xi_i))$, and it is convex by Paper I, Lemma (Properties of $R_j$)(ii).*

*Proof.* Put $w_i = \xi_{i+1} - \xi_i$, $v_i = f(\xi_i)$, $s_i = (v_{i+1} - v_i)/w_i$, $c_i = v_i - s_i\xi_i$, and let $d_i(b)$ be the length of $\{v \in [\xi_i, \xi_{i+1}] : s_i v + c_i \leq b\}$: for $s_i > 0$ it is $\max(0, \min(\xi_{i+1}, (b-c_i)/s_i) - \xi_i)$, for $s_i < 0$ it is $\max(0, \xi_{i+1} - \max(\xi_i, (b-c_i)/s_i))$, and for $s_i = 0$ it is $w_i \mathbf{1}_{\{b \geq c_i\}}$. Since overlaps are endpoints, $\mathcal{L}_f = \sum_i d_i$, with rational constants. By convexity a segment with $s_i = 0$ lies at level $f_{\min}$, contributing the constant $w_i$ on $[f_{\min}, \max f]$; every other contribution is continuous and

affine between $v_i$ and $v_{i+1}$. Hence $\mathcal{L}_f$ is continuous, piecewise affine with knots among the $v_i$, and the $\mathcal{L}_f(v_i)$ are rational.

For strict monotonicity, let $f_{\min} \le b < b' \le \max f$ and pick $v_0$ with $f(v_0) = \frac{1}{2}(b+b')$. Since $\{f \le b\} \ne \emptyset$ and $\{f \le b'\}$ are closed intervals and $v_0 \in \{f \le b'\} \setminus \{f \le b\}$, the convex hull of $\{f \le b\} \cup \{v_0\}$ lies in $\{f \le b'\}$ and is strictly longer than $\{f \le b\}$. Also $\mathcal{L}_f(\max f) = h$.

Lemma A.6(a) gives the formula for $R_1 f$. Between consecutive $v_i$, $\mathcal{L}_f(b) = C + \Sigma b$ with rational $C$ and $\Sigma > 0$, with inverse $x \mapsto (x - C)/\Sigma$; this gives the knots and rationality. Convexity is Paper I, Lemma (Properties of $R_j$)(ii). □

*Remark* 2.8 (An unequal input without linear comparison function). In the cubic model (Paper I, § "Dimension four: the cubic model"), given capacities $\rho_i$ are realised through auxiliary capacities $a_i > \rho_i$ and a height $h$ with $a_i < h < \frac{1}{3}$ (Lemma 5.1); the profile is $P(p) = 3(1-3p) - \sum_i (a_i - p)_+$ on $\Delta = [0, h]$, and by Theorem M2.6 any continuous minorant of $P$ may replace it. For $K(p) = \kappa p$ put $F = K - P = (\kappa + 9)p - 3 + \sum_i (a_i - p)_+$. If $(K, (q_t))$ is a comparison datum for $(\Delta, P)$ with gap $\gamma > 0$, then, evaluating condition (2) of Definition M2.4 at $w = 0$ and at a preimage under $q_1$ of a maximum point of $F$ (the map $q_1$ sends $\bar{D}(h)$ onto itself),

$$\min_{[0,h]} F < K(0) = 0, \qquad \max_{[0,h]} F < \max_{[0,h]} K. \tag{1}$$

Replacing $P$ by a smaller function increases $F$, so both inequalities only become harder to satisfy, and what follows applies to every minorant.

Consider first the limiting data $a^*$: nine capacities $8/25$ and one capacity $7/25$, so that $\sum_i a_i^2 = 1$, and $h = \frac{1}{3}$, where $\int_0^h P = \frac{1}{2}[1 - \sum_i a_i^2 - (1-3h)^2] = 0$. Then $F(0) = \sum_i a_i - 3 = \frac{4}{25}$, and the slopes of $F$ on $[0, \frac{7}{25}]$, $[\frac{7}{25}, \frac{8}{25}]$, $[\frac{8}{25}, \frac{1}{3}]$ are $\kappa - 1$, $\kappa$, $\kappa + 9$. If $\kappa < 0$, then $\max F \ge F(0) > 0 = \max K$. If $\kappa \ge 1$, then $\min F = F(0) = \frac{4}{25} > 0$. If $0 \le \kappa < 1$, then $\min F = F(\frac{7}{25}) = \frac{7\kappa - 3}{25}$ and $\max F - \max K \ge \frac{4}{25} - \frac{\kappa}{3}$; the larger of these two numbers is at least $\frac{9}{1150}$ (equality at $\kappa = \frac{21}{46}$). So for every $\kappa \in \mathbb{R}$ one of the inequalities (1) fails by at least $\frac{9}{1150}$.

The bound survives perturbation. Let $\eta_0 = 1/2000$, $|a_i - a_i^*| \le \eta_0$ and $\frac{1}{3} - \eta_0 \le h < \frac{1}{3}$. Then $F$ differs from its limiting value by at most $10\eta_0$ at every $p \in [0, h]$, uniformly in $\kappa$; minima over $[0, h] \subset [0, \frac{1}{3}]$ are at least the limiting minima, $F(0)$ is still among the values, and $\max_{[0,h]} K = \kappa h \le \kappa/3$ for $\kappa \ge 0$. The three cases therefore hold with every bound lowered by $10\eta_0 = \frac{1}{200} < \frac{9}{1150}$, and (1) fails again. Such auxiliary data are forced by the given capacities. A comparison datum forces $\int_0^h P > 0$ (integrate condition (2) of Definition M2.4 over $\bar{D}(h)$, using that $q_1$ preserves area), i.e. $\sum_i a_i^2 + (1-3h)^2 < 1$. With $D = 1 - \sum_i \rho_i^2$ and $a_i > \rho_i$ this gives $a_i^2 - \rho_i^2 < D$, hence $a_i - \rho_i < D/(2\rho_i)$, and $(1-3h)^2 < D$, hence $\frac{1}{3} - h < \sqrt{D}/3$. Therefore, if $|\rho_i - a_i^*| \le \eta_0/2$ for all $i$ (so $\rho_i > \frac{1}{4}$) and $D \le 9\eta_0^2$, every admissible choice of $(a, h)$ satisfies $|a_i - a_i^*| < \eta_0$ and $h > \frac{1}{3} - \eta_0$, and no linear $K$ serves; for instance $\rho_i = (1 - 10^{-7})a_i^*$. The scalar construction of Proposition 4.8 applies to all of these inputs.

## 3. Rearranging convex functions by explicit isotopies

This section proves Theorem M2.10: one coordinate rearrangement $R_j$ of a convex polyhedral function with real coefficients is realised, up to an arbitrarily small error, by an explicit Hamiltonian isotopy of the toric domain $V_\Delta$ (Theorem 3.13); Theorem 3.7 is the planar core. On one slice, rescaled to $[0, 1]$, $f$ is replaced by a proper strictly convex potential $U$ and the angle by a wall $V$, and the construction of Paper I, § "Area-preserving disc maps from Morse charts" turns this pair into a planar isotopy; the only estimate specific to convex $f$ is the area estimate (Lemma 3.5). All constants are uniform in the other moments (Theorem 3.9), and the family is lifted to $\mathbb{C}^m$ by the rescaling of [Ope26, §2.3] (Definition 3.10). There the planar maps come

from an existence lemma on nested discs (compare [Sch99, Lemma 3.2]); here they are given by closed formulas, jointly smooth in parameter, time and point.

Fibre Hamiltonians follow $\iota_{X_Q}\omega = -dQ$ (Paper I, § "Conventions"). We use the conventions of Paper I, § "Area-preserving disc maps from Morse charts": $D(A) = \{z \in \mathbb{C} : \pi|z|^2 < A\}$, $\bar{D}(A)$ its closure, and the standard area form on every plane. The notation of Paper I, § "Coordinate rearrangement and the equal-ball identities" is in force: $\Delta = \{p \in \mathbb{R}^m_{\geq 0} : \langle b_\nu, p\rangle \leq \ell_\nu\}$, the toric domain $V_\Delta = \{w \in \mathbb{C}^m : p(w) \in \Delta\}$ with moments $p(w) = (\pi|w_i|^2)_i$ and form $\omega_V = \sum_i dp_i \wedge d\theta_i$, the sum of the standard area forms of the coordinate planes, the slice lengths $L_\nu$, $L$, the projection $\Delta_j$, the coordinate rearrangement $R_j$ (Definition M2.8) and the slice description of §M2.3.

### 3.1. **Planar data and potentials.**

Planar data. Let $d \geq 0$ and let $\mathcal{O} \subset \mathbb{R}^d$ be open ($d = 0$ means that $\mathcal{O}$ is one point and the parameter is absent). Let $\mathcal{I}$ be a finite nonempty index set and $a_r, b_r \in C^\infty(\mathcal{O})$, $r \in \mathcal{I}$, and put

$$f_y(p) = \max_{r\in\mathcal{I}}\big(a_r(y) + b_r(y)p\big) \qquad (y \in \mathcal{O},\ p \in \mathbb{R}). \tag{2}$$

Let $\Lambda \geq 1$ and $M_0$ be real numbers with

$$|b_r(y)| \leq \Lambda \quad \text{and} \quad |a_r(y)| + |b_r(y)| \leq M_0 \qquad (r \in \mathcal{I},\ y \in \mathcal{O}).$$

We call $(\mathcal{O}, (a_r, b_r)_{r\in\mathcal{I}}, \Lambda, M_0)$ *planar data*. The second bound implies $|f_y| \leq M_0$ on $[0,1]$ and $|b_r(y)| \leq M_0$. For $y \in \mathcal{O}$, $g_y$ denotes the increasing rearrangement of $f_y|_{[0,1]}$,

$$g_y(x) = \min_{0\leq a\leq 1-x} \max\big(f_y(a), f_y(a+x)\big) \qquad (0 \leq x \leq 1),$$

which is nondecreasing with $g_y(0) = \min_{[0,1]} f_y$ and $g_y(1) = \max_{[0,1]} f_y$ (Lemma A.6). Since $f_y$ is $\Lambda$-Lipschitz on $[0,1]$, Lemma 2.6 gives

$$0 \leq g_y(x') - g_y(x) \leq \Lambda(x' - x) \qquad (0 \leq x \leq x' \leq 1). \tag{3}$$

Below, $\eta$ is the width of the sacrificed boundary layer of $[0,1]$, $\beta_0$ that of the angular strips, $\sigma_0$ the size of each smoothing error, $\kappa_0$ and $\nu_0$ the strengths of barrier and wall, and $r_0$ brackets the minimum point of $U_y$.

**Definition 3.1** (Planar constants and potentials)**.** Given planar data and a real $\xi > 0$, put

$$\eta = \frac{1}{2}\min\Big(\frac{1}{16}, \frac{\xi}{32(\Lambda+1)}\Big), \quad \beta_0 = \frac{\eta}{8}, \quad \sigma_0 = \frac{\xi}{8}, \quad \nu_0 = 4\sigma_0\beta_0(1-\beta_0), \quad \kappa_0 = \frac{\sigma_0\eta}{8},$$
$$c_0 = \frac{8(\#\mathcal{I}+1)}{\sigma_0}, \qquad r_0 = \min\Big(\frac{1}{8}, \sqrt{\frac{\kappa_0}{4(\Lambda+2\kappa_0+1)}}\Big), \qquad \Lambda_2 = c_0\Lambda^2,$$

and define, for $y \in \mathcal{O}$,

$$\tilde{f}_y(p) = \frac{1}{c_0}\log\sum_{r\in\mathcal{I}} \exp\big(c_0(a_r(y) + b_r(y)p)\big) \quad (p \in \mathbb{R}),$$
$$U_y(p) = \tilde{f}_y(p) + \kappa_0\Big(\frac{1}{p} + \frac{1}{1-p}\Big) \quad (0 < p < 1),$$
$$V(\theta) = \frac{\nu_0\theta^2}{\frac{1}{4} - \theta^2} \quad \big(-\tfrac{1}{2} < \theta < \tfrac{1}{2}\big).$$

The barrier makes $U_y$ proper on $(0,1)$ even where $f_y$ is affine; the wall $V$ does the same in the angle.

**Lemma 3.2** (The potentials)**.** *For planar data and $\xi > 0$, with the constants of Definition 3.1:*

(a) *$\tilde f_y(p)$ is $C^\infty$ in $(y,p)\in\mathcal O\times\mathbb R$ and real-analytic and convex in $p$; $U$ is $C^\infty$ on $\mathcal O\times(0,1)$ and real-analytic in $p$; $V$ is even and real-analytic on $(-\frac12,\frac12)$ with $V(0)=0$.*

(b) *(Approximation.) $f_y\le\tilde f_y<f_y+\sigma_0/4$ on $\mathbb R$; $U_y>f_y$ on $(0,1)$; $0\le U_y-f_y<\sigma_0/2$ on $[\eta,1-\eta]$; $0\le V\le\sigma_0$ on $[-\frac12+\beta_0,\frac12-\beta_0]$.*

(c) *(Convexity and properness.) $|\tilde f_y'|\le\Lambda$ and $0\le\tilde f_y''\le\Lambda_2$ on $\mathbb R$; $U_y''\ge32\kappa_0$ on $(0,1)$; $U_y(p)\to+\infty$ as $p\downarrow0$ or $p\uparrow1$; $V''>0$ and $V(\theta)\to+\infty$ as $|\theta|\uparrow\frac12$.*

(d) *(Minimum.) The minimum point $p^*(y)$ of $U_y$ satisfies $r_0<p^*(y)<1-r_0$; with $U_{\min}(y)=U_y(p^*(y))$,*

$$32\kappa_0\le U_y''(p^*(y))\le\Lambda_2+\frac{4\kappa_0}{r_0^3},\qquad a_{\min}:=\Big(\frac{2}{\Lambda_2+4\kappa_0/r_0^3}\Big)^{1/2}\le a_0(y)\le a_{\max}:=\frac{1}{4\sqrt{\kappa_0}},$$

*where $a_0(y)=\sqrt{2/U_y''(p^*(y))}$; and $b_0=\sqrt{2/V''(0)}=1/(2\sqrt{\nu_0})$.*

(e) *(Angular inverse.) The inverse $Y$ of the signed square root of $V$ is $Y(v)=v/(2\sqrt{v^2+\nu_0})$.*

(f) *(Energy.) Let $\delta_E=\eta/8$ and $E_{\rm box}=\Lambda+\sigma_0/4+16\kappa_0/\eta+\nu_0/(4\delta_E(1-\delta_E))$. For $E\ge E_{\rm box}$, the area $\mathcal A_y(E)$ of $\{(p,\theta)\in(0,1)\times(-\frac12,\frac12):U_y(p)-U_{\min}(y)+V(\theta)\le E\}$ is at least $A_{\min}=(1-2\delta_E)^2$, and $A_{\min}>1-2\eta$.*

(g) *(Margins.) Let $E>0$. If $0<p<1$ and $U_y(p)-U_{\min}(y)<E$, then $\min(p,1-p)>\kappa_0/(M_0+4\kappa_0+E)$. If $|\theta|<\frac12$ and $V(\theta)<E$, then $|\theta|<\frac12\sqrt{E/(E+\nu_0)}$.*

*Proof.* (a) For fixed $y$, $\tilde f_y$ is a log-sum-exp of affine functions, hence real-analytic and convex (Lemma 2.4(b)); in $(y,p)$ it is the composite of $(y,p)\mapsto(a_r(y)+b_r(y)p)_r$ with the real-analytic $x\mapsto c_0^{-1}\log\sum_r e^{c_0x_r}$. The barrier and $V$ are rational without poles on the stated intervals.

(b) By Lemma 2.4(a), $f_y\le\tilde f_y\le f_y+\log(\#\mathcal I)/c_0$, and $\log(\#\mathcal I)/c_0<\#\mathcal I/c_0<\sigma_0/4$. The barrier is positive, so $U_y>f_y$, and on $[\eta,1-\eta]$ it is at most $2\kappa_0/\eta=\sigma_0/4$. The function $V$ is nondecreasing in $|\theta|$; at $|\theta|=\frac12-\beta_0$, $\theta^2\le\frac14$ and $\frac14-\theta^2=(\frac12-|\theta|)(\frac12+|\theta|)=\beta_0(1-\beta_0)$, so $V\le\nu_0/(4\beta_0(1-\beta_0))=\sigma_0$.

(c) By Lemma 2.4(c), $\tilde f_y'=\sum_r\varpi_rb_r(y)$ with convex weights $\varpi_r$, so $|\tilde f_y'|\le\Lambda$, and $\tilde f_y''$ is $c_0$ times the variance of the slopes $b_r(y)$ under these weights (proof of Lemma 2.4(b)), which lies in $[0,c_0\Lambda^2]$. The barrier has second derivative $2\kappa_0(p^{-3}+(1-p)^{-3})$, convex and symmetric about $\frac12$ with minimum $32\kappa_0$ there; as $\tilde f_y$ is bounded on $[0,1]$, $U_y$ is proper. From $V(\theta)=\nu_0\big(\frac14(\frac14-\theta^2)^{-1}-1\big)$, $V''(\theta)=\nu_0(1+12\theta^2)/(8(\frac14-\theta^2)^3)>0$, and $V$ is proper. So $U_y'$ is strictly increasing and tends to $\mp\infty$ at 0 and 1; it has a unique zero $p^*(y)\in(0,1)$, changes sign there, and $U_y''(p^*)>0$. $V$ is even and strictly convex with $V(0)=0$, so $V'(\theta)\theta>0$ for $\theta\ne0$, and $V''(0)=8\nu_0>0$.

(d) Since $r_0<\frac14$ and $|\tilde f_y'|\le\Lambda$,

$$U_y'(r_0)\le\Lambda-\frac{\kappa_0}{r_0^2}+\frac{\kappa_0}{(1-r_0)^2}<\Lambda+2\kappa_0-\frac{\kappa_0}{r_0^2}<0,$$

because $\kappa_0/r_0^2\ge4(\Lambda+2\kappa_0+1)>\Lambda+2\kappa_0$; symmetrically $U_y'(1-r_0)>0$. Hence $r_0<p^*(y)<1-r_0$. The lower bound for $U_y''(p^*)$ is (c); the upper bound is $\tilde f_y''\le\Lambda_2$ plus $2\kappa_0((p^*)^{-3}+(1-p^*)^{-3})\le4\kappa_0/r_0^3$. The bounds for $a_0$ follow, and $V''(0)=8\nu_0$ gives $b_0$.

(e) $Y$ is odd, real-analytic and increasing, with limits $\pm\frac12$. With $\vartheta=Y(v)$, $\vartheta^2=v^2/(4(v^2+\nu_0))$ and $\frac14-\vartheta^2=\nu_0/(4(v^2+\nu_0))$, so $V(\vartheta)=v^2$, and $Y$ has the sign of $v$.

(f) On $Q_E=[\delta_E,1-\delta_E]\times[-\frac12+\delta_E,\frac12-\delta_E]$, by (b), $U_y(p)\le\max_{[0,1]}f_y+\sigma_0/4+2\kappa_0/\delta_E$, $U_{\min}(y)\ge\min_{[0,1]}f_y$, $\max_{[0,1]}f_y-\min_{[0,1]}f_y\le\Lambda$, and $V\le\nu_0/(4\delta_E(1-\delta_E))$ as in (b). So $U_y-U_{\min}(y)+V\le E_{\rm box}$ on $Q_E$, a set of area $A_{\min}=1-\eta/2+\eta^2/16>1-2\eta$.

(g) By (c), $|\tilde{f}_y'| \le \max_r |b_r(y)| \le M_0$, so $|\tilde{f}_y(\frac{1}{2}) - \tilde{f}_y(p)| \le M_0/2$ for $p \in (0,1)$; and $U_{\min}(y) \le U_y(\frac{1}{2}) = \tilde{f}_y(\frac{1}{2}) + 4\kappa_0$. If $U_y(p) - U_{\min}(y) < E$, then

$$\frac{\kappa_0}{p} < U_y(p) - \tilde{f}_y(p) < U_{\min}(y) + E - \tilde{f}_y(p) \le \tilde{f}_y(\tfrac{1}{2}) - \tilde{f}_y(p) + 4\kappa_0 + E \le M_0 + 4\kappa_0 + E,$$

and likewise for $1-p$. If $V(\theta) < E$, then $\nu_0\theta^2 < E(\frac{1}{4} - \theta^2)$, so $\theta^2 < E/(4(E+\nu_0))$. □

3.2. **Isotopy data.** Theorem A.13 needs an admissible potential pair and isotopy data (Definition A.7); for the convex potentials both are closed formulas in the planar constants. Here $r_B < 1$ bounds the support, and $K_{\min}$, $A_{\max}$ are auxiliary.

**Definition 3.3** (Modified wells and isotopy data of the convex pair)**.** Given planar data and $\xi > 0$, with the constants of Definition 3.1, let $\psi$ and $C_\psi$ be the flat cutoff and its derivative bound of §4.3, and put $\delta = \xi/64$, $\alpha_y = U_y''(p^*(y))$, $\alpha_{\min} = 32\kappa_0$, $\alpha_{\max} = c_0\Lambda^2 + 4\kappa_0/r_0^3$,

$$K_3 = 8c_0^2\Lambda^3 + 192\kappa_0/r_0^4 + 1, \qquad \epsilon_U = \min\left(\frac{r_0}{8}, \frac{\alpha_{\min}}{32K_3(1+C_\psi)}, \sqrt{\frac{\delta}{\alpha_{\max}}}\right),$$

$$g_y(x) = U_y(p^*(y) + x) - U_y(p^*(y)),$$

$$\widetilde{U}_y(x) = g_y(x) + \psi(x/\epsilon_U)\left(\frac{\alpha_y x^2}{2} - g_y(x)\right) \qquad (-p^*(y) < x < 1 - p^*(y)).$$

For the angle, apply (20) to $(f, A, B, c) = (V, -\frac{1}{2}, \frac{1}{2}, 0)$ with $\delta$ in place of $\delta_{\mathrm{w}}$: $r_V = \frac{1}{8}$, $\alpha_V = V''(0) = 8\nu_0$, $G_V = \psi(x/r_V)V$, $K_V = 1 + \max_x |G_V'''(x)|$,

$$\epsilon_V = \min\left(\frac{r_V}{4}, \frac{\alpha_V}{32K_V(1+C_\psi)}, \sqrt{\delta/\alpha_V}\right),$$

$$\widetilde{V}(x) = V(x) + \psi(x/\epsilon_V)\left(\frac{\alpha_V x^2}{2} - G_V(x)\right) \qquad (|x| < \tfrac{1}{2}).$$

The *modified convex pair* is $U_y(p) := \widetilde{U}_y(p - p^*(y))$ on $I_U = (0,1)$ and $V := \widetilde{V}$ on $I_V = (-\frac{1}{2}, \frac{1}{2})$, with disc area $h = 1$; $X_y$ and $Y$ denote its Morse inverses (Lemma A.9). Its slopes at zero are $a_0(y) = \sqrt{2/\alpha_y}$ and $b_0 = 1/(2\sqrt{\nu_0})$.

The isotopy data are $p^*_{\mathrm{lo}} = r_0$, $p^*_{\mathrm{hi}} = 1 - r_0$ and

$$\delta_E = \eta/8, \quad A_{\min} = (1-2\delta_E)^2, \quad p^\dagger = 1 - 5\eta, \quad e_q = 2\eta,$$
$$p_{\mathrm{on}} = 1 - 31\eta/4, \quad p_{\mathrm{zero}} = 1 - 61\eta/8,$$
$$E_{\mathrm{rect}} = 2M_0 + 2 + 2\kappa_0/\delta_E + \nu_0/(2\delta_E) + 2\delta,$$
$$a_{\min} = \sqrt{2/\alpha_{\max}}, \quad a_{\max} = \sqrt{2/\alpha_{\min}},$$
$$K_{\min} = \pi a_{\min} b_0,$$
$$\bar{E} = E_{\mathrm{rect}} + \frac{p^\dagger A_{\min}}{K_{\min}(A_{\min} - p^\dagger)} + 1, \quad E_c = \bar{E},$$
$$\bar{X} = a_{\max}\sqrt{\bar{E}} + 1, \qquad \bar{Y} = b_0\sqrt{\bar{E}} + \tfrac{1}{2},$$
$$\Delta P = \frac{\kappa_0}{2M_0 + 2 + 4\kappa_0 + \bar{E} + \delta},$$
$$\Delta\theta = \frac{1}{2}\left(1 - \sqrt{\frac{\bar{E}+\delta+1}{\bar{E}+\delta+1+\nu_0}}\right),$$
$$\delta_x = \bar{X}, \quad \delta_\theta = \bar{Y}, \quad \delta_s = \min\left(\frac{1}{4}, \frac{\Delta P}{2\bar{X}}, \frac{\Delta\theta}{2\bar{Y}}\right),$$
$$\lambda_0 = \min\left(\frac{1}{2}, \frac{r_0}{2\bar{X}}, \frac{1}{4\bar{Y}}\right),$$
$$P_{\min} = \tfrac{1}{4}\min(r_0, \Delta P), \quad P_{\max} = \tfrac{3}{2}, \quad M_C = \frac{1}{2\sqrt{\pi P_{\min}}} + 2\sqrt{\pi P_{\max}},$$
$$R^\dagger = \sqrt{p^\dagger/\pi}, \quad R_h = \sqrt{1/\pi}, \quad r_b^+ = \sqrt{(1-r_0)/\pi},$$
$$A_{\max} = \frac{\sqrt{a_{\max}/b_0}}{2\sqrt{\pi r_0}} + 2\sqrt{\pi}\sqrt{b_0/a_{\min}},$$
$$B_{\mathrm{lin}} = R^\dagger(1 + A_{\max}), \quad B^* = M_C\sqrt{\bar{X}^2 + \bar{Y}^2}, \quad R_q = \sqrt{(1 - \Delta P/2)/\pi},$$
$$c_{\mathrm{sh}} = \min\left(\frac{1}{4}, \frac{R_h - r_b^+}{8\max(R^\dagger, B_{\mathrm{lin}}, B^*)}\right),$$
$$R_{\mathrm{supp}} = \max(R^\dagger, r_b^+ + c_{\mathrm{sh}}B_{\mathrm{lin}}, r_b^+ + c_{\mathrm{sh}}B^*, R_q),$$
$$r_B = P_{\mathrm{supp}} = \pi R_{\mathrm{supp}}^2.$$

All entries are independent of $y$. Apart from $K_V$, the maximum of an explicit smooth compactly supported function, they are finite expressions in $\Lambda$, $M_0$, $\#\mathcal{I}$, $\xi$ and $C_\psi$.

**Lemma 3.4** (The modified convex pair and its isotopy data)**.** *For planar data and $\xi > 0$:*

*(a) $|\widetilde{U}_y(x) - g_y(x)| \le \delta$ and $|\widetilde{V} - V| \le \delta$; $\widetilde{U}_y(x) = \frac{1}{2}\alpha_y x^2$ for $|x| \le \epsilon_U$ and $\widetilde{V}(x) = \frac{1}{2}\alpha_V x^2$ for $|x| \le \epsilon_V$; $x\widetilde{U}_y'(x) > 0$ and $x\widetilde{V}'(x) > 0$ for $x \neq 0$; $\widetilde{U}_y \ge 0$, $\widetilde{V} \ge 0$.*

*(b) The modified convex pair is an admissible potential pair over $\mathcal{Y} = \mathcal{O}$ with $h = 1$, exactly quadratic near the minimum, and $r_0 < p^*(y) < 1 - r_0$.*

*(c) The numbers of Definition 3.3 are isotopy data for this pair (using the uniform-energy alternative in (D4)); in particular $P_{\mathrm{supp}} = r_B < 1$.*

*Proof.* (a) By Lemma 3.2(c),(d), $U_y$ is smooth and strictly convex on $(0,1)$, proper, with unique critical point $p^*(y) \in (r_0, 1-r_0)$ and $\alpha_{\min} \le \alpha_y \le \alpha_{\max}$. We apply Lemma 4.4 to $f = U_y$, $c = p^*(y)$, $\varepsilon = \epsilon_U$, $K = K_3$. First $2\epsilon_U \le r_0/4 < \min(p^*, 1-p^*)$. For $|x| \le 2\epsilon_U$ the point

$p = p^* + x$ lies in $[\frac{3}{4}r_0, 1 - \frac{3}{4}r_0]$. There $g_y'''(x) = \tilde{f}_y'''(p) + 6\kappa_0((1-p)^{-4} - p^{-4})$, and $\tilde{f}_y'''$ is $c_0^2$ times the third central moment of the slopes $b_r(y)$ under the weights of Lemma 2.4(c), so $|\tilde{f}_y'''| \le c_0^2(2\Lambda)^3$; and $6\kappa_0 \max(p^{-4}, (1-p)^{-4}) \le 6\kappa_0(4/(3r_0))^4 < 192\kappa_0/r_0^4$. Hence $|g_y'''| \le K_3$ there. Finally $\epsilon_U \le \alpha_{\min}/(32K_3(1+C_\psi)) \le \alpha_y/(32K_3(1+C_\psi))$ and $\epsilon_U \le \sqrt{\delta/\alpha_{\max}} \le \sqrt{\delta/\alpha_y}$. Lemma 4.4 gives the assertions for $\widetilde{U}_y$ (with error $\delta/24$). It applies again to $V$, which is smooth, even, strictly convex and proper with $V(0) = 0$ (Lemma 3.2(c)), with $\chi_V g_V = G_V$ and $\epsilon_V$ the width of (20). Nonnegativity follows from the sign of the derivatives and the value 0 at 0.

(b) $p^*$ is smooth on $\mathcal{O}$ by the implicit function theorem ($U_y'(p^*(y)) = 0$, $U_y'' > 0$), so $\alpha_y$, $(y,x) \mapsto g_y(x)$ and $(y,p) \mapsto \widetilde{U}_y(p - p^*(y))$ are smooth; $\widetilde{V}$ does not depend on $y$. Paper I, condition (P1) follows from (a): $U_y$ has its minimum at $p^*(y) \in (0,1)$ with second derivative $\alpha_y > 0$, and $V$ has minimum 0 at 0 with $V''(0) = 8\nu_0 > 0$. Paper I, condition (P2): near the ends of their intervals the modified wells equal the proper functions $g_y$ and $V$, and they are exactly quadratic on the $y$-independent neighbourhoods $|x| \le \epsilon_U$, $|x| \le \epsilon_V$, with coefficients $\frac{1}{2}\alpha_y$ and $4\nu_0$.

(c) Write $\mathcal{A}_y$, $\mathcal{A}_{y,s}$ for the area functions of Lemma A.10 for this pair. All numbers are positive; in particular $A_{\min} = 1 - \frac{\eta}{2} + \frac{\eta^2}{16} > 1 - 5\eta = p^\dagger > 0$ since $\eta \le \frac{1}{32}$.

*Area gate.* Let $Q_E = [\delta_E, 1-\delta_E] \times [-\frac{1}{2} + \delta_E, \frac{1}{2} - \delta_E]$. For $(p,\theta) \in Q_E$, $|\tilde{f}_y(p) - \tilde{f}_y(p^*)| \le M_0$ (since $|\tilde{f}_y'| \le M_0$) and the barrier at $p$ is at most $2\kappa_0/\delta_E$, so $g_y(p - p^*) = U_y(p) - U_{\min}(y) \le M_0 + 2\kappa_0/\delta_E$; and $V(\theta) \le \nu_0/(4\delta_E(1-\delta_E)) \le \nu_0/(2\delta_E)$. By (a), $\widetilde{U}_y(p-p^*) + \widetilde{V}(\theta) \le M_0 + 2\kappa_0/\delta_E + \nu_0/(2\delta_E) + 2\delta \le E_{\rm rect}$ on $Q_E$. By Lemma A.9(iv), $\mathcal{A}_y(E) \ge \operatorname{area} Q_E = A_{\min}$ for $E \ge E_{\rm rect}$. Since $j_0(y) = a_0(y)b_0 \ge a_{\min}b_0$, Lemma A.10(b) gives, for $s \in [0,1]$,

$$\mathcal{A}_{y,s}(\bar{E}) \ge \alpha(1-s)^2 + \beta s^2 \ge \frac{\alpha\beta}{\alpha+\beta}, \qquad \alpha = K_{\min}\bar{E}, \quad \beta = A_{\min}.$$

As $\alpha > p^\dagger\beta/(\beta - p^\dagger)$, $\alpha(\beta - p^\dagger) > p^\dagger\beta$, i.e. $\alpha\beta/(\alpha+\beta) > p^\dagger$. Hence $\mathcal{A}_{y,s}(E_c) > p^\dagger$ for all $s \in [0,1]$ (the uniform-energy alternative of (D4), $E_c = \bar{E}$), and (D5) holds for the areas.

*(D1).* $1 - \frac{31}{4}\eta < 1 - \frac{61}{8}\eta < 1 - 5\eta$, and $r_0 < p^*(y) < 1 - r_0 < 1$ by Lemma 3.2(d).

*(D2).* Let $u^2, v^2 < E_c$, $x = X_y(u)$, $\theta = Y(v)$ and $p = p^* + x$. Then $\widetilde{U}_y(x) = u^2 < \bar{E}$, so $U_y(p) - U_{\min}(y) = g_y(x) < \bar{E} + \delta$, and Lemma 3.2(g) gives $\min(p, 1-p) > \kappa_0/(M_0 + 4\kappa_0 + \bar{E} + \delta) > \Delta P$. Thus $\Delta P < p < 1 - \Delta P \le \pi R_q^2 < 1$. Likewise $V(\theta) \le \widetilde{V}(\theta) + \delta < \bar{E} + \delta$, so $|\theta| < \frac{1}{2}\sqrt{(\bar{E}+\delta)/(\bar{E}+\delta+\nu_0)} \le \frac{1}{2} - \Delta\theta$, the function $E \mapsto E/(E+\nu_0)$ being increasing.

*(D3) and (D5).* For $u^2 \le \bar{E}$: $a_0(y)|u| \le a_{\max}\sqrt{\bar{E}}$ and $|X_y(u)| < 1$ (as $X_y(u) \in (-p^*, 1-p^*)$), so $\max(a_0|u|, |X_y(u)|) \le \bar{X}$ and $|a_0 u - X_y(u)| \le \bar{X} = \delta_x$. For $v^2 \le \bar{E}$: $b_0|v| \le b_0\sqrt{\bar{E}}$ and $|Y(v)| < \frac{1}{2}$, so similarly with $\bar{Y} = \delta_\theta$.

*(D4) and (D6).* By the definitions of $\delta_s$ and $\lambda_0$ (with $p^*_{\rm lo} = r_0$), $\delta_s \le \frac{1}{4}$, $\delta_s\delta_x \le \frac{1}{2}\Delta P$ and $\delta_s\delta_\theta \le \frac{1}{2}\Delta\theta$; the energy condition was shown above.

*(D7).* $P_{\min} = \frac{1}{4}\min(r_0, \Delta P) \le \frac{1}{2}\min(p^*_{\rm lo}, \Delta P)$ and $P_{\max} = \frac{3}{2} \ge \max(\frac{3}{2}p^*_{\rm hi}, 1)$. By (90), with $c_x = \sqrt{a_0/b_0}$ and $r_b = \sqrt{p^*/\pi}$,

$$|(\mathbf{A}_y)_{11}| = \frac{\sqrt{a_0/b_0}}{2\sqrt{\pi p^*}} \le \frac{\sqrt{a_{\max}/b_0}}{2\sqrt{\pi r_0}}, \qquad |(\mathbf{A}_y)_{22}| = 2\sqrt{\pi p^*}\sqrt{b_0/a_0} \le 2\sqrt{\pi}\sqrt{b_0/a_{\min}};$$

so both are at most $A_{\max}$ and $R^\dagger \max(1, |(\mathbf{A}_y)_{11}|, |(\mathbf{A}_y)_{22}|) \le B_{\rm lin}$. The bounds for $B^*$ and $c_{\rm sh}$ hold by definition. Lemma A.8(b) now gives $P_{\rm supp} < h = 1$. □

### 3.3. **The area estimate.**

**Lemma 3.5** (Area estimate)**.** *Assume planar data, $\xi > 0$ and the constants of Definition 3.1, and write $\check{\mathcal{A}}_y(E)$ for the area of $\{(p,\theta) \in (0,1) \times (-\frac{1}{2},\frac{1}{2}) : U_y(p) - U_{\min}(y) + V(\theta) \le E\}$. Let $y \in \mathcal{O}$ and $0 \le I \le 1 - 3\eta$. Then*

$$\text{(4)}\quad \operatorname{area}\big\{(p,\theta) \in (0,1) \times (-\tfrac{1}{2},\tfrac{1}{2}) : U_y(p) + V(\theta) \le g_y(I+3\eta) + 2\sigma_0\big\} \ge (I+\eta)(1-2\beta_0) > I.$$

*Consequently, if $E \ge 0$ and $\check{\mathcal{A}}_y(E) \le I$, then $U_{\min}(y) + E < g_y(I+3\eta) + 2\sigma_0$, and every $(p,\theta)$ with $U_y(p) + V(\theta) \le U_{\min}(y) + E$ satisfies*

$$\text{(5)}\qquad f_y(p) < g_y(I) + 3\Lambda\eta + 2\sigma_0 < g_y(I) + \tfrac{11}{32}\xi.$$

*Proof.* Put $b = g_y(I+3\eta)$ $(I + 3\eta \le 1)$. By Lemma A.6(a), $\{p \in [0,1] : f_y(p) \le b\}$ is a closed interval $J$ of length at least $I + 3\eta$, so $J' = J \cap [\eta, 1-\eta]$ has length at least $I + \eta$. By Lemma 3.2(b), $U_y(p) < b + \sigma_0/2$ on $J'$ and $V(\theta) \le \sigma_0$ for $|\theta| \le \frac{1}{2} - \beta_0$. So the set of (4) contains $J' \times [-\frac{1}{2} + \beta_0, \frac{1}{2} - \beta_0]$, and its area is at least $(I+\eta)(1-2\beta_0) \ge I + \eta - 2\beta_0 = I + \frac{3}{4}\eta > I$, as $I + \eta \le 1$ and $2\beta_0 = \eta/4$.

If $\check{\mathcal{A}}_y(E) \le I$ but $U_{\min}(y) + E \ge b + 2\sigma_0$, the sublevel set at level $U_{\min}(y) + E$ contains the set of (4), so $\check{\mathcal{A}}_y(E) > I$, a contradiction. For $(p,\theta)$ with $U_y(p) + V(\theta) \le U_{\min}(y) + E$, Lemma 3.2(b) and $V \ge 0$ give $f_y(p) < U_y(p) + V(\theta) < b + 2\sigma_0$, and $b \le g_y(I) + 3\Lambda\eta$ by (3). Finally $3\Lambda\eta + 2\sigma_0 \le 3\xi/64 + \xi/4 < 11\xi/32$. □

The case $I = 0$ is included: the centre of the disc goes to the minimum of $U_y + V$ (Theorem A.12).

**Lemma 3.6** (Area estimate after quadratic modification)**.** *In the setting of Lemma 3.5, put $H_0(p,\theta) = U_y(p) - U_{\min}(y) + V(\theta)$ and $\widetilde{H}(p,\theta) = \widetilde{U}_y(p - p^*(y)) + \widetilde{V}(\theta)$ on $(0,1) \times (-\frac{1}{2},\frac{1}{2})$, with the modified wells of Definition 3.3, so that $|\widetilde{H} - H_0| \le 2\delta$, $\delta = \xi/64$ (Lemma 3.4(a)). Let $\widetilde{\mathcal{A}}_y(E)$ be the area of $\{\widetilde{H} \le E\}$. If $0 \le I \le 1 - 3\eta$, $E \ge 0$ and $\widetilde{\mathcal{A}}_y(E) \le I$, then every point with $\widetilde{H}(p,\theta) \le E$ satisfies*

$$f_y(p) < g_y(I) + 3\Lambda\eta + 2\sigma_0 + 5\delta \le g_y(I) + \frac{3}{8}\xi < g_y(I) + \xi.$$

*Proof.* Set $T = g_y(I+3\eta) + 2\sigma_0 - U_{\min}(y)$. By Lemma 3.5 the set $\{H_0 \le T\}$, which is contained in $\{\widetilde{H} \le T + 2\delta\}$, has area greater than $I$; as $\widetilde{\mathcal{A}}_y(E) \le I$, $E < T + 2\delta$. At a point with $\widetilde{H} \le E$,

$$f_y(p) \le U_y(p) + V(\theta) = U_{\min}(y) + H_0(p,\theta) \le U_{\min}(y) + E + 2\delta < g_y(I+3\eta) + 2\sigma_0 + 4\delta.$$

By (3), $g_y(I+3\eta) \le g_y(I) + 3\Lambda\eta$, and $3\Lambda\eta + 2\sigma_0 + 5\delta \le \frac{3\xi}{64} + \frac{16\xi}{64} + \frac{5\xi}{64} = \frac{3}{8}\xi$. □

### 3.4. **One convex function.**

**Theorem 3.7** (Planar rearrangement of a convex function)**.** *Let $f(p) = \max_{r\in\mathcal{I}}(a_r + b_r p)$ with finitely many real $a_r, b_r$, let $\Lambda \ge 1$ and $M_0$ be real with $|b_r| \le \Lambda$ and $|a_r| + |b_r| \le M_0$ for every $r$, let $g$ be the increasing rearrangement of $f$ on $[0,1]$, and let $\xi > 0$ be real. Let $Q^{\mathrm{pl}} = Q$ and $q_t = \hat{q}_t$ be the Hamiltonian and the isotopy of Theorem A.13 for the modified pair of Definition 3.3 $(d = 0)$ and the isotopy data of Definition 3.3, and let $r_B$ be the number defined there. Then:*

- *(a) $Q^{\mathrm{pl}}$ is $C^\infty$ on $\mathbb{R} \times \mathbb{C}$, and $Q^{\mathrm{pl}}(t,u) = 0$ if $\pi|u|^2 \ge r_B$ or $t \notin (0,1)$;*
- *(b) $q_t$ is generated by $Q^{\mathrm{pl}}$ $(\iota_X\omega = -dQ_t^{\mathrm{pl}})$, $q_t = \mathrm{id}$ for $t \le 0$, $q_t = q_1$ for $t \ge 1$, $q_t = \mathrm{id}$ on $\{\pi|u|^2 \ge r_B\}$, and $q_t(\bar{D}(A)) = \bar{D}(A)$ for every $A \ge r_B$, in particular for $A = 1$;*
- *(c) for every $z \in \bar{D}(1)$,*

$$f\big(\pi|q_1(z)|^2\big) \le g(\pi|z|^2) + \xi.$$

*Proof.* By Lemma 3.4, Theorem A.13 applies with $h = 1$ and a one-point parameter space. Its items (a), (b), (d) give (a) and (b): $P_{\mathrm{supp}} = r_B$, and $Q$ vanishes on $\{\pi|u|^2 > P_{\mathrm{supp}}\}$, hence by continuity on $\{\pi|u|^2 \geq P_{\mathrm{supp}}\}$; likewise $q_t = \mathrm{id}$ there.

(c) Put $I = \pi|z|^2$. If $I \leq 1 - 8\eta < p_{\mathrm{on}}$, Theorem A.13(c) gives $q_1(z) = q_{\mathrm{in}}(z) = C(f_{\mathrm{in}}(z))$, with $f_{\mathrm{in}}$ the map $f_y$ of Definition A.11 and $C$ the polar chart (89). By Theorem A.12(b),(c), with $f_{\mathrm{in}}(z) = (x, \theta)$ and $p = p^* + x = \pi|q_1(z)|^2$, $\widetilde{H}(p, \theta) = E(I)$, the level whose closed sublevel has area exactly $I$ (Lemma A.9(iv)). As $I \leq 1 - 3\eta$, Lemma 3.6 gives $f(\pi|q_1(z)|^2) < g(I) + \frac{3}{8}\xi$. If $1 - 8\eta < I \leq 1$, then $\pi|q_1(z)|^2 \in [0, 1]$ by (b), so by (3)

$$f(\pi|q_1(z)|^2) \leq \max_{[0,1]} f = g(1) \leq g(I) + \Lambda(1 - I) < g(I) + 8\Lambda\eta \leq g(I) + \tfrac{1}{8}\xi. \qquad \square$$

*Remark* 3.8 (Dimension four). For $m = 1$ there is no parameter: $\Delta = [0, h]$ with $h = L$ a positive real, $\ell$ is a constant in $[h - \zeta/2, h - \delta_1]$, and $\chi$ is a constant, equal to 1 when $h \geq \frac{7}{4}\eta_1$ (Definition 3.10, Lemma 3.11). If $\chi \neq 0$, Theorem 3.13 is Theorem 3.7 run with the clock $\chi S(t)$ and rescaled to the disc of area $\ell$; if $\chi = 0$, the isotopy is the identity. Outside the equal-capacity case the four-dimensional branch instead acts directly on continuous convex functions on $[0, h]$, with one planar path on the whole disc of area $h$ (§4).

3.5. **Families.**

**Theorem 3.9** (Planar rearrangement with parameters)**.** *Let $(\mathcal{O}, (a_r, b_r)_{r\in\mathcal{I}}, \Lambda, M_0)$ be planar data and $\xi > 0$ real. Let $Q^{\mathrm{pl}}(y, t, u) = Q_{y,t}(u)$ and $q_{y,t} = \hat{q}_{y,t}$ be the Hamiltonian and the isotopy of Theorem A.13 for the modified potential pair of Definition 3.3 over $\mathcal{O}$ and the isotopy data of Definition 3.3. Then $Q^{\mathrm{pl}}$ and $(y, t, u) \mapsto q_{y,t}(u)$ are $C^\infty$ on $\mathcal{O} \times \mathbb{R} \times \mathbb{C}$; for every $y \in \mathcal{O}$, $Q^{\mathrm{pl}}(y, \cdot, \cdot)$ and $q_{y,\cdot}$ have properties (a) and (b) of Theorem 3.7 with the same $r_B$; and for every $y \in \mathcal{O}$ and $z \in \bar{D}(1)$,*

$$f_y\big(\pi|q_{y,1}(z)|^2\big) \leq g_y(\pi|z|^2) + \xi.$$

*Proof.* The constants do not depend on $y$, and Lemmas 3.2 and 3.4 hold for every $y \in \mathcal{O}$; so Theorem A.13 applies over $\mathcal{Y} = \mathcal{O}$, and Theorem A.14(a) gives joint smoothness. For fixed $y$ the proof of Theorem 3.7 applies verbatim with $f_y, g_y, U_y, \mathcal{A}_y$. $\square$

3.6. **Slice lengths, cutoffs and the parameter domain.** Fix a convex polyhedral function $F = \max_{r\in\mathcal{I}} \phi_r$ ($\phi_r$ real affine on $\mathbb{R}^m$, $\mathcal{I}$ finite), a coordinate $j$ and a real $e > 0$. Write $p = (y, x)$ with $x = p_j$, $y \in \mathbb{R}^{m-1}$, let $\mathcal{N}_j = \{\nu : b_{\nu j} > 0\}$, which is nonempty, $N_j = \#\mathcal{N}_j$, and let $N_\Delta \geq N_j$ be the number of faces of $\Delta$ (Paper I, § "Setting"). For $m = 1$, $y$ is absent, every $L_\nu$ is a positive constant and the coordinate-box conditions below are void.

**Definition 3.10** (Slice data and the lifted Hamiltonian)**.** (1) (Constants.)

$$\Lambda_F = 1 + \sum_r |\partial_j \phi_r|, \quad \eta_1 = \frac{e}{64\Lambda_F}, \quad \zeta = \frac{\eta_1}{8}, \quad \xi = \frac{e}{4}, \quad \delta_1 = \frac{\zeta}{4}, \quad s_1 = \frac{\zeta}{4(N_\Delta + 1)},$$

and

$$\varrho = 2 + \sum_i \sum_{\nu:\, b_{\nu i} > 0} \frac{\ell_\nu}{b_{\nu i}}.$$

(2) (Slice length.) For $y \in \mathbb{R}^{m-1}$,

$$\ell(y) = -s_1 \log \sum_{\nu\in\mathcal{N}_j} \exp\big(-L_\nu(y)/s_1\big) - \delta_1.$$

(3) (Cutoffs.) With the flat step $S$ of Definition A.1,

$$\chi_L(y)=\prod_{\nu\in\mathcal{N}_j} S\Big(\frac{L_\nu(y)-5\eta_1/4}{\eta_1/2}\Big),\quad \rho_{\mathrm{box}}(y)=\prod_{i\neq j}\Big(1-S\Big(\frac{y_i^2-\varrho^2}{3\varrho^2}\Big)\Big),\quad \chi=\rho_{\mathrm{box}}\,\chi_L.$$

(4) (Parameter domain.) $\mathcal{O}_j=\{y\in\mathbb{R}^{m-1}:\|y\|_\infty<3\varrho,\ L_\nu(y)>\frac34\eta_1\ (\nu\in\mathcal{N}_j)\}$, an open polyhedron containing $\{y\in\Delta_j: L(y)>\frac34\eta_1\}$ (Lemma 3.11(c)).

(5) (Planar data on slices.) $a_r(y)=\phi_r(y,0)$ and $b_r(y)=\ell(y)\,\partial_j\phi_r$, so that $f_y(p)=F(y,\ell(y)p)$ in (2);

$$\bar\ell=1+\sum_{\nu\in\mathcal{N}_j}\frac{|\ell_\nu|+4\varrho\sum_i|b_{\nu i}|}{b_{\nu j}},\qquad \Lambda=1+\bar\ell\sum_r|\partial_j\phi_r|,$$

$$M_0=1+\sum_r\left(|\phi_r(0)|+4\varrho\sum_i|\partial_i\phi_r|+|\partial_j\phi_r|\bar\ell\right).$$

Let $Q^{\mathrm{pl}}$, $q_{y,t}$ and $r_B$ be those of Theorem 3.9 for the planar data $(\mathcal{O}_j,(a_r,b_r),\Lambda,M_0)$ and $\xi$.

(6) (Formula and zero sets.) Let $\mathcal{Z}_j,\mathcal{F}_j\subset\mathbb{R}^{m-1}$ be the open sets

$$\mathcal{Z}_j=\big\{y:|y_i|>\tfrac94\varrho \text{ for some } i\neq j,\ \text{or } L_\nu(y)<\tfrac98\eta_1 \text{ for some } \nu\in\mathcal{N}_j\big\},$$
$$\mathcal{F}_j=\big\{y:|y_i|<\tfrac52\varrho \text{ for all } i\neq j,\ \text{and } L_\nu(y)>\eta_1 \text{ for all } \nu\in\mathcal{N}_j\big\}.$$

(7) (Lifted Hamiltonian.) For $t\in\mathbb{R}$ and $w\in\mathbb{C}^m$, with $y=y(w)=(\pi|w_i|^2)_{i\neq j}$,

$$Q^{(j)}(t,w)=\begin{cases}\chi(y)S'(t)\,\ell(y)\,Q^{\mathrm{pl}}\big(y,\chi(y)S(t),\,w_j/\sqrt{\ell(y)}\big) & \text{if } y\in\mathcal{F}_j,\\ 0 & \text{if } y\in\mathcal{Z}_j.\end{cases}\tag{6}$$

By Lemma 3.11(e) the two cases of (6) cover $\mathbb{R}^{m-1}$, agree on their overlap, and the formula is used only on $\mathcal{F}_j\subset\mathcal{O}_j$. If $\mathcal{O}_j=\emptyset$, then $\mathcal{F}_j=\emptyset$ and $Q^{(j)}=0$.

**Lemma 3.11** (Properties of the slice data)**.** *In the situation of Definition 3.10:*

*(a)* $\ell$ *is real-analytic on* $\mathbb{R}^{m-1}$ *and* $\delta_1\le L(y)-\ell(y)\le\zeta/2$ *for all* $y$.

*(b)* $\chi$ *is* $C^\infty$ *with values in* $[0,1]$. $\rho_{\mathrm{box}}(y)=1$ *if* $\|y\|_\infty\le\varrho$, *and* $\rho_{\mathrm{box}}=0$ *on the open set* $\{|y_i|>2\varrho$ *for some* $i\}$; $\chi_L=0$ *on the open set* $\{L_\nu<\frac54\eta_1$ *for some* $\nu\in\mathcal{N}_j\}$, *and* $\chi_L(y)=1$ *if* $L(y)\ge\frac74\eta_1$.

*(c)* $\Delta_j\subset\{\|y\|_\infty\le\varrho-2\}$; *hence* $\{y\in\Delta_j:L(y)>\frac34\eta_1\}\subset\mathcal{O}_j$, *and* $\chi(y)=1$ *for* $y\in\Delta_j$ *with* $L(y)\ge2\eta_1$.

*(d) On* $\mathcal{O}_j$: $\ell>\frac58\eta_1$, $|b_r|\le\Lambda$, *and* $|f_y(p)|\le M_0$ *for* $0\le p\le1$; *thus* $(\mathcal{O}_j,(a_r,b_r),\Lambda,M_0)$ *are planar data. Moreover* $\ell\le\bar\ell$ *on* $\mathcal{O}_j$, *and* $f_y$ *is* $\Lambda_F\ell(y)$*-Lipschitz on* $[0,1]$.

*(e)* $\mathcal{Z}_j\cup\mathcal{F}_j=\mathbb{R}^{m-1}$, $\mathcal{F}_j\subset\mathcal{O}_j$, $\ell>\frac78\eta_1$ *on* $\mathcal{F}_j$, *and* $\chi=0$ *on the open set* $\mathcal{Z}_j$.

*(f) If* $\chi(y)\neq0$, *then* $|y_i|<2\varrho$ *for all* $i\neq j$, $L_\nu(y)>\frac54\eta_1$ *for all* $\nu\in\mathcal{N}_j$, $y\in\mathcal{F}_j$, *and* $\ell(y)>\eta_1$.

*Proof.* (a) $\ell+\delta_1=-s_1\log\sum_\nu e^{-L_\nu/s_1}$ is the soft minimum of the affine $L_\nu$, $\nu\in\mathcal{N}_j$; Lemma 2.4(a),(b), applied to $-L_\nu$, shows that it is real-analytic with values in $[L-s_1\log N_j,\,L]$. As $\log N_j<N_j\le N_\Delta<N_\Delta+1$, $s_1\log N_j<\zeta/4$, and $L-\ell\in[\delta_1,\delta_1+\zeta/4]\subset[\delta_1,\zeta/2]$.

(b) Smoothness and range follow from Lemma A.2. The $i$-th factor of $\rho_{\mathrm{box}}$ is 1 if $|y_i|\le\varrho$ (argument $\le0$) and 0 if $|y_i|\ge2\varrho$ (argument $\ge1$). The $\nu$-th factor of $\chi_L$ vanishes if $L_\nu<\frac54\eta_1$; if $L\ge\frac74\eta_1$, every $L_\nu\ge L$ gives argument $\ge1$.

(c) Let $y\in\Delta_j$ and $i\neq j$. Some $\nu$ has $b_{\nu i}>0$ (by compactness of $\Delta$, as for $j$ in Paper I, § "Setting"), and $b_{\nu i}y_i\le\langle b_\nu,(y,0)\rangle\le\ell_\nu$ with $y_i\ge0$; so $0\le y_i\le\varrho-2$. Then $\rho_{\mathrm{box}}(y)=1$ and

$\|y\|_\infty < 3\varrho$. If $L(y) > \frac{3}{4}\eta_1$, then $L_\nu(y) \geq L(y) > \frac{3}{4}\eta_1$ for all $\nu \in \mathcal{N}_j$, so $y \in \mathcal{O}_j$; if $L(y) \geq 2\eta_1$, moreover $\chi_L(y) = 1$ by (b).

(d) On $\mathcal{O}_j$ every $L_\nu > \frac{3}{4}\eta_1$, so $L > \frac{3}{4}\eta_1$ and $\ell \geq L - \zeta/2 > \frac{3}{4}\eta_1 - \frac{1}{16}\eta_1 > \frac{5}{8}\eta_1$. For $\nu \in \mathcal{N}_j$ and $\|y\|_\infty < 3\varrho$, $L_\nu(y) \leq (\ell_\nu + 3\varrho\sum_{i\neq j} b_{\nu i})/b_{\nu j}$, so $0 < \ell < L \leq \bar{\ell}$ and $|b_r| = \ell|\partial_j\phi_r| \leq \bar{\ell}\sum_r |\partial_j\phi_r| < \Lambda$. For $0 \leq p \leq 1$, $|a_r(y)| + |b_r(y)| \leq M_0$ and therefore $|f_y(p)| \leq M_0$, since $|\phi_r(y,0)| \leq |\phi_r(0)| + 3\varrho\sum_{i\neq j}|\partial_i\phi_r|$. The functions $a_r$ are affine and $b_r$ real-analytic. The Lipschitz bound is the slope bound $\max_r |b_r(y)| \leq \Lambda_F \ell(y)$.

(e) If $y \notin \mathcal{Z}_j$, then $|y_i| \leq \frac{9}{4}\varrho < \frac{5}{2}\varrho$ and $L_\nu(y) \geq \frac{9}{8}\eta_1 > \eta_1$, so $y \in \mathcal{F}_j$. If $y \in \mathcal{F}_j$, then $\|y\|_\infty < \frac{5}{2}\varrho < 3\varrho$ and $L_\nu > \eta_1 > \frac{3}{4}\eta_1$, so $y \in \mathcal{O}_j$, and $\ell \geq L - \zeta/2 > \eta_1 - \frac{1}{16}\eta_1$. If $y \in \mathcal{Z}_j$, either some $|y_i| > \frac{9}{4}\varrho > 2\varrho$, and $\rho_{\mathrm{box}}(y) = 0$ by (b), or some $L_\nu(y) < \frac{9}{8}\eta_1 < \frac{5}{4}\eta_1$, and $\chi_L(y) = 0$ by (b).

(f) If $\chi(y) \neq 0$, every factor of $\chi_L(y)$ and of $\rho_{\mathrm{box}}(y)$ is nonzero, so $L_\nu(y) > \frac{5}{4}\eta_1$ and $y_i^2 < 4\varrho^2$. Hence $|y_i| < 2\varrho < \frac{5}{2}\varrho$ and $L_\nu(y) > \frac{5}{4}\eta_1 > \eta_1$, so $y \in \mathcal{F}_j$, and $\ell \geq L - \zeta/2 > \frac{5}{4}\eta_1 - \frac{1}{16}\eta_1 > \eta_1$. □

### 3.7. **The lift.**

**Lemma 3.12** (Support and outer faces)**.** *Let $Q^{(j)}$ be as in Definition 3.10 and put*

$$c_j = \tfrac{1}{2}(1 - r_B)\,\eta_1 \min_{\nu\in\mathcal{N}_j} b_{\nu j} > 0.$$

*(a) If $Q^{(j)}(t,w) \neq 0$, then, with $y = y(w)$, $\chi(y) \neq 0$, $\|y\|_\infty < 2\varrho$ and $p_j(w) < r_B\,\ell(y)$. In particular $Q^{(j)}$ vanishes outside $\mathbb{R} \times K_j$, $K_j = \{w : \|y(w)\|_\infty \leq 2\varrho,\ p_j(w) \leq r_B\bar{\ell}\}$, a compact set.*

*(b) For $\nu \in \mathcal{N}_j$ and every $t$: if $Q^{(j)}(t,w) \neq 0$, then*

$$\ell_\nu - \langle b_\nu, p(w)\rangle = b_{\nu j}\big(L_\nu(y) - p_j\big) \geq b_{\nu j}\big((L(y) - \ell(y)) + (1 - r_B)\ell(y)\big) > b_{\nu j}(1 - r_B)\eta_1. \tag{7}$$

*Hence $Q_t^{(j)}$ vanishes on the open set $\{w : \langle b_\nu, p(w)\rangle > \ell_\nu - c_j\}$.*

*(c) For $\nu \notin \mathcal{N}_j$ and every $t$, $\{Q_t^{(j)}, \langle b_\nu, p\rangle\} = 0$ on $\mathbb{C}^m$.*

*Proof.* (a) By (6), $Q^{(j)}(t,w) \neq 0$ forces $y \in \mathcal{F}_j$, $\chi(y) \neq 0$ and $Q^{\mathrm{pl}}(y, \chi(y)S(t), w_j/\sqrt{\ell(y)}) \neq 0$. Lemma 3.11(f) gives $\|y\|_\infty < 2\varrho$, and $\pi|w_j|^2/\ell(y) < r_B$ since $Q^{\mathrm{pl}}(y,\cdot,u) = 0$ for $\pi|u|^2 \geq r_B$ (Theorem 3.9); and $\ell(y) \leq \bar{\ell}$ (Lemma 3.11(d),(e)). $K_j$ is closed, and bounded by $|w_i|^2 \leq 2\varrho/\pi$ $(i \neq j)$, $|w_j|^2 \leq r_B\bar{\ell}/\pi$.

(b) $\langle b_\nu, p\rangle = \sum_{i\neq j} b_{\nu i}y_i + b_{\nu j}p_j$ and $L_\nu = (\ell_\nu - \sum_{i\neq j} b_{\nu i}y_i)/b_{\nu j}$ give the identity; $p_j < r_B\ell$ (by (a)) and $L_\nu \geq L$ give the first inequality, and $L - \ell \geq 0$, $\ell > \eta_1$ (Lemma 3.11(a),(f)) the second. As $c_j < b_{\nu j}(1 - r_B)\eta_1$, $Q_t^{(j)} = 0$ where $\ell_\nu - \langle b_\nu, p\rangle < c_j$.

(c) $\langle b_\nu, p\rangle = \sum_{i\neq j} b_{\nu i}p_i$. The field of $p_i = \pi|w_i|^2$ rotates the $i$-th coordinate plane and fixes the other coordinates, and $Q_t^{(j)}$ depends on $w_i$, $i \neq j$, only through $p_i$; so $\{Q_t^{(j)}, p_i\} = \pm dQ_t^{(j)}(X_{p_i}) = 0$. □

**Theorem 3.13** (Realisation of one coordinate rearrangement)**.** *Let $F = \max_{r\in\mathcal{I}} \phi_r$ be convex polyhedral with real coefficients on $\Delta$, $j \in \{1,\dots,m\}$ and $e > 0$ real, and let $Q^{(j)}$ be the function of Definition 3.10. Then $Q^{(j)}$ is $C^\infty$ on $\mathbb{R}\times\mathbb{C}^m$, vanishes outside $\mathbb{R}\times K_j$ with $K_j$ compact, and $Q^{(j)}(t,\cdot) = 0$ for $t \leq 0$ and $t \geq 1$. Its flow $(q_t^{(j)})_{t\in[0,1]}$ ($\iota_X\omega_V = -dQ_t^{(j)}$) is a smooth Hamiltonian isotopy of $\mathbb{C}^m$ with:*

*(a) $q_0^{(j)} = \mathrm{id}$; each $q_t^{(j)}$ preserves the moments $p_i$, $i \neq j$, and maps $V_\Delta$ onto $V_\Delta$; explicitly, for $w$ with $\chi(y(w)) \neq 0$ the $j$-th coordinate of $q_t^{(j)}(w)$ is $\sqrt{\ell(y)}\,q_{y,\chi(y)S(t)}(w_j/\sqrt{\ell(y)})$, $y = y(w)$, and for $\chi(y(w)) = 0$ it is $w_j$; the coordinates $w_i$, $i \neq j$, are rotated in their planes, so their moduli are preserved;*

*(b) with the positive $c_j$ of Lemma 3.12, for every $\nu \in \mathcal{N}_j$ and every $t$, $Q_t^{(j)}$ vanishes on $\{\langle b_\nu, p\rangle > \ell_\nu - c_j\}$, and for every $\nu \notin \mathcal{N}_j$, $Q_t^{(j)}$ Poisson-commutes with $\langle b_\nu, p\rangle$;*

*(c) $F\big(p(q_1^{(j)}(w))\big) \le (R_jF)(p(w)) + e$ for every $w \in V_\Delta$.*

*In particular Theorem M2.10 holds, with $Q_t^{(j)} = Q^{(j)}(t,\cdot)$.*

*Proof. Smoothness.* The sets $\Omega_Z = \{w : y(w) \in \mathcal{Z}_j\}$ and $\Omega_F = \{w : y(w) \in \mathcal{F}_j\}$ are open and cover $\mathbb{C}^m$ (Lemma 3.11(e)). On $\mathbb{R}\times\Omega_Z$, $Q^{(j)} = 0$. On $\mathbb{R}\times\Omega_F$, $y(w) \in \mathcal{O}_j$ and $\ell(y(w)) > \frac{7}{8}\eta_1$ (Lemma 3.11(e)), so (6) is a composite of $C^\infty$ maps: $w \mapsto y(w)$ is polynomial, $\chi$ and $\ell$ are smooth, $w \mapsto w_j/\sqrt{\ell(y(w))}$ is smooth (also at $w_j = 0$ and on every axis $w_i = 0$), and $Q^{\mathrm{pl}}$ is smooth on $\mathcal{O}_j \times \mathbb{R}\times\mathbb{C}$ (Theorem 3.9). On the overlap the formula vanishes since $\chi(y) = 0$ there (Lemma 3.11(e)). Hence $Q^{(j)}$ is well defined and $C^\infty$; by Lemma 3.12(a) it vanishes outside $\mathbb{R}\times K_j$, and $Q^{(j)}(t,\cdot) = 0$ for $t \le 0$ and $t \ge 1$ since $S'(t) = 0$. So its flow exists for all times and is a smooth isotopy of $\mathbb{C}^m$.

*The flow (a).* Since $\omega_V$ is the sum of the area forms of the coordinate planes, the $w_i$-component of the field at $w$ is the planar Hamiltonian field of $w_i \mapsto Q^{(j)}(t,w)$, the other coordinates being fixed. For $i \ne j$ this function depends on $w_i$ only through $p_i$, so the component is $\partial Q_t^{(j)}/\partial p_i$ times the field of $p_i$, which rotates the $i$-th plane; hence $p_i$ and $y$ are constant along the flow. If $\chi(y) = 0$, the function $w_j \mapsto Q^{(j)}(t,w)$ vanishes identically (for $y \in \mathcal{Z}_j$ by definition, for $y \in \mathcal{F}_j$ through the factor $\chi(y)$), so $w_j$ is constant. If $\chi(y) \ne 0$, then $y \in \mathcal{F}_j$ and $Q^{(j)}(t,w) = \ell\, G_t(w_j/\sqrt{\ell})$ with $G_t(u) = \chi S'(t) Q^{\mathrm{pl}}(y, \chi S(t), u)$, $\ell = \ell(y)$, $\chi = \chi(y)$. The field of a planar Hamiltonian $\ell\, G(w/\sqrt{\ell})$ is $\sqrt{\ell}\, X_G(w/\sqrt{\ell})$, since its gradient is $\sqrt{\ell}(\nabla G)(w/\sqrt{\ell})$ and the field is a fixed linear map of the gradient. Hence $u = w_j/\sqrt{\ell}$ solves $\dot u = \chi S'(t) X_{Q^{\mathrm{pl}}(y,\chi S(t),\cdot)}(u)$, as does $t \mapsto q_{y,\chi S(t)}(u_0)$, with the same initial value; so $u(t) = q_{y,\chi S(t)}(u_0)$. This is the description in (a), and $q_0^{(j)} = \mathrm{id}$.

By the slice description of §M2.3, $w \in V_\Delta$ if and only if $y(w) \in \Delta_j$ and $p_j(w) \le L(y(w))$. The flow preserves $y$, and fixes $w_j$ if $\chi(y) = 0$. If $\chi(y) \ne 0$, the planar map $q_{y,\chi S(t)}$ maps $\bar D(1)$ onto itself and is the identity on $\{\pi|u|^2 \ge r_B\}$ (Theorem 3.9); in the variable $w_j = \sqrt{\ell}u$ it maps $\{p_j \le \ell\}$ onto itself and fixes every point with $p_j \ge r_B\ell$. Since $\ell < L$ (Lemma 3.11(a)), it maps $\{p_j \le L(y)\}$ into itself and fixes every point with $p_j > L(y)$. Thus the bijection $q_t^{(j)}$ of $\mathbb{C}^m$ maps $V_\Delta$ and its complement into themselves, hence $V_\Delta$ onto $V_\Delta$.

*Faces (b).* This is Lemma 3.12(b),(c).

*The estimate (c).* Let $w \in V_\Delta$, $p(w) = (y,x)$, so $y \in \Delta_j$ and $0 \le x \le L(y)$; by (a), $p(q_1^{(j)}(w)) = (y,x')$ with $0 \le x' \le L(y)$. $F(y,\cdot)$ is $\Lambda_F$-Lipschitz, and by Lemmas A.6 and 2.2(iii) $(R_jF)(y,x)$ is the increasing rearrangement of $F(y,\cdot)$ on $[0,L(y)]$ at $x$; so $(R_jF)(y,x) \ge \min_{[0,L(y)]} F(y,\cdot)$.

*Case $L(y) < 2\eta_1$.* Then

$$F(y,x') \le \max_{[0,L(y)]} F(y,\cdot) \le \min_{[0,L(y)]} F(y,\cdot) + \Lambda_F L(y) < (R_jF)(y,x) + 2\Lambda_F\eta_1,$$

and $2\Lambda_F\eta_1 = e/32$.

*Case $L(y) \ge 2\eta_1$.* Then $\chi(y) = 1$ and $y \in \mathcal{O}_j$ (Lemma 3.11(c)). Write $\ell = \ell(y)$, $L = L(y)$, and let $g_y$ be the increasing rearrangement of $f_y(p) = F(y,\ell p)$ on $[0,1]$. For $0 \le x'' \le \ell$, the substitution $a \mapsto \ell a$ gives

$$g_y(x''/\ell) = \min_{0\le a\le \ell - x''} \max\big(F(y,a), F(y,a+x'')\big), \tag{8}$$

the rearrangement of $F(y,\cdot)$ on the shortened slice $[0,\ell]$.

If $x \le \ell$: by (a), $x' = \ell\,\pi|q_{y,1}(u)|^2$ with $u = w_j/\sqrt{\ell} \in \bar D(1)$, $\pi|u|^2 = x/\ell$. Theorem 3.9 gives $F(y,x') = f_y(\pi|q_{y,1}(u)|^2) \le g_y(x/\ell) + \xi$. Let $[a, a+x] \subset [0,L]$ be a minimising interval

for $(R_jF)(y,x)$ and $a' = \min(a, \ell - x) \geq 0$. Then $[a', a'+x] \subset [0,\ell]$ and $0 \leq a - a' = \max(0, a+x-\ell) \leq L - \ell$, so both endpoints move by at most $L-\ell$, and by (8)

$$g_y(x/\ell) \leq \max\big(F(y,a'), F(y,a'+x)\big) \leq (R_jF)(y,x) + \Lambda_F(L-\ell).$$

If $\ell < x \leq L$: then $\pi|u|^2 = x/\ell > 1 \geq r_B$ for $u = w_j/\sqrt{\ell}$, so $x' = x$ by (a). For a minimising interval $[a, a+x]$ of $(R_jF)(y,x)$, $0 \leq a \leq L - x < L - \ell$, hence $F(y,x) \leq F(y,a+x) + \Lambda_F a \leq (R_jF)(y,x) + \Lambda_F(L-\ell)$.

In both subcases, by Lemma 3.11(a),

$$F(y,x') \leq (R_jF)(y,x) + \xi + \Lambda_F\frac{\zeta}{2} = (R_jF)(y,x) + \frac{e}{4} + \frac{e}{1024} < (R_jF)(y,x) + e.$$

This proves (c). Items (a)–(c) are the conclusions of Theorem M2.10: the isotopy is defined on all of $\mathbb{C}^m$, hence near $V_\Delta$, and $c_j > 0$; a positive dyadic number below $c_j$ gives a rational collar if required. □

## 4. The scalar comparison datum for the cubic model

We give the scalar comparison iteration on an interval (the scalar instance of [Ope26, §2]), a fixed mollifier, the planar potentials and the Hamiltonian realiser, and specialise them to the four-dimensional cubic model for all $k \geq 1$ and all capacities. We use $D(A) = \{z \in \mathbb{C} : \pi|z|^2 < A\}$, the area form $\omega_{\mathbb{C}}$, the flat step $S$ of Definition A.1, and $\iota_{X_Q}\omega_{\mathbb{C}} = -dQ$.

4.1. **Finite iteration and fixed mollification.** Let $h > 0$ and let $P\colon \mathbb{R} \to \mathbb{R}$ be continuous and concave on $[0,h]$, with $\bar{P} > 0$, $L_P \geq 0$ and

$$\int_0^h P(p)\,dp = \mu h, \qquad 0 < \tau < 1, \qquad P(p) \geq \beta > 0 \quad (\tau h \leq p \leq h), \tag{9}$$
$$0 < d < \min(\mu, \beta), \qquad P_{\min} \leq P(p), \qquad |P(p)| \leq \bar{P},$$
$$|P(p) - P(q)| \leq L_P|p-q| \quad (p,q \in [0,h]).$$

For continuous convex $f$ on $[0,h]$ write

$$(Rf)(I) = \min_{0 \leq x \leq h-I} \max\big(f(x), f(x+I)\big), \qquad 0 \leq I \leq h, \tag{10}$$

the case $m = 1$, $\Delta = [0,h]$ of §2 (whose proofs do not use rationality). By Lemma A.6, Paper I, Lemma (Properties of $R_j$) and Lemma 2.6, $Rf$ is continuous, convex, nondecreasing and $L$-Lipschitz if $f$ is, $Rf(0) = \min f$, $Rf(h) = \max f$, $\int_0^h Rf = \int_0^h f$, $R(f+c) = Rf + c$, and $R$ is order preserving and sup-norm nonexpansive. Set

$$\eta_{\mathrm{c}} = (1-\tau)(\mu - d), \qquad B_P = \max(1, d - P_{\min} + 1), \qquad \lambda = 1 - \frac{\eta_{\mathrm{c}}}{2(B_P + \eta_{\mathrm{c}})}, \tag{11}$$
$$K_0 = 0, \qquad K_{r+1} = R(\lambda K_r - P), \qquad M = \frac{L_P}{1-\lambda},$$
$$N = \left\lfloor \frac{\log^+\big(16\bar{P}/[d(1-\lambda)]\big)}{1-\lambda} \right\rfloor + 1, \qquad \log^+ x = \max(0, \log x).$$

Then $0 < 1 - \lambda < \frac{1}{2}$. For piecewise affine $P$ (§4.5) each $K_r$ is piecewise affine with finitely many knots, computed from $K_{r-1}$ by Lemma 2.7 (valid verbatim for real data).

**Lemma 4.1** (Finite scalar gap)**.** *Each $K_r$ is continuous, convex, nondecreasing and $M$-Lipschitz on $[0,h]$, and*

$$R(K_N - P)(I) \leq K_N(I) - \frac{7}{8}d \qquad (0 \leq I \leq h). \tag{12}$$

*Proof.* By induction, $\lambda K_r - P$ is continuous, convex and $(\lambda M + L_P)$-Lipschitz, and $\lambda M + L_P = M$; so $K_{r+1} = R(\lambda K_r - P)$ is continuous, convex, nondecreasing and $M$-Lipschitz. Nonexpansiveness gives $\|K_1 - K_0\| = \|R(-P) - R(0)\| \le \bar{P}$ and $\|K_{r+1} - K_r\| \le \lambda \|K_r - K_{r-1}\|$, hence

$$\|K_{r+1} - K_r\|_{[0,h]} \le \lambda^r \bar{P}. \tag{13}$$

Integral preservation gives $\int_0^h K_{r+1} = \lambda \int_0^h K_r - \mu h$, so

$$\int_0^h K_r = -\frac{\mu h(1-\lambda^r)}{1-\lambda}, \qquad (1-\lambda)K_r(0) \le -\mu(1-\lambda^r),$$

the second because the nondecreasing $K_r$ is at most its mean at 0.

We next prove

$$(1-\lambda)K_r(h) \le \lambda^r \bar{P} - d. \tag{14}$$

We have $\mu \le \bar{P}$ by integration, and

$$\frac{\lambda \eta_{\mathrm{c}}}{1-\lambda} = 2B_P + \eta_{\mathrm{c}} > d - P_{\min}, \qquad \eta_{\mathrm{c}} \le \mu - d.$$

Consequently $(1-\lambda)(d - P_{\min}) \le \lambda(\mu - d)$, or $\lambda\mu + (1-\lambda)P_{\min} \ge d$. The case $r = 0$ of (14) follows from $d < \mu \le \bar{P}$. For the induction step, $Rf(h) = \max f$ and convexity of $\lambda K_r - P$ give

$$K_{r+1}(h) = \max\{\lambda K_r(0) - P(0), \lambda K_r(h) - P(h)\}.$$

After multiplication by $1-\lambda$, the first entry is bounded above by $-\lambda\mu + \lambda^{r+1}\mu - (1-\lambda)P_{\min} \le \lambda^{r+1}\bar{P} - d$. The second is bounded above by $\lambda^{r+1}\bar{P} - \lambda d - (1-\lambda)P(h) \le \lambda^{r+1}\bar{P} - d$, since $P(h) \ge \beta > d$. This proves (14).

Since $\lambda \le e^{-(1-\lambda)}$ and $(1-\lambda)N > \log^+\big(16\bar{P}/[d(1-\lambda)]\big)$,

$$\lambda^N \bar{P} < \bar{P} \exp\Big(-\log^+ \frac{16\bar{P}}{d(1-\lambda)}\Big) \le \frac{d(1-\lambda)}{16} \le \frac{d}{16}.$$

Finally, $(1-\lambda)K_N \le (1-\lambda)K_N(h)$, order preservation and translation invariance give

$$\begin{aligned} R(K_N - P) &\le R(\lambda K_N - P) + (1-\lambda)K_N(h) \\ &= K_{N+1} + (1-\lambda)K_N(h) \le K_N + 2\lambda^N \bar{P} - d \le K_N - \frac{7}{8}d, \end{aligned}$$

using (13) and (14). □

Define

$$K^{\mathrm{ext}}(p) = \min_{0 \le x \le h} \big(K_N(x) + M|p - x|\big), \qquad b(t) = \begin{cases} \exp\big(-1/(1-t^2)\big), & |t| < 1, \\ 0, & |t| \ge 1, \end{cases} \tag{15}$$

$$b_0 = \int_{-1}^1 b(t)\,dt, \qquad w = \frac{d}{32(M+1)}, \qquad K(p) = \frac{1}{b_0}\int_{-1}^1 K^{\mathrm{ext}}(p - wt)b(t)\,dt.$$

**Proposition 4.2** (Smooth scalar comparator)**.** *The function $K$ of* (15) *is smooth, convex and $M$-Lipschitz on $\mathbb{R}$. Moreover,*

$$\|K - K_N\|_{[0,h]} \le \frac{d}{32}, \qquad R(K-P)(I) \le K(I) - \frac{13}{16}d \quad (0 \le I \le h). \tag{16}$$

*Proof.* $K^{\mathrm{ext}} = K_N$ on $[0,h]$ is $M$-Lipschitz and convex (partial minimum of a jointly convex function); mollification by $b/b_0$ preserves both, is smooth, and moves values by at most $Mw < d/32$. Nonexpansiveness and Lemma 4.1 give $R(K-P) \le K_N - 7d/8 + d/32 \le K - 13d/16$. □

4.2. **The scalar planar potentials.** Let $F$ be continuous, convex and $L$-Lipschitz on $[0,h]$ ($L \ge 0$), $e > 0$, and $U_0 \colon \mathbb{R} \to \mathbb{R}$ smooth and convex with

$$F \le U_0 \le F + \sigma_{\mathrm{in}} \quad \text{on } [0,h], \qquad 0 \le \sigma_{\mathrm{in}} \le e/16. \tag{17}$$

Define

(18)

$$\eta = \frac{1}{2}\min\left(\frac{h}{16}, \frac{e}{32(L+1)}\right), \qquad \beta_{\mathrm{a}} = \frac{\eta}{8h}, \qquad \sigma = \frac{e}{8}, \qquad \nu = 4\sigma\beta_{\mathrm{a}}(1-\beta_{\mathrm{a}}), \qquad \kappa = \frac{\sigma\eta}{8},$$

$$U(p) = U_0(p) + \kappa\left(\frac{1}{p} + \frac{1}{h-p}\right) \quad (0<p<h), \qquad V(y) = \frac{\nu y^2}{\frac{1}{4} - y^2} \quad (|y| < \tfrac{1}{2}).$$

Then $0 < \eta \le h/32$, $L\eta \le e/64$, $0 < \beta_{\mathrm{a}} \le 1/256$, $\kappa, \nu > 0$; $p$ is the moment and $y$ the angle on $(0,h) \times (-\frac{1}{2}, \frac{1}{2})$.

**Lemma 4.3** (Area below a scalar potential level)**.** *$U$ and $V$ are smooth and strictly convex on their intervals and tend to $+\infty$ at both endpoints. Hence $U$ has a unique critical point $p^* \in (0,h)$, its minimum, with $U''(p^*) > 0$; we write $m_* = U(p^*)$. Also $V(0) = 0$ is the minimum of $V$ and $V''(0) = 8\nu$. For $0 \le I \le h - 4\eta$, put $b_I = RF(I + 3\eta) + 2\sigma$. Then*

$$\mathrm{area}\{(p,y) \in (0,h) \times (-\tfrac{1}{2}, \tfrac{1}{2}) : U(p) + V(y) \le b_I\} > I. \tag{19}$$

*Proof.* $U'' = U_0'' + 2\kappa(p^{-3} + (h-p)^{-3}) > 0$ and $U \to +\infty$ at $0, h$, giving $p^*$. Also $V(y) = \frac{\nu}{4}\big(\frac{1}{y+1/2} + \frac{1}{1/2-y}\big) - \nu$ and $V'(y) = (\nu/2)y/(\frac{1}{4} - y^2)^2$, giving the claims on $V$.

On $[\eta, h-\eta]$ the barrier is at most $\sigma/4$, so $U \le F + 3\sigma/4$; for $|y| \le \frac{1}{2} - \beta_{\mathrm{a}}$, $V(y) \le \sigma$. Lemma A.6(a) gives $[a, a+I+3\eta] \subset [0,h]$ with $F \le RF(I+3\eta)$ there; its intersection with $[\eta, h-\eta]$, times $[-\frac{1}{2}+\beta_{\mathrm{a}}, \frac{1}{2}-\beta_{\mathrm{a}}]$, lies in the set of (19) and has area at least $(I+\eta)(1-2\beta_{\mathrm{a}}) > I$, since $I + \eta < 4h$. □

4.3. **Quadratic wells and the endpoint map.** Let $\psi(t) = 1 - S((t^2-1)/3)$: smooth, even, $0 \le \psi \le 1$, $\psi = 1$ on $[-1,1]$, $\psi = 0$ outside $(-2,2)$, $C_\psi = \max_{[-2,2]} |\psi'|$. Let $f$ be smooth on $(A,B)$, $c \in (A,B)$, $f'(c) = 0$, $\alpha = f''(c) > 0$, $(x-c)f'(x) > 0$ for $x \ne c$, $f \to +\infty$ at $A, B$. Put

(20)

$$\delta_{\mathrm{w}} = e/64, \qquad g_f(x) = f(c+x) - f(c), \qquad r_f = \tfrac{1}{4}\min(c-A, B-c), \qquad \chi_f(x) = \psi(x/r_f),$$

$$K_{3,f} = 1 + \max_{x \in [-2r_f, 2r_f]} |(\chi_f g_f)'''(x)|, \qquad \varepsilon_f = \min\left(\frac{r_f}{4}, \frac{\alpha}{32K_{3,f}(1+C_\psi)}, \sqrt{\frac{\delta_{\mathrm{w}}}{\alpha}}\right),$$

$$\widetilde{f}(x) = g_f(x) + \psi(x/\varepsilon_f)\big(\alpha x^2/2 - \chi_f(x) g_f(x)\big) \qquad (A - c < x < B - c),$$

with $\chi_f g_f$ extended by zero outside $[-2r_f, 2r_f]$. As $2\varepsilon_f \le r_f/2$, $\widetilde{f} = g_f + \psi(x/\varepsilon_f)(\alpha x^2/2 - g_f)$ and $K_{3,f} > \max_{|x| \le 2\varepsilon_f} |g_f'''|$.

**Lemma 4.4** (Quadratic modification of a well)**.** *Let $f, c, \alpha$ be as above, and let $\varepsilon > 0$, $K > 0$, $\delta > 0$ satisfy $2\varepsilon < \min(c-A, B-c)$, $|g_f'''| \le K$ on $[-2\varepsilon, 2\varepsilon]$, $\varepsilon \le \alpha/(32K(1+C_\psi))$ and $\varepsilon \le \sqrt{\delta/\alpha}$. Put $\widetilde{f}(x) = g_f(x) + \psi(x/\varepsilon)(\alpha x^2/2 - g_f(x))$ on $(A-c, B-c)$. Then $\widetilde{f}$ is smooth and*

$$|\widetilde{f} - g_f| \le \delta/24, \qquad \widetilde{f}(x) = \alpha x^2/2 \quad (|x| \le \varepsilon), \qquad x\widetilde{f}'(x) > 0 \quad (x \ne 0), \qquad \widetilde{f} = g_f \quad (|x| \ge 2\varepsilon).$$

*The map $\Sigma_f(x) = \mathrm{sign}(x)\sqrt{\widetilde{f}(x)}$ is an increasing smooth diffeomorphism of $(A-c, B-c)$ onto $\mathbb{R}$, and its inverse $X_f$ satisfies*

$$\widetilde{f}(X_f(u)) = u^2, \qquad X_f'(u) > 0, \qquad X_f(u) = u/\sqrt{\alpha/2} \quad (|u| \le \sqrt{\alpha/2}\,\varepsilon).$$

*In particular this applies to* (20) *with* $(\varepsilon, K, \delta) = (\varepsilon_f, K_{3,f}, \delta_{\mathrm{w}})$*, and to* $(f, A, B, c) = (U, 0, h, p^*)$ *and* $(V, -\frac{1}{2}, \frac{1}{2}, 0)$*; we write* $\widetilde{U}$*,* $\widetilde{V}$ *for the two modifications.*

*Proof.* Since $g_f(0) = g_f'(0) = 0$ and $g_f''(0) = \alpha$, Taylor's theorem gives, for $|x| \le 2\varepsilon$,

$$|g_f(x) - \alpha x^2/2| \le K|x|^3/6, \qquad |g_f'(x) - \alpha x| \le Kx^2/2.$$

Hence $|\widetilde{f} - g_f| \le K(2\varepsilon)^3/6 = \frac{4}{3}K\varepsilon \cdot \varepsilon^2 \le \frac{4}{3} \cdot \frac{\alpha}{32} \cdot \frac{\delta}{\alpha} = \delta/24$. Differentiating, $\widetilde{f}'(x) - \alpha x = (1 - \psi(x/\varepsilon))(g_f'(x) - \alpha x) + \varepsilon^{-1}\psi'(x/\varepsilon)(\alpha x^2/2 - g_f(x))$, and for $0 < |x| < 2\varepsilon$ (where $|x|/\varepsilon < 2$)

$$|\widetilde{f}'(x) - \alpha x| \le Kx^2\Big(\frac{1}{2} + \frac{C_\psi}{3}\Big) \le K|x| \cdot 2\varepsilon(1 + C_\psi) \le \frac{\alpha|x|}{16}.$$

So $x\widetilde{f}'(x) > 0$ there; for $|x| \ge 2\varepsilon$, $\widetilde{f} = g_f$ and $xg_f'(x) > 0$ by hypothesis. As $\widetilde{f}(0) = 0$, $\widetilde{f} > 0$ for $x \ne 0$. On $|x| < \varepsilon$, $\Sigma_f(x) = \sqrt{\alpha/2}\,x$; on $x \ne 0$, $\Sigma_f$ is smooth with $\Sigma_f' = \widetilde{f}'/(2\Sigma_f) > 0$. Since $\widetilde{f} = g_f \to +\infty$ at both ends, $\Sigma_f$ is an increasing bijection onto $\mathbb{R}$ with positive derivative, hence a diffeomorphism; the identities for $X_f = \Sigma_f^{-1}$ follow. For (20), $2\varepsilon_f \le r_f/2$, and the hypotheses on $U$ and $V$ hold by Lemma 4.3: strict convexity gives $(x - c)f'(x) > 0$. □

Let $X_U, X_V$ be the inverses of Lemma 4.4 for $f = U, V$, with widths $\varepsilon_U, \varepsilon_V$; they map $\mathbb{R}$ onto $(-p^*, h - p^*)$, $(-\frac{1}{2}, \frac{1}{2})$. Put

$$b_{\mathrm{top}} = RF(h - \eta) + 2\sigma, \qquad \bar{E} = b_{\mathrm{top}} - m_* + 2\delta_{\mathrm{w}}, \qquad u_1 = \sqrt{\bar{E}} + 1, \tag{21}$$
$$\varphi(t) = tS(t), \qquad Z(u) = \varphi(u - u_1) - \varphi(-u - u_1),$$
$$X(u) = X_U(u) + Z(u), \qquad Y(v) = X_V(v) + Z(v), \qquad a = \sqrt{2/U''(p^*)}, \qquad b = \sqrt{2/V''(0)},$$
$$\varepsilon_{\mathrm{M}} = \min\big(\sqrt{U''(p^*)/2}\,\varepsilon_U, \sqrt{V''(0)/2}\,\varepsilon_V, 1\big).$$

By (19) with $I = h - 4\eta$, $\bar{E} > 0$. Since $\varphi$ is smooth, 0 on $(-\infty, 0]$, $t$ on $[1, \infty)$ and $\varphi' \ge 0$, $Z$ is smooth, nondecreasing, zero on $[-u_1, u_1]$ and unbounded both ways; so $X, Y$ are increasing diffeomorphisms of $\mathbb{R}$ with $X = X_U$, $Y = X_V$ on $[-u_1, u_1]$ and $X(u) = au$, $Y(v) = bv$ on $[-\varepsilon_{\mathrm{M}}, \varepsilon_{\mathrm{M}}]$.

With the normalisations of Paper I, Definition (Rotation and area normalisation) and $\mathrm{Rot}_\theta\, w = e^{i\theta}w$,

$$\bar{J}(t) = \frac{1}{2\pi}\int_0^{2\pi} J(\mathrm{Rot}_\theta\,\sqrt{t})\,d\theta, \qquad B_J(w) = \frac{1}{2\pi}\int_0^{2\pi}(\theta - \pi)J(\mathrm{Rot}_\theta\, w)\,d\theta, \tag{22}$$
$$T_J(w) = \mathrm{Rot}_{B_J(w)/\bar{J}(|w|^2)}\,w, \qquad \mathcal{A}_J(E) = \pi\int_0^E \bar{J}(t)\,dt, \qquad E_J = \mathcal{A}_J^{-1},$$
$$D_J(z) = \sqrt{\frac{\pi E_J(I)}{I}}\,z \quad (I = \pi|z|^2 > 0), \qquad D_J(0) = 0.$$

If $J$ is smooth, positive, constant near 0 and $\mathcal{A}_J \to \infty$, Paper I, Lemma (Rotation normalisation) and Paper I, Lemma (Area normalisation) show that $T_J$, $D_J$ are diffeomorphisms, linear near 0, $T_J$ preserves circles about 0 with $T_J^*(\bar{J}(|w|^2)\,du \wedge dv) = J\,du \wedge dv$, and $|D_J(z)|^2 = E_J(\pi|z|^2)$, $D_J^*(\bar{J}(|w|^2)\,du \wedge dv) = \omega_{\mathbb{C}}$. Define

$$F_{\mathrm{M}}(u + iv) = X(u) + iY(v), \qquad J(u + iv) = X'(u)Y'(v), \qquad f = F_{\mathrm{M}} \circ T_J^{-1} \circ D_J. \tag{23}$$

Then $J > 0$ is smooth, $J = ab$ for $|u|, |v| \le \varepsilon_{\mathrm{M}}$, $F_{\mathrm{M}}^*(dx \wedge dy) = J\,du \wedge dv$, and with $\varrho_X = (X^{-1})^2$, $\varrho_Y = (Y^{-1})^2$,

$$\mathcal{A}_J(E) = \mathrm{area}\{\varrho_X(x) + \varrho_Y(y) \le E\} \longrightarrow \infty, \qquad \varrho_X(\mathrm{Re}\, f(z)) + \varrho_Y(\mathrm{Im}\, f(z)) = E_J(\pi|z|^2). \tag{24}$$

Hence $f$ is an area-preserving diffeomorphism of $\mathbb{C}$ (Paper I, Lemma (Area normalisation)(iv)). Also $\varrho_X(x) < u_1^2$ iff $\widetilde{U}(x) < u_1^2$, and then $\varrho_X = \widetilde{U}$; likewise for $\varrho_Y$, $\widetilde{V}$. The endpoint is

$$P_- = p^* + X_U(-u_1), \qquad P_+ = p^* + X_U(u_1), \qquad a_{\mathrm{cut}} = P_-/4, \tag{25}$$
$$s_a(p) = p\,S((p-a)/a) + \tfrac{a}{2}\big(1 - S((p-a)/a)\big),$$
$$C_a(x+iy) = -\sqrt{s_a(p^*+x)/\pi}\,e^{2\pi i y}, \qquad \Phi = C_{a_{\mathrm{cut}}} \circ f.$$

Here $0 < P_- < p^* < P_+ < h$; $s_a = a/2$ for $p \le a$, $s_a = p$ for $p \ge 2a$, $s_a > 0$, so $C_a$ is smooth.

**Lemma 4.5** (Scalar endpoint and its error)**.** *The map $\Phi$ is smooth on $\mathbb{C}$, is injective and area preserving on $D(h-4\eta)$, and satisfies*

$$F(\pi|\Phi(z)|^2) < RF(\pi|z|^2) + e \qquad (\pi|z|^2 \le h - 4\eta). \tag{26}$$

*On the latter closed disc, $P_- < \pi|\Phi(z)|^2 < P_+$.*

*Proof.* Let $0 \le I \le h-4\eta$, $b_I = RF(I+3\eta) + 2\sigma \le b_{\mathrm{top}}$, $E' = b_I - m_* + 2\delta_{\mathrm{w}} \le \bar{E}$. By Lemma 4.4, $U(p) + V(y) \le b_I$ implies $\varrho_X(p-p^*) + \varrho_Y(y) \le E'$, so Lemma 4.3 and (24) give $\mathcal{A}_J(E') > I$, i.e.

$$E_J(I) < b_I - m_* + 2\delta_{\mathrm{w}} \le \bar{E} < u_1^2. \tag{27}$$

For $\pi|z|^2 = I$, $(x,y) = f(z)$, (24) gives $|X^{-1}(x)|, |Y^{-1}(y)| < u_1$, so $p^* + x \in (P_-, P_+)$, $|y| < \frac{1}{2}$, $\widetilde{U}(x) + \widetilde{V}(y) = E_J(I)$, hence $U(p^*+x) + V(y) < b_I + 4\delta_{\mathrm{w}}$. As $F < U$, $V \ge 0$ and $RF$ is $L$-Lipschitz (Lemma 2.6), $F(p^*+x) < RF(I) + 3L\eta + 2\sigma + 4\delta_{\mathrm{w}} \le RF(I) + \frac{23}{64}e$. Since $p^* + x > 4a_{\mathrm{cut}}$, $\pi|\Phi(z)|^2 = p^* + x$, giving (26). There $C_{a_{\mathrm{cut}}}$ is the injective polar chart $(x,y) \mapsto -\sqrt{(p^*+x)/\pi}\,e^{2\pi i y}$ pulling $\omega_{\mathbb{C}}$ back to $dx \wedge dy$ (Paper I, Lemma (The physical chart)), so $\Phi$ is injective and area preserving on $D(h-4\eta)$, and smooth on $\mathbb{C}$. □

4.4. **The Alexander path and its Hamiltonian.** The Alexander family joins the endpoint to its linearisation:

$$b_* := \Phi(0) = -r_b, \qquad r_b = \sqrt{p^*/\pi}, \qquad q_\alpha(z) = b_* + \int_0^1 D\Phi(t\alpha z)[z]\,dt \quad (0 \le \alpha \le 1), \tag{28}$$

where $\Phi(0) = -r_b$ since $f(0) = F_{\mathrm{M}}(0) = 0$ and $s_{a_{\mathrm{cut}}}(p^*) = p^*$. Thus $q_\alpha = b_* + (\Phi(\alpha z) - b_*)/\alpha$ for $\alpha > 0$, $q_1 = \Phi$, and $q_0 = b_* + A_0 z$ with $A_0 = D\Phi(0)$. Near 0 we have $J = ab$, so $T_J = \mathrm{id}$ and $D_J = (ab)^{-1/2}\mathrm{id}$ (Paper I, Lemma (Rotation normalisation)(iv) and Paper I, Lemma (Area normalisation)(v)), whence $Df(0) = \mathrm{diag}(c_x, c_y)$ with $c_x = a/\sqrt{ab}$, $c_y = b/\sqrt{ab}$. With $DC_{a_{\mathrm{cut}}}(0) = \mathrm{diag}(-1/(2\pi r_b), -2\pi r_b)$ (Paper I, Lemma (The physical chart)) this gives $A_0 = \mathrm{diag}(A_{11}, A_{22})$, $A_{11} = -c_x/(2\pi r_b)$, $A_{22} = -2\pi r_b c_y$, $A_{11}A_{22} = 1$. Let

$$p^\dagger = h - 5\eta, \qquad R^\dagger = \sqrt{p^\dagger/\pi}, \qquad R_h = \sqrt{h/\pi}, \qquad R_q = \sqrt{P_+/\pi}, \tag{29}$$
$$M_\Phi = \max_{\pi|z|^2 \le p^\dagger} \|D\Phi(z)\|, \qquad B_* = M_\Phi R^\dagger, \qquad B_3 = R^\dagger \max(1, |A_{11}|, |A_{22}|),$$
$$c_{\mathrm{sh}} = \min\left(\frac{1}{4}, \frac{R_h - r_b}{8\max(R^\dagger, B_3, B_*)}\right),$$
$$R_{\mathrm{supp}} = \max(R^\dagger, r_b + c_{\mathrm{sh}}B_3, r_b + c_{\mathrm{sh}}B_*, R_q), \qquad P_{\mathrm{supp}} = \pi R_{\mathrm{supp}}^2.$$

Put $M_\alpha = \mathrm{Rot}_{\pi\alpha}\,\mathrm{diag}(|A_{11}|^\alpha, |A_{22}|^\alpha)$; then $M_0 = \mathrm{Id}$, $M_1 = A_0$ (both entries of $A_0$ are negative), $\det M_\alpha = 1$, $\|M_\alpha\| \le \max(1, |A_{11}|, |A_{22}|)$. The five segments on $D(p^\dagger)$ are

$$\begin{aligned} &\mathfrak{e}^0_\alpha(z) = c_{\mathrm{sh}}^\alpha z, && \mathfrak{e}^1_\alpha(z) = \alpha b_* + c_{\mathrm{sh}} z,\\ &\mathfrak{e}^2_\alpha(z) = b_* + c_{\mathrm{sh}} M_\alpha z,\\ &\mathfrak{e}^3_\alpha(z) = b_* + c_{\mathrm{sh}}(q_\alpha(z) - b_*),\\ &\mathfrak{e}^4_\alpha(z) = b_* + c_{\mathrm{sh}}^{1-\alpha}(\Phi(z) - b_*). \end{aligned} \tag{30}$$

Put $\mathfrak{e}_t = \mathfrak{e}^j_{S(5t-j)}$ for $j/5 \le t \le (j+1)/5$ ($j = 0, \dots, 4$), $\mathfrak{e}_t = \mathrm{id}$ for $t \le 0$, $\mathfrak{e}_t = \Phi$ for $t \ge 1$ (Paper I, Construction (The five-segment path) with inner family $q_\alpha$).

**Lemma 4.6** (The scalar ordinary path)**.** *$(\mathfrak{e}_t)$ is an ordinary path in the sense of Paper I, Definition (Ordinary path) (with a one-point parameter space), with constants $p^\dagger$, $P_{\mathrm{supp}}$, $h$, $c_{\mathrm{sh}}$ and*

$$\gamma_t = c_{\mathrm{sh}}^{2S(5t)-2S(5t-4)}, \qquad c_{\mathrm{sh}}^2 \le \gamma_t \le 1, \qquad \gamma_1 = 1, \qquad P_{\mathrm{supp}} < h.$$

*Proof.* $q_\alpha(z)$ is smooth on $[0,1] \times D(p^\dagger)$ (smooth integrand). For $\alpha > 0$ it is $\Phi \circ (\alpha\,\cdot)$ followed by an affine bijection, so by Lemma 4.5 ($\Phi$ injective with $\det D\Phi = 1$ on $D(p^\dagger) \subset D(h - 4\eta)$) it is injective with $\det Dq_\alpha = 1$; $q_0$ is affine with $\det A_0 = 1$; and $|q_\alpha(z) - b_*| \le M_\Phi |z| \le B_*$.

(Paper I, condition (O1)): the segments agree at the seams ($M_0 = \mathrm{Id}$, $M_1 = A_0$, $q_0 - b_* = A_0 z$, $\mathfrak{e}^4_1 = \Phi$) and all derivatives of $S$ vanish at $0, 1$, so Paper I, Lemma (Flat concatenation) applies. (Paper I, condition (O2)): the segments are injective with constant Jacobians $c_{\mathrm{sh}}^{2\alpha}, c_{\mathrm{sh}}^2, c_{\mathrm{sh}}^2, c_{\mathrm{sh}}^2, c_{\mathrm{sh}}^{2(1-\alpha)}$, i.e. $\gamma_t$. (Paper I, condition (O3)): for $|z| < R^\dagger$ the images of $\mathfrak{e}^0, \dots, \mathfrak{e}^3$ have modulus at most $R^\dagger$, $r_b + c_{\mathrm{sh}} B_3$, $r_b + c_{\mathrm{sh}} B_3$, $r_b + c_{\mathrm{sh}} B_*$, and $\mathfrak{e}^4_\alpha(z)$ is a convex combination of $b_*$ and $\Phi(z)$, both of modulus $\le R_q$ (Lemma 4.5, $p^* < P_+$). Finally $R^\dagger, R_q < R_h$ (as $P_+ < h$) and $r_b + c_{\mathrm{sh}} \max(B_3, B_*) \le r_b + \frac{1}{8}(R_h - r_b) < R_h$ (as $p^* < h$), so $P_{\mathrm{supp}} < h$. □

The ambient realiser is that of Paper I, § "The cut-off ambient flow" and Paper I, § "The Hamiltonian". With $e_q = 2\eta$ put

$$\begin{aligned} &p_{\mathrm{on}} = h - 31e_q/8, \qquad p_{\mathrm{zero}} = h - 61e_q/16, \qquad \chi(I) = 1 - S\!\left(\frac{I - p_{\mathrm{on}}}{p_{\mathrm{zero}} - p_{\mathrm{on}}}\right),\\ &W_t(y) = \chi(\pi|\mathfrak{e}_t^{-1}(y)|^2)\,\partial_t\mathfrak{e}_t(\mathfrak{e}_t^{-1}(y)) \quad \text{on } \mathfrak{e}_t(D(p^\dagger)), \qquad W_t = 0 \quad \text{elsewhere}, \end{aligned} \tag{31}$$

with $\mathfrak{e}_t^{-1}(y)$ the unique preimage in $D(p^\dagger)$. Since $h - 4e_q < p_{\mathrm{on}} < p_{\mathrm{zero}} < h - 15e_q/4 < p^\dagger$, these are the field and cutoff of Paper I, Lemma (The ambient flow). Let $\phi_t$ be the flow of $W_t$, $\phi_0 = \mathrm{id}$, and

$$G_t(w) = \phi_t(w/\sqrt{\gamma_t}), \qquad \widehat{J}_t = \det DG_t, \qquad \widehat{q}_t = G_t \circ T^{-1}_{\widehat{J}_t} \circ D_{\widehat{J}_t}, \tag{32}$$

with the normalisations (22) (the $\hat{q}$ of Paper I, Lemma (Normalisation of the whole path)). As in Paper I, Construction (The Hamiltonian),

$$\begin{aligned} &a^q_t(z)[v] = \omega_{\mathbb{C}}\big(\partial_t\widehat{q}_t(z), D\widehat{q}_t(z)[v]\big), \qquad h^q_t(z) = \int_0^1 a^q_t(rz)[z]\,dr,\\ &z_{\mathrm{bd}} = \sqrt{(h + P_{\mathrm{supp}})/(2\pi)}, \qquad Q_t(y) = -h^q_t(\widehat{q}_t^{-1}(y)) + h^q_t(z_{\mathrm{bd}}). \end{aligned} \tag{33}$$

**Proposition 4.7** (Scalar Hamiltonian realisation)**.** *The maps $\widehat{q}_t$ ($t \in \mathbb{R}$) are symplectic diffeomorphisms of $\mathbb{C}$, smooth in $(t, z)$ with smooth inverses; $\widehat{q}_t = \mathrm{id}$ for $t \le 0$ and $\widehat{q}_t = \widehat{q}_1$ for $t \ge 1$; each $\widehat{q}_t$ is the identity on $\{\pi|z|^2 > P_{\mathrm{supp}}\}$ and maps $\bar{D}(h)$ onto itself. The function $Q$ of (33) is*

*smooth on $\mathbb{R}\times\mathbb{C}$, vanishes for $t\notin(0,1)$ and on $\{\pi|y|^2>P_{\mathrm{supp}}\}$, and generates $\widehat{q}$: $\partial_t\widehat{q}_t=\Xi_t\circ\widehat{q}_t$ with $\iota_{\Xi_t}\omega_{\mathbb{C}}=-dQ_t$. The endpoint satisfies*

$$F(\pi|\widehat{q}_1(z)|^2)\le RF(\pi|z|^2)+e\qquad(\pi|z|^2\le h). \tag{34}$$

*Proof.* By Lemma 4.6, Paper I, Lemma (The ambient flow), Paper I, Lemma (Normalisation of the whole path) and Paper I, Proposition ($Q$ generates $\hat{q}$) apply with levels $p_{\mathrm{on}}<p_{\mathrm{zero}}<p^{\dagger}$, giving everything except (34), and $\widehat{q}_1=\Phi$ on $\bar{D}(p_{\mathrm{on}})\supset\bar{D}(h-8\eta)$ (Paper I, Lemma (Normalisation of the whole path)(f)). For $I=\pi|z|^2\le h-8\eta$, (34) is Lemma 4.5. For $h-8\eta<I\le h$, $\pi|\widehat{q}_1(z)|^2\in[0,h]$, and since $RF(h)=\max F$ and $RF$ is $L$-Lipschitz, $F(\pi|\widehat{q}_1(z)|^2)\le RF(h)\le RF(I)+8L\eta\le RF(I)+e/8$. □

4.5. **Substitution of the cubic parameters.** Let $k\ge1$, $0<\rho_i<1/3$, $\sum_i\rho_i^2<1$. The auxiliary quantities of Lemma 5.1 are

(35)

$$D_\rho=1-\sum_i\rho_i^2,\quad S_\rho=\sum_i\rho_i,\quad \rho_{\max}=\max_i\rho_i,\qquad \delta_a=\min\left(\frac{1/3-\rho_{\max}}{8},\frac{D_\rho}{8(S_\rho+k)},\frac{1}{8}\right),$$

$$a_i=\rho_i+\delta_a,\qquad a_{\max}=\rho_{\max}+\delta_a,\qquad D_a=1-\sum_i a_i^2,\qquad \varsigma=\min\left(\frac{1/3-a_{\max}}{2},\frac{D_a}{12}\right),$$

$$h=\frac{1}{3}-\varsigma,\qquad \tau=\frac{a_{\max}+h}{2h},\qquad H_{\mathrm{b}}(p)=3(1-3p),\qquad P_{\mathrm{b}}(p)=H_{\mathrm{b}}(p)-\sum_i(a_i-p)_+,$$

$$\beta=3(1-3h),\qquad \mu=\frac{1-\sum_i a_i^2-(1-3h)^2}{2h},\qquad d=\tfrac12\min(\mu,\beta),\qquad d_n=\tfrac12(h+1/3).$$

For (9) take

$$P=P_{\mathrm{b}},\qquad P_{\min}=-\sum_i a_i,\qquad \bar{P}=3+\sum_i a_i,\qquad L_P=9+k, \tag{36}$$

construct $K$ by (11) and (15), and set

$$F(p)=K(p)-P_{\mathrm{b}}(p),\qquad e=d/16,\qquad L=M+9+k,\qquad \varepsilon_{\mathrm{cap}}=d/(128k), \tag{37}$$

$$U_0(p)=K(p)-H_{\mathrm{b}}(p)+\sum_i\frac{a_i-p+\sqrt{(a_i-p)^2+\varepsilon_{\mathrm{cap}}^2}}{2},\qquad \sigma_{\mathrm{in}}=d/256.$$

Let $\widehat{q}$, $Q$ be (32), (33) for these data, and $g=3d/4$.

**Proposition 4.8** (The cubic comparison datum)**.** *For every $k\ge1$ and all $\rho_1,\dots,\rho_k\in(0,\frac13)$ with $\sum_i\rho_i^2<1$, the data* (35)–(37) *satisfy the hypotheses* (9) *of Lemma 4.1 and* (17) *of §4.2, and*

$$K(I)-F(\pi|\widehat{q}_1(z)|^2)\ge g\ge\tfrac{31}{32}g\qquad(I=\pi|z|^2\le h). \tag{38}$$

*Consequently $(k,\rho_i,a_i,h,d_n,g,K,\widehat{q},Q)$ are valid cubic data in the sense of Paper I, Definition (Cubic data).*

*Proof. Auxiliary capacities.* By $\delta_a\le\frac18$ and $\delta_a\le D_\rho/(8(S_\rho+k))$, $\sum_i a_i^2-\sum_i\rho_i^2=2\delta_aS_\rho+k\delta_a^2\le D_\rho/4$, so $0<\frac34D_\rho\le D_a\le1$. As $a_{\max}<\frac13$: $\varsigma>0$, $a_{\max}<h<\frac13$, $0<\tau<1$, $\tau h>a_{\max}$; also $(1-3h)^2=9\varsigma^2\le D_a^2/16<D_a$.

*Hypotheses* (9). From $\int_0^h H_{\mathrm{b}}=\frac12[1-(1-3h)^2]$ and $\int_0^h(a_i-p)_+\,dp=\frac12a_i^2$, $\int_0^h P_{\mathrm{b}}=\frac12[D_a-(1-3h)^2]=\mu h>0$. For $p\ge\tau h>a_{\max}$ the caps vanish, so $P_{\mathrm{b}}(p)=H_{\mathrm{b}}(p)\ge\beta>0$; hence

$0 < d < \min(\mu, \beta)$. $P_{\mathrm{b}}$ is continuous, concave, $(9+k)$-Lipschitz, and $-\sum_i a_i \le P_{\mathrm{b}} \le 3$ on $[0,h]$ (as $H_{\mathrm{b}} > 0$), giving $P_{\min}, \bar{P}, L_P$.

*Hypotheses* (17). By Proposition 4.2, $K$ is smooth, convex, $M$-Lipschitz, so $F = K - H_{\mathrm{b}} + \sum_i (a_i - p)_+$ is convex and $L$-Lipschitz on $[0,h]$. The soft cap $s_\varepsilon(x) = \frac{1}{2}(x + \sqrt{x^2+\varepsilon^2})$ is smooth, convex, with $0 \le s_\varepsilon(x) - x_+ \le \varepsilon/2$; so $U_0 = K - H_{\mathrm{b}} + \sum_i s_{\varepsilon_{\mathrm{cap}}}(a_i - p)$ is smooth, convex, and $F \le U_0 \le F + k\varepsilon_{\mathrm{cap}}/2 = F + d/256$, with $\sigma_{\mathrm{in}} = d/256 = e/16$.

*Gap.* By (34) and (16), $F(\pi|\widehat{q}_1(z)|^2) \le RF(I) + \frac{d}{16} \le K(I) - \frac{13d}{16} + \frac{d}{16} = K(I) - g$.

*Validity.* Paper I, Definition (Cubic data)(a): $0 < \rho_i < a_i \le a_{\max} < h < d_n < \frac{1}{3}$ and $0 < g < \beta = H_{\mathrm{b}}(h)$; (b) is Proposition 4.7 with $P_{\mathrm{supp}} < h$; (c) is (38). □

The handle construction of Paper I, § "The four-dimensional handle" uses $e_h = g/16$, $H^* = H_{\mathrm{b}} - e_h$, $\mu_h = g/(16k)$, $J = K - H^*$, $h_i(p) = \frac{1}{2}\big(a_i - p + \sqrt{(a_i-p)^2 + \mu_h^2}\big)$; the width $\mu_h$ is distinct from $\varepsilon_{\mathrm{cap}}$. The source capacities of the general cubic embeddings are

$$\bar{\rho}_i = \frac{\rho_i + a_i}{2}, \qquad \delta_i = a_i - \bar{\rho}_i, \qquad \hat{\rho}_i = \bar{\rho}_i + \frac{\delta_i}{4}, \tag{39}$$

so $\rho_i < \bar{\rho}_i < \hat{\rho}_i < a_i$. With the data of Proposition 4.8, the maps $f_i^{\mathrm{cap}} = b_{\mathrm{ball}} \circ \mathcal{T}_{\mathrm{u}} \circ \mathsf{N} \circ f_i^N$ of Paper I, Remark (The four-dimensional embeddings) are smooth symplectic embeddings $B^\circ(\hat{\rho}_i) \to B^\circ(1)$ with pairwise disjoint images (Paper I, Proposition (The four-dimensional embeddings into the cover), Paper I, Theorem (Transfer on the cover) and Paper I, Theorem (Packing in dimension four)). For $C > 0$ and $c_i$ with $3c_i < C$, $\sum_i c_i^2 < C^2$, put $\rho_i = c_i/C$ and

$$f_i(u) = \sqrt{C}\, f_i^{\mathrm{cap}}(u/\sqrt{C}), \qquad c_i' = C\bar{\rho}_i, \qquad \bar{c}_i = C\hat{\rho}_i. \tag{40}$$

Since $u \mapsto u/\sqrt{C}$ maps $B^\circ(\bar{c}_i)$ onto $B^\circ(\hat{\rho}_i)$ and multiplies $\omega_0$ by $1/C$, the $f_i$ are smooth symplectic embeddings $B^\circ(\bar{c}_i) \to B^\circ(C)$ with pairwise disjoint images (so the $f_i(B(c_i'))$ are disjoint compacta), and $c_i < c_i' < \bar{c}_i$, as required in Definition M2.13.

## 5. Dimension four: general capacities

Paper I proves the packing in the cubic model for valid cubic data (Paper I, Theorem (Packing in dimension four)) and supplies such data for $k \ge 10$ equal capacities with $\frac{3}{4} < \rho\sqrt{k} < 1$. The scalar comparison datum supplies them for all other admissible capacities. We record the auxiliary capacities, the resulting theorem and the obstruction showing that the condition $3\rho_i < 1$ cannot be replaced by the volume and two-ball inequalities.

We enlarge each $\rho_i$ by a slack $\delta$ to an auxiliary capacity $a_i$ and choose $h$ so that the budget $\int_0^h P_{\mathrm{b}}$ left after subtracting the enlarged balls (total volume $\frac{1}{2}\sum_i a_i^2$) from the integrated profile $\frac{1}{2} - \frac{1}{2}(1-3h)^2$ (Paper I, Lemma (Integrated profile)) stays positive. As $\sum_i \rho_i^2 \uparrow 1$, $\delta$, $\frac{1}{3} - h$ and the mean of $P_{\mathrm{b}}$ tend to zero.

**Lemma 5.1** (Auxiliary capacities)**.** *Let $k \ge 1$ and $0 < \rho_1, \dots, \rho_k < \frac{1}{3}$ with $\sum_i \rho_i^2 < 1$. Put $S_\rho = \sum_i \rho_i$, $D = 1 - \sum_i \rho_i^2$, $\varpi = \frac{1}{3} - \max_i \rho_i$,*

$$\delta = \min\Big(\frac{\varpi}{8},\ \frac{D}{8(S_\rho + k)},\ \frac{1}{8}\Big), \qquad a_i = \rho_i + \delta, \qquad D_a = 1 - \sum_i a_i^2,$$

$$\varsigma = \min\Big(\frac{\frac{1}{3} - a_{\max}}{2},\ \frac{D_a}{12}\Big), \qquad h = \frac{1}{3} - \varsigma, \qquad \tau_\Delta = \frac{a_{\max} + h}{2h},$$

*with $a_{\max} = \max_i a_i$, and let $P_{\mathrm{b}}(p) = H_{\mathrm{b}}(p) - \sum_i (a_i - p)_+$ on $\Delta = [0,h]$. Then:*

*(i) $\rho_i < a_i \le a_{\max} < \tau_\Delta h < h < \frac{1}{3}$, and $0 < \tau_\Delta < 1$;*

*(ii) $\sum_i a_i^2 + (1-3h)^2 < 1$;*

*(iii)* $P_{\mathrm{b}}$ *is continuous, concave and piecewise affine with breakpoints in* $\{a_1,\dots,a_k\}$*, and*

$$\int_0^h P_{\mathrm{b}}(p)\,dp = \frac{1-\sum_i a_i^2-(1-3h)^2}{2} > 0;$$

*(iv)* $P_{\mathrm{b}}(p) = H_{\mathrm{b}}(p) \geq 3(1-3h) > 0$ *for* $p \in [\tau_\Delta h, h]$.

*If the* $\rho_i$ *are rational, so are* $\delta$*,* $a_i$*,* $\varsigma$*,* $h$*,* $\tau_\Delta$ *and the breakpoints and values of* $P_{\mathrm{b}}$*.*

*Proof.* (ii) As $\delta \leq \frac{1}{8}$, $\sum_i a_i^2 - \sum_i \rho_i^2 = 2S_\rho\delta + k\delta^2 \leq 2\delta(S_\rho + k) \leq D/4$, so $0 < \frac{3}{4}D \leq D_a \leq 1$ and $(1-3h)^2 = 9\varsigma^2 \leq D_a^2/16 < D_a$.

(i) $a_i > \rho_i$ since $\delta > 0$, and $a_{\max} \leq \max_i \rho_i + \varpi/8 < \frac{1}{3}$, so $\varsigma > 0$, $h < \frac{1}{3}$ and $h \geq \frac{1}{2}(\frac{1}{3} + a_{\max}) > a_{\max}$. Hence $\tau_\Delta h = \frac{1}{2}(a_{\max} + h)$ lies strictly between $a_{\max}$ and $h$, and $0 < \tau_\Delta < 1$.

(iii) $H_{\mathrm{b}}$ is affine and each $-(a_i - p)_+$ is concave, piecewise affine with breakpoint $a_i < h$ and $\int_0^h (a_i - p)_+\,dp = a_i^2/2$; also $\int_0^h H_{\mathrm{b}} = \frac{1}{2}\big(1-(1-3h)^2\big)$ by Paper I, Lemma (Integrated profile). Positivity is (ii).

(iv) For $p \geq \tau_\Delta h > a_{\max}$ all $(a_i - p)_+$ vanish and $H_{\mathrm{b}}(p) = 3(1-3p) \geq 3(1-3h)$. Rationality is clear. □

**Theorem 5.2** (Packing in dimension four)**.** *Let* $k \geq 1$ *and* $0 < \rho_1, \dots, \rho_k < \frac{1}{3}$ *with* $\sum_i \rho_i^2 < 1$*. Then there are explicit numbers* $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}}$ *and smooth maps* $f_i \colon B^\circ(\rho_i^{\mathrm{dom}}) \to \mathbb{C}^2$ *with symplectic derivative, injective on* $B(\rho_i^+)$*, such that the sets* $f_i(B(\rho_i^+))$ *are pairwise disjoint and contained in* $B(1-\mu_{\mathrm{ball}})$ *for some* $\mu_{\mathrm{ball}} \in (0,1)$*. In the linear branch (*$k \geq 10$*,* $\rho_i = \rho$*,* $\frac{3}{4} < \rho\sqrt{k} < 1$*) the maps and capacities are those of Paper I; otherwise the maps are those of Paper I, Construction (Data of the four-dimensional packing) for the valid cubic data of Proposition 4.8, and* $(\rho_i^+, \rho_i^{\mathrm{dom}}) = (\bar\rho_i, \hat\rho_i)$ *as in* (39)*.*

*Proof.* In the linear branch the data of the equal case of Paper I, Construction (Data of the four-dimensional packing) are valid cubic data with $d_n = \frac{1}{2}(h + \frac{1}{3})$ (Paper I, Remark (The equal-capacity branch)). Otherwise the data (35)–(37), whose auxiliary capacities are those of Lemma 5.1, are valid cubic data with $d_n = \frac{1}{2}(h + \frac{1}{3})$ by Proposition 4.8, and $\rho_i < \bar\rho_i < \hat\rho_i$ by (39). In both cases Paper I, Theorem (Packing in dimension four) applies, with the recorded capacities of Paper I, Remark (Recorded capacities). □

For $k = 9$ the formulas of the linear branch would give $a = t_0/3 = h$, violating $a < h$; so $k = 9$ always uses the scalar datum.

*Remark* 5.3 (Obstructions). The condition $3\rho_i < 1$ of Theorem 5.2 is not necessary (one ball of capacity $\rho < 1$ embeds by inclusion), but it cannot be replaced by the volume and two-ball inequalities: five balls of capacity $\frac{11}{25} > \frac{1}{3}$ and five of capacity $\frac{1}{100}$ satisfy $\sum_i \rho_i^2 = \frac{1937}{2000} < 1$ and $\rho_i + \rho_j \leq \frac{22}{25} < 1$, yet by the disjoint-union property of ECH capacities [Hut11] the fifth ECH capacity of their union is at least $5 \cdot \frac{11}{25} = \frac{11}{5}$, while that of $B^\circ(1)$ is 2; so they do not embed.

## 6. The bordered handle construction

This section gives the handle construction for the higher-dimensional model inputs (Definition M2.3) and proves the packing theorem in a model. It is an explicit version of the surface packing lemma of [Ope26, §3.2]. Beyond the general construction, it supplies the finite primitive, the handle data of the diagonal quadric, and the nondegeneracy and flow statements; the bump forms, tree primitives, normalisation, Moser fields and the maps $\psi_i$ are fixed in §8, the layer maps in §7. The periodic four-dimensional instance is Paper I, § "The four-dimensional handle".

6.1. **The handle construction.** We give the handle construction for $n \geq 3$, $m = n-1$, in the form used by Theorem M2.6; it follows the outline of Paper I, § "Packing over a handle", whose twist and suspension lemma are recalled below. Write $\Omega = \omega_V + d\Gamma_{\mathrm{phys}}$ on the explicit affine product, with compact model region $V_\Delta(E) = \Sigma_\ell \times V_\Delta$, physical base curvature $\kappa(p)$ and profile $H_{\mathrm{b}}(p) = \int_{\Sigma_\ell} \kappa(p)$. The comparison datum has gap $\gamma$ and reserve $0 < \gamma_* \leq \gamma$. A horizontal one-form annihilates fibre tangent vectors; a family of base one-forms $\alpha(p)$ depending smoothly on $p$ gives a toric horizontal form, with differential $d_\Sigma \alpha(p) + \sum_j dp_j \wedge \partial_j \alpha(p)$.

*The scalar budget.* For a small-ball input, with $\mathcal{B}_\Delta$ as in (M1), the cap identity gives

$$\frac{1 - \sum_i \rho_i^n}{n!} = \left(\frac{1}{n!} - \mathcal{B}_\Delta\right) + \int_\Delta P_{\mathrm{b}} + \sum_i \frac{a_i^n - \rho_i^n}{n!}. \tag{41}$$

The model parameters make the sum of the first and last terms on the right smaller than the left side, so $\int_\Delta P_{\mathrm{b}} > 0$. In the quadric input the reference-area tail controls the difference from the limiting affine profile; the six-dimensional input is in §11.2. The comparison uses a minorant with positive mean that is positive near the outer faces. Its reserve pays for the background and collar mass, and $a_i - \rho_i$ pays for the strip and the source neighbourhood. For a large-ball input the limiting budget for the remaining balls is $(1 - a^n)/n!$, where $a$ is the reserved capacity; the parameters are in §11. In all cases the transfer is applied to the already constructed compact family of model images.

**Definition 6.1** (Handle data)**.** Each higher-dimensional model supplies the following data.

(a) An explicit annular coordinate map on $|s_{\mathrm{a}}| < w$, with working band $|s_{\mathrm{a}}| \leq b < w$. We use $s = (s_{\mathrm{a}} + b)/(2b)$, so the working band is $[0, 1]$. For the diagonal-quadric presentation the model uses $w = 1/8$ and $b = 1/16$ inside a double cylinder of half-width $1/4$ on which the annular coordinate is defined.
(b) A fixed compact support set $\mathcal{B} \subset \operatorname{int} \Sigma_\ell$.
(c) A smooth closed one-form $\beta$, supported in $\mathcal{B}$, whose annular expression is $b(v)\, dv$, with $b \geq 0$, $b > 0$ on $(v_-, v_+)$ and $\int_0^1 b(v)\, dv = 1$.
(d) An explicit reference primitive $\Gamma_{\mathrm{ref}}$ with positive curvature $\Lambda = d_\Sigma \Gamma_{\mathrm{ref}}$, a reference-area upper bound $A_{\mathrm{ref}} > 0$, and a positive physical density floor $\kappa_-$ relative to $\Lambda$ on the correction region: $\kappa(p) \geq \kappa_- \Lambda$ for $p \in \Delta$.
(e) A finite coordinate-box presentation of that region: explicit chart maps and inverses, partition weights, unit-mass coordinate bumps, and a rooted tree on the boxes in which every box other than the root has an overlapping parent of smaller index. Rectangular integration along this tree defines the operator $\mathcal{P}$ of §8.
(f) The product fibre coordinates of the affine model.
(g) A collar cutoff $\chi_\partial$, equal to one near the boundary and zero on the inner region containing the annulus, together with the model's proved tail bound. A second, larger finite atlas covers the whole closed model region for the Moser flows.

The fixed annulus and dual form are constructed in §10; the finite atlas and tree are in §8. The annular gauge is identically one on $|s_{\mathrm{a}}| < b_1 = (b + w)/2$. We write $\mathcal{A}^+$ for this open band and $\mathcal{A}$ for the smaller open band $|s_{\mathrm{a}}| < (b + b_1)/2$; both contain the working band $0 \leq s \leq 1$ in their interior, and the clock $f(s) = S(s)$ is constant on collars of the boundary of $\mathcal{A}$. Put $\eta = df \wedge \beta$ and write $\Lambda = a_\Lambda(s, v)\, ds \wedge dv$, $a_\Lambda > 0$.

**Lemma 6.2** (Top power; Paper I)**.** *(a) In local product coordinates let $\kappa'$ be a base two-form and $a'_1, \ldots, a'_m$ base one-forms, with coefficients depending on the base point and on $p$. Then $(\omega_V + \kappa' + \sum_j dp_j \wedge a'_j)^n = n\, \omega_V^m \wedge \kappa'$.*

*(b) On $E|_{\mathcal{A}^+}$, in the explicit product coordinates, let $\sigma_\Sigma = a_\Sigma(s,v)\,ds\wedge dv$ be positive, let $G(s,w)$ be smooth and independent of $v$, and $e>0$. Then $\big(\omega_V + e\sigma_\Sigma + d(G\beta)\big)^n = n\,(ea_\Sigma(s,v) + b(v)\,\partial_s G)\,\omega_V^m \wedge ds\wedge dv$.*

*In particular the forms in (a) are nondegenerate where $\kappa'>0$; when closed, they are symplectic there. The forms in (b) are symplectic where $\partial_s G \geq 0$.*

6.1.1. *The first path.* Set

$$e_\Lambda = \min\left(\frac{\gamma_*}{32A_{\mathrm{ref}}}, \frac{\kappa_-}{2}\right) > 0, \qquad \epsilon_0 = \frac{\gamma_*}{16}. \tag{42}$$

The model tail estimate and the reference-area bound give

$$\int_{\Sigma_\ell} \chi_\partial \kappa(p) \leq \frac{\gamma_*}{32}, \qquad e_\Lambda \int_{\Sigma_\ell} \Lambda \leq \frac{\gamma_*}{32} \qquad (p\in\Delta). \tag{43}$$

Apply the finite primitive operator $\mathcal{P}$ of §8, a finite sum of rectangular integrals over the boxes corrected along the tree so that $d\mathcal{P}\omega = \omega$ for every two-form $\omega$ of total mass zero, to $(1-\chi_\partial)(\kappa(p) - e_\Lambda\Lambda)$, then subtract its annular period and an exact annular gauge. The resulting primitive $\Gamma_n$ is the formula in Construction 8.2. It has

$$\begin{aligned} &\kappa_{\mathrm{sm}}(p) = \chi_\partial\kappa(p) + (1-\chi_\partial)e_\Lambda\Lambda, \qquad \epsilon_{\mathrm{sm}}(p) = \int_{\Sigma_\ell}\kappa_{\mathrm{sm}}(p),\\ &H^*(p) = H_{\mathrm{b}}(p) - \epsilon_{\mathrm{sm}}(p), \qquad d_\Sigma\Gamma_n(p) = \kappa_{\mathrm{sm}}(p) + H^*(p)\eta. \end{aligned} \tag{44}$$

The mass equality is proved by calibration with the root bump (Lemma 8.1). Since $H^*(p) = \int_{\Sigma_\ell}(1-\chi_\partial)(\kappa(p) - e_\Lambda\Lambda)$ and $\kappa(p) \geq \kappa_-\Lambda \geq 2e_\Lambda\Lambda$ on the correction region, $H^* \geq 0$ on $\Delta$; and by (43)

$$0 \leq \epsilon_{\mathrm{sm}} \leq \epsilon_0 \leq \gamma/16, \qquad H^* \geq H_{\mathrm{b}} - \gamma_*/16. \tag{45}$$

On the annulus the period and gauge corrections give $\Gamma_n = e_\Lambda\Gamma_{\mathrm{ref}} + f(s)H^*(p)\beta$ on an open neighbourhood of the working band. Define

$$\alpha_1 = \Gamma_n - \Gamma_{\mathrm{phys}}. \tag{46}$$

**Proposition 6.3** (First path)**.** *The form $\Omega_\lambda = \Omega + \lambda d\alpha_1$, $0\leq\lambda\leq 1$, is symplectic on a neighbourhood of the model region. Its base curvature is the positive convex combination of $\kappa$ and $d_\Sigma\Gamma_n$, and on the annulus its endpoint is*

$$\Omega_1 = \omega_V + e_\Lambda\Lambda + d\big(f(s)H^*(p)\beta\big).$$

*The variation primitive is horizontal and toric, is smooth in the moment variables, and vanishes near the outer base boundary.*

*Proof.* The base curvature and annular normal form follow from the normalisation identities. Toric mixed terms do not affect the top power (Lemma 6.2(a)), so the positive convex combination is nondegenerate. The difference $\Gamma_n - \Gamma_{\mathrm{phys}}$ is compactly supported in the interior of the base. Smoothness in the moment variables follows from the explicit primitive formulas. □

6.1.2. *The twist.* On a collar of each outer face the generators $Q_t$ of the comparator Poisson-commute with $\mu_\nu$, so $q_t(V_\Delta) = V_\Delta$, $q_t(\operatorname{int} V_\Delta) = \operatorname{int} V_\Delta$, and $q_t$ maps each outer face into itself. We write $q = q_1$. For $\lambda\in[0,1]$ define on $E|_{\mathcal{A}}$

$$\mathcal{D}_\lambda(s,v,w) = \big(s, v, q_{\lambda f(s)}(w)\big), \qquad Q_s^\lambda = \lambda f'(s)\,Q_{\lambda f(s)}, \qquad G_\lambda(s,w) = f(s)\,H^*\big(p(q_\lambda^{-1}(w))\big), \tag{47}$$

and $\mathcal{D} = \mathcal{D}_1$; then $\partial_s q_{\lambda f(s)} = X_{Q_s^\lambda}\circ q_{\lambda f(s)}$.

**Lemma 6.4** (Suspension; Paper I)**.** *On $E|_{\mathcal{A}}$, for $\lambda\in[0,1]$:*

*(i)* $\mathcal{D}_\lambda^*\omega_V = \omega_V + d\big[(Q_s^\lambda \circ q_{\lambda f(s)})\,ds\big]$;
*(ii)* *with the horizontal one-form*

$$\eta_\lambda = (Q_s^\lambda \circ q_{\lambda f(s)})\,ds + f(s)\Big[H^*\big(p(q_\lambda^{-1} q_{\lambda f(s)}(w))\big) - H^*\big(p(w)\big)\Big]\beta,$$

*one has* $\mathcal{D}_\lambda^*\big(\omega_V + e_\Lambda\Lambda + d(G_\lambda\beta)\big) = \omega_V + e_\Lambda\Lambda + d\big(f(s)H^*(p)\beta\big) + d\eta_\lambda$;
*(iii)* $\eta_0 = 0$, $\eta_\lambda$ *is smooth in* $(\lambda, s, v, w)$, *and* $\eta_\lambda = 0$ *for* $s \notin (0,1)$. *Hence* $\eta_\lambda$ *extends by zero to a horizontal one-form on a neighbourhood of* $V_\Delta(E)$, *supported over* $\mathcal{A}$, *and* $\Omega_1 + d\eta_\lambda = \mathcal{D}_\lambda^*\big(\omega_V + e_\Lambda\Lambda + d(G_\lambda\beta)\big)$ *on* $E|_{\mathcal{A}}$.

The second path is $\Omega_\lambda^{(1)} = \Omega_1 + d\eta_\lambda$; by Lemma 6.2(b) and Lemma 6.13 it is an exact symplectic path on a neighbourhood of $V_\Delta(E)$ (Proposition 6.15). The map $\mathcal{D}_\lambda$ itself is defined only over $\mathcal{A}$; only the forms are glued to $E$.

6.1.3. *Layers and the Moser map.* Fix source capacities $\rho_i^\sharp < a_i$ and put $\sigma_i = (a_i - \rho_i^\sharp)/2 > 0$. Define the smooth caps and the collar coefficient by

$$h_i(p) = (a_i - |p|)S\big((a_i - |p|)/\sigma_i\big) \quad (1 \le i \le k), \qquad J = K - H^* = K - H_{\mathrm{b}} + \epsilon_{\mathrm{sm}}. \tag{48}$$

Put $N = k+1$ and $D_s = 1/(3N+1)$. For $j = 1, \dots, N$ set $l_j = (3j-2)D_s$, $r_j = (3j-1)D_s$, $I_j = [l_j, r_j]$ and $\varsigma_j(s) = S\big(2(s - l_j - D_s/4)/D_s\big)$; the last index $j = N$ is denoted by $*$. The intervals are ordered and disjoint in $(0,1)$, and each clock is constant on collars of its interval endpoints, with $\varsigma_j' \ge 0$.

**Proposition 6.5** (Layers)**.** *Under the hypotheses of Theorem M2.6, with* $\epsilon_{\mathrm{sm}}$ *from* (44) *and* $h_i, J$ *from* (48)*:*

*(i)* *each* $h_i$ *is smooth on* $\mathbb{R}^m$, *and* $0 \le h_i(p) \le (a_i - |p|)_+$ *and* $(a_i + \rho_i^\sharp)/2 - |p| \le h_i(p)$; *it vanishes for* $|p| \ge a_i$ *and equals* $a_i - |p|$ *for* $|p| \le (a_i + \rho_i^\sharp)/2$;
*(ii)* *the residual* $R^*(w) = K\big(p(q^{-1}w)\big) - J\big(p(w)\big) - \sum_{i=1}^k h_i\big(p(w)\big)$ *satisfies, for* $w \in V_\Delta$,

$$R^*(w) \;\ge\; \gamma - \sup_\Delta \epsilon_{\mathrm{sm}} \;\ge\; \frac{15\gamma}{16} > 0;$$

*(iii)* *the final coefficient* $G_{\mathrm{f}}(s,w) = J(p(w)) + \sum_{i=1}^k \varsigma_i(s)h_i(p(w)) + \varsigma_*(s)R^*(w)$ *satisfies* $\partial_s G_{\mathrm{f}} \ge 0$ *on* $\mathcal{A} \times V_\Delta$; $G_{\mathrm{f}} = J \circ p$ *for* $s \le l_1$; $G_{\mathrm{f}} = K \circ p \circ q^{-1}$ *for* $s \ge r_*$; *and on* $I_i \times V_\Delta$, $1 \le i \le k$,

$$G_{\mathrm{f}}(s,w) = A_i(p(w)) + \varsigma_i(s)h_i(p(w)), \qquad A_i = J + \sum_{j<i} h_j.$$

*Proof.* (i) Write $x = a_i - |p|$, so $h_i = x\,S(x/\sigma_i)$ is a smooth function of $p$, and $(a_i + \rho_i^\sharp)/2 - |p| = x - \sigma_i$. Recall $0 \le S \le 1$, $S = 0$ on $(-\infty, 0]$ and $S = 1$ on $[1, \infty)$. If $x \le 0$, then $h_i = 0 = x_+$ and $x - \sigma_i < 0$. If $0 < x < \sigma_i$, then $0 \le h_i \le x = x_+$ and $x - \sigma_i < 0 \le h_i$. If $x \ge \sigma_i$, then $h_i = x = x_+$. This gives all claims of (i).

(ii) For $w \in V_\Delta$, put $z = q^{-1}(w)$. By $J = K - H_{\mathrm{b}} + \epsilon_{\mathrm{sm}}$, the hypothesis $H_{\mathrm{b}} - \sum_i (a_i - |p|)_+ \ge P$ and (i), $R^*(w) \ge K(p(z)) - (K - P)(p(qz)) - \sup_\Delta \epsilon_{\mathrm{sm}} \ge \gamma - \sup_\Delta \epsilon_{\mathrm{sm}}$ by Definition M2.4 at $z$. Finally $\epsilon_{\mathrm{sm}} \le \gamma/16$ by (45).

(iii) $\partial_s G_{\mathrm{f}} = \sum_i \varsigma_i' h_i + \varsigma_*' R^* \ge 0$ by (i) and (ii). For $s \le l_1$ all clocks vanish; for $s \ge r_*$ all equal one, and $J + \sum_i h_i + R^* = K \circ p \circ q^{-1}$. On $I_i$, $\varsigma_j = 1$ for $j < i$ and $\varsigma_j = 0$ for $j > i$, including $j = *$. □

*The third path.* Keep $\mathcal{D}$ fixed and interpolate the coefficient from $G_1 = f(s)H^* \circ p \circ q^{-1}$ to $G_{\mathrm{f}}$. On $E|_{\mathcal{A}}$ put $\xi_{\mathcal{A}} = ((G_{\mathrm{f}} - G_1) \circ \mathcal{D})\,\beta$; it equals $J(p)\beta$ on both collars of $\mathcal{A}$, and

$$\xi = J(p)\,\mathrm{pr}^*\beta + \mathrm{ext}_0\left(\xi_{\mathcal{A}} - J(p)\beta\right) \tag{49}$$

is a horizontal one-form on a neighbourhood of $V_\Delta(E)$. Put $\Omega_\tau^{(2)} = \Omega_1^{(1)} + \tau\, d\xi$ and $\Omega_{\mathrm{final}} = \Omega_1^{(2)}$. On $E|_{\mathcal{A}}$,

$$\begin{aligned} \Omega_\tau^{(2)} &= \mathcal{D}^*\big(\omega_V + e_\Lambda\Lambda + d(\hat{G}_\tau\beta)\big), \qquad \hat{G}_\tau = (1-\tau)G_1 + \tau G_{\mathrm{f}}, \\ \Omega_{\mathrm{final}} &= \mathcal{D}^*\Omega_{\mathrm{pre}}, \qquad\qquad\qquad\quad \Omega_{\mathrm{pre}} = \omega_V + e_\Lambda\Lambda + d(G_{\mathrm{f}}\beta); \end{aligned} \tag{50}$$

off $E|_{\mathcal{A}}$, $\Omega_\tau^{(2)} = \Omega_1 + \tau\, dJ\wedge\beta$ differs from $\Omega_1$ by a toric mixed term. Every $\Omega_\tau^{(2)}$ is symplectic on a neighbourhood of $V_\Delta(E)$ (Proposition 6.16).

For each of the three paths write its primitive as $\lambda_{j,t} = \lambda_V + Q_{j,t}$ and its variation as $\nu_{j,t} = \partial_t Q_{j,t}$; these are $\alpha_1$, $\partial_t\eta_t$ and $\xi$, respectively. In the larger finite atlas use the whole-path Gram cutoff $\chi_j$ of Construction 8.4. Define the cut field by $\iota_{Y_{j,t}} d\lambda_{j,t} = -\chi_j\nu_{j,t}$, computed in the charts of the finite atlas, and let $\Phi_j$ be its time-one map. Put $\Phi = \Phi_2\circ\Phi_1\circ\Phi_0$ and $\Phi^{-1} = \Phi_0^{-1}\circ\Phi_1^{-1}\circ\Phi_2^{-1}$.

**Proposition 6.6** (Moser maps)**.** *The three cut fields are smooth with common compact spatial support. Their time-one maps are diffeomorphisms, preserve the model region and its interior in both directions, and satisfy $\Phi_j^*\Omega_{j,1} = \Omega_{j,0}$ there. Consequently $\Phi^*\Omega_{\mathrm{final}} = \Omega$.*

*Proof. Faces.* Near each outer face, every path form is invariant under the flow of the face moment $\mu_\nu$: the toric terms are invariant, and the comparator isotopy commutes with this flow because $Q_t$ Poisson-commutes with $\mu_\nu$ on the collar (Definition M2.4(1)). Hence $\iota_{X_{\mu_\nu}}\Omega_{j,t} = -d\mu_\nu$ there. The variation primitives annihilate $X_{\mu_\nu}$, so $d\mu_\nu(Y_{j,t}) = 0$ near the boundary value, and no trajectory crosses a face level, or an intersection of faces, in either direction.

*Completeness.* The variation primitives vanish near the outer base boundary and the fields have compact support, so the flows are complete, with the backward flows as inverses.

*Pullback.* On the model region $\chi_j = 1$, and Cartan's formula gives $\frac{d}{dt}\Phi_{j,t}^*\Omega_{j,t} = \Phi_{j,t}^* d(\iota_{Y_{j,t}}\Omega_{j,t} + \nu_{j,t}) = 0$; the resulting coordinate ODEs and inverse endpoints are the maps of the construction. □

*Layer coordinates.* On $\Lambda_i = \{l_i < s < r_i,\ v_- < v < v_+\}\times \operatorname{int} V_\Delta$ let $T(v) = \int_{v_-}^{v} b$, a diffeomorphism $(v_-, v_+)\to(0,1)$ with $dT = \beta$. Define

$$\ell(s,T) = \frac{e_\Lambda a_\Lambda(s, v(T))}{b(v(T))}, \qquad x_i(s,T,p) = \int_{l_i}^{s} \ell(u,T)\, du + \varsigma_i(s)h_i(p). \tag{51}$$

Then $\partial_s x_i = \ell + \varsigma_i' h_i > 0$, $x_i = 0$ at $s = l_i$, and $x_i > h_i(p)$ at $s = r_i$, so for $0 < x < h_i(p)$ and $0 < T < 1$ there is a unique root $s = s_i(x,T,p) \in (l_i, r_i)$, smooth by the implicit function theorem. Since $e_\Lambda\Lambda = \ell\, ds\wedge dT$ and, by Proposition 6.5(iii), $d(G_{\mathrm{f}}\beta) = (\varsigma_i' h_i\, ds + dA_i + \varsigma_i\, dh_i)\wedge dT$ on $\Lambda_i$,

$$\Omega_{\mathrm{pre}} = \omega_V + dx_i\wedge dT + dA_i(p)\wedge dT \qquad \text{on } \Lambda_i. \tag{52}$$

The multi-angle shear $\mathrm{Sh}_A$ (Lemma A.15) removes $dA_i(p)\wedge dT$, and the disc-to-strip map $\Psi_\delta$ (Proposition A.16) supplies $(x,T)$.

## 6.2. **The finite primitive.**

**Construction 6.7** (Coordinate primitive)**.** Let $r\, dx\wedge dy$ be a smooth two-form with compact support in an open coordinate rectangle $I\times I'$, put $M(x) = \int r(x,y)\, dy$ and fix a smooth bump $b$ with compact support in $I'$ and $\int b = 1$. With integrals starting at the lower endpoints of $I$ and $I'$, put

$$P(r) = -\left[\int^{y}\big(r(x,t) - b(t)M(x)\big)\, dt\right] dx + b(y)\left[\int^{x} M(t)\, dt\right] dy. \tag{53}$$

For a finite presentation (Definition 6.9) with partition weights $\zeta_i$, unit-mass bumps $\eta_i$ and tree primitives $T_i$ (§8), the operator is

$$\mathcal{P}\omega = \sum_i \left[P_i(\zeta_i\omega - m_i(\omega)\eta_i) + m_i(\omega)T_i\right], \qquad m_i(\omega) = \int \zeta_i\omega. \tag{54}$$

Each local coefficient is extended by zero only after multiplication by its compactly supported weight.

**Lemma 6.8** (Primitive identity)**.** *If $\int_{I\times I'} r\,dx\wedge dy = 0$, then $P(r)$ has compact support in $I\times I'$ and $dP(r) = r\,dx\wedge dy$. Let $\omega(p)$ be a smooth family of two-forms supported in the correction core with $m(\omega(p)) = \sum_i m_i(\omega(p)) = 0$. Then $d_\Sigma\mathcal{P}\omega(p) = \omega(p)$, and $\mathcal{P}\omega(p)$ has fixed compact support in the interior of the model base and depends smoothly on $(p, x)$.*

*Proof.* The differential of the first term of (53) is $(r - bM)\,dx\wedge dy$, that of the second is $bM\,dx\wedge dy$. The $dx$-coefficient vanishes for $y$ below $\operatorname{supp} r\cup\operatorname{supp} b$, and above it because $\int(r(x,t) - b(t)M(x))\,dt = 0$; it vanishes for $x$ outside the $x$-projection of $\operatorname{supp} r$. The $dy$-coefficient vanishes outside $\operatorname{supp} b$ in $y$, below $\operatorname{supp} M$ in $x$, and above it because $\int M = 0$. For the operator, every argument of $P_i$ has zero mass, so $dP_i(\cdot)$ returns it; the tree primitives satisfy $dT_i = \eta_i - \eta_0$ (§8); hence $d\mathcal{P}\omega = \omega - m(\omega)\eta_0 = \omega$. All integrals are over fixed compact coordinate domains with smooth integrands, which gives the support and smoothness statements. □

### 6.3. **The handle data of the diagonal quadric.**

**Definition 6.9** (Finite presentation)**.** A finite presentation consists of coordinate maps with explicit inverses, compact inner and outer coordinate boxes, partition weights $\zeta_i$ ($\zeta_i$ supported in outer box $i$, $\sum_i\zeta_i = 1$ on the union of the inner boxes), unit-mass bumps in the boxes and in the overlaps of parent and child, and a rooted tree on the boxes in which every box $i\neq 0$ has a parent $j < i$ whose overlap with box $i$ contains a coordinate box.

**Construction 6.10** (Handle and normalisation data)**.** For the diagonal affine curve $\Sigma$ of Definition 9.1 take the transverse cylinder $e(s,\theta)$, the dual form $\beta = \beta_0 - d(e_*(\chi F))$ and the clock profile $b$ of Proposition 10.4. The annular coordinate of Definition 6.1(a) is $s_{\mathrm{a}} = s$ on $|s| < w = \frac{1}{8}$, with working half-width $b = \frac{1}{16}$; on $|s| < \frac{5}{32}$, which contains this annulus, $e^*\beta = b(\theta)\,d\theta$. The reference primitive is $\Gamma_{\mathrm{ref}} = \lambda = d^c\log(1+|z|^2)$, with $\Lambda = \sigma$, and $\Gamma_{\mathrm{phys}}(p) = (1-2|p|)\lambda$, so $\kappa(p) = (1-2|p|)\sigma$. On $\Sigma_\ell$ with $R_\ell\geq 5$ the correction atlas is the family of Lemma 10.3 with $R = R_\ell - 1$, and the flow atlas is the same family with $R = R_\ell$, restricted to a neighbourhood of $\Sigma_\ell$. The collar cutoff is $\chi_\partial = S\big((|t|^2-(R_\ell-2)^2)/((R_\ell-1)^2-(R_\ell-2)^2)\big)$. The mixed model of §12.4 uses the same base data, and the six-dimensional model uses them with its physical primitive $\Gamma(p)$ of Definition 13.3.

**Lemma 6.11** (Handle certificates)**.** *The form $\beta$ is smooth, closed and supported in $\{|t|\leq 3\}\subset\operatorname{int}\Sigma_\ell$; on the annulus it equals $b(\theta)\,d\theta$ with $b\geq 0$, $b > 0$ on $(\frac14,\frac34)$ and $\int_0^1 b = 1$. The inner boxes of the correction atlas cover $\{|t|\leq R_\ell - 1\}$, which contains the support of $1-\chi_\partial$ and the handle, and all its outer boxes lie in $\operatorname{int}\Sigma_\ell$. For $p\in\Delta_h$, $\kappa(p)\geq(1-2h)\Lambda$ on all of $\Sigma_\ell$, so the correction region of Definition 6.1(d) is $\Sigma_\ell$ with $\kappa_- = 1-2h$. The inner boxes of the flow atlas cover $\Sigma_\ell$.*

*Proof.* The statements on $\beta$ and $b$ are Proposition 10.4; the annuli $A$, $C$ lie in $|t|\leq\frac53$ (proof of Lemma 10.5). The covering and containment statements are Lemma 10.3 with $R = R_\ell - 1$ and $R = R_\ell$: its branch-square condition $i + \frac{1}{16} < (R+1)^2$ holds because $i\leq m < R_\ell - 1$. The cutoff $\chi_\partial$ vanishes for $|t|\leq R_\ell - 2$, so $\operatorname{supp}(1-\chi_\partial)\subset\{|t|\leq R_\ell - 1\}$, and the handle lies in $|t|\leq 3 < R_\ell - 1$. Finally $1-2|p|\geq 1-2h > 0$ on $\Delta_h$. □

6.4. **Normalisation and the budget.**

**Lemma 6.12** (Normalised datum)**.** *Let $\gamma_*$ be the reserve of the comparison datum. The primitive $\Gamma_n$ of Construction 8.2 agrees with $\Gamma_{\mathrm{phys}}$ near $\partial\Sigma_\ell$, its base curvature $d_\Sigma\Gamma_n(p) = \kappa_{\mathrm{sm}}(p)+H^*(p)\eta$ is positive for $p \in \Delta$, and on the annulus*

$$\Gamma_n(p) = e_\Lambda\Gamma_{\mathrm{ref}} + f(s)H^*(p)\beta. \tag{55}$$

*The mass $H^*$ is smooth, nonnegative on $\Delta$, and $H^* \geq H_{\mathrm{b}} - \gamma_*/16$.*

*Proof.* This is Proposition 8.3. Agreement near the boundary holds because the corrections $\mathcal{P}(\cdot)$, $c_p\beta$ and $d_\Sigma(\zeta u_p)$ are supported in the correction core, in $\operatorname{supp}\beta$ and in the annulus, all compact in $\operatorname{int}\Sigma_\ell$ (Lemma 6.11). □

For given capacities $\rho_i < a_i$ the layers use

$$\rho_i^\sharp = \frac{3a_i+\rho_i}{4}, \qquad \varepsilon_i = \frac{a_i-\rho_i^\sharp}{4}. \tag{56}$$

The collar bounds (43) and the definition (44) of $H^*$ give, for $w \in V_\Delta$, by Proposition 6.5(ii),

$$H^* \geq H_{\mathrm{b}} - \gamma_*/16, \tag{57}$$
$$R^*(w) = K(p(q^{-1}w)) - K(p(w)) + H^*(p(w)) - \sum_i h_i(p(w)) \geq \gamma - \gamma_*/16 > 0.$$

**Lemma 6.13** (Nondegeneracy and the Gram cutoffs)**.** *Let $K = V_\Delta(E)$. For $j = 0,1,2$ and $t \in [0,1]$ the forms $\Omega_{j,t}$ of §8 are symplectic at every point of $K$, hence on an open neighbourhood of $[0,1]\times K$ in $[0,1]\times E$. The cutoff $\chi_j$ of §8 equals one on a neighbourhood of $[0,1]\times K$, and $\chi_j(t,\cdot)$ has compact support contained in the set where $\Omega_{j,t}$ is nondegenerate.*

*Proof.* All three paths are closed, being exact. By Lemma 6.2(a), for a toric primitive $Q$ the top power of $\omega_V + dQ$ is $n\,\omega_V^m \wedge d_\Sigma Q$; mixed terms $dp_j \wedge (\cdot)$ do not contribute.

*Path* 0. $d_\Sigma Q_{0,t} = (1-t)\kappa(p)+t(\kappa_{\mathrm{sm}}(p)+H^*\eta)$. For $p \in \Delta$, $\kappa(p) > 0$ (Definition M2.3), $\kappa_{\mathrm{sm}}(p)$ is a convex combination of $\kappa(p)$ and $e_\Lambda\Lambda > 0$, and $H^*\eta \geq 0$ since $H^* \geq 0$ and $\eta = f'b\,ds\wedge dv \geq 0$. So the base form is positive.

*Path* 1. Off $E|_{\mathcal{A}}$, $\Omega_{1,t} = \Omega_{0,1}$. On $E|_{\mathcal{A}}$, $\Omega_{1,t} = \mathcal{D}_t^*(\omega_V + e_\Lambda\Lambda + d(G_t\beta))$ (Lemma 6.4(ii)), whose top power at $\mathcal{D}_t(x)$ is, by Lemma 6.2(b), $n(e_\Lambda a_\Lambda + b\,f'(s)H^*(p(q_t^{-1}w')))\,\omega_V^m \wedge ds \wedge dv$ with $w'$ the fibre point of $\mathcal{D}_t(x)$. Since $q_\tau(V_\Delta) = V_\Delta$ for all $\tau$ (Definition M2.4), $x \in K$ gives $w' \in V_\Delta$ and $q_t^{-1}w' \in V_\Delta$, so $H^* \geq 0$ there; $e_\Lambda a_\Lambda > 0$, $b \geq 0$, $f' \geq 0$, and $\mathcal{D}_t$ is a diffeomorphism.

*Path* 2. On $E|_{\mathcal{A}}$, $\Omega_{2,t} = \mathcal{D}^*(\omega_V + e_\Lambda\Lambda + d(\hat{G}_t\beta))$ by (50), with top-power coefficient $e_\Lambda a_\Lambda + b\,\partial_s\hat{G}_t$ at $\mathcal{D}(x)$; here $\partial_s\hat{G}_t = (1-t)f'H^*(p(q^{-1}w')) + t\,\partial_s G_{\mathrm{f}} \geq 0$ on $\mathcal{A}\times V_\Delta$ by the previous case and Proposition 6.5(iii). Off $E|_{\mathcal{A}}$, $\Omega_{2,t} = \Omega_{0,1} + t\,dJ\wedge\beta$ differs from $\Omega_{0,1}$ by a mixed term and has the same top power.

Thus each $\Omega_{j,t}^n \neq 0$ on the compact set $[0,1]\times K$; since nondegeneracy is an open condition and the forms depend smoothly on $(t,x)$, it holds on an open neighbourhood. For the cutoff, at each point of $K$ some flow-atlas bump $w_i$ equals one and the determinant of $\Omega_{j,t}$ in chart $i$ is nonzero, so $g_j > 0$ on $[0,1]\times K$ and $\mu_j > 0$. The factor $S(4g_j/\mu_j - 1)$ equals one where $g_j \geq \mu_j/2$, and $S(2(R+1-|w|^2))$ equals one where $|w|^2 \leq R+\frac{1}{2}$; both sets contain an open neighbourhood of $[0,1]\times K$. Conversely $\chi_j(t,z) \neq 0$ forces $g_j(t,z) > \mu_j/4 > 0$, so some $w_i(z) \neq 0$ with nonzero determinant in chart $i$, and $|w|^2 < R+1$; the set $\{g_j(t,\cdot) \geq \mu_j/4,\ |w|^2 \leq R+1\}$ is closed, its base projection lies in the compact union of the supports of the $w_i$, and it is therefore compact. □

6.5. **The three paths and the Moser maps.**

**Proposition 6.14** (First path)**.** *The path $\Omega_{0,t} = \omega_V + d((1-t)\Gamma_{\mathrm{phys}} + t\Gamma_n)$ has the curvature, positivity and annular endpoint stated in Proposition 6.3. Its variation primitive $\alpha_1$ of* (46) *is supported in the union of the correction core,* $\operatorname{supp}\beta$ *and the annulus.*

*Proof.* The base curvature and positivity are Path 0 of Lemma 6.13; the endpoint is (55); the support statement is the proof of Lemma 6.12. □

**Proposition 6.15** (Second path)**.** *The path $\Omega_{1,t} = \omega_V + d(\Gamma_n + \eta_t)$, with $\eta_t$ the suspension primitive of Lemma 6.4, is closed, equals $\Omega_{0,1}$ at $t = 0$, and is symplectic on a neighbourhood of $V_\Delta(E)$. Over $\mathcal{A}$ it is the pullback by $\mathcal{D}_t$ of $\omega_V + e_\Lambda\Lambda + d(G_t\beta)$, $G_t = f(s)H^*(p(q_t^{-1}w))$.*

*Proof.* The pullback identity is Lemma 6.4(ii). The coefficient correction and the suspension term,

$$f(s)\big[H^*(p(q_t^{-1}q_{tf(s)}w)) - H^*(p(w))\big] \qquad \text{and} \qquad (Q_s^t \circ q_{tf(s)})\,ds,$$

vanish for $s \le 0$ and $s \ge 1$ (Lemma 6.4(iii)), so extension by zero is smooth and $\eta_0 = 0$. Nondegeneracy is Path 1 of Lemma 6.13. □

**Proposition 6.16** (Third path)**.** *The path $\Omega_{2,t} = \omega_V + d(\Gamma_n + \eta_1 + t\xi)$ is closed and symplectic on a neighbourhood of $V_\Delta(E)$. Over the annulus it is the pullback by $\mathcal{D}$ of the interpolation from $G_1 = fH^* \circ p \circ q^{-1}$ to*

$$G_{\mathrm{f}} = J \circ p + \sum_i \varsigma_i h_i \circ p + \varsigma_* R^*, \qquad J = K - H^*, \tag{58}$$

*whose $s$-derivative is nonnegative on $\mathcal{A} \times V_\Delta$. On ball layer $i$ only the toric coefficient $A_i(p) + \varsigma_i(s)h_i(p)$ remains. Off the annulus the correction is the toric form $J(p)\beta$.*

*Proof.* The clocks have disjoint ordered transition intervals, and $h_i \ge 0$, $R^* > 0$ on $V_\Delta$ by (57); this gives Proposition 6.5(iii). On both end collars of $\mathcal{A}$ the annular correction equals $J(p)\beta$, so $\xi$ is smooth (49). Nondegeneracy is Path 2 of Lemma 6.13. □

**Proposition 6.17** (The Moser maps)**.** *For $j = 0, 1, 2$ the cut field $Y_j$, defined by $\iota_{Y_{j,t}}\Omega_{j,t} = -\chi_j\,\partial_t Q_{j,t}$ on the nondegenerate locus and by zero elsewhere, is smooth on $[0,1]\times E$ with compact spatial support. Its time-one map $\Phi_j$ is a diffeomorphism of $E$ with inverse the backward flow, $\Phi_j^{\pm1}(K) = K$, $\Phi_j^{\pm1}(\operatorname{int} K) = \operatorname{int} K$, and $\Phi_j^*\Omega_{j,1} = \Omega_{j,0}$ on $K$. These are the identities of Proposition 6.6.*

*Proof.* By Lemma 6.13, $\chi_j(t,\cdot)$ is supported in a compact subset of the nondegenerate locus, where the defining equation has the smooth solution $-\Omega_{j,t}^{-1}(\chi_j\partial_t Q_{j,t})$ (Cramer's rule in each chart); this solution vanishes near the boundary of the support, so the extension by zero is smooth. A smooth field with compact spatial support on $[0,1]\times E$ has a complete flow, and its time-one map is a diffeomorphism whose inverse is the backward flow.

On a neighbourhood $U$ of $[0,1]\times K$ we have $\chi_j = 1$, so $Y_j$ is there the Moser field of the pair $(\Omega_{j,t}, \partial_t Q_{j,t})$, which satisfies $\partial_t\Omega_{j,t} = d\,\partial_t Q_{j,t}$. Lemma 8.5 shows that every integral curve of $Y_j$ on a subinterval of $[0,1]$ which meets $K$ (resp. $\operatorname{int} K$) stays in $K$ (resp. $\operatorname{int} K$): near $\partial\Sigma_\ell$ the variation primitives vanish, so $Y_j = 0$ there; and near each outer face $\{\mu_\nu = \ell_\nu\}$ the face moment satisfies $\iota_{X_\nu}\Omega_{j,t} = -d\mu_\nu$ and $\partial_t Q_{j,t}(X_\nu) = 0$, so $d\mu_\nu(Y_{j,t}) = 0$ by Lemma A.4(d). The same holds in reverse time. Lemma A.4(e) applied to $\mathcal{K} = K$ then gives $\Phi_j(K) = K$ and $\Phi_j^*\Omega_{j,1} = \Omega_{j,0}$ on $K$; applied to the backward flow it gives $\Phi_j^{-1}(K) = K$, and the statement for $\operatorname{int} K$ follows in the same way, since every integral curve through a point of $\operatorname{int} K$ stays in it. □

6.6. **Proof of the packing theorem in a model (Theorem M2.6).** Let $a_i$, $P$, $(K,q)$ and $\gamma$ be as in Theorem M2.6, with (43), and let $\rho_i < a_i$. Write $\mathbb{C}^n = \mathbb{C}\times\mathbb{C}^m \ni (z_{\mathrm{b}}, z_{\mathrm{f}})$, $\varpi = \pi|z_{\mathrm{b}}|^2$ and $p = p(z_{\mathrm{f}})$, so $\omega_0 = \omega_{\mathbb{C}} + \omega_V$ and $B(c) = \{\varpi + |p| \le c\}$. For $n = 2$ the proof is completed in Paper I, Remark (Dimension four) and Paper I, § "The four-dimensional handle".

*Parameters.* Put $\hat\rho_i = (3a_i + \rho_i)/4$ and $\varepsilon_i = (a_i - \hat\rho_i)/4$, so $B(\rho_i) \subset B^\circ(\hat\rho_i)$; in (48) take $\rho_i^\sharp = \hat\rho_i$.

*Forms and Moser map.* Construct $\Omega_\tau^{(2)}$, $\mathcal{D}$, $G_{\mathrm{f}}$ and $\Phi$ as above. Then $\Phi^*\Omega_{\mathrm{final}} = \Omega$ by Proposition 6.6, and $\Omega_{\mathrm{final}} = \mathcal{D}^*\Omega_{\mathrm{pre}}$ on $E|_{\mathcal{A}}$ by (50); $\mathcal{D}$ fixes base points, and its inverse $(s,v,w) \mapsto (s, v, q_{f(s)}^{-1}(w))$ preserves $\operatorname{int} V_\Delta$.

*Layer maps.* For $(z_{\mathrm{b}}, z_{\mathrm{f}}) \in B^\circ(\hat\rho_i)$ let $(x,T) = \Psi_{\varepsilon_i}(z_{\mathrm{b}})$. Then $|p| < a_i$, so $z_{\mathrm{f}} \in \operatorname{int} V_\Delta$, and by Proposition 6.5(i),

$$0 < x < \varpi + \tfrac{\varepsilon_i}{2} < \hat\rho_i - |p| + \tfrac{a_i - \hat\rho_i}{8} < \tfrac{a_i + \hat\rho_i}{2} - |p| \le h_i(p).$$

Hence

$$\mathcal{E}_i(z_{\mathrm{b}}, z_{\mathrm{f}}) = \Big(s_i\big(x,T,p\big),\ v(T),\ \mathrm{Sh}_{A_i,T}(z_{\mathrm{f}})\Big) \in \Lambda_i$$

is defined, a composite of diffeomorphisms onto open sets. By (52), Lemma A.15 and Proposition A.16, $\mathcal{E}_i^*\Omega_{\mathrm{pre}} = (\Psi_{\varepsilon_i}\times\mathrm{id})^*(\omega_V + dx\wedge dT) = \omega_0$. Put

$$\psi_i = \Phi^{-1}\circ\mathcal{D}^{-1}\circ\mathcal{E}_i \colon B^\circ(\hat\rho_i) \to E.$$

This is the map of Construction 8.4; its factors $\Phi_j^{-1}$ are the inverse flows of $Q_0 = (1-t)\Gamma_{\mathrm{phys}} + t\Gamma_n$, $Q_1 = \Gamma_n + \eta_t$ and $Q_2 = \Gamma_n + \eta_1 + t\xi$.

*Conclusions.*

(a) $\psi_i^*\Omega = \mathcal{E}_i^*\Omega_{\mathrm{pre}} = \omega_0$, and $\psi_i$ is an injective immersion of an open set, hence a symplectic embedding of $B^\circ(\hat\rho_i) \supset B(\rho_i)$.

(b) $\psi_i$ takes values in $\operatorname{int} V_\Delta(E)$, since $\mathcal{E}_i$ maps there and $\mathcal{D}^{-1}$, $\Phi^{-1}$ preserve it (Proposition 6.6).

(c) The images are pairwise disjoint, because the intervals $(l_i, r_i)$ are disjoint and the common maps $\mathcal{D}^{-1}$, $\Phi^{-1}$ are injective.

Every ingredient is a formula, a definite integral, the solution of an explicit ODE, or a unique root, so $\psi_i$ is explicit in the sense of Definition M2.18; details are in §8. □

6.7. **Assembly.**

**Theorem 6.18** (Packing in the model)**.** *For the higher-dimensional model inputs, the maps $\psi_i$ of Construction 8.4 satisfy Theorem M2.6.*

*Proof.* By Proposition 6.17, $\Phi = \Phi_2\circ\Phi_1\circ\Phi_0$ satisfies $\Phi^*\Omega_{\mathrm{final}} = \Omega$ on $K$ and $\Phi^{-1}$ maps $\operatorname{int} K$ onto itself. The layer maps $E_i$ satisfy $E_i^*\Omega_{\mathrm{pre}} = \omega_0$ and have pairwise disjoint images in $\operatorname{int} K$ (Proposition 7.7), and $\mathcal{D}^{-1}$ pulls $\Omega_{\mathrm{final}}$ back to $\Omega_{\mathrm{pre}}$ (50). The composite $\psi_i = \Phi^{-1}\circ\mathcal{D}^{-1}\circ E_i$ is therefore a smooth symplectic embedding of $B^\circ(\rho_i^\sharp)$, a neighbourhood of $B(\rho_i)$, and the images are pairwise disjoint because $\mathcal{D}^{-1}$ and $\Phi^{-1}$ are single injective maps (Proposition 7.9). □

## 7. The layer maps

This section proves the layer step of the handle construction: one ball is placed in each layer of the handle, and the maps $\psi_i = \Phi^{-1}\circ\mathcal{D}^{-1}\circ E_i$ are symplectic embeddings with pairwise disjoint images in the interior of the model (Proposition 7.9). The disc-to-strip map and the enlarged capacities are constructed in Paper I, § "The disc-to-strip map and the layer maps" and recalled in §A; the ball estimate is §7.1.

*Conventions.* On $\mathbb{C} \ni z$ we write $\omega_{\mathbb{C}} = dx\wedge dy$ for $z = x+iy$ and $I = \pi|z|^2$; recall $dx\wedge dy = dI\wedge d\theta$ with $\theta = \arg z/2\pi$. The strip $(0,\infty)\times(0,1)$ carries coordinates $(x,T)$ and the area form $dx\wedge dT$. We write $\mathbb{C}^n = \mathbb{C}\times\mathbb{C}^m \ni (z_{\mathrm{b}}, z_{\mathrm{f}})$, $\varpi = \pi|z_{\mathrm{b}}|^2$ and $p = p(z_{\mathrm{f}})$, so that $\omega_0 = \omega_{\mathbb{C}}(z_{\mathrm{b}}) + \omega_V(z_{\mathrm{f}})$ and $B(c) = \{\varpi + |p| \le c\}$, $B^\circ(c) = \{\varpi + |p| < c\}$.

7.1. **The ball estimate.** We work in the situation of Theorem M2.6 with $n \ge 3$: a model input (Definition M2.3) of rank $m = n-1$ with handle data (Definition 6.1), capacities $a_1,\dots,a_k > 0$ with $\{p \in \mathbb{R}^m_{\ge 0} : |p| \le a_i\} \subset \tau_\Delta\Delta$ for some $\tau_\Delta < 1$, a function $P$, a comparison datum $(K,q)$ with gap $\gamma > 0$, and $\rho_i < a_i$. The outer faces of $\Delta$ are $\{\langle b_\nu, p\rangle = \ell_\nu\}$, $1 \le \nu \le N$, $\ell_\nu > 0$. The caps $h_i$ are those of (48) with $\rho_i^\sharp = \hat\rho_i$ (Definition A.17); by Proposition 6.5(i) each $h_i$ is smooth on $\mathbb{R}^m$ and

$$0 \le h_i(p), \qquad (a_i + \hat\rho_i)/2 - |p| \le h_i(p) \qquad (p \in \mathbb{R}^m). \tag{59}$$

**Lemma 7.1** (Interior of the fibre polytope)**.** *Let $V_\Delta = \{w \in \mathbb{C}^m : \langle b_\nu, p(w)\rangle \le \ell_\nu \text{ for all } \nu\}$. Then*

$$\operatorname{int} V_\Delta = \{w \in \mathbb{C}^m : \langle b_\nu, p(w)\rangle < \ell_\nu \text{ for all } \nu\}, \tag{60}$$

*and $\{w : |p(w)| \le a_i\} \subset \operatorname{int} V_\Delta$ for $1 \le i \le k$.*

*Proof.* The right side of (60) is open and contained in $V_\Delta$. If $\langle b_\nu, p(w)\rangle = \ell_\nu > 0$, then $\langle b_\nu, p((1+t)w)\rangle = (1+t)^2\ell_\nu > \ell_\nu$ for $t > 0$, so $w$ is not interior. If $|p(w)| \le a_i$, then $p(w) \in \tau_\Delta\Delta$, so $\langle b_\nu, p(w)\rangle \le \tau_\Delta\ell_\nu < \ell_\nu$. □

**Lemma 7.2** (Ball estimate)**.** *Let $1 \le i \le k$, $(z_{\mathrm{b}}, z_{\mathrm{f}}) \in \mathbb{C}\times\mathbb{C}^m$, $\varpi = \pi|z_{\mathrm{b}}|^2$, $p = p(z_{\mathrm{f}})$, and let $x$ be the first component of $\Psi_{\varepsilon_i}(z_{\mathrm{b}})$.*

*(a) If $\varpi + |p| \le a_i - \delta_i$, then*

$$x < \varpi + \frac{\delta_i}{16} \le a_i - |p| - \frac{15\delta_i}{16} \le h_i(p) - \frac{11\delta_i}{16} < h_i(p) - \frac{\delta_i}{2}. \tag{61}$$

*(b) If $\varpi + |p| \le \bar\rho_i$, then $x < a_i - |p| - \frac{11\delta_i}{16} \le h_i(p) - \frac{7\delta_i}{16} < h_i(p) - \frac{\delta_i}{4}$.*

*(c) If $\varpi + |p| < \hat\rho_i$, then $x < a_i - |p| - \frac{7\delta_i}{16} \le h_i(p) - \frac{3\delta_i}{16} < h_i(p)$.*

*In each case $|p| < a_i$, hence $z_{\mathrm{f}} \in \operatorname{int} V_\Delta$.*

*Proof.* By (91), $x < \varpi + \varepsilon_i/2 = \varpi + \delta_i/16$, and in (a), (b), (c) respectively $\varpi \le a_i - |p| - \delta_i$, $\varpi \le a_i - |p| - \frac34\delta_i$, $\varpi < a_i - |p| - \frac12\delta_i$. The cap comparison is (59) in the form $a_i - \delta_i/4 - |p| \le h_i(p)$. Finally $|p| \le \varpi + |p| < a_i$; apply Lemma 7.1. □

7.2. **Layer coordinates.** We use the objects of §6.1 and §6. In the product coordinates of Definition 6.1, $E|_{\mathcal{A}^+} \cong \mathcal{A}^+ \times \mathbb{C}^m$ with base coordinates $(s,v)$ and fibre coordinate $w$, and $\omega_V = \sum_j dp_j \wedge d\theta_j$. The one-form $\beta$ equals $b(v)\,dv$ on $\mathcal{A}^+$, where $b \ge 0$ is smooth, $b > 0$ exactly on $(v_-, v_+)$ and $\int b = 1$. The layer intervals $I_j = [l_j, r_j] \subset (0,1)$, $1 \le j \le k$, and the residual interval $I_*$ are pairwise disjoint, with flat steps $\varsigma_j$ rising from 0 to 1 across $I_j$ ($\varsigma_j' \ge 0$). By Proposition 6.16, the final coefficient $G_{\mathrm{f}}$ satisfies, on $I_i \times \mathbb{C}^m$,

$$G_{\mathrm{f}}(s,w) = A_i(p(w)) + \varsigma_i(s)h_i(p(w)), \qquad A_i = J + \sum_{j<i} h_j, \tag{62}$$

where $J = K - H^* = K - H_{\mathrm{b}} + \epsilon_{\mathrm{sm}}$ is the collar coefficient of (48). $K$ is smooth near $\Delta$ (Definition M2.4) and the $h_j$ are smooth on $\mathbb{R}^m$. So are, near $\Delta$, $H_{\mathrm{b}}(p) = \int_{\Sigma_\ell}\kappa(p)$ and $\epsilon_{\mathrm{sm}}(p) = \int_{\Sigma_\ell}\kappa_{\mathrm{sm}}(p)$: $\kappa(p)$ extends smoothly in $p$ (Definition M2.3), $\kappa_{\mathrm{sm}}(p)$ is $\chi_\partial\kappa(p)$ plus the

$p$-independent form $(1-\chi_\partial)e_\Lambda\Lambda$ by (44), and $\Sigma_\ell$ is compact, so one may differentiate under the integral sign. Hence $A_i$ is smooth on an open neighbourhood $W$ of $\Delta$ in $\mathbb{R}^m$. Write $\Lambda = a_\Lambda(s,v)\,ds\wedge dv$ with $a_\Lambda>0$ on $\mathcal{A}^+$ and put $\ell_0(s,v)=e_\Lambda a_\Lambda(s,v)>0$. On $E|_{\mathcal{A}}$, over a neighbourhood of $V_\Delta$, the twisted form is given by (50) (Proposition 6.16),

$$\Omega_{\mathrm{pre}} = \omega_V + e_\Lambda\Lambda + d(G_{\mathrm{f}}\beta) = \omega_V + \ell_0(s,v)\,ds\wedge dv + d(G_{\mathrm{f}}\beta), \qquad \Omega_{\mathrm{final}} = \mathcal{D}^*\Omega_{\mathrm{pre}}. \tag{63}$$

The $i$-th layer is the open set

$$\Lambda_i = \{(s,v,w) : l_i < s < r_i,\ v_- < v < v_+,\ w\in \operatorname{int} V_\Delta\} \subset E|_{\mathcal{A}}.$$

On $\Lambda_i$ only the $i$-th step moves; $T$ integrates $\beta$, and $x_i$ integrates the background area in $s$ plus the height $\varsigma_i h_i$ of that step.

**Construction 7.3** (Layer coordinates)**.** Let $T(v)=\int_{v_-}^{v} b(t)\,dt$ for $v_-<v<v_+$, an increasing diffeomorphism onto $(0,1)$ with $dT=\beta$, and $v(T)$ its inverse. For $0<s<1$, $0<T<1$ and $p\in\mathbb{R}^m$ put

$$\ell(s,T)=\frac{\ell_0(s,v(T))}{b(v(T))}, \qquad L_i(s,T)=\int_{l_i}^{s}\ell(u,T)\,du,$$
$$x_i(s,T,p)=L_i(s,T)+\varsigma_i(s)h_i(p), \qquad x_i^+(T,p)=x_i(r_i,T,p).$$

On $\Lambda_i$ the layer chart is

$$\kappa_i(s,v,w)=\big(x_i(s,T(v),p(w)),\,T(v),\,w\big),$$

and

$$\mathcal{U}_i=\big\{(x,T,w)\in\mathbb{R}\times(0,1)\times\operatorname{int}V_\Delta:\ 0<x<x_i^+(T,p(w))\big\}.$$

**Lemma 7.4** (Layer root)**.** *(i) $x_i$ is smooth on $(0,1)\times(0,1)\times\mathbb{R}^m$, and $\partial_s x_i=\ell(s,T)+\varsigma_i'(s)h_i(p)\ge\ell(s,T)>0$. Moreover $x_i(l_i,T,p)=0$ and $x_i^+(T,p)=L_i(r_i,T)+h_i(p)>h_i(p)$.*

*(ii) For $0<T<1$, $w\in\operatorname{int}V_\Delta$ and $0<x<x_i^+(T,p(w))$ there is a unique $s=\sigma_i(x,T,p(w))\in(l_i,r_i)$ with $x_i(s,T,p(w))=x$. This root is smooth as a function of $(x,T,w)$ on that open domain, including the fibre coordinate axes.*

*(iii) $\kappa_i$ is a diffeomorphism of $\Lambda_i$ onto $\mathcal{U}_i$, with inverse $(x,T,w)\mapsto\big(\sigma_i(x,T,p(w)),\,v(T),\,w\big)$.*

*Proof.* (i) $v(T)$ is smooth because $T'(v)=b(v)>0$ on $(v_-,v_+)$, so $L_i$ is smooth, as are $\varsigma_i$ and $h_i$; and $\varsigma_i'\ge0$, $h_i\ge0$ by (59). At $s=l_i$, $L_i=\varsigma_i=0$; at $s=r_i$, $\varsigma_i=1$ and $L_i(r_i,T)>0$ because $\ell>0$ and $r_i>l_i$. (ii) By (i), $s\mapsto x_i(s,T,p)$ is strictly increasing on $[l_i,r_i]$ from 0 to $x_i^+(T,p)$, which gives existence and uniqueness of the root in $(l_i,r_i)$. The domain is open since $x_i^+$ and $p(w)$ are continuous, and near each point the implicit function theorem ($\partial_s x_i>0$) gives a smooth local solution, equal to the root by uniqueness; $p(w)$ is smooth in Cartesian coordinates, so this holds at the axes too. (iii) $\kappa_i$ is smooth since $p(w)$ is smooth in Cartesian coordinates. For fixed $(v,w)$ the first component increases strictly from 0 to $x_i^+(T(v),p(w))$ as $s$ runs over $(l_i,r_i)$, and $T$ is a bijection $(v_-,v_+)\to(0,1)$; hence $\kappa_i$ is a bijection $\Lambda_i\to\mathcal{U}_i$, $\mathcal{U}_i$ is open, and the inverse is the stated map, which is smooth by (ii). □

**Lemma 7.5** (The form on a layer)**.** *On $\Lambda_i$, $\Omega_{\mathrm{pre}}=\omega_V+dx_i\wedge dT+dA_i(p)\wedge dT$, where $x_i=x_i(s,T(v),p(w))$; equivalently $(\kappa_i^{-1})^*\Omega_{\mathrm{pre}}=\omega_V+dx\wedge dT+dA_i(p)\wedge dT$ on $\mathcal{U}_i$.*

*Proof.* On $\Lambda_i$ we have $\beta=dT$ and $ds\wedge dv=b^{-1}ds\wedge dT$. Since $\beta$ is closed and $G_{\mathrm{f}}$ does not depend on $v$, (63) and (62) give

$$\Omega_{\mathrm{pre}}=\omega_V+\ell(s,T)\,ds\wedge dT+\big(\varsigma_i'h_i\,ds+dA_i+\varsigma_i\,dh_i\big)\wedge dT,$$

with $A_i$, $h_i$ evaluated at $p(w)$. On the other hand $dx_i\wedge dT=\big(\partial_sL_i+\varsigma_i'h_i\big)\,ds\wedge dT+\varsigma_i\,dh_i\wedge dT$, because the $dT$-component of $dx_i$ drops out, and $\partial_sL_i=\ell(s,T)$. Subtracting gives the claim. □

7.3. **The layer maps.** The term $dA_i(p)\wedge dT$ is removed by the multi-angle shear $\mathrm{Sh}_{A_i}(x,T,w) = (x,T,\mathrm{Sh}_{A_i,T}(w))$ of Lemma A.15, applied with $A = A_i$ on $W$:

$$\mathrm{Sh}_{A_i,T}(w)_j = \exp\big(-2\pi i\, T\, \partial_j A_i(p(w))\big)\, w_j \qquad (1 \le j \le m).$$

It preserves $x$, $T$ and $p$, hence maps $\mathcal{U}_i$ diffeomorphically onto itself (its inverse, the same formula with $+2\pi i$, also preserves $\mathcal{U}_i$), and $\mathrm{Sh}_{A_i}^*\big(\omega_V + dx\wedge dT + dA_i(p)\wedge dT\big) = \omega_V + dx\wedge dT$.

**Construction 7.6** (Layer maps)**.** For $1 \le i \le k$ and $(z_\mathrm{b}, z_\mathrm{f}) \in B^\circ(\hat\rho_i)$ put $(x,T) = \Psi_{\varepsilon_i}(z_\mathrm{b})$, $p = p(z_\mathrm{f})$ and

$$E_i(z_\mathrm{b}, z_\mathrm{f}) = \Big(\sigma_i(x,T,p),\ v(T),\ \mathrm{Sh}_{A_i,T}(z_\mathrm{f})\Big) = \kappa_i^{-1} \circ \mathrm{Sh}_{A_i} \circ (\Psi_{\varepsilon_i} \times \mathrm{id})\,(z_\mathrm{b}, z_\mathrm{f}).$$

**Proposition 7.7** (Layer maps)**.** *For $1 \le i \le k$:*

*(i) $E_i$ is well defined and smooth on $B^\circ(\hat\rho_i)$, which contains $B(\bar\rho_i) \supset B(a_i - \delta_i)$, and is a diffeomorphism onto an open subset of $\Lambda_i$, with inverse*

$$E_i^{-1}(s,v,w) = \Big(\Psi_{\varepsilon_i}^{-1}\big(x_i(s,T,p(w)),\,T\big),\ \mathrm{Sh}_{A_i,T}^{-1}(w)\Big), \qquad T = T(v);$$

*(ii) $E_i^*\Omega_\mathrm{pre} = \omega_0$;*

*(iii) $E_i(B^\circ(\hat\rho_i))$ lies over the base rectangle $(l_i, r_i) \times (v_-, v_+) \subset \mathcal{A}$ and in $\{|p| < a_i\} \subset \operatorname{int} V_\Delta$;*

*(iv) on $B(\bar\rho_i)$ the coordinate $x = x_i \circ E_i$ satisfies $x < h_i(p) - \delta_i/4$, and on $B(a_i - \delta_i)$ it satisfies $x < h_i(p) - \delta_i/2$; the set $E_i(B(\bar\rho_i))$ is a compact subset of $\Lambda_i$;*

*(v) the images $E_i(B^\circ(\hat\rho_i))$, $1 \le i \le k$, are pairwise disjoint.*

*Proof.* (i) By Lemma 7.2(c), for $(z_\mathrm{b}, z_\mathrm{f}) \in B^\circ(\hat\rho_i)$ we have $z_\mathrm{f} \in \operatorname{int} V_\Delta$ and $0 < x < h_i(p) < x_i^+(T,p)$ (Lemma 7.4(i)), and $0 < T < 1$ by (91). Hence the diffeomorphism $\Psi_{\varepsilon_i} \times \mathrm{id}$ of $\mathbb{C} \times \mathbb{C}^m$ onto $(0,\infty) \times (0,1) \times \mathbb{C}^m$ (Proposition A.16(i)) maps $B^\circ(\hat\rho_i)$ onto an open subset of $\mathcal{U}_i$; $\mathrm{Sh}_{A_i}$ is a diffeomorphism of $\mathcal{U}_i$ onto itself, and $\kappa_i^{-1}$ one of $\mathcal{U}_i$ onto $\Lambda_i$ (Lemma 7.4(iii)). Composing the three inverses gives the formula, as $\mathrm{Sh}_{A_i,T}^{-1}$ preserves $p$. The inclusions of balls hold because $a_i - \delta_i < \bar\rho_i < \hat\rho_i$.

(ii) By Lemmas 7.5 and A.15 and Proposition A.16(i),

$$\begin{aligned} E_i^*\Omega_\mathrm{pre} &= (\Psi_{\varepsilon_i} \times \mathrm{id})^* \mathrm{Sh}_{A_i}^*\big(\omega_V + dx\wedge dT + dA_i(p)\wedge dT\big) \\ &= (\Psi_{\varepsilon_i} \times \mathrm{id})^*\big(\omega_V + dx\wedge dT\big) \\ &= \omega_V(z_\mathrm{f}) + \omega_\mathbb{C}(z_\mathrm{b}) = \omega_0. \end{aligned}$$

(iii) $\sigma_i(x,T,p) \in (l_i, r_i)$, $v(T) \in (v_-, v_+)$, and the fibre component has the same $p$ as $z_\mathrm{f}$, with $|p| < a_i$ by Lemma 7.2.

(iv) The two bounds are Lemma 7.2(b) and (a), since $x_i \circ E_i$ is the first component of $\Psi_{\varepsilon_i}(z_\mathrm{b})$ and the shear preserves $p$. The closed ball $B(\bar\rho_i)$ is compact and contained in the domain of the continuous map $E_i$, whose values lie in $\Lambda_i$.

(v) The $s$-components of points of different images lie in the disjoint open intervals $(l_i, r_i)$. □

For $(z_\mathrm{b}, z_\mathrm{f}) \in B(\bar\rho_i)$, Proposition A.16(iv) with $I_\mathrm{max} = \bar\rho_i$ and $b_\mathrm{st} = \varepsilon_i/\pi$ bounds $\Psi_{\varepsilon_i}(z_\mathrm{b})$ away from $x = 0$ and $T \in \{0,1\}$ by explicit constants, and $|p| \le \bar\rho_i < a_i$; these margins enter the effective data of §15.

7.4. **Undoing the twist and the Moser map.** To pass from $(\Lambda_i, \Omega_\mathrm{pre})$ to $(E, \Omega)$ we undo the twist $\mathcal{D}$ and the Moser map $\Phi$.

**Lemma 7.8** (The twist and the Moser map)**.** *(D) The twist $\mathcal{D}(s,v,w) = (s,v,q_{f(s)}(w))$ restricts to a diffeomorphism of $\{(s,v,w) \in E|_\mathcal{A} : w \in \operatorname{int} V_\Delta\}$ onto itself which fixes base points, with inverse $\mathcal{D}^{-1}(s,v,w) = (s,v,q_{f(s)}^{-1}(w))$, and $(\mathcal{D}^{-1})^*\Omega_\mathrm{final} = \Omega_\mathrm{pre}$ there.*

*(Φ)* $\Phi = \Phi_2 \circ \Phi_1 \circ \Phi_0$ *is a diffeomorphism of* $E$ *which maps* $V_\Delta(E)$ *and its interior* $\operatorname{int} V_\Delta(E)$ *onto themselves, and* $\Phi^*\Omega_{\mathrm{final}} = \Omega$ *on* $V_\Delta(E)$.

*Proof.* (D) The map $(t, w) \mapsto q_t(w)$ is smooth and each $q_t$ is a diffeomorphism of $\mathbb{C}^m$ with $q_t(V_\Delta) = V_\Delta$ (Definition M2.4), hence $q_t(\operatorname{int} V_\Delta) = \operatorname{int} V_\Delta$. So $\mathcal{D}$ and the displayed inverse are mutually inverse smooth maps of the stated set into itself fixing base points, and $(\mathcal{D}^{-1})^*\Omega_{\mathrm{final}} = \Omega_{\mathrm{pre}}$ since $\Omega_{\mathrm{final}} = \mathcal{D}^*\Omega_{\mathrm{pre}}$ by (63).

(Φ) Each $\Phi_j$ is the time-one map of a compactly supported smooth vector field, hence a diffeomorphism of $E$. By Propositions 6.6 and 6.17, each $\Phi_j$ maps $V_\Delta(E)$ and its interior onto themselves and satisfies $\Phi_j^*\Omega_{j,1} = \Omega_{j,0}$ there, where $\Omega_{0,0} = \Omega$, $\Omega_{j,1} = \Omega_{j+1,0}$ and $\Omega_{2,1} = \Omega_{\mathrm{final}}$. Composing gives $\Phi^*\Omega_{\mathrm{final}} = \Phi_0^*\Phi_1^*\Phi_2^*\Omega_{2,1} = \Omega$ on $V_\Delta(E)$. □

**Proposition 7.9** (Embeddings into the model)**.** *For* $1 \le i \le k$ *let*

$$\psi_i = \Phi^{-1} \circ \mathcal{D}^{-1} \circ E_i \colon B^\circ(\hat\rho_i) \to E.$$

*(i)* $\psi_i$ *is a smooth symplectic embedding of* $(B^\circ(\hat\rho_i), \omega_0)$ *onto an open subset of* $(\operatorname{int} V_\Delta(E), \Omega)$*; in particular it restricts to a symplectic embedding of a neighbourhood of* $B(\bar\rho_i)$*, and* $\psi_i(B(\bar\rho_i))$ *is a compact subset of* $\operatorname{int} V_\Delta(E)$*;*

*(ii) the images* $\psi_i(B^\circ(\hat\rho_i))$*,* $1 \le i \le k$*, are pairwise disjoint.*

*Proof.* (i) By Proposition 7.7(i),(iii), $E_i$ is a diffeomorphism of $B^\circ(\hat\rho_i)$ onto an open subset of $\Lambda_i$. By Lemma 7.8(D), $\mathcal{D}^{-1}$ maps it diffeomorphically onto an open subset of $E|_{\mathcal{A}}$ lying over $(l_i, r_i) \times (v_-, v_+)$ with fibre component in $\operatorname{int} V_\Delta$. On this rectangle $b(v) > 0$, so it lies in the support of $\beta$, contained in $\mathcal{B} \subset \operatorname{int} \Sigma_\ell$ (Definition 6.1); hence the image lies in $\operatorname{int} \Sigma_\ell \times \operatorname{int} V_\Delta = \operatorname{int} V_\Delta(E)$, which $\Phi^{-1}$ maps diffeomorphically onto itself (Lemma 7.8(Φ)). So $\psi_i$ is a diffeomorphism onto an open subset of $\operatorname{int} V_\Delta(E)$, and

$$\psi_i^*\Omega = E_i^*(\mathcal{D}^{-1})^*(\Phi^{-1})^*\Omega = E_i^*(\mathcal{D}^{-1})^*\Omega_{\mathrm{final}} = E_i^*\Omega_{\mathrm{pre}} = \omega_0$$

by Lemma 7.8 and Proposition 7.7(ii). Since $\bar\rho_i < \hat\rho_i$, $B^\circ(\hat\rho_i)$ is a neighbourhood of the compact set $B(\bar\rho_i)$, whose image is compact.

(ii) The sets $E_i(B^\circ(\hat\rho_i))$ are pairwise disjoint (Proposition 7.7(v)), and $\mathcal{D}^{-1}$ and $\Phi^{-1}$ are injective maps independent of $i$. □

These are the higher-dimensional embeddings of Theorem 6.18; for the periodic four-dimensional input see Paper I, Remark (The four-dimensional embeddings), whose images avoid the exceptional fibres by the equivariant handle and layer identities.

## 8. The finite-atlas toric construction

This section fixes the finite primitive operator, the normalised primitive $\Gamma_n$, the three Moser fields and the maps $\psi_i$ behind the model maps of §6.1, for a higher-dimensional model input (Definition M2.3) with the handle data and notation of Definition 6.1, all forms being evaluated in the explicit coordinates of the model curve. The *correction atlas* is the finite family of rational coordinate boxes of Definition 6.9; its inner boxes cover the *correction core*, a compact subset of $\operatorname{int} \Sigma_\ell$ that contains the handle and the support of $1 - \chi_\partial$ in $\Sigma_\ell$ and lies in the correction region of Definition 6.1(d), where $\kappa(p) \ge \kappa_- \Lambda$ for $p \in \Delta$. Its overlap tree has root the box 0, and every other box $i$ has a parent $j < i$. A second, larger finite atlas, the *flow atlas*, covers the whole closed model base.

8.1. **A fixed finite primitive operator.** Let $\zeta_i$ be the partition weights of the correction atlas ($\sum_i \zeta_i = 1$ on the correction core, $\zeta_i$ supported in box $i$), $\eta_i$ a unit-mass bump in box $i$, and $\eta_{ij}$ a unit-mass bump in the overlap of a child $i$ and its parent $j$. Write $P_i$ for the rectangular primitive (53) in box $i$; thus $dP_i\alpha = \alpha$ for every two-form $\alpha$ supported in box $i$ with zero coordinate mass, and $P_i\alpha$ is supported in box $i$ (Lemma 6.8). Set

$$e_i = P_i(\eta_i - \eta_{ij}) + P_j(\eta_{ij} - \eta_j), \qquad T_0 = 0, \qquad T_i = e_i + T_j.$$

For a two-form $\omega$ supported in the correction core put

$$m_i(\omega) = \int \zeta_i\omega, \quad m(\omega) = \sum_i m_i(\omega), \qquad \mathcal{P}\omega = \sum_i \big[P_i(\zeta_i\omega - m_i(\omega)\eta_i) + m_i(\omega)T_i\big],$$

where $\int \zeta_i\omega$ is the coordinate integral in box $i$. Each argument of $P_i$ has zero coordinate mass, and $de_i = \eta_i - \eta_j$; by induction along the tree, $dT_i = \eta_i - \eta_0$. Hence

$$d\mathcal{P}\omega = \sum_i \big[\zeta_i\omega - m_i(\omega)\eta_i + m_i(\omega)(\eta_i - \eta_0)\big] = \omega - m(\omega)\eta_0,$$

so $d\mathcal{P}\omega = \omega$ when $m(\omega) = 0$. Every sum is finite and every integral has a fixed coordinate domain, so a smooth family $\omega(p)$ gives a smooth family $\mathcal{P}\omega(p)$ with fixed compact support in $\operatorname{int}\Sigma_\ell$.

**Lemma 8.1** (Intrinsic calibration)**.** *Let $\eta = df \wedge \beta$. Then $\int_{\Sigma_\ell} \eta = 1$, $m(\eta) \neq 0$, and for every smooth two-form $\omega$ supported in the correction core,*

$$\frac{m(\omega)}{m(\eta)} = \int_{\Sigma_\ell} \omega,$$

*where the right side is the intrinsic surface integral.*

*Proof.* If $\alpha$ is supported in the correction core and $m(\alpha) = 0$, then $\alpha = d\mathcal{P}\alpha$ with $\mathcal{P}\alpha$ compactly supported in $\operatorname{int}\Sigma_\ell$, so $\int_{\Sigma_\ell} \alpha = 0$ by Stokes' theorem. The bump $\eta_0$ has intrinsic integral 1 (change of variables in the oriented root chart); hence $m(\eta_0) \neq 0$. For $\omega$ as in the statement the form $\alpha = m(\omega)\eta_0 - m(\eta_0)\omega$ has $m(\alpha) = 0$, so $m(\omega) = m(\eta_0)\int \omega$. The form $\eta = f'(s)b(v)\,ds \wedge dv$ is supported in the handle, and $\int \eta = (f(1) - f(0))\int_0^1 b = 1$; so $m(\eta) = m(\eta_0) \neq 0$, and dividing gives the identity. □

**Construction 8.2** (The normalised primitive)**.** Let $\gamma_*$ be the reserve of the comparison datum, $A_{\mathrm{ref}}$ the reference-area bound and $\kappa_- > 0$ the density floor of Definition 6.1, and $e_\Lambda = \min\big(\gamma_*/(32A_{\mathrm{ref}}), \kappa_-/2\big)$ as in (42). Put

$$\omega_p = (1 - \chi_\partial)\big(\kappa(p) - e_\Lambda\Lambda\big), \qquad H(p) = \frac{m(\omega_p)}{m(\eta)}, \qquad \Gamma_r(p) = \Gamma_{\mathrm{phys}}(p) + \mathcal{P}\big(H(p)\eta - \omega_p\big),$$

and, on the annulus $|s_{\mathrm{a}}| < w$, $N_p = e_\Lambda\Gamma_{\mathrm{ref}} + H(p)f(s)\beta$. In the annular coordinates $(s_{\mathrm{a}}, v)$, $v \in \mathbb{R}/\mathbb{Z}$, write $\Gamma_r(p) - N_p = A_p\,ds_{\mathrm{a}} + B_p\,dv$, and set

$$c_p = \int_0^1 B_p(0, v)\,dv, \qquad u_p(s_{\mathrm{a}}, v) = \int_0^{s_{\mathrm{a}}} A_p(\sigma, 0)\,d\sigma + \int_0^v \big(B_p(s_{\mathrm{a}}, \nu) - c_p b(\nu)\big)\,d\nu.$$

With $b_1 = (b + w)/2$ and $b_2 = (b_1 + w)/2$ let $\zeta = \chi^{\mathrm{I}}_{-b_2, b_2, b_2 - b_1}(s_{\mathrm{a}})$ (Definition A.3), which equals 1 for $|s_{\mathrm{a}}| \le b_1$ and 0 for $|s_{\mathrm{a}}| \ge b_2$, and put

$$\Gamma_n(p) = \Gamma_r(p) - c_p\beta - d_\Sigma\big(\zeta u_p\big),$$

where $\zeta u_p$ is extended by zero outside the annulus and $d_\Sigma$ differentiates in the base variables at fixed $p$. The first variation primitive is $\alpha_1 = \Gamma_n - \Gamma_{\mathrm{phys}}$.

**Proposition 8.3** (Normalisation identities)**.** *$H(p)$, $c_p$, $u_p$ and $\Gamma_n(p)$ are smooth in $p$ and the base variables, and for $p$ in a neighbourhood of $\Delta$:*

*(i)* $H(p) = H_{\mathrm{b}}(p) - \epsilon_{\mathrm{sm}}(p) = H^*(p)$ *in the notation of* (44)*, and* $0 \le H(p)$ *and* $H(p) \ge H_{\mathrm{b}}(p) - \gamma_*/16$ *on* $\Delta$*;*

*(ii)* $d_\Sigma \Gamma_n(p) = \kappa_{\mathrm{sm}}(p) + H(p)\eta$ *with* $\kappa_{\mathrm{sm}} = \chi_\partial \kappa + (1-\chi_\partial) e_\Lambda \Lambda$*; in particular* $d_\Sigma \Gamma_n(p) = \kappa(p)$ *where* $\chi_\partial = 1$ *outside the handle, and* $d_\Sigma \Gamma_n(p)$ *is positive on* $\Sigma_\ell$ *for* $p \in \Delta$*;*

*(iii)* $\Gamma_n(p) = N_p = e_\Lambda \Gamma_{\mathrm{ref}} + H(p) f(s) \beta$ *on* $\{|s_{\mathrm{a}}| < b_1\} = \mathcal{A}^+$*.*

*Proof.* Smoothness: $\kappa(p)$ is smooth in $p$ (Definition M2.3), and $m$, $\mathcal{P}$, $c_p$ and $u_p$ are finite sums of integrals of smooth integrands over fixed compact domains.

(i) By Lemma 8.1, $H(p) = \int_{\Sigma_\ell} (1-\chi_\partial)(\kappa(p) - e_\Lambda \Lambda) = H_{\mathrm{b}}(p) - \int_{\Sigma_\ell} \kappa_{\mathrm{sm}}(p)$. The lower bound is $\epsilon_{\mathrm{sm}} \le \int \chi_\partial \kappa + e_\Lambda \int_{\Sigma_\ell} \Lambda \le \gamma_*/16$ by (43). For $H \ge 0$: where $1 - \chi_\partial \ne 0$ we are in the correction region, so for $p \in \Delta$, $\kappa(p) - e_\Lambda \Lambda \ge (\kappa_- - e_\Lambda)\Lambda \ge \frac{\kappa_-}{2}\Lambda \ge 0$ because $e_\Lambda \le \kappa_-/2$; thus $\omega_p \ge 0$ and $H(p) = \int \omega_p \ge 0$.

(ii) $H\eta - \omega_p$ is supported in the correction core and has $m(H\eta - \omega_p) = 0$ by the choice of $H$; so $d_\Sigma \Gamma_r = \kappa + H\eta - \omega_p = \kappa_{\mathrm{sm}} + H\eta$. The terms $c_p \beta$ and $d_\Sigma(\zeta u_p)$ are closed. On $\Delta$, $\kappa_{\mathrm{sm}}$ is a convex combination of the positive forms $\kappa(p)$ and $e_\Lambda \Lambda$, and $H\eta = H f' b \, ds \wedge dv \ge 0$ by (i).

(iii) The cutoff $\chi_\partial$ vanishes on the annulus (Definition 6.1(g)), so there $d_\Sigma \Gamma_r = e_\Lambda \Lambda + H \, df \wedge \beta = d_\Sigma N_p$, and $\Gamma_r - N_p$ is closed: $\partial_v A_p = \partial_{s_{\mathrm{a}}} B_p$. By Stokes' theorem its period $\int_0^1 B_p(s_{\mathrm{a}}, v)\, dv$ does not depend on $s_{\mathrm{a}}$ and equals $c_p$. Hence $\partial_v u_p = B_p - c_p b$ and $\partial_{s_{\mathrm{a}}} u_p = A_p(s_{\mathrm{a}}, 0) + \int_0^v \partial_\nu A_p(s_{\mathrm{a}}, \nu)\, d\nu = A_p$, and $u_p(s_{\mathrm{a}}, v+1) - u_p(s_{\mathrm{a}}, v) = c_p - c_p \int_0^1 b = 0$; so $u_p$ is a smooth function on the annulus with $d_\Sigma u_p = \Gamma_r - N_p - c_p \beta$. Where $\zeta = 1$ this gives $\Gamma_n = N_p$. □

8.2. **The three Moser fields.** For given closed source capacities $\rho_i < a_i$ put $\rho_i^\sharp = (3a_i + \rho_i)/4$. Take the caps $h_i$ and clocks $\varsigma_j$ of (48) with this $\rho_i^\sharp$, and the strip parameter $\varepsilon_i = (a_i - \rho_i^\sharp)/4$; these are the constants $\hat\rho_i = \rho_i^\sharp$ and $\varepsilon_i$ of Definition A.17. The three primitive families, without the vertical Liouville form $\lambda_V$, are

$$Q_{0,t} = (1-t)\Gamma_{\mathrm{phys}} + t\Gamma_n, \qquad Q_{1,t} = \Gamma_n + \eta_t, \qquad Q_{2,t} = \Gamma_n + \eta_1 + t\xi,$$

with $\eta_t$ the suspension primitive of Lemma 6.4 (built from the twist (47)) and $\xi$ the one-form (49), formed with these caps and clocks and with $H^* = H$. Put $\lambda_{j,t} = \lambda_V + Q_{j,t}$ and $\Omega_{j,t} = d\lambda_{j,t}$; then $\Omega_{0,0} = \Omega$, $\Omega_{0,1} = \Omega_{1,0} = \Omega_1$, $\Omega_{1,1} = \Omega_{2,0}$ and $\Omega_{2,1} = \Omega_{\mathrm{final}}$, and $\partial_t \Omega_{j,t} = d(\partial_t Q_{j,t})$. By §6.1 each $\Omega_{j,t}$, $t \in [0,1]$, is symplectic on a neighbourhood of the closed model region $K = V_\Delta(E)$.

Let $w_i$ be the bumps of the flow atlas ($w_i = 1$ on the inner box of chart $i$, compactly supported in chart $i$; the inner boxes cover the model base) and put

$$g_j(t,z) = \sum_i w_i(z) \det\big(\Omega_{j,t} \text{ in chart } i\big)^2, \qquad \mu_j = \min\Big(1, \min_{[0,1]\times K} g_j\Big).$$

At each point of $K$ some $w_i$ equals 1 and the corresponding determinant is nonzero, so $g_j > 0$ on the compact set $[0,1]\times K$ and $\mu_j > 0$. Conversely, $g_j(t,z) > 0$ implies that $\Omega_{j,t}$ is nondegenerate at $z$. With $|w|^2$ the sum of squares of the real fibre coordinates and $R = \max_K |w|^2$, set

$$\chi_j(t,z) = S\big(4g_j(t,z)/\mu_j - 1\big)\, S\big(2(R + 1 - |w|^2)\big) \qquad (t \in [0,1]).$$

Then $\chi_j = 1$ on the open set $\{g_j > \mu_j/2,\ |w|^2 < R + \frac12\} \supset [0,1]\times K$; and $\chi_j(t,\cdot)$ is supported in the compact set $\{g_j(t,\cdot) \ge \mu_j/4,\ |w|^2 \le R+1\}$, on which $\Omega_{j,t}$ is nondegenerate (the base part is compact because the $w_i$ have compact support). Define $Y_{j,t}$ by $\iota_{Y_{j,t}} \Omega_{j,t} = -\chi_j \, \partial_t Q_{j,t}$ on the nondegenerate locus and $Y_{j,t} = 0$ off the support of $\chi_j$; the two definitions agree on the overlap, so $Y_j$ is smooth on $[0,1]\times E$ with compact spatial support. Let $\Phi_j$ be its time-one map, a diffeomorphism of $E$ whose inverse is the backward flow.

**Construction 8.4** (The canonical toric map)**.** Let $E_i$ be the layer map of Construction 7.6 for the constants above, and $\mathcal{D}^{-1}(s,v,w) = (s,v,q_{f(s)}^{-1}(w))$ the inverse twist. The canonical map is

$$\psi_i = \Phi_0^{-1} \circ \Phi_1^{-1} \circ \Phi_2^{-1} \circ \mathcal{D}^{-1} \circ E_i \colon B^\circ(\rho_i^\sharp) \to E.$$

This formula defines the maps of the quadric, mixed and six-dimensional models.

**Lemma 8.5** (Invariance of the model region)**.** *For $j = 0,1,2$, every integral curve $\gamma\colon J \to E$ of $Y_j$, $J \subset [0,1]$ an interval, which meets $K$ (resp.* $\operatorname{int} K$*) stays in $K$ (resp.* $\operatorname{int} K$*).*

*Proof.* Write $K = \Sigma_\ell \times V_\Delta$ with $V_\Delta = \{\mu_\nu \le \ell_\nu \text{ for all } \nu\}$, $\mu_\nu = \langle b_\nu, p\rangle$; the conditions $p_j \ge 0$ hold everywhere.

*Base boundary.* The variation primitives vanish over a neighbourhood $N$ of $\partial\Sigma_\ell$ in the base: $\partial_t Q_{0,t} = \alpha_1 = \mathcal{P}(H\eta - \omega_p) - c_p\beta - d_\Sigma(\zeta u_p)$ is supported in the union of the correction core, the support of $\beta$ and the annulus; $\partial_t Q_{1,t} = \partial_t \eta_t$ is supported over the annulus (Lemma 6.4(iii)); and $\partial_t Q_{2,t} = \xi$ is supported in the support of $\beta$ and over the annulus (equation 49). These are compact subsets of $\operatorname{int}\Sigma_\ell$, so $Y_j$ vanishes over $N$ (Lemma A.4(c)); by uniqueness an integral curve meeting a point over $N$ is constant, and no integral curve crosses $\partial\Sigma_\ell$.

*Outer fibre faces.* Fix $\nu$ and let $X_\nu = \sum_j b_{\nu,j}\partial_{\theta_j}$, so that $\iota_{X_\nu}\omega_V = -d\mu_\nu$; its flow $\phi_r$ is a torus rotation of the fibre, preserves $p$, and commutes with every map that depends on $w$ only through $p$. By Definition M2.4(1), $Q_t$ Poisson-commutes with $\mu_\nu$ on the collar $C_\nu = \{\mu_\nu > \ell_\nu - \varsigma\}$. Hence $\mu_\nu$ is constant along trajectories of $X_{Q_t}$ while they stay in $C_\nu$; since $C_\nu$ is a superlevel set of $\mu_\nu$, a trajectory starting in $C_\nu$ never leaves it, so $q_t$ and $q_t^{-1}$ map $C_\nu$ into itself and preserve $\mu_\nu$ there. Also $Q_t$ is constant along $\phi_r$ in $C_\nu$, so $\phi_r^* X_{Q_t} = X_{Q_t}$ there (the form $\omega_V$ is $\phi_r$-invariant). The curves $t \mapsto \phi_r(q_t(w))$ and $t \mapsto q_t(\phi_r(w))$, $w \in C_\nu$, solve the same ODE with the same initial value; hence $\phi_r \circ q_t = q_t \circ \phi_r$ on $C_\nu$, and likewise for $q_t^{-1}$.

Each $Q_{j,t}$ is a horizontal one-form whose coefficients are built from the toric forms $\Gamma_{\text{phys}}(p)$, $\Gamma_n(p)$, $J(p)\beta$, from the functions $Q_{\lambda f(s)} \circ q_{\lambda f(s)}$ and $H^*(p(q_\lambda^{-1} q_{\lambda f(s)} w))$ of Lemma 6.4, and from $G_{\mathrm{f}} \circ \mathcal{D}$ and $G_1 \circ \mathcal{D}$, where $G_{\mathrm{f}}$ involves $R^*(w) = K(p(q^{-1}w)) - J(p) - \sum_i h_i(p)$. By the commutation just proved, each of these is invariant under $\phi_r$ over $C_\nu$; for instance $Q_t(q_t(\phi_r w)) = Q_t(\phi_r(q_t w)) = Q_t(q_t w)$. Hence, over $C_\nu$: (a) $\partial_t Q_{j,t}(X_\nu) = 0$, since $Q_{j,t}$ is horizontal and $X_\nu$ is vertical; (b) $\mathcal{L}_{X_\nu} Q_{j,t} = 0$, so $\iota_{X_\nu} dQ_{j,t} = \mathcal{L}_{X_\nu} Q_{j,t} - d\big(Q_{j,t}(X_\nu)\big) = 0$ and $\iota_{X_\nu}\Omega_{j,t} = \iota_{X_\nu}\omega_V = -d\mu_\nu$. Lemma A.4(d), with $\mathfrak{v}_t = \chi_j \partial_t Q_{j,t}$, $V = X_\nu$ and $\mu = \mu_\nu$, shows that $\mu_\nu$ is constant along every integral curve of $Y_j$ while the curve stays in $C_\nu$. An integral curve therefore cannot pass from $\{\mu_\nu \le \ell_\nu\}$ to $\{\mu_\nu > \ell_\nu\}$, nor from $\{\mu_\nu < \ell_\nu\}$ to $\{\mu_\nu \ge \ell_\nu\}$, in either time direction: near a time at which it is on the level $\ell_\nu$ it lies in $C_\nu$, where $\mu_\nu$ is constant along it. Applying this to every $\nu$ and combining with the base boundary gives the claim for $K$ and for $\operatorname{int} K = \operatorname{int}\Sigma_\ell \times \{\mu_\nu < \ell_\nu \text{ for all } \nu\}$ (Lemma 7.1). □

*Correctness of the construction.* Since $\chi_j = 1$ near $[0,1]\times K$, Lemma 8.5 and Lemma A.4(b),(e) with $\mathfrak{v}_t = \partial_t Q_{j,t}$ give $\Phi_j(K) = K$, $\Phi_j(\operatorname{int} K) = \operatorname{int} K$ and $\Phi_j^*\Omega_{j,1} = \Omega_{j,0}$ on $K$. Hence $\Phi = \Phi_2 \circ \Phi_1 \circ \Phi_0$ satisfies $\Phi^*\Omega_{\text{final}} = \Omega$ on $K$ (Lemma 7.8), and Proposition 7.9 shows that the $\psi_i$ are smooth symplectic embeddings of $B^\circ(\rho_i^\sharp)$ into $(\operatorname{int} K, \Omega)$ with pairwise disjoint images. Since $\rho_i < \rho_i^\sharp$, $B^\circ(\rho_i^\sharp)$ is a neighbourhood of $B(\rho_i)$. By Lemma A.4(f), running the three fields consecutively with flat clocks gives one Moser isotopy with time-one map $\Phi$ and the same variation primitives. □

## 9. Dimension at least six: capacities below one half

Let $n \ge 3$, $m = n-1$, and $\rho_i = c_i/C$. The model is a fixed affine curve times $m$ complex fibre planes. Its horizontal curvature has coefficient $1 - 2|p|$, so the fibre simplex stops below

$|p| = 1/2$; compact exhaustions of the curve have area tending to $2^m$, the integrated profiles tend to $1/n!$, and the truncation loss is subtracted from the comparison profile. The transfer into $B^\circ(1)$ realises the normal-transfer mechanism of [Ope26] in affine coordinates.

9.1. **The affine curve and the finite model.**

**Definition 9.1** (The diagonal affine curve)**.** Let $z = (t, y_1, \dots, y_m) \in \mathbb{C}^{m+1}$ and put

$$F_j(z) = y_j^2 - t^2 + j, \qquad \Sigma = \{z : F_1(z) = \cdots = F_m(z) = 0\}, \qquad D(z) = 1 + |z|^2.$$

On $\Sigma$ set $\lambda = d^c \log D$ and $\sigma = d\lambda$, using $d^c = i(\bar\partial - \partial)/(4\pi)$. For an integer $\ell \geq 0$ put

$$R_\ell = 2 + \sum_{j=1}^{m} j + \ell, \qquad \Sigma_\ell = \{z \in \Sigma : |t| \leq R_\ell\}, \qquad A_\ell = \int_{\Sigma_\ell} \sigma.$$

**Lemma 9.2** (Regularity and exhaustion)**.** *The affine locus $\Sigma$ is a smooth complex curve, $\sigma$ is positive, and $t : \Sigma \to \mathbb{C}$ is proper. Each $\Sigma_\ell$ is a compact connected surface with smooth boundary and connected nonempty interior. The fixed annular supports of Proposition 10.4 lie in its interior, and*

$$0 \leq 2^m - A_\ell \leq \frac{2^m(m+2)}{R_\ell^2} \leq \frac{2^m(m+2)}{\ell+2}.$$

*Proof.* At most one $y_j$ vanishes. If none vanishes, the $y$-columns of $dF$ are invertible; if $y_i = 0$, then $t^2 = i \neq 0$ and the columns $t, (y_j)_{j\neq i}$ are invertible. The implicit coordinate formulas are given in Lemma 10.2. Positivity follows by restricting the affine Fubini–Study form, and $|y_j|^2 \leq |t|^2 + j$ proves properness. All critical values of $t$ lie at $\pm\sqrt{j}$, strictly inside $|t| < R_\ell$, so the boundary is smooth. Loops about these branch values independently reverse the root signs and connect the sheets over the disc; thus the interior is connected. The area estimate, including its limiting constant $2^m$, is proved from the explicit affine ends in Lemma 10.5. □

**Definition 9.3** (The finite quadric model)**.** For $0 < h < 1/2$ take the product $E_\ell = \Sigma_\ell \times \mathbb{C}^m$, with $p_j = \pi|U_j|^2$, $|p| = \sum_j p_j$,

$$\Delta_h = \{p \geq 0 : |p| \leq h\}, \qquad \Omega_0 = \omega_V + d[(1 - 2|p|)\lambda], \qquad H_\ell(p) = A_\ell(1 - 2|p|).$$

The differential includes the fibre variables, and the marked set is empty. The limiting profile $H_{\mathrm{b}}(p) = 2^m(1 - 2|p|)$ is a comparison budget, not the area of a finite base. All forms and maps are defined on $\Sigma \times \mathbb{C}^m$.

**Lemma 9.4** (The bordered toric model)**.** *For $0 < h < 1/2$, this is a bordered toric model of rank $m$. Its horizontal form at fixed moment is $(1 - 2|p|)\sigma > 0$, its base area is exactly $H_\ell$, and its fixed handle data are those of Proposition 10.4. The finite primitive operator is applied to forms supported away from the outer collar.*

*Proof.* The form is closed and torus invariant, and restricts to $\omega_V$ on each fibre. In product coordinates it is

$$\Omega_0 = \omega_V + (1 - 2|p|)\sigma - 2\sum_j dp_j \wedge \lambda.$$

Its horizontal coefficient is positive on a neighbourhood of $\Delta_h$, and the compact base and its boundary have the properties of Lemma 9.2. The annuli, closed dual, collar cutoff and finite coordinate primitive are supplied by Proposition 10.4; the mass budget is checked in the proof of Theorem 9.7. □

**Lemma 9.5** (Simplex integrals)**.** *Let $m \geq 1$ be an integer and $\Delta_h = \{p \in \mathbb{R}^m_{\geq 0} : |p| \leq h\}$.*

*(a) For $0 < h < \infty$ and continuous $f\colon [0, h] \to \mathbb{R}$, $\int_{\Delta_h} f(|p|)\, dp = \frac{1}{(m-1)!} \int_0^h f(t)\, t^{m-1}\, dt$.*

*(b) For $0 < h \le \frac{1}{2}$,*

$$\int_{\Delta_h} 2^m(1-2|p|)\,dp = J_m(h) := \frac{2^m h^m}{m!}\Big[1 - \frac{2mh}{m+1}\Big], \qquad J_m(\tfrac{1}{2}) = \frac{1}{(m+1)!}.$$

*(c) For $h = \frac{1}{2} - \delta$ with $0 < \delta < \frac{1}{2}$, $\frac{1}{(m+1)!} - J_m(h) \le \frac{2\delta^2}{(m-1)!}$.*

*Proof.* (a) The simplex $\{p \in \mathbb{R}^m_{\ge 0} : |p| \le t\}$ has volume $t^m/m!$, so the distribution of $|p|$ on $\Delta_h$ has density $t^{m-1}/(m-1)!$. (b) With $f(t) = 2^m(1-2t)$, (a) gives $\frac{2^m}{(m-1)!}\big(\frac{h^m}{m} - \frac{2h^{m+1}}{m+1}\big) = J_m(h)$, and $J_m(\frac{1}{2}) = \frac{1}{m!}(1 - \frac{m}{m+1}) = \frac{1}{(m+1)!}$. (c) Using $t^{m-1} \le 2^{-(m-1)}$ on $[h, \frac{1}{2}]$,

$$\frac{1}{(m+1)!} - J_m(h) = \frac{2^m}{(m-1)!}\int_h^{1/2}(1-2t)t^{m-1}\,dt \le \frac{2}{(m-1)!}\int_h^{1/2}(1-2t)\,dt = \frac{2\delta^2}{(m-1)!}. \quad \square$$

For the finite model the integrated profile is $(A_\ell/2^m)J_m(h)$, which tends to $1/n!$ as $\ell \to \infty$ and $h \uparrow 1/2$.

9.2. **The affine transfer.** For $0 < b < 1/2$ the transfer $\mathcal{T}_b = \mathrm{ch} \circ \psi_1^{-1} \circ \mathrm{pr} \circ \tau \circ \iota_b$ of §10 is the composite of the radial injection $\iota_b$ (Construction 10.7), the horizontal transport $\tau$ along the graph family $X_T = \{F(z) = TW\}$ from $T = 0$ to $T = 1$ (Proposition 10.10), the inverse Moser return $\psi_1^{-1}$ (Proposition 10.11) and the ball chart $\mathrm{ch}(z) = z/\sqrt{\pi D(z)}$. It is a symplectic embedding of the tube $\{|p| < b\}$ into $B^\circ(1)$ (Theorem 10.12).

9.3. **Parameters and packing.**

**Lemma 9.6** (Parameters for the quadric branch)**.** *Let $n \ge 3$ and let $a_1, \dots, a_k$ be real numbers with $0 < a_i < \frac{1}{2}$ and $\sum_i a_i^n < 1$. Put $a_{\max} = \max_i a_i$ and*

$$\varsigma = 1 - \sum_i a_i^n, \quad \delta = \min\Big(\frac{1/2 - a_{\max}}{2},\ \frac{\varsigma}{4n(n-1)}\Big),$$

$$h = \tfrac{1}{2} - \delta, \quad b = \frac{h + 1/2}{2}, \quad \tau_\Delta = \frac{a_{\max} + h}{2h}.$$

*These numbers are real (rational for rational $a$), and:*

*(i) $a_{\max} < h < b < \frac{1}{2}$, $\tau_\Delta < 1$, and $\{|p| \le a_i\} \subset \tau_\Delta \Delta_h$ for every $i$;*
*(ii) $P_{\mathrm{b}}(p) = H_{\mathrm{b}}(p) - \sum_i (a_i - |p|)_+$ is a continuous, concave piecewise affine function (rational when $a$ is rational) on $\Delta_h$;*
*(iii) $\int_{\Delta_h} P_{\mathrm{b}}\,dp > 7\varsigma/(8\,n!)$, and the mean $\mu$ of $P_{\mathrm{b}}$ on $\Delta_h$ satisfies $\mu > 7\varsigma/(8\,n!)$;*
*(iv) $P_{\mathrm{b}} \ge P_{\min} := 2^m(1-2h) = 2^{m+1}\delta > 0$ on $\Delta_h \setminus \tau_\Delta \Delta_h$;*
*(v) the number $d = \frac{1}{2}\min\big(\varsigma/(2\,n!),\ 2^{m+1}\delta\big)$ satisfies $0 < d < \min(\mu, P_{\min})$.*

*The factor $\frac{7}{8}$ leaves room for the profile-budget defect $\frac{1}{n!} - J_m(h) < \varsigma/(8\,n!)$ of the proof of (iii). If $0 < \rho_i < \frac{1}{2}$ and $\sum_i \rho_i^n < 1$, rational numbers $a_i > \rho_i$ with $a_i < \frac{1}{2}$ and $\sum_i a_i^n < 1$ exist, and any enumeration of $\mathbb{Q}^k$ reaches one after finitely many steps.*

*Proof.* (i) $\delta > 0$, so $h < b < \frac{1}{2}$; and $h \ge \frac{1}{2} - \frac{1/2 - a_{\max}}{2} = \frac{1/2 + a_{\max}}{2} > a_{\max}$. Hence $\tau_\Delta h = \frac{a_{\max} + h}{2} \in (a_{\max}, h)$, which gives $\tau_\Delta < 1$ and $\{|p| \le a_i\} \subset \{|p| \le \tau_\Delta h\}$. (ii) $H_{\mathrm{b}}$ is affine with integer coefficients and each $(a_i - |p|)_+$ is convex and piecewise affine (rational for rational $a_i$). (iii) Since $a_i < h$, the support of $(a_i - |p|)_+$ in $\mathbb{R}^m_{\ge 0}$ lies in $\Delta_h$, so by Lemma M2.2 its integral over $\Delta_h$ is $a_i^n/n!$. With Lemma 9.5(b),(c) for $m = n - 1$,

$$\int_{\Delta_h} P_{\mathrm{b}}\,dp = J_m(h) - \frac{1-\varsigma}{n!} = \frac{\varsigma}{n!} - \Big(\frac{1}{n!} - J_m(h)\Big) \ge \frac{\varsigma}{n!} - \frac{2\delta^2}{(n-2)!}.$$

As $\frac{1}{(n-2)!} = \frac{n(n-1)}{n!}$ and $\delta \le \varsigma/(4n(n-1))$, $\frac{2\delta^2}{(n-2)!} \le \frac{\varsigma^2}{8n(n-1)\,n!} < \frac{\varsigma}{8\,n!}$, because $0 < \varsigma < 1$ and $n(n-1) \ge 6$. This gives the bound on the integral. Since $\operatorname{vol}\Delta_h = h^m/m! < 1$ and the integral is positive, $\mu$ exceeds the integral. (iv) On $\Delta_h \setminus \tau_\Delta \Delta_h$ we have $|p| > \tau_\Delta h > a_{\max}$, so all caps vanish and $P_{\mathrm{b}} = H_{\mathrm{b}} \ge 2^m(1-2h)$. (v) follows from (iii) and (iv).

For the last assertion, the set $\{a \in \mathbb{R}^k : \ a_i > \rho_i,\ a_i < \frac{1}{2},\ \sum_i a_i^n < 1\}$ is open and contains $\rho + t(1,\dots,1)$ for all small $t > 0$, so it contains rational points, which any enumeration of $\mathbb{Q}^k$ reaches. □

**Theorem 9.7** (Packing by small balls in dimension $2n \ge 6$)**.** *Let $n \ge 3$ and $0 < \rho_1, \dots, \rho_k < \frac{1}{2}$ with $\sum_i \rho_i^n < 1$, and let $a_1, \dots, a_k$ be real numbers with $\rho_i < a_i < \frac{1}{2}$ and $\sum_i a_i^n < 1$. Put $\rho_i^+ = (2\rho_i + a_i)/3$ and $\rho_i^{\mathrm{dom}} = (\rho_i + 2a_i)/3$. The comparator construction with real coefficients, with closed source capacities $\rho_i^{\mathrm{dom}}$ (Construction 8.4), defines from these data smooth symplectic embeddings $f_i$ of open neighbourhoods of the closed balls $B(\rho_i^{\mathrm{dom}}) \subset \mathbb{C}^n$ into $B^\circ(1)$ and a rational $\zeta > 0$, such that the sets $f_i(B(\rho_i^{\mathrm{dom}}))$ are pairwise disjoint and contained in $B(1-\zeta)$.*

*Proof of Theorem 9.7. Parameters and comparison.* Let $\delta, h, b, \tau_\Delta, d$ be given by Lemma 9.6 for $a$, put $P = P_{\mathrm{b}}$. By parts (i)–(v) of that lemma, $\Delta_h$, $\tau_\Delta$ and $P$ satisfy the hypotheses of Proposition M2.9 with $\beta = P_{\min}$ and with this $d$, and Theorem M2.11 provides a comparison datum $(K, q)$ for $(\Delta_h, P_{\mathrm{b}})$ with gap $\gamma = 11d/16 > 0$.

*Truncation and packing.* Put $m = n - 1$ and

$$l = \left\lceil \frac{2^{m+6}(m+2)}{d} \right\rceil, \qquad R_l = 2 + \frac{m(m+1)}{2} + l, \qquad \Sigma_l = \{|t| \le R_l\} \subset \Sigma.$$

By Lemma 10.5, its exact profile is $H_l(p) = A_l(1-2|p|)$, with $0 \le H_{\mathrm{b}} - H_l \le d/64$. Lower the comparison profile to $P_l = P_{\mathrm{b}} - 11d/32$; the same comparison datum has gap $\gamma_l = 11d/32$, because replacing $P$ by $P - c$ reduces its gap by $c$. Also

$$P_l \le H_l - \sum_i (a_i - |p|)_+.$$

The handle data are those of Construction 6.10 on $\Sigma_l$: by Lemma 6.11 the inner boxes of the correction atlas cover $\{|t| \le R_l - 1\}$, which contains the handle and the support of $1 - \zeta_\partial$, and the correction region is all of $\Sigma_l$ with $\kappa_- = 1 - 2h$, as required in §8. Set $\gamma_* = 11d/64$ and

$$c_{\mathrm{bg}} = \min\left(\frac{\gamma_*}{32\,2^m}, \frac{\kappa_-}{2}\right), \qquad \zeta_\partial(t) = S\left(\frac{|t|^2 - (R_l-2)^2}{(R_l-1)^2 - (R_l-2)^2}\right).$$

The background form $\big[\zeta_\partial(1-2|p|) + (1-\zeta_\partial)c_{\mathrm{bg}}\big]\sigma$ agrees with the original curvature near the boundary and is positive, since $c_{\mathrm{bg}} \le \kappa_-/2$. The removed density $(1-\zeta_\partial)(1-2|p|-c_{\mathrm{bg}})\sigma$ has compact support in $\{|t| \le R_l - 1\}$; its integral is the handle mass. The same end estimate as in Lemma 10.5, at radius $R_l - 2$, gives

$$\int_{|t| > R_l - 2} \sigma \le \frac{2^m(m+2)}{(R_l-2)^2} \le \frac{d}{65536} < \frac{\gamma_*}{32}.$$

Here $d < 1$, $m \ge 2$ and $R_l - 2 \ge l \ge 64\,2^m(m+2)/d$. Since $c_{\mathrm{bg}} \int_{\Sigma_l} \sigma \le \gamma_*/32$, the whole background mass is less than $\gamma_*/16$. Also $\min H_l > d$, so $\epsilon_0 = \gamma_*/16$ in (42). Apply the layer construction with the further interior source capacity $u_i = (3a_i + \rho_i^{\mathrm{dom}})/4 < a_i$. Theorem M2.6, with the exact truncated profile and its positive remaining gap, gives smooth symplectic embeddings $\psi_i$ on a neighbourhood of $B(\rho_i^{\mathrm{dom}})$ into $\operatorname{int} V_{\Delta_h}(E|_{\Sigma_l})$, with pairwise disjoint images.

*Transfer.* By Theorem 10.12, $\mathcal{T}_b$ is a symplectic embedding of a neighbourhood of $V_{\Delta_h}(E_l)$ into $B^\circ(1)$. Proposition M2.7 gives $f_i = \mathcal{T}_b \circ \psi_i$, smooth and symplectic on $B^\circ(\rho_i^{\mathrm{dom}})$, with pairwise disjoint images of $B(\rho_i^{\mathrm{dom}})$, since $\mathcal{T}_b$ is injective.

*Margin.* Let $\mathcal{K} = \bigcup_i \psi_i(B(\rho_i^{\mathrm{dom}}))$ and $m_{\mathcal{K}} = \min_{\mathcal{K}}(1 - \pi|\mathcal{T}_b|^2) > 0$. The positive rational $\zeta = (2 + \lceil 1/m_{\mathcal{K}} \rceil)^{-1} < m_{\mathcal{K}}$ puts all images in $B(1-\zeta)$. □

If some normalised capacity is at least $\frac{1}{2}$, the cap $\{|p| \le a_i\}$ does not fit in $\Delta_h$ for any $h < \frac{1}{2}$. This case is treated in Section 11.

## 10. The affine quadric transfer

We use the affine locus $\Sigma \subset \mathbb{C}^{m+1}$, the forms $\lambda, \sigma$ and the exhaustion $\Sigma_\ell$ of Definition 9.1, and write $F(z) = (y_j^2 - t^2 + j)_{j=1}^m$. In §10.1 $m = n - 1 \ge 2$; the transfer of §§10.2–10.4 is valid for every $m \ge 1$.

### 10.1. **Sheets and explicit coordinates.**

**Lemma 10.1** (The affine sheets)**.** *The projection $t : \Sigma \to \mathbb{C}$ has $2^m$ points in every fibre away from $\{\pm\sqrt{j} : 1 \le j \le m\}$. Over $t = \pm\sqrt{i}$ it has $2^{m-1}$ points, with simple branching in coordinate $y_i$. Outside a sufficiently large disc its $2^m$ ends are indexed by signs $\epsilon_j \in \{\pm 1\}$ and have coordinates*

$$t = s^{-1}, \qquad y_j = \epsilon_j s^{-1}\sqrt{1 - js^2}, \qquad 0 < |s| < 1/R_0,$$

*where $R_0 = 2 + \sum_j j$. Each root is the branch equal to one at zero; the inverse is $s = 1/t$.*

*Proof.* The independent root signs count the fibres. At a branch value only one root vanishes, and the local equation is $y_i^2 = t^2 - i$, with derivative $2t \ne 0$ in $t$. The formula at the ends follows by factoring $t^2$ from every radicand; on $|s| < 1/R_0$ all perturbations have modulus less than one, so these roots and signs are unambiguous and account for every point. □

**Lemma 10.2** (Explicit coordinate boxes)**.** *Let $c = (t_c, y_c) \in \Sigma$.*

*(i) If every $y_{c,j} \ne 0$, a coordinate near $c$ is $u = t$, with inverse*

$$t = u, \qquad y_j = y_{c,j}\sqrt{1 + (u^2 - t_c^2)/y_{c,j}^2}.$$

*(ii) If $y_{c,i} = 0$, a coordinate near $c$ is $v = y_i$, with inverse*

$$t = t_c\sqrt{1 + v^2/t_c^2}, \qquad y_i = v, \qquad y_j = y_{c,j}\sqrt{1 + v^2/y_{c,j}^2} \quad (j \ne i).$$

*Here $t_c^2 = i$, and $y_{c,j}^2 = i - j \ne 0$ for $j \ne i$.*

*The square roots are the branches equal to one at zero. Restricting a chart to a coordinate square on which every perturbation of its radicand has modulus less than one gives an injective holomorphic chart. The inverse coordinates are ambient coordinate projections, and transition functions are compositions of these projections with the roots above.*

*Proof.* Squaring the coordinates verifies all equations. At the centre the ratios $y_j/y_{c,j}$ (in (i)), respectively $t/t_c$ and $y_j/y_{c,j}$, $j \ne i$ (in (ii)), equal one, and requiring positive real part selects these branches uniquely. The coordinate projection composed with the inverse chart is the identity, and the branch conditions give the reverse identity. Holomorphicity follows from the binomial series on the prescribed radicand domains. □

**Lemma 10.3** (A finite rational family on a compact projection domain)**.** *Let $R \ge 0$ be rational and put*

$$\eta = \frac{1}{64}, \quad r = \frac{1}{262144(R+1)}, \quad h = \frac{r}{32}, \quad N = \lceil R/h \rceil + 2.$$

*Consider the grid $q = h(a + ib)$ with $-N \le a, b \le N$. Retain the ordinary centres*

$$c(q, \epsilon) = \big(q, (\epsilon_j\sqrt{q^2 - j})_{j=1}^m\big)$$

*that pass the rational tests*

$$|q|^2 < (R+\tfrac{1}{2})^2, \qquad |q^2-j|^2 > (\eta^2/2)^2 \quad (1 \le j \le m).$$

*Use all signs, ordinary coordinate squares of outer half-width $r$ and inner half-width $r/4$, and the branch charts centred at every $t_c = \pm\sqrt{i}$, $y_{c,i} = 0$, $y_{c,j} = \pm\sqrt{i-j}$, with outer half-width $1/8$ and inner half-width $1/32$. The inner open squares cover $\{|t| \le R\} \subset \Sigma$. Every ordinary closed outer square lies in $\{|t| < R+1\}$. The same holds for the branch squares provided $i + 1/16 < (R+1)^2$ for all $i$. All index tests are comparisons of rational polynomials in* $\operatorname{Re} q$, $\operatorname{Im} q$.

*Proof.* The chosen constants satisfy

$$2r\big(2(R+1)+2r\big) < \eta^2/2, \qquad 4r(2R+4r) < \eta^2, \qquad 2h \le r/8, \qquad R+h < Nh.$$

If every $|y_j| \ge \eta$, choose a grid point with $|q-t| < 2h \le r/8$. Then

$$|q^2 - t^2| \le |q-t|(|q|+|t|) < \eta^2/2,$$

so $|q^2 - j| > \eta^2/2$ for each $j$, and the centre passes the tests. For one of its root signs all ratios to the coordinates of the point have positive real part, and the ordinary chart formula puts the point in the inner square. The first displayed inequality bounds every radicand perturbation on the outer square strictly below one; also $|q| + 2r < R+1$. If some $|y_i| < \eta = (1/8)/8$, the separated parameters give $|t|^2$ close to $i$ and $|y_j|^2$ close to $|i-j|$, and the branch formula with the corresponding root signs applies; its perturbations have modulus at most $4(1/8)^2 < 1$, and the point is in the inner square. On its closed outer square, $|t|^2 = |i + y_i^2| \le i + 1/16 < (R+1)^2$. □

**Proposition 10.4** (Fixed handle data)**.** *On each compact truncation containing the fixed supports, the quadric model has the annular data below, chosen independently of $h$; its primitive operator acts on zero-mass forms with compact support in the affine interior. Put $P(z) = (z+z^{-1})/2$ and $M(z) = (z - z^{-1})/2$. Two annuli, initially defined near $|z| = 1$, are*

$$A(z) = \big(\sqrt{1+P(z)^2},\ P(z),\ M(z),\ (i\sqrt{j-1-P(z)^2})_{j=3}^{m}\big),$$
$$C(z) = \big(P(z),\ M(z),\ (i\sqrt{j-P(z)^2})_{j=2}^{m}\big).$$

*Their inverses are respectively $z = y_1 + y_2$ and $z = t + y_1$. The transverse cylinder is $e(s,\theta) = C(\exp(s + 2\pi i\theta))$. Fix $\varepsilon = 1/8$ and $w = 1/4$, and let*

$$T_\varepsilon(z) = S\left(\frac{|z|^2 - (1-\varepsilon)^2}{(1+\varepsilon)^2 - (1-\varepsilon)^2}\right), \qquad \beta_0 = A_* dT_\varepsilon,$$

*with zero extension outside $A$. Then $\beta_0$ is smooth, closed, compactly supported and has period one on the core of $C$. With $b(\theta) = 2S'(2\theta - 1/2)$ on $[0,1]$, extended periodically, the fixed smooth circle clock supported in $[1/4, 3/4]$ with integral one, write*

$$e^*\beta_0 = A_0(s,\theta)\, ds + B_0(s,\theta)\, d\theta.$$

*The periodic function*

$$F(s,\theta) = \int_0^s A_0(v,0)\, dv + \int_0^\theta (B_0(s,v) - b(v))\, dv$$

*satisfies $dF = e^*\beta_0 - b(\theta)d\theta$. With*

$$\chi(s) = S\big((7w/8 - s)4/w\big)S\big((7w/8 + s)4/w\big), \qquad \beta = \beta_0 - d\big(e_*(\chi F)\big),$$

*the function $e_*(\chi F)$ is extended by zero. The form $\beta$ equals $b(\theta)d\theta$ on $|s| < 5w/8$. The outer cylinder has width $w = 1/4$ and the annular handle width $w/2 = 1/8$; the normalization and layer paths use the working half-width $b_H = w/4 = 1/16$ and the radial clock $S((s+b_H)/(2b_H))$.*

*In the product fibre coordinates,*

$$\Omega_0 = \omega_V + d\Gamma_{\mathcal{A}}, \qquad \Gamma_{\mathcal{A}} = (2|p|-1)\vartheta, \qquad \vartheta = -d^c \log(1+|t|^2+\textstyle\sum_j |y_j|^2).$$

*The auxiliary area form and the zero-mass primitive operator are obtained from the fixed finite coordinate families of Lemma 10.3, the chart bump partition and local integral primitive of Construction 6.7; their annular correction is the same two-segment formula for $F$.*

*Proof.* The identity $P^2 - M^2 = 1$ verifies the two annulus formulas. On $|z| = 1$, $P(z)$ is real in $[-1, 1]$ and $M(z)$ is purely imaginary, so every remaining radicand, including $1 + P^2$ and $j - 1 - P^2$ for $j \geq 3$, is positive and the formulas extend to annuli of positive width, on which the displayed inverses prove injectivity. The two core circles meet once: $y_1$ is real on the first and imaginary on the second, hence zero at an intersection; the positive choice of $t$ then gives $t = 1$, $y_2 = i$. The radial coordinate crosses the transition in $T_\varepsilon$ with positive orientation at this point. Integrating its derivative gives one, so $\int_{C(|z|=1)} \beta_0 = 1$. The derivative of $T_\varepsilon$ vanishes near both boundary circles, so extension by zero is smooth and closed. Closedness gives $\partial_\theta A_0 = \partial_s B_0$; differentiating the two integrals proves the formula for $dF$, and the period-one identity makes $F$ periodic. The cutoff equals one on $|s| \leq 5w/8$ and vanishes for $|s| \geq 7w/8$, which proves the assertions about $\beta$ and smooth zero extension. All these supports are compact in the affine curve.

The fixed widths are admissible uniformly in the number of quadrics. On $3/4 \leq |z| \leq 5/4$ one has $|P(z)| \leq 4/3$ and $|\operatorname{Im} P(z)| \leq 7/24$. Hence $\operatorname{Re}(1 + P(z)^2) > 0$ and $\operatorname{Re}(j - 1 - P(z)^2) > 0$ for $j \geq 3$, so $A$ is defined on the whole closed annulus $\frac{3}{4} \leq |z| \leq \frac{5}{4}$, which contains the support $\frac{7}{8} \leq |z| \leq \frac{9}{8}$ of $dT_\varepsilon$. For $|s| \leq 1/2$, the inequality $e^{1/2} < 7/4$ gives $|P(e^{s+2\pi i\theta})| \leq 11/8 < \sqrt{2}$; thus every $j - P^2$, $j \geq 2$, has positive real part, and $C$ is defined on that whole closed cylinder. The chosen widths leave a neighbourhood of the supports inside both coordinate domains. The finite chart partition and zero-mass local primitives give the remaining scalar handle data by Construction 6.7; the initial boxes cover the required compact support. Their overlap graph is connected by additional rational subboxes: an open cover of the connected open affine base has finite overlap chains between any two nonempty members, and each nonempty overlap contains a rational subbox with compact closure in both charts, so enumerating these subboxes and their strict inclusion certificates finds finite chains. The normalized chart bump weights give the partition, and the chains give the remaining finite data of that construction. Finally $\vartheta = -\lambda$, so $(2|p| - 1)\vartheta = (1 - 2|p|)\lambda$ gives the stated horizontal primitive directly. □

**Lemma 10.5** (Explicit truncation error)**.** *Put $R_l = 2 + \sum_{j=1}^m j + l$, $\Sigma_l = \{|t| \leq R_l\} \subset \Sigma$, and $A_l = \int_{\Sigma_l} \sigma$. Then $\Sigma_l$ is a compact connected surface with smooth boundary, and*

$$0 \leq 2^m - A_l \leq \frac{2^m(m+2)}{R_l^2} \leq \frac{2^m(m+2)}{l+2}.$$

*In particular, for $d > 0$ and*

$$l = \left\lceil \frac{2^{m+6}(m+2)}{d} \right\rceil, \qquad H_l(p) = A_l(1 - 2|p|),$$

*one has $0 \leq H_{\mathrm{b}}(p) - H_l(p) \leq d/64$ on $\Delta_h$. The handle and dual supports of Proposition 10.4 lie in the interior of $\Sigma_l$.*

*Proof.* The projection $t$ is proper on $\Sigma$, since $|y_j|^2 = |t^2 - j| \leq |t|^2 + j$. Its critical points have some $y_j = 0$, i.e. $t^2 = j$, so the circle $|t| = R_l$ contains no critical value and $\Sigma_l$ has smooth boundary. The interior is connected: analytic continuation around a small circle about $\sqrt{j}$ changes the sign of $y_j$ and of no other coordinate, so the monodromy acts transitively on the $2^m$ sheets over a

regular point of the disc. The handle supports lie in $|t| \le \frac{5}{3}$ (proof of Proposition 10.4: $|P| \le \frac{4}{3}$ on $A$ and $|P| \le \frac{11}{8}$ on $C$), inside $\Sigma_0$ since $R_l \ge 5$.

Let $R \ge R_0 = 2 + \frac{m(m+1)}{2}$. By Lemma 10.1, $\{|t| > R\}$ is the disjoint union of $2^m$ punctured discs $0 < |s| < 1/R$, on each of which $[1 : t : y] = [s : 1 : v]$ with $F(s) = (s, 1, v_1(s), \dots, v_m(s))$, $v_j(s) = \epsilon_j\sqrt{1 - js^2}$. Since $\lambda = d^c \log D$ and $\log D = \log|F(s)|^2 - \log|s|^2$ with $\log|s|^2$ harmonic, $\sigma = dd^c \log|F(s)|^2$ there, and its density with respect to Lebesgue measure in $s$ is $\pi^{-1}(|F|^2|F'|^2 - |F^\dagger F'|^2)/|F|^4 \le |F'|^2/(\pi|F|^2)$. For $|s| \le 1/R$ we have $|v_j|^2 = |1 - js^2| \ge 1 - j/R^2 \ge \frac{1}{2}$ and $|v_j'|^2 = j^2|s|^2/|v_j|^2 \le 2j^2/R^2$, and $\sum_j j^2 \le R_0^2 \le R^2$, so $|F'|^2 \le 3$; with $|F| \ge 1$ the density is at most $3/\pi$, and each end contributes at most $3/R^2$. Hence $\int_{|t|>R} \sigma \le 2^m \cdot 3/R^2 \le 2^m(m+2)/R^2$.

On the other hand, by Stokes' theorem $A_{R'} = \int_{|t|=R'} \lambda$ for $\Sigma_{R'} = \{|t| \le R'\}$. On each end, $\lambda = d^c \log|t|^2 + d^c \log|F(s)|^2 = \frac{d\arg t}{2\pi} + d^c \log|F(s)|^2$, and $|\partial \log|F|^2| = |F^\dagger F'|/|F|^2 = O(|s|)$, so the second term contributes $O(R'^{-2})$ on the circle of length $2\pi/R'$ in $s$. Thus $A_{R'} \to 2^m$, and since $\sigma > 0$, $2^m - A_R = \int_{|t|>R} \sigma \in [0, 2^m(m+2)/R^2]$. For $R = R_l \ge l + 2 \ge 2$ we get $R_l^2 \ge l + 2$, which is the second bound. The ceiling choice of $l$ gives $2^m - A_l \le d/64$, and $0 < 1 - 2|p| \le 1$ on $\Delta_h$ gives the profile estimate. □

10.2. **The affine potentials.** Write $D = 1 + |z|^2$ and $u = |W|^2/D^2$. On $\mathbb{C}^{m+1} \times \mathbb{C}^m$ put

$$\Phi_b = \log D + b \log(1 + u), \qquad \Omega_b = dd^c \Phi_b.$$

**Lemma 10.6** (Positivity and the graph forms)**.** *For $0 < b < 1/2$, $\Omega_b$ is a smooth Kähler form. For $T > 0$, its restriction to the graph $j_T(z) = (z, F(z)/T)$ is*

$$\omega_T = dd^c \left[ \log D + b \log\left(1 + \frac{|F|^2}{T^2 D^2}\right)\right].$$

*The path $\omega_s^{\mathrm{ret}} = dd^c[\log D + sb \log(1 + |F|^2/D^2)]$, $0 \le s \le 1$, is also Kähler.*

*Proof.* The identity

$$\Phi_b = (1 - 2b) \log D + b \log(D^2 + |W|^2)$$

exhibits its Hessian as a positive combination of Fubini–Study Hessians: $D^2 + |W|^2$ is the squared norm of the vector of ordered quadratic monomials of $(1, z)$ and the coordinates $W$, whose constant entry is one and whose linear $z$ entries and $W$ entries separate tangent vectors. The first summand is positive on nonzero base vectors and the second on nonzero pure fibre vectors, so the sum is positive definite. The graph identity is pullback by the holomorphic map $j_T$, and the same decomposition with $sb$ proves positivity of the return path. □

**Construction 10.7** (Radial injection)**.** For $0 < b < 1/2$, let

$$\iota_b(z, U) = \left(z, \frac{D(z)\sqrt{\pi}\, U}{\sqrt{b - |p|}}\right), \qquad \iota_b^{-1}(z, W) = \left(z, \sqrt{\frac{b}{\pi}} \frac{W}{\sqrt{D(z)^2 + |W|^2}}\right).$$

The first map has domain $|p| < b$ over $\Sigma$, and the second is defined on $\Sigma \times \mathbb{C}^m$.

**Lemma 10.8** (The radial pullback)**.** *These maps are mutually inverse smooth maps, and $\iota_b^* \Omega_b = \Omega_0$ on the tube $|p| < b$. Moreover $u \circ \iota_b = |p|/(b - |p|)$.*

*Proof.* With $|W|^2 = D^2\pi|U|^2/(b - |p|)$ we get $u = |p|/(b - |p|)$, $1 + u = b/(b - |p|)$ and $|p| = bu/(1+u)$, which verifies the inverse formulas; both denominators are positive on their domains. On $\{W \ne 0\}$, $d^c \log(1 + u) = \frac{u}{1+u}(d^c \log|W|^2 - 2\lambda)$ and $d^c \log|W|^2 = \mathrm{Im}(W^\dagger dW)/(2\pi|W|^2)$.

Along $\iota_b$, $W = cU$ with $c > 0$ a function of $(z, |p|)$, so $\operatorname{Im}(W^\dagger dW) = c^2 \operatorname{Im}(U^\dagger dU)$ and $b\frac{u}{1+u} d^c \log|W|^2 = |p|\frac{\operatorname{Im}(U^\dagger dU)}{2\pi|U|^2} = \frac{1}{2}\operatorname{Im}(U^\dagger dU) = \sum_j p_j d\theta_j$. Hence

$$\iota_b^* d^c \Phi_b = \lambda + \sum_j p_j d\theta_j - 2|p|\lambda = \sum_j p_j d\theta_j + (1-2|p|)\lambda,$$

which is smooth on the axes. Its exterior derivative is $\omega_V + d[(1-2|p|)\lambda] = \Omega_0$. □

10.3. **Complete transport in affine coordinates.** Let $J_z = dF_z$. At $(z, W) \in X_T = \{F(z) = TW\}$ define $V_T = (\dot z, \dot W)$ by

$$J_z \dot z - T\dot W = W, \qquad \Omega_b(V_T, k) = 0 \quad (k \in \ker(J_z, -T)). \tag{64}$$

Here $\ell = (J_z, -T)$ is complex linear and onto: at $T = 0$ because $J_z$ is onto at points of $\Sigma$ (Lemma 9.2), for $T \neq 0$ because of the fibre term. The $\Omega_b$-orthogonal complement of the complex subspace $\ker \ell$ is its orthogonal complement for the Kähler metric, on which $\ell$ is an isomorphism; so $V_T$ exists, is unique, and depends smoothly on $(z, W, T)$ on the set $\mathcal{X} = \{F(z) = TW\}$, which is a smooth submanifold of $\mathbb{C}^{m+1} \times \mathbb{C}^m \times \mathbb{R}$ near $T = 0$ and the graph of $W = F(z)/T$ for $T > 0$.

**Lemma 10.9** (Extension to projective space)**.** *Let $\overline{\mathbb{C}^{m+1}} = \mathbb{CP}^{m+1}$ with homogeneous coordinates $Z = (Z_*, Z')$, $z = Z'/Z_*$, and hyperplane at infinity $H_\infty = \{Z_* = 0\}$. For $T > 0$ and $0 \le s \le 1$ the forms*

$$\omega_T = dd^c\Big[\log D + b\log\Big(1 + \frac{|F|^2}{T^2D^2}\Big)\Big], \qquad \omega_s^{\mathrm{ret}} = dd^c\Big[\log D + sb\log\Big(1 + \frac{|F|^2}{D^2}\Big)\Big]$$

*extend to Kähler forms on $\mathbb{CP}^{m+1}$, and the functions $g_T = \frac{|F|^2}{T^2D^2+|F|^2}$ and $\log(1+|F|^2/D^2)$ extend smoothly. Hence the Moser fields $N_T^{\mathrm{gr}}$, $\iota_{N_T^{\mathrm{gr}}}\omega_T = \frac{2b}{T}\, d^c g_T$, and $Y_s$, $\iota_{Y_s}\omega_s^{\mathrm{ret}} = -b\, d^c \log(1 + |F|^2/D^2)$, extend to smooth vector fields on $\mathbb{CP}^{m+1}$, smooth in $T > 0$ and $s$, and tangent to $H_\infty$. Their flows over any compact parameter interval are complete on $\mathbb{CP}^{m+1}$ and preserve $\mathbb{C}^{m+1}$.*

*Proof.* Let $F_j^{\mathrm{hom}}(Z) = Z_{y_j}^2 - Z_t^2 + jZ_*^2$ and let $V_2(Z)$ be the vector of ordered quadratic monomials $Z_aZ_b$, so $|V_2(Z)|^2 = |Z|^4$. Put $P_T(Z) = (V_2(Z), F^{\mathrm{hom}}(Z)/T)$, a homogeneous quadratic map without nontrivial zero. At $Z = (1, z)$, $D^2 + |F|^2/T^2 = |P_T(Z)|^2$, so

$$\omega_T = dd^c\big[(1-2b)\log|Z|^2 + b\log|P_T(Z)|^2\big] = (1-2b)\,\omega_{\mathrm{FS}} + b\,P_T^*\omega_{\mathrm{FS}},$$

which is a Kähler form on $\mathbb{CP}^{m+1}$ because $1-2b > 0$ and the second term is semipositive; likewise $\omega_s^{\mathrm{ret}} = (1-2sb)\omega_{\mathrm{FS}} + sb\,P_1^*\omega_{\mathrm{FS}}$ with $1-2sb \ge 1-2b > 0$. The functions $g_T = |F^{\mathrm{hom}}/T|^2/|P_T|^2$ and $\log(1+|F|^2/D^2) = \log(|P_1|^2/|Z|^4)$ are quotients of homogeneous functions of equal degree with nonvanishing denominators, hence smooth on $\mathbb{CP}^{m+1}$. The Moser fields are therefore smooth on the compact manifold $\mathbb{CP}^{m+1}$, and their flows are complete.

The involution $\iota(Z_*, Z') = (-Z_*, Z')$ is unitary, $F^{\mathrm{hom}} \circ \iota = F^{\mathrm{hom}}$ and $|V_2 \circ \iota| = |V_2|$, so it preserves all the forms and functions above, hence the fields (Lemma A.4(g)). At a point $x \in H_\infty$, which is fixed by $\iota$, a $\iota$-invariant vector lies in the $+1$ eigenspace of $d\iota_x$, which is $T_xH_\infty$, since $d\iota_x$ acts by $-1$ on the normal direction. So the fields are tangent to $H_\infty$, their flows preserve $H_\infty$, and hence its complement $\mathbb{C}^{m+1}$. □

**Proposition 10.10** (Complete horizontal transport)**.** *For every point of $X_0 = \Sigma \times \mathbb{C}^m$ the field (64) has a unique trajectory to $T = 1$. Its time-one map $\tau$ is smooth and injective and satisfies $\tau^*(\Omega_b|_{X_1}) = \Omega_b|_{X_0}$. Along each trajectory the normalized fibre norm $u = |W|^2/D^2$ is nonincreasing for $T \ge 0$; more precisely,*

$$\frac{du}{dT} = -\frac{2\pi T}{b}(1+u)^2\Omega_b(V_T, iV_T) \le 0.$$

*Proof. Constraint.* Along a trajectory, $\frac{d}{dT}(F(z)-TW) = J_z\dot z - W - T\dot W = 0$, so the trajectory stays in $X_T$.

*The graph equation.* For $T>0$ identify $X_T$ with $\mathbb{C}^{m+1}$ by $j_T(z) = (z, F(z)/T)$; then $j_T^*\Omega_b = \omega_T$ (Lemma 10.6). Write $V_T = dj_T(\dot z) - (0, W/T)$, using $\dot W = (J_z\dot z - W)/T$. For $v\in\mathbb{C}^{m+1}$, $dj_T(v)\in\ker\ell$, so $\omega_T(\dot z, v) = \Omega_b((0,W/T), dj_T(v))$. The fibre circle action is generated by $X = 2\pi i(0,W)$ and preserves $\Phi_b$, and $d^c\Phi_b(X) = bu/(1+u)$; hence $\iota_X\Omega_b = -d(bu/(1+u))$. Using $J$-invariance of $\Omega_b$ and holomorphy of $j_T$, $\Omega_b((0,W), dj_T(v)) = \Omega_b(i(0,W), dj_T(iv)) = -\frac{1}{2\pi}d\big(b\frac{u}{1+u}\big)(dj_T(iv))$, and for real $\phi$, $d\phi(iv) = -4\pi\, d^c\phi(v)$. Since $u\circ j_T = |F|^2/(T^2D^2)$, so that $\frac{u}{1+u}\circ j_T = g_T$,

$$\iota_{\dot z}\omega_T = \frac{2b}{T}\, d^c g_T,$$

that is, $\dot z = N_T^{\mathrm{gr}}(z)$ is the field of Lemma 10.9. (Equivalently $\dot z = -r_T N_T/T$ with $\iota_{N_T}\omega_T = -b\, d^c\log(1+u\circ j_T)$ and $r_T = 2/(1+u\circ j_T)$.)

*Completeness.* Let $x\in X_0$. Since $V_T$ is smooth on $\mathcal{X}$, the trajectory exists on $[0,\delta]$ for some $\delta\in(0,1)$. For $T\in[\delta,1]$ it is, in the graph coordinate, an integral curve of $N_T^{\mathrm{gr}}$, whose flow is complete in $\mathbb{C}^{m+1}$ by Lemma 10.9; so it extends to $[0,1]$, and it is unique. The time-$T$ maps are restrictions of the flow of the smooth field $V_T+\partial_T$ on $\mathcal{X}$, hence smooth and injective (backward uniqueness).

*Norm.* By the circle identity above, $\frac{d}{dT}\frac{bu}{2\pi(1+u)} = \Omega_b(V_T, i(0,W))$. Since $\ell(i(0,W)) = -iTW$ and $\ell(iV_T) = iW$, the vector $i(0,W) + T\, iV_T$ lies in $\ker\ell$, so $\Omega_b(V_T, i(0,W)) = -T\,\Omega_b(V_T, iV_T)$. This gives the displayed derivative, which is $\le 0$ for $T\ge 0$ because $\Omega_b(V, iV)$ is the Kähler norm of $V$.

*Pullback.* Let $\Phi(\sigma,T)$ be the transported image of a smooth family of initial points. Its tangent vectors are $(\kappa,t)$ with $\ell(\kappa) = tW$, and $\partial_T\Phi = (V_T, 1)$. Writing $\kappa = tV_T + (\kappa - tV_T)$ with $\kappa - tV_T\in\ker\ell$ gives $\Omega_b(V_T,\kappa) = 0$, so $\iota_{\partial_T}\Phi^*\Omega_b = 0$. As $\Phi^*\Omega_b$ is closed, Cartan's formula gives $\partial_T(\Phi_T^*\Omega_b) = 0$, which is the pullback identity. □

## 10.4. **Moser return and the ball chart.**

**Proposition 10.11** (Affine return)**.** *Let*

$$\omega_s^{\mathrm{ret}} = dd^c\left[\log D + sb\log\left(1+\frac{|F|^2}{D^2}\right)\right], \qquad \iota_{Y_s}\omega_s^{\mathrm{ret}} = -b\, d^c\log\left(1+\frac{|F|^2}{D^2}\right).$$

*The smooth field $Y_s$ has a flow through every affine point for all $s\in[0,1]$, in both directions. If $\psi_s$ starts at $s=0$, then $\psi_s^*\omega_s^{\mathrm{ret}} = dd^c\log D$.*

*Proof.* By Lemma 10.9, $Y_s$ is a smooth field on $\mathbb{CP}^{m+1}$ tangent to $H_\infty$, so its flow is defined for all $s$ in both directions and preserves $\mathbb{C}^{m+1}$. Since $\partial_s\omega_s^{\mathrm{ret}} = d\big(b\, d^c\log(1+|F|^2/D^2)\big)$, Cartan's formula gives $\frac{d}{ds}\psi_s^*\omega_s^{\mathrm{ret}} = \psi_s^* d\big(\iota_{Y_s}\omega_s^{\mathrm{ret}} + b\, d^c\log(1+|F|^2/D^2)\big) = 0$, so $\psi_s^*\omega_s^{\mathrm{ret}} = \omega_0^{\mathrm{ret}} = dd^c\log D$. Finally $j_1^*\Omega_b = \omega_1^{\mathrm{ret}}$ by Lemma 10.6. □

**Theorem 10.12** (The affine quadric transfer)**.** *For $0<b<1/2$, the map*

$$\mathcal{T}_b(z,U) = \mathrm{ch}\big(\psi_1^{-1}(\mathrm{pr}(\tau(\iota_b(z,U))))\big), \qquad \mathrm{ch}(z) = \frac{z}{\sqrt{\pi(1+|z|^2)}},$$

*is defined on the whole tube $U_b = \{(z,U)\in\Sigma\times\mathbb{C}^m : |p(U)|<b\}$ and is a smooth symplectic embedding of $(U_b,\Omega_0)$ into $(B^\circ(1),\omega_0)$, for every $m\ge 1$. It agrees with* ch *on the zero section. For every compact $\mathcal{K}\subset U_b$ there is $m_{\mathcal{K}}>0$ with $\mathcal{T}_b(\mathcal{K})\subset B(1-m_{\mathcal{K}})$. For $0<h<b$ the map is defined on a neighbourhood of every finite model $V_{\Delta_h}(E_\ell)$; it depends on $b$ and the fixed curve, and not on $h$, $\ell$ or $\mathcal{K}$.*

*Proof.* By Lemma 10.8, Proposition 10.10 and Proposition 10.11,

$$\begin{aligned}\mathcal{T}_b^*\omega_0 &= \iota_b^*\tau^*\mathrm{pr}^*(\psi_1^{-1})^*\mathrm{ch}^*\omega_0 = \iota_b^*\tau^*\mathrm{pr}^*\omega_1^{\mathrm{ret}}\\ &= \iota_b^*\tau^*(\Omega_b|_{X_1}) = \iota_b^*\Omega_b = \Omega_0,\end{aligned}$$

using $\mathrm{ch}^*\omega_0 = dd^c\log D$ (Lemma A.5), $(\psi_1^{-1})^*dd^c\log D = \omega_1^{\mathrm{ret}}$, and $\mathrm{pr}^*\omega_1^{\mathrm{ret}} = \mathrm{pr}^*j_1^*\Omega_b = \Omega_b|_{X_1}$ since $j_1\circ\mathrm{pr} = \mathrm{id}$ on $X_1$. Each factor is injective: $\iota_b$ has an inverse, $\tau$ and $\psi_1^{-1}$ are time maps of flows, pr is injective on the graph $X_1$, and ch is a bijection onto $B^\circ(1)$ (Lemma A.5(b)). A symplectic map between manifolds of equal dimension $2n$ is an immersion, so $\mathcal{T}_b$ is an injective local diffeomorphism, hence an open embedding, with values in $B^\circ(1)$ because $\pi|\mathrm{ch}(z)|^2 = 1-D^{-1} < 1$. At $U=0$, $W=0$ and $V_T=0$, so $\tau(z,0)=(z,0)$; and $Y_s$ vanishes on $\Sigma=\{F=0\}$, where $d|F|^2=0$; so $\mathcal{T}_b(z,0)=\mathrm{ch}(z)$. A compact source has compact image in the open ball, at positive distance from its boundary sphere. For $0<h<b$ the closed set $V_{\Delta_h}(E_\ell)$ lies in the open tube $U_b$, and no choice in the construction involves $h$, $\ell$ or $\mathcal{K}$. □

## 11. One large ball: the models

Throughout this section $n\ge 3$, the target is $B^\circ(1)$, and the normalised input consists of one *distinguished* capacity $A$ and further capacities $\rho_1,\dots,\rho_k$ $(k\ge 0)$ with

$$A\ge\tfrac12,\qquad A^n+\sum_{i=1}^k\rho_i^n<1,\qquad A+\rho_i<1\quad(1\le i\le k). \tag{65}$$

We write $\rho_0 = A$ when all balls are treated alike. The quadric model of Section 9 accepts only capacities below one half, and at most one capacity is $\ge\frac12$. The distinguished ball is placed by inclusion in a reserved ball $B(a)$, $a>A$, and the others are packed into the shell $S_a=\{a<\pi|y|^2<1\}$, $\mathrm{vol}\,S_a=(1-a^n)/n!$; a toric ball stacked in the normal direction needs capacity below $s=1-a$, which $A+\rho_i<1$ allows with $a>A$. The two-stage mechanism follows [Ope26]: for $n\ge4$ a mixed model over the affine quadric curve returned by the slit-shell map (Lemma 12.7); for $n=3$ the polynomial cubic model and a polynomial bubble.

**Definition 11.1** (Shell and reserved ball)**.** For $0<a<1$ put $s=1-a$, $S_a=\{a<\pi|y|^2<1\}\subset\mathbb{C}^n$, and call $B(a)$ the *reserved ball*.

The auxiliary capacities $r_i$ below play the role of the capacities $a_i$ of Theorem M2.6; the letter $a$ is kept for the reserved capacity.

**Lemma 11.2** (Auxiliary capacities)**.** *Let $n\ge3$ and let $A,\rho_1,\dots,\rho_k>0$. Call a tuple $(a;r_1,\dots,r_k)\in\mathbb{R}^{k+1}$ valid auxiliary data for $(A;\rho_1,\dots,\rho_k)$ if*

$$\max(A,\tfrac12)<a<1,\qquad \rho_i<r_i<s=1-a\quad(1\le i\le k),\qquad a^n+\sum_{i=1}^k r_i^n<1. \tag{66}$$

*If the input satisfies* (65)*, the valid tuples form a nonempty open subset of $\mathbb{R}^{k+1}$; in particular rational valid tuples exist. For a rational tuple, validity is a finite conjunction of strict inequalities, decided by rational arithmetic for rational input and semi-decided from names otherwise (Lemma 15.2).*

*Proof.* Under (65) the point $(a,r)=(A,\rho)$ satisfies the strict inequalities $a<1$, $a^n+\sum r_i^n<1$ and $a+r_i<1$, and $A\ge\frac12$; they persist on an open neighbourhood $U\subset\mathbb{R}^{k+1}$ of $(A,\rho)$. Put

$$V=U\cap\{(a,r): a>\max(A,\tfrac12),\ r_i>\rho_i\ (1\le i\le k)\}.$$

For small $\eta>0$ the point $(A+\eta,\rho_1+\eta,\dots,\rho_k+\eta)$ lies in $U$ and, since $A\ge\frac12$, in $V$; so $V$ is nonempty and contains a rational tuple by density of $\mathbb{Q}^{k+1}$. Every point of $V$ is valid, because

$a+r_i<1$ is $r_i<s$; the set of valid tuples is open, being defined by strict continuous inequalities. The last assertion is Lemma 15.2. □

In dimension six the reserved capacity $a$ must also be rational, so that the polynomial construction has integer weights; the other auxiliary capacities may be real. Construction M3.4 fixes these data by explicit formulas.

11.1. **Dimension at least eight.** Let $n\ge 4$ and $m=n-2$. Use the affine curve $\Sigma$ of Definition 9.1, with $m$ equations, its positive form $\sigma=d\lambda$ and the compact domains $\Sigma_\ell$. The second-stage transfer $\varphi$ of Theorem 10.12 in dimension $2(n-1)$ maps a tube in $\Sigma\times\mathbb{C}^m$ into $B^\circ(1)\subset\mathbb{C}^{n-1}$ and agrees with the affine ball chart on the zero section.

**Definition 11.3** (The affine mixed model)**.** Take $E_0=\Sigma\times\mathbb{C}^{m+1}$, with fibre coordinates $(z_1,\dots,z_{n-1})$,

$$p_j=\pi|z_j|^2,\qquad u=p_1,\qquad w=\sum_{j=2}^{n-1}p_j,\qquad \Omega_{\mathrm{mix}}=\omega_V+d[(1-u-2w)\lambda].$$

For $0<h_1<1$ and $0<h_2<1/2$ put

$$\Delta(h_1,h_2)=\{p\ge 0: u\le h_1,\ w+h_2u\le h_2\},\qquad H_{\mathrm{b}}(p)=2^m(1-u-2w).$$

The finite model is the restriction to $\Sigma_R\times\{p\in\Delta\}$, where $\Sigma_R$ and $A_R=\int_{\Sigma_R}\sigma$ are as in Proposition 11.5; it has empty marked set and base profile $H_b^R(p)=A_R(1-u-2w)$.

The full moment-dependent form has horizontal coefficient $1-u-2w$ and mixed terms $-du\wedge\lambda-2\sum_{j\ge2}dp_j\wedge\lambda$. On $\Delta$,

$$1-u-2w\ge(1-u)(1-2h_2)\ge(1-h_1)(1-2h_2)>0. \tag{67}$$

So the finite product is a bordered toric model, with the annuli and primitive operator of Proposition 12.6 and the mass budget of Proposition 11.5. The return target has product coordinates $(z,\xi)\in\mathbb{C}\times B^\circ_{n-1}(1)$ and the coupled form

$$\Omega_c=d[\lambda_{\mathrm{fib}}+(1-u)\lambda_{\mathrm{std}}],\qquad \lambda_{\mathrm{std}}(\xi)=\tfrac12\operatorname{Im}(\bar\xi^{\mathsf T}d\xi),\quad u=\pi|z|^2;$$

the phase preserving this form is a radial integral, smooth in the affine base and fibre coordinates.

**Lemma 11.4** (Integrated mixed profile)**.** *For $0<h_1<1$ and $0<h_2\le\frac12$,*

$$\int_{\Delta(h_1,h_2)}H_{\mathrm{b}}\,dp=J(h_1,h_2):=J_m(h_2)\,\frac{1-(1-h_1)^n}{n},\qquad J_m(h)=\frac{2^mh^m}{m!}\Big[1-\frac{2mh}{m+1}\Big],$$

*and $J(h_1,h_2)\to J(s,\frac12)=(1-a^n)/n!$ as $(h_1,h_2)\to(s,\frac12)$, $s=1-a$. For $r>0$, $\int_{\mathbb{R}^{n-1}_{\ge0}}(r-|p|)_+\,dp=r^n/n!$.*

*Proof.* Fix $u\in[0,h_1]$. The remaining variables $p'=(p_2,\dots,p_{n-1})$ range over $\{p'\ge0,\ w\le h_2(1-u)\}$. With the simplex formula of Lemma 9.5(a) and the substitution $w=(1-u)t$,

$$\begin{aligned}\int 2^m(1-u-2w)\,dp'&=\frac{2^m}{(m-1)!}\int_0^{h_2(1-u)}(1-u-2w)w^{m-1}\,dw\\&=(1-u)^{m+1}\frac{2^m}{(m-1)!}\int_0^{h_2}(1-2t)t^{m-1}\,dt,\end{aligned}$$

and the last integral times $2^m/(m-1)!$ is $J_m(h_2)$. Integrating in $u$ gives $\int_0^{h_1}(1-u)^{m+1}du=\frac{1-(1-h_1)^{m+2}}{m+2}$ and $m+2=n$. By Lemma 9.5(b), $J_m(\frac12)=1/(m+1)!=1/(n-1)!$, so $J(s,\frac12)=$

$(1-a^n)/n!$; $J$ is continuous. The cap integral is Lemma 9.5(a) in dimension $n-1$ with $f(t)=(r-t)_+$. □

**Proposition 11.5** (The affine model and its area budget)**.** *Let $n\ge 4$, $m=n-2$, and let $(a;r)$ be a valid auxiliary tuple. Take $h_1,h_2,d,P$ from Lemma 12.1 and $(K,q)$ from Theorem M2.11 for this profile. Put $A_m=2^m(m+2)$, $\ell=\lceil 64A_m/d\rceil$ and $R=2+\sum_{j=1}^m j+\ell$. In the affine quadric coordinate $t$, let $\Sigma_R=\{|t|\le R\}$ and $A_R=\int_{\Sigma_R}\sigma$. The restricted mixed model has base area $H_b^R(p)=A_R(1-u-2w)$ and satisfies*

$$0\le H_{\mathrm{b}}(p)-H_b^R(p)\le d/64\quad(p\in\Delta).$$

*The same comparison datum works for $P_R=P-11d/32$, with gap $11d/32$ and reserve $11d/64$; moreover $P_R\le H_b^R-\sum_i(r_i-|p|)_+$. The boundary cutoff and primitive operator on $\Sigma_R$ satisfy the handle hypotheses with this reserve.*

*Proof.* The quadric tail bound gives $2^m-A_R\le A_m/R^2\le d/64$, and $0<1-u-2w\le 1$ on $\Delta$. Subtracting $11d/32$ from the comparison profile subtracts the same amount from its gap, and $d/64<11d/32$. The boundary and background estimates are Proposition 12.5. □

The finite base profile has $\int_\Delta H_b^R=(A_R/2^m)J(h_1,h_2)$; its difference from $J(h_1,h_2)$ is absorbed by the reserve of Proposition 11.5. The packing for $n\ge 4$ is Theorem 12.9 in §12.

11.2. **Dimension six.** For $n=3$ the second stage of §11.1 would have a conic as its base; instead we use the polynomial cubic model of [Ope26], whose area profile still approaches the volume of the shell, and place the large ball by a polynomial bubble. All sections used below are given by explicit formulas.

**Definition 11.6** (The cubic model)**.** Put $s=1-a$ and $b=(2-a)/3$, where $(a;r)$ is a valid auxiliary tuple with $a\in\mathbb{Q}$, and let $A<r_0<a$. Use the homogeneous polynomial $F(u,v,w)=vw(v-w)+u^2(w-2v)$ and its affine restriction $f(v,w)=vw(v-w)+w-2v$. The model base is $C=\{vw\ne 0, f(v,w)=0\}$, with four ends of homogeneous limits $[0:1:0],[0:0:1],[0:1:1],[1:0:0]$, parametrised by

$$t\in\mathbb{C},\quad y_1^2=t^2-1,\quad y_2^2=t^2-2,\qquad (v,w)=(y_1-t,y_2-t).$$

Take the rational parameters $h_1,e,d_1,h_2,d_2$ of Lemma 13.2; they satisfy

$$\max_i r_i<h_1<e<s<d_1<h_2<d_2<b,\qquad \Delta=\{p\ge 0: p_1\le h_1,\ p_1+p_2\le h_2\}. \tag{68}$$

Put $Q=\lfloor 3/(a-r_0)\rfloor+1$ and

$$L=(Q+1)\operatorname{den}(a)\operatorname{den}(e)\operatorname{den}(d_2),\qquad A_*=Le,\quad B_*=Ld_2,\quad S_*=L(1-a),\quad D=L+S_*,$$

where den is the reduced denominator; these are integers (the weights $A_*,B_*$ are not the input capacity $A$) and $L>Q$. Let $\Lambda=\{(j_1,j_2)\in\mathbb{Z}^2_{\ge 0}: j_1\le A_*,\ j_1+j_2\le B_*\}$ and put

$$d_j=D-3(j_1+j_2),\qquad m_j=(S_*-j_1-j_2)_+,$$

$$q_{d,m}(v,w)=(1,v^{d-m},w^{d-m},v^{d-m}w^m,v^mw^{d-m}),\qquad a_j=\|q_{d_j,m_j}\|^2.$$

The central potential is

$$\phi_L(y,z)=\frac{1}{L}\log\sum_{j\in\Lambda}a_j(y)|z_1|^{2j_1}|z_2|^{2j_2}. \tag{69}$$

Its radial moment coordinates identify a neighbourhood of $\Delta$ with a toric normal model on $C\times\mathbb{C}^2$; the primitive and the identification are given in Definition 13.3.

The integer inequalities $0 < A_* < S_* \le B_*$, $3B_* < D$, $2S_* < L$ follow from (68) and $a > 1/2$. The five entries of $q_{d,m}$ are fixed before the transport parameters.

**Lemma 11.7** (Limiting profile and its integral)**.** *The limiting base area at physical moment $p$ is*

$$H_0(p) = 9(b - |p|) - 2(s - |p|)_+, \qquad |p| = p_1 + p_2.$$

*For $0 < h_1 \le s \le h_2 \le b$,*

$$\int_\Delta H_0 \, dp = \frac{3}{2}[b^3 - (b - h_1)^3] - \frac{9}{2}h_1(b - h_2)^2 - \frac{1}{3}[s^3 - (s - h_1)^3].$$

*As $(h_1, h_2) \to (s, b)$ this tends to $(1-a^3)/6$. Moreover $H_0 \ge 9(b-h_2)$ on $\Delta$ and $\int_{p\ge 0}(r-|p|)_+ \, dp = r^3/6$.*

*Proof.* Put $d(p) = D - 3L|p|$ and $h(p) = (S_* - L|p|)_+$. By Proposition 13.6 (the four end limits in its Step 1 and the marked-height estimate of its Step 4), the limiting pole orders at $[0:1:0], [0:0:1], [0:1:1], [1:0:0]$ are $d(p) - h(p)$, $d(p) - h(p)$, $d(p)$ and 0, and Stokes' theorem gives $H_0 = L^{-1}(3d(p) - 2h(p))$. For fixed $p_1 = t$, integration in $|p| \in [t, h_2]$ gives $\frac{9}{2}[(b-t)^2 - (b-h_2)^2] - (s-t)^2$; integrate in $t \in [0, h_1]$ and use $3b = 1 + s$. The lower bound holds because $H_0$ decreases with $|p|$ and $h_2 \ge s$; the cap integral is $\int_0^r (r-t)t \, dt$. □

**Proposition 11.8** (Polynomial lifts)**.** *For every $j \in \Lambda$ the five polynomials $f^{j_2} q_{d_j, m_j, i}$ have explicit homogeneous lifts to the quadric $\mathcal{Q}(X) = X_0X_1 - X_2X_3 = 0$, described by the homogeneous substitution*

$$(u, v, w) \longmapsto (vw, u^2, uv, uw).$$

*Multiplication by $\mathcal{Q}^{j_1}$ gives homogeneous polynomials $P_{j,i}$ of degree $L$, each vanishing to order at least $L - S_*$ at $[1:0:0:0]$. For $t, \delta > 0$ set*

$$P_{t,\delta}(X)_{j,i} = \delta^{-j_1} t^{-j_2} P_{j,i}(X).$$

*On the tube $(v, w, z) \mapsto (v, w, vw + \delta z)$ in the chart $X_1 = 1$ the factors $\delta^{-j_1}$ cancel exactly, and the resulting polynomial family extends smoothly to $\delta = 0$, with value*

$$z^{j_1} (f(v, w)/t)^{j_2} q_{d_j, m_j, i}(v, w).$$

*Proof.* For a monomial $v^J w^K$ and $\bar{n} = L - 2j_1$ put $\alpha_0 = (J + K - \bar{n})_+$, $\alpha_1 = \bar{n} + \alpha_0 - J - K$, $\alpha_2 = J - \alpha_0$, $\alpha_3 = K - \alpha_0$. Then $X_0^{\alpha_0} X_1^{\alpha_1} X_2^{\alpha_2} X_3^{\alpha_3}$ pulls back to $u^{2\bar{n} - J - K} v^J w^K$. The inequalities $J, K \le \bar{n}$ and the required order at the distinguished point follow from $j_1 < S_*$, $3(j_1 + j_2) < L + S_*$ and the degree bounds $\deg_v, \deg_w (f^{j_2} q_i) \le 2j_2 + d_j - m_j$, $\deg (f^{j_2} q_i) \le 3j_2 + d_j$. Finally $\mathcal{Q}(vw + \delta z, 1, v, w) = \delta z$. The complete inequalities are Lemma 13.5. □

*Packing the smaller balls.* By Lemma 13.2, $\int_\Delta H_0 > \sum_i r_i^3/6$ for the parameters (68). The function $G = H_0 - \sum_i (r_i - |p|)_+$ is concave, has positive integral, and is positive off a smaller homothetic copy of $\Delta$ containing all the cap supports. The comparison datum of Theorem M2.11 for a lowered copy of $G$ and the handle construction over a compact piece $\Sigma_l$ of $C$ give the inner packing (Proposition 14.1).

On the compact union of these ball images, a short exact cubic transfer gives the cubic ambient form and a second exact transfer along the tube gives the form defined by $P_{t,\delta}$; in the chart $X_0 = 1$ this compact set avoids the origin. A polynomial bubble inserts the distinguished ball near the origin while moving the smaller images within a disjoint open set, and a common global Moser map and the Fubini–Study ball chart put all the images into $B^\circ(1)$. The formulas, the two compact transfer estimates and the packing theorem (Theorem 14.7) are in §14.

In Construction M3.4(iv) these data are $r_0 = A+2\eta$, $\rho_0^+ = A+\eta$, and $r_i = \rho_i+3\eta$, $r_i' = \rho_i+2\eta$, $\rho_i^+ = \rho_i + \eta$ for the smaller balls; the auxiliary ball with index $j$ has $r_j = \eta$, $r_j' = \eta/2$, and its map is replaced by $f_0$.

*Example.* For $A = 59/100$ and eighteen smaller capacities $7/20 > 1/3$, $A^3 + 18(7/20)^3 = 0.977129$ and $A + 7/20 = 0.94$; so Theorem 14.7 applies beyond the range of a cubic used directly as a second stage.

## 12. One large ball in dimension at least eight: the shell return and the packing

Let $n \geq 4$, $m = n-2$. The base is the affine locus $\Sigma = \{y_j^2 = t^2 - j\}$ with $\lambda = d^c \log(1+|t|^2 + \sum_j |y_j|^2)$ and $\sigma = d\lambda$. Throughout $(a; r)$ is valid auxiliary data for $(A; \rho)$, with $1/2 < a < 1$ and $s = 1-a$.

12.1. **Parameters.** Recall $H_{\mathrm{b}} = 2^m(1-u-2w)$, $\Delta(h_1, h_2)$ (Definition 11.3) and $J(h_1, h_2) = \int_{\Delta(h_1,h_2)} H_{\mathrm{b}}$, with $J(s, \frac{1}{2}) = (1-a^n)/n!$ (Lemma 11.4). The parameters of Theorem 12.9 are:

**Lemma 12.1** (Parameters of the large branch)**.** *Let $n \geq 4$, $k \geq 1$, and let $(a; r_1, \dots, r_k)$ be a valid auxiliary tuple. Put $s = 1-a$, $r_* = \max_i r_i$, $\varsigma = 1 - a^n - \sum_i r_i^n$,*

$$\delta = \min\Big(\frac{s - r_*}{2},\ \frac{\varsigma}{2n!(4+2^n)}\Big), \qquad h_1 = s - \delta, \qquad h_2 = \tfrac{1}{2} - \delta,$$

$$b_2 = \tfrac{1}{2}\big(h_2 + \tfrac{1}{2}\big), \qquad h_2' = \tfrac{1}{2}(h_2 + b_2), \qquad \tau_\Delta = \frac{r_* + h_1}{2h_1},$$

$$\beta_\Delta = 2^m(1-h_1)(1-2h_2), \qquad d = \tfrac{1}{2}\min\Big(\beta_\Delta, \frac{J(h_1,h_2) - \sum_i r_i^n/n!}{2}\Big), \qquad \Delta = \Delta(h_1, h_2),$$

*and $P = H_{\mathrm{b}} - \sum_i (r_i - |p|)_+$ on $\Delta$. All these numbers are positive, and rational when the auxiliary data are rational, and:*

- *(i) $r_* < h_1 < s$ and $r_* < h_2 < h_2' < b_2 < \frac{1}{2}$;*
- *(ii) $J(h_1, h_2) \geq \big(1 - a^n - \frac{1}{2}\varsigma\big)/n!$, hence $\int_\Delta P\, dp = J(h_1,h_2) - \frac{1}{n!}\sum_i r_i^n \geq \varsigma/(2\,n!) > 0$;*
- *(iii) $\tau_\Delta < 1$ and $\{p \in \mathbb{R}^{n-1}_{\geq 0} : |p| \leq r_i\} \subset \tau_\Delta \Delta$ for every $i$;*
- *(iv) $P$ is a concave, piecewise affine function with real coefficients on $\Delta$, and $P = H_{\mathrm{b}} \geq \beta_\Delta > 0$ on $\Delta \setminus \tau_\Delta \Delta$;*
- *(v) $\operatorname{vol}\Delta < 1$, and the mean $\bar{P}$ of $P$ on $\Delta$ satisfies $\bar{P} > \int_\Delta P\, dp$;*
- *(vi) $0 < d < \min(\bar{P}, \beta_\Delta)$.*

*In particular $(\Delta, P, \tau_\Delta, d)$ satisfies the hypotheses of Proposition M2.9, in the real-coefficient form stated after it (with rational coefficients when the auxiliary data are rational).*

*Proof.* The auxiliary inequalities give $r_i < s < \frac{1}{2}$ and $\varsigma > 0$, so $\delta > 0$. Rational auxiliary data give rational parameters.

(i) $r_* < (s + r_*)/2 \leq h_1 < s$, and $h_2 - h_1 = 1/2 - s > 0$ gives $r_* < h_2$; $h_2 < h_2' < b_2 < 1/2$ by definition.

(ii) Put $F = n!$ and $B = (1-2\delta)^m(1+2m\delta)$. Substituting the two heights in the formula for $J$ of Lemma 11.4 gives $FJ(h_1,h_2) = B\big(1 - (a+\delta)^n\big)$. Here $0 < \delta < 1/4$; Bernoulli's inequality and $(1-2\delta)^m \leq 1$ give $1 - 4m^2\delta^2 \leq B \leq 1 + 2m\delta$, and expanding the difference of powers with $0 \leq a, \delta \leq 1$ gives $0 \leq (a+\delta)^n - a^n \leq \delta n 2^n$. Therefore

$$\begin{aligned} 1 - a^n - FJ(h_1,h_2) &= (1-B)(1-a^n) + B\big((a+\delta)^n - a^n\big) \\ &\leq 4m^2\delta^2 + (1+2m\delta)\delta n 2^n \\ &\leq F(4+2^n)\delta \leq \varsigma/2. \end{aligned}$$

For the penultimate inequality use $m^2 \le n!$, $\delta \le 1/4$ and $(1+2m\delta)n \le (m+1)(m+2) \le n!$. The caps lie in $\Delta$ by (iii), and their integrals are $r_i^n/n!$, giving the formula for $\int_\Delta P$.

(iii) $\tau_\Delta < 1$ since $r_* < h_1$ and $r_* < h_2$. Let $|p| = u+w \le r_i$. Then $u \le r_* < \frac{1}{2}(h_1+r_*) \le \tau_\Delta h_1$, and $w + h_2 u \le w + u \le r_* < \tau_\Delta h_2$: here $h_2 < 1$ and $\tau_\Delta h_2 > \tau_\Delta h_1 > r_*$. Hence $p \in \tau_\Delta\Delta = \{u \le \tau_\Delta h_1,\ w + h_2 u \le \tau_\Delta h_2\}$.

(iv) $H_{\mathrm{b}}$ is affine with integer coefficients and each $-(r_i - |p|)_+$ is concave and piecewise affine. If $p \in \Delta \setminus \tau_\Delta\Delta$, then $|p| > r_i$ for every $i$ by (iii), so all caps vanish and $P = H_{\mathrm{b}}$, and $H_{\mathrm{b}} \ge \beta_\Delta$ by (67).

(v) $\Delta \subset [0, h_1] \times \{p' \in \mathbb{R}^m_{\ge 0} : |p'| \le h_2\}$, so $\operatorname{vol}\Delta \le h_1 h_2^m/m! < 1$; since $\int_\Delta P > 0$, its mean exceeds its integral.

(vi) follows because $d \le \beta_\Delta/2$ and $d \le \frac{1}{4}\int_\Delta P < \bar{P}$. The last assertion is (iii)–(vi) with $\beta_P = \beta_\Delta$; $\Delta$ is a polytope and $P$ is concave and piecewise affine. □

12.2. **The second stage and radial phase.** Put $b_2 = (2h_2+1)/4$ and let $\varphi = \mathcal{T}_{b_2}$ be the affine quadric transfer in dimension $2(n-1)$. Write $v = (v_2, \dots, v_{n-1})$ for its fibre moments and

$$\ell = \sum_{j=2}^{n-1} v_j\, d\theta_j + (1 - 2|v|)\lambda, \qquad d\ell = \Omega_X,$$

a smooth primitive across the axes that vanishes on radial fibre tangents. The fibre contraction $\mathsf{r}_t(x, z') = (x, tz')$, $0 \le t \le 1$, preserves $\mathcal{N}_{b_2} = \{(x, z') : |v(z')| < b_2\}$.

**Lemma 12.2** (The zero section)**.** *The second transfer satisfies $\varphi(x, 0) = \mathrm{ch}(x)$ and $i^*(\varphi^*\lambda_{\mathrm{std}} - \ell) = 0$, where $i(x) = (x, 0)$.*

*Proof.* The first identity is part of Theorem 10.12. The chart ch pulls the standard Liouville form back to $\lambda = d^c \log D$ (Lemma A.5), and $\ell$ also restricts to $\lambda$ on the zero section. □

12.3. **The phase identity.**

**Lemma 12.3** (The radial phase and its covector)**.** *Put $\Xi = \varphi^*\lambda_{\mathrm{std}} - \ell$ and*

$$\Psi(x, z') = \int_0^1 \lambda_{\mathrm{std}}(\varphi(x, tz'))[\partial_t \varphi(x, tz')]\, dt.$$

*Then $\Psi$ is smooth on $\mathcal{N}_{b_2}$ and*

$$d\Psi = \Xi = \varphi^*\lambda_{\mathrm{std}} - \ell. \tag{70}$$

*For $\hat{\mathcal{J}}_{x,z'}(z_1) = e^{2\pi i \Psi(x,z')} z_1$ and $\Phi_2(x, z_1, z') = (\hat{\mathcal{J}}_{x,z'}(z_1), \varphi(x, z'))$, one has*

$$\Phi_2^* \Omega_c = d[\lambda_{\mathrm{fib}} + (1-u)\ell]. \tag{71}$$

*Proof.* The transfer pullback gives $d\Xi = \varphi^*\omega_0 - d\ell = 0$, and $i^*\Xi = 0$ by Lemma 12.2. Write $\mathsf{r}^*\Xi = a_t + g\, dt$ on $[0,1] \times \mathcal{N}_{b_2}$. Closedness gives $\partial_t a_t = dg$; integrating from zero to one gives $\Xi = d\int_0^1 g(t, \cdot)dt$. Since $\ell$ is zero on radial fibre tangents, the last integral is exactly $\Psi$, smooth (including at $z' = 0$) by smooth parameter integration.

A phase rotation of $z_1$ pulls its Liouville form back to $\lambda_{\mathrm{fib}} + u\, d\Psi$, so the primitive of the target coupled form pulls back to

$$\lambda_{\mathrm{fib}} + u\, d\Psi + (1-u)\varphi^*\lambda_{\mathrm{std}} = \lambda_{\mathrm{fib}} + \varphi^*\lambda_{\mathrm{std}} - u\ell.$$

whose exterior derivative is $d[\lambda_{\mathrm{fib}} + (1-u)\ell]$ because $d\varphi^*\lambda_{\mathrm{std}} = d\ell$. This proves (71). □

12.4. **The finite mixed model.** Use the affine product and form of Definition 11.3. For the moment change put

$$\chi(x, z_1, z') = (x, z_1, z'/\sqrt{1-u}), \qquad \mathcal{U} = \{u < 1,\ w < b_2(1-u)\}, \qquad \mathcal{R} = \Phi_2 \circ \chi.$$

**Lemma 12.4** (Mixed form and return)**.** *On $u < 1$ the smooth inverse of $\chi$ is $(x, z_1, v) \mapsto (x, z_1, \sqrt{1-u}\, v)$. It preserves $u$ and fibre angles, with tail moments $v_j = p_j/(1-u)$. On the interior of the finite mixed region,*

$$\mathcal{R}^*\Omega_c = \Omega_{\mathrm{mix}} = \omega_V + d[(1-u-2w)\lambda].$$

*On the finite region $\mathcal{M}_R = \Sigma_R \times \{z : p(z) \in \Delta\}$, the map $\mathcal{R}$ is injective, smooth on the interior, preserves $u$ and has base moment $\mu < 1$. This compact region lies in $\mathcal{U}$; its base profile is $A_R(1-u-2w)$ and its scalar integral is $(A_R/2^m)J(h_1, h_2)$.*

*Proof.* The square root is positive on $u < 1$, which proves both smooth coordinate formulas across the axes. Using $v_j = p_j/(1-u)$ gives

$$\chi^*\{\lambda_{\mathrm{fib}} + (1-u)\ell\} = \sum_{j=1}^{n-1} p_j d\theta_j + (1-u-2w)\lambda.$$

Take the full exterior derivative and use (71). Injectivity follows from injectivity of $\varphi$, cancellation of the common nonzero phase in the first coordinate, and the inverse of $\chi$. The base moment bound is the open-ball containment of $\varphi$. On $\Delta$, $u \le h_1 < 1$ and $w \le h_2(1-u) < b_2(1-u)$. The horizontal curvature at fixed moment is $(1-u-2w)\sigma$, with base integral $A_R(1-u-2w)$; the scalar formula for $J$ gives the last assertion. □

**Proposition 12.5** (Affine mixed-model budget)**.** *Let $m = n-2 \ge 2$, and take $\Delta, d$ from Lemma 12.1. Put $A_m = 2^m(m+2)$, $\ell = \lceil 64A_m/d \rceil$, $R = 2 + \sum_{j=1}^{m} j + \ell$, $\Sigma_R = \{|t| \le R\}$ and $A_R = \int_{\Sigma_R} \sigma$. Set*

$$\gamma_* = \frac{11d}{64}, \qquad \kappa_{\min} = (1-h_1)(1-2h_2), \qquad c_{\mathrm{bg}} = \min\left(\frac{\gamma_*}{32\,2^m}, \frac{\kappa_{\min}}{2}\right).$$

*Use the flat cutoff $\chi_R = \chi_{R-2,R-1}(t)$ of Definition A.3, $\chi_R(t) = 1 - S\big((|t|^2 - (R-2)^2)/((R-1)^2 - (R-2)^2)\big)$, which equals one for $|t| \le R-2$ and zero for $|t| \ge R-1$. Set $c(p) = 1-u-2w$, $H_{\mathrm{b}}^R(p) = \int_{\Sigma_R} c(p)\sigma = A_R c(p)$ and*

$$\Omega_{\mathrm{rem}}(p) = \chi_R(c(p) - c_{\mathrm{bg}})\sigma, \quad H_R(p) = \int_{\Sigma_R} \Omega_{\mathrm{rem}}(p), \quad \kappa_{\mathrm{bg}}(p) = c(p)\sigma - \Omega_{\mathrm{rem}}(p).$$

*For every $p \in \Delta$, $\kappa_{\mathrm{bg}}(p) > 0$, it equals $c(p)\sigma$ near $\partial\Sigma_R$ and $c_{\mathrm{bg}}\sigma$ on the handle, and*

$$0 < \int_{\Sigma_R} \kappa_{\mathrm{bg}}(p) < \gamma_*/16, \qquad H_R(p) > H_{\mathrm{b}}^R(p) - \gamma_*/16.$$

*If $\eta_{\mathrm{h}} = df \wedge \beta$ is the unit-integral handle density, the form $H_R(p)\eta_{\mathrm{h}} - \Omega_{\mathrm{rem}}(p)$ has integral zero and is supported in $\Sigma_{R-1} \subset \operatorname{int}\Sigma_R$, and the finite coordinate primitive of Lemma 6.8 applied to it depends smoothly on $p$.*

*Proof.* On $\Delta = \{u \le h_1,\ w \le h_2(1-u)\}$ we have $c(p) \ge (1-u)(1-2h_2) \ge (1-h_1)(1-2h_2) = \kappa_{\min} > 0$ and $c(p) \le 1$. Hence $\kappa_{\mathrm{bg}} = [c(p)(1-\chi_R) + c_{\mathrm{bg}}\chi_R]\sigma$ is positive, equals $c(p)\sigma$ for $|t| \ge R-1$, and equals $c_{\mathrm{bg}}\sigma$ for $|t| \le R-2$, which contains the handle supports ($|t| \le 3$, Lemma 10.5). Since $R - 2 = R_{\ell-2}$ in the notation of Lemma 10.5 and $\int_\Sigma \sigma = 2^m$, that lemma and $c \le 1$ give

$$\int_{\Sigma_R \setminus \Sigma_{R-2}} c(p)\sigma \le 2^m - A_{\ell-2} \le \frac{2^m(m+2)}{(R-2)^2} \le \frac{d^2}{4096 \cdot 2^m(m+2)} \le \frac{d}{65536} < \frac{\gamma_*}{32},$$

because $R-2 \geq \ell \geq 64\cdot 2^m(m+2)/d$, $2^m(m+2)\geq 16$, and $d\leq 1$: indeed $d \leq J(h_1,h_2)/4 \leq J_m(h_2)/4 \leq 1/4$, as $2^m h_2^m \leq 1$. Also $A_R \leq 2^m$, so $c_{\mathrm{bg}}A_R \leq \gamma_*/32$. Adding, $\int_{\Sigma_R}\kappa_{\mathrm{bg}} \leq \int_{\Sigma_R\setminus\Sigma_{R-2}} c\sigma + c_{\mathrm{bg}}A_R < \gamma_*/16$, and the integral is positive because $\kappa_{\mathrm{bg}}>0$. Integrating $c(p)\sigma = \kappa_{\mathrm{bg}}+\Omega_{\mathrm{rem}}$ gives $H_R = H_{\mathrm{b}}^R - \int\kappa_{\mathrm{bg}} > H_{\mathrm{b}}^R - \gamma_*/16$. Both $\eta_{\mathrm{h}}$ and $\Omega_{\mathrm{rem}}$ are supported in $\Sigma_{R-1}$, and $\int(H_R\eta_{\mathrm{h}} - \Omega_{\mathrm{rem}}) = H_R - H_R = 0$. The primitive is a finite sum of coordinate integrals over fixed compact boxes of smooth functions of $(p,x)$, hence smooth in $p$ (Lemma 6.8). □

**Proposition 12.6** (Handle data for the affine mixed model)**.** *Use the base $\Sigma_R$ of Proposition 12.5, the fixed annuli and dual form of Proposition 10.4, the physical primitive $\Gamma_{\mathrm{phys}}(p) = (1-u-2w)\lambda$ with curvature $\kappa(p) = (1-u-2w)\sigma$, the reference primitive $\lambda$ with $\Lambda=\sigma$ and $A_{\mathrm{ref}} = 2^m$, the density floor $\kappa_- = (1-h_1)(1-2h_2)$, and the collar cutoff $\chi_\partial = 1-\chi_R$. Take as correction core $\Sigma_{R-1}$, covered by the inner boxes of the quadric presentation of Construction 6.10, and as correction region the union of these boxes, contained in $\operatorname{int}\Sigma_R$. Then:*

- *(a) the support of $1-\chi_\partial = \chi_R$ in $\Sigma_R$ and the handle are contained in the correction core;*
- *(b) $\kappa(p) \geq \kappa_-\Lambda$ on all of $\Sigma_R$ for $p\in\Delta$, in particular on the correction region;*
- *(c) with $P_R = P - 11d/32$, gap $11d/32$ and reserve $\gamma_* = 11d/64$, these data satisfy the bordered handle hypotheses of Theorem M2.6, and Construction 8.4 applies to them.*

*Proof.* (a) $\chi_R$ vanishes for $|t|\geq R-1$, and the annular supports lie in $|t|<3\leq R-2$ (Lemma 10.5). (b) On $\Delta$, $1-u-2w \geq (1-h_1)(1-2h_2)$ by (67). (c) The product form is $\omega_V + d\Gamma_{\mathrm{phys}}$, including all moment derivatives. Proposition 12.5, whose background coefficient is $c_{\mathrm{bg}} = \min(\gamma_*/(32A_{\mathrm{ref}}), \kappa_-/2) = e_\Lambda$, gives the positive background, the collar bound and the compactly supported zero-mass defect, whose finite coordinate primitive depends smoothly on $p$. Hence the normalisation and layer construction apply with reserve $\gamma_*$. □

12.5. **The slit-shell map.** The coupled target is the explicit product $(\xi,z)\in\{\pi|\xi|^2<1\}\times\mathbb{C}$. Write $\mu = \pi|\xi|^2$, $u=\pi|z|^2$, $\lambda_{\mathrm{base}} = \lambda_{\mathrm{std}}(\xi)$ and $\lambda_{\mathrm{fib}} = \frac12\operatorname{Im}(\bar z\,dz)$, so

$$\Omega_1 = \Omega_c = d[\lambda_{\mathrm{fib}} + (1-u)\lambda_{\mathrm{base}}]. \tag{72}$$

The strip map moves the first moment from $u$ to $x>0$; the factor $\sqrt{1-u}$ in (73) is the compensating base dilation, and the zero section lands at depth $x>0$ inside the shell.

**Lemma 12.7** (Slit-shell map)**.** *Let $0<a<1$, $s=1-a$, and let $\hat h, \hat\mu, \delta$ be real numbers with*

$$0<\hat h<s, \qquad 0<\hat\mu<1, \qquad 0<\delta<\tfrac14\min\bigl(s-\hat h,\ (1-\hat h)(1-\hat\mu)\bigr).$$

*Let $\Psi=\Psi_{2\delta}\colon z\mapsto(x,T)$ be the strip map of Proposition A.16 (Proposition A.16) with parameter $2\delta$, put $b_\delta = 2\delta/\pi$ and $U=\{(\xi,z):\mu<\hat\mu,\ u<\hat h\}$, and define $\mathrm{Sl}(\xi,z) = (y_1,y_2,\dots,y_n)\in\mathbb{C}^n$ on $U$ by*

$$y_1 = e^{-2\pi iT}\sqrt{\frac{1-x-(1-u)\mu}{\pi}}, \qquad (y_2,\dots,y_n) = e^{-2\pi iT}\sqrt{1-u}\,\xi. \tag{73}$$

*For $I\geq 0$ put $\mathcal{E} = I + b_\delta(\frac{\pi}{2}-1)$ and*

$$x_*(I) = \frac{b_\delta^2}{2\bigl(\mathcal{E}+b_\delta+\sqrt{\mathcal{E}(\mathcal{E}+2b_\delta)}\bigr)}. \tag{74}$$

*Then:*

- *(i) On $U$, $0<x<u+\delta$, and $1-x-(1-u)\mu > (1-\hat h)(1-\hat\mu)-\delta > 3\delta$; $\mathrm{Sl}$ is smooth on $U$, including along $z=0$, and $y_1\neq 0$.*
- *(ii) $\mathrm{Sl} = \mathrm{Hopf}\circ\mathrm{Dil}$, where $\mathrm{Dil}(\xi,z) = (\xi',x,T)$ with $\xi' = \sqrt{(1-u)/(1-x)}\,\xi$, $\pi|\xi'|^2<1$, and $\mathrm{Hopf}(\xi',x,T) = \sqrt{(1-x)/\pi}\,e^{-2\pi iT}\bigl(\sqrt{1-\pi|\xi'|^2},\sqrt{\pi}\,\xi'\bigr)$.*

*(iii) With* $\lambda_0 = \frac{1}{2}\operatorname{Im}(\bar{y}^{\mathsf{T}} dy)$,

$$\mathrm{Sl}^*\lambda_0 = (1-u)\lambda_{\mathrm{base}} - (1-x)\,dT, \qquad \mathrm{Sl}^*\omega_0 = \Omega_1.$$

*(iv)* $\pi|\mathrm{Sl}(\xi,z)|^2 = 1-x$, *and* $a < 1-\hat{h}-\delta < \pi|\mathrm{Sl}|^2 < 1$ *on* $U$.

*(v)* Sl *is injective, with inverse on its image* $x = 1-\pi|y|^2$, $e^{-2\pi i T} = y_1/|y_1|$ *with* $T \in (0,1)$, $z = \Psi^{-1}(x,T)$, $\xi = e^{2\pi i T}(y_2,\dots,y_n)/\sqrt{1-u}$. *Hence* Sl *is a symplectic embedding of* $(U,\Omega_1)$ *onto an open subset of* $S_a$.

*(vi) If* $I \geq 0$, $\mathcal{K} \subset U$ *is compact and* $u \leq I$ *on* $\mathcal{K}$, *then* $x \geq x_*(I) > 0$ *on* $\mathcal{K}$, *with* $x_*(I)$ *given by* (74), *and* $|T-\frac{1}{2}| \leq \frac{1}{2}\sqrt{1-(1+\mathcal{E}/b_\delta)^{-2}} < \frac{1}{2}$ *on* $\mathcal{K}$, *where* $\mathcal{E} = I + b_\delta(\frac{\pi}{2}-1)$. *Hence* $\mathrm{Sl}(\mathcal{K})$ *lies in the compact set* $\{1-\hat{h}-\delta \leq \pi|y|^2 \leq 1-x_*(I)\} \subset S_a$.

*Proof.* (i) Proposition A.16(iii) with parameter $2\delta$ gives $0 < x < \pi|z|^2 + \delta = u + \delta$. Hence on $U$

$$1-x-(1-u)\mu > (1-u)(1-\mu) - \delta > (1-\hat{h})(1-\hat{\mu}) - \delta > 4\delta - \delta = 3\delta.$$

The functions $x$, $T$ are smooth in $z$ (Proposition A.16(i)), $u$ and $\mu$ are polynomials, and the radicands $1-x-(1-u)\mu$ and $1-u$ are bounded below by positive constants on $U$; so the formula is smooth on $U$, and $|y_1|^2 > 3\delta/\pi > 0$.

(ii) $1-\pi|\xi'|^2 = \big(1-x-(1-u)\mu\big)/(1-x) > 0$ by (i). The components of Hopf $\circ$ Dil are

$$e^{-2\pi i T}\sqrt{(1-x)/\pi}\,\sqrt{1-\pi|\xi'|^2} = y_1,$$
$$e^{-2\pi i T}\sqrt{(1-x)/\pi}\sqrt{\pi}\sqrt{(1-u)/(1-x)}\,\xi = e^{-2\pi i T}\sqrt{1-u}\,\xi.$$

(iii) Write $y = cV$ with $c = \sqrt{(1-x)/\pi}\,e^{-2\pi i T}$ and $V = (\sqrt{1-\pi|\xi'|^2}, \sqrt{\pi}\,\xi')$, so $|V| = 1$. Then $\bar{y}^{\mathsf{T}} dy = \bar{c}\,dc + |c|^2 \bar{V}^{\mathsf{T}} dV$. Here $\operatorname{Im}(\bar{c}\,dc) = -2\pi|c|^2 dT$, and $\operatorname{Im}(\bar{V}^{\mathsf{T}} dV) = \pi \operatorname{Im}(\bar{\xi'}^{\mathsf{T}} d\xi') = 2\pi\lambda_{\mathrm{base}}(\xi')$ since the first entry of $V$ is real. As $\pi|c|^2 = 1-x$, $\mathrm{Sl}^*\lambda_0 = (1-x)\big(\lambda_{\mathrm{base}}(\xi') - dT\big)$. For a positive real function $c'$, $\lambda_{\mathrm{base}}(c'\xi) = \frac{1}{2}\operatorname{Im}\big(c'\bar{\xi}^{\mathsf{T}}(\xi\,dc' + c'd\xi)\big) = c'^2\lambda_{\mathrm{base}}(\xi)$, because $\bar{\xi}^{\mathsf{T}}\xi\,c'dc'$ is real. With $c'^2 = (1-u)/(1-x)$ this gives $\mathrm{Sl}^*\lambda_0 = (1-u)\lambda_{\mathrm{base}}(\xi) - (1-x)\,dT$. Its differential is $d[(1-u)\lambda_{\mathrm{base}}] + dx \wedge dT$, and $dx \wedge dT = \Psi^*(dx \wedge dT) = \omega_{\mathbb{C}} = d\lambda_{\mathrm{fib}}$ (Proposition A.16(i)). By (72), $\mathrm{Sl}^*\omega_0 = \Omega_1$.

(iv) $\pi|y|^2 = \pi|c|^2|V|^2 = 1-x$. By (i), $0 < x < u+\delta < \hat{h}+\delta$, so $1-\hat{h}-\delta < \pi|y|^2 < 1$, and $1-\hat{h}-\delta > 1-s+3\delta > a$ because $s-\hat{h} > 4\delta$.

(v) Let $y = \mathrm{Sl}(\xi,z)$. Then $x = 1-\pi|y|^2$; since $y_1 \neq 0$, $y_1/|y_1| = e^{-2\pi i T}$ determines $T \in (0,1)$; $\Psi$ is a bijection of $\mathbb{C}$ onto $(0,\infty)\times(0,1)$, so $z = \Psi^{-1}(x,T)$, hence $u$; and $\xi$ is read off from the last coordinates. So Sl is injective. $\Omega_1 = \omega_{\mathbb{C}} + (1-u)\omega_{\mathrm{base}} - du \wedge \lambda_{\mathrm{base}}$ is symplectic on $\{u<1\}$: $du \wedge \omega_{\mathbb{C}} = 0$, and $\lambda_{\mathrm{base}} \wedge \omega_{\mathrm{base}}^{n-1} = 0$ for degree reasons, so $\Omega_1^n = n(1-u)^{n-1}\omega_{\mathbb{C}} \wedge \omega_{\mathrm{base}}^{n-1} \neq 0$. Since $\mathrm{Sl}^*\omega_0 = \Omega_1$, $d\,\mathrm{Sl}$ is injective; $\dim_{\mathbb{R}} U = 2n$, so Sl is a local diffeomorphism, and an injective local diffeomorphism is a diffeomorphism onto its open image. The image lies in $S_a$ by (iv).

(vi) If $u \leq I$ on $\mathcal{K}$, Proposition A.16(iv) with parameter $2\delta$ (so $b = b_\delta$) gives $x \geq \frac{1}{2}\big[\mathcal{E} + b_\delta - \sqrt{\mathcal{E}(\mathcal{E}+2b_\delta)}\big]$, which equals $x_*(I)$ because $(\mathcal{E}+b_\delta)^2 - \mathcal{E}(\mathcal{E}+2b_\delta) = b_\delta^2$, and gives the bound on $T$. Then $\pi|y|^2 = 1-x \leq 1-x_*(I)$ on $\mathrm{Sl}(\mathcal{K})$, and with (iv) this set lies in the closed annulus $\{1-\hat{h}-\delta \leq \pi|y|^2 \leq 1-x_*(I)\}$, which is compact and contained in $S_a$ because $1-\hat{h}-\delta > a$ and $x_*(I) > 0$. □

**Proposition 12.8** (Natural slit domain)**.** *For every* $\delta > 0$, *formula* (73), *using* $\Psi_{2\delta}$, *defines a smooth injective symplectic map on* $V_\delta = \{(\xi,z) : u < 1,\ 1-x-(1-u)\mu > 0\}$, *with capacity* $1-x$. *Let* $0 < h < s < 1$ *and* $0 \leq M < 1$. *If* $\delta < (1-h)(1-M)$ *and* $h+\delta < s$, *every compact set with* $u \leq h$ *and* $\mu \leq M$ *lies in* $V_\delta \cap \{x < s\}$, *and its image is compact in* $S_a$, $a = 1-s$. *The bound* (74) *still holds on such a set whenever* $I \geq 0$ *and* $u \leq I$.

*Proof.* The smoothness, pullback and inverse calculations in the proof of Lemma 12.7 use only $u < 1$ and positivity of the radicand. On the stated compact set the strip inequality gives $1 - x - (1-u)\mu > (1-h)(1-M) - \delta > 0$ and $x < h + \delta < s$. The set $V_\delta \cap \{x < s\}$ is open and its image has capacity in $(a, 1)$. The lower bound for $x$ uses only the strip map and the bound for $u$, so is unchanged. □

12.6. **Packing.** Below, $\rho_i^{\mathrm{dom}}$ is the closed source capacity to which Theorem M2.6 is applied, and $\rho_i^+$ the enlarged closed capacity of Construction M3.4(iii).

**Theorem 12.9** (One large ball, $n \geq 4$)**.** *Let $n \geq 4$, $k \geq 0$, and let $(A; \rho_1, \dots, \rho_k)$ admit the valid auxiliary tuple $(a; r_1, \dots, r_k)$. Put $\rho_0 = A$, $\rho_0^+ = \frac{1}{2}(A + a)$, and let $f_0$ be the identity map of $\mathbb{C}^n$. If $k \geq 1$, let:*

- *(a) $h_1, h_2, b_2, h_2', \tau_\Delta, d, \Delta, P$ be as in Lemma 12.1;*
- *(b) $(K, q)$ be the comparison datum of Theorem M2.11 for $(\Delta, P, \tau_\Delta, d)$, with gap $\gamma = \frac{11}{16}d$;*
- *(c) $\rho_i^{\mathrm{dom}} = (3\rho_i + 5r_i)/8$, $\rho_i^+ = \frac{1}{2}(\rho_i + \rho_i^{\mathrm{dom}}) = (11\rho_i + 5r_i)/16$, and $\psi_i$ the maps of Construction 8.4 for the mixed model over $\Sigma_R$ with the handle data of Proposition 12.6, the profile $P_R = P - 11d/32$ with the datum $(K, q)$, auxiliary capacities $r_i$, closed source capacities $\rho_i^{\mathrm{dom}}$ and internal capacities $(3r_i + \rho_i^{\mathrm{dom}})/4$;*
- *(d) $\mathcal{K} = \bigcup_{i \geq 1} \psi_i(B(\rho_i^{\mathrm{dom}}))$, $\mathcal{K}' = \Phi_2\chi(\mathcal{K})$, $\mu_{\mathcal{K}} = \max_{\mathcal{K}'} \mu$, and*

$$\hat{\mu} = \tfrac{1}{2}(1 + \mu_{\mathcal{K}}), \qquad D = \min\big((1-h_1)(1-\hat{\mu}), s - h_1\big), \qquad \delta_{\mathrm{sl}} = D/2;$$

- *(e) $f_i = \mathrm{Sl} \circ \Phi_2 \circ \chi \circ \psi_i$ for $1 \leq i \leq k$, with Sl the map of Proposition 12.8 for $\delta = \delta_{\mathrm{sl}}$, that is, built from $\Psi_D$.*

*Put $\zeta = x_*(h_1)$ of (74) with $\delta = \delta_{\mathrm{sl}}$ if $k \geq 1$, and $\zeta = 1 - \rho_0^+$ if $k = 0$. Then $\mu_{\mathcal{K}} < 1$, $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}}$ for $1 \leq i \leq k$, $A < \rho_0^+ < a$, $0 < \zeta \leq 1 - \rho_0^+$, and:*

- *(i) $f_0$ is a symplectic embedding of $\mathbb{C}^n$, and $f_i$ $(i \geq 1)$ is a smooth symplectic embedding of an open neighbourhood of $B(\rho_i^{\mathrm{dom}})$ into $B^\circ(1)$;*
- *(ii) $f_0(B(\rho_0^+)) = B(\rho_0^+) \subset B^\circ(a)$, and $f_i(B(\rho_i^{\mathrm{dom}})) \subset \mathcal{C} = \{1 - h_1 - \delta_{\mathrm{sl}} \leq \pi|y|^2 \leq 1 - \zeta\} \subset S_a$ for $i \geq 1$;*
- *(iii) the sets $f_0(B(\rho_0^+))$ and $f_i(B(\rho_i^{\mathrm{dom}}))$, $1 \leq i \leq k$, are pairwise disjoint and contained in $B(1 - \zeta)$.*

*The same holds with $\rho_i^+$ in place of $\rho_i^{\mathrm{dom}}$ $(i \geq 1)$, since $B(\rho_i^+) \subset B(\rho_i^{\mathrm{dom}})$. Moreover $f_0$ restricts to an embedding of every ball $B^\circ(\rho)$, $\rho \leq 1$, into $B^\circ(1)$; in particular it is defined on $B^\circ(a)$, the open domain $B^\circ(\rho_j^{\mathrm{dom}})$ of Construction M3.4(iii).*

*Proof. The distinguished ball.* The auxiliary inequalities give $A < a$, so $A < \rho_0^+ < a$, and the identity maps $B(\rho_0^+)$ into $B^\circ(a) \subset B^\circ(1)$. If $k = 0$ this proves everything, since $\zeta = 1 - \rho_0^+$.

*Comparison and packing in the model.* Let $k \geq 1$. By Lemma 12.1(iii)–(vi), $(\Delta, P, \tau_\Delta, d)$ satisfies the hypotheses of Proposition M2.9, so Theorem M2.11 provides the comparison datum $(K, q)$ for $(\Delta, P)$ with gap $\gamma = \frac{11}{16}d > 0$. By Lemma 12.4 and Proposition 12.6 the restricted mixed model on $\Sigma_R$ is a bordered toric model with handle data, including the correction-core and density-floor conditions of Section 8. By Proposition 11.5, the same datum compares $P_R$ with gap $11d/32$, reserve $11d/64$, and $P_R \leq H_b^R - \sum_i (r_i - |p|)_+$; and $\{|p| \leq r_i\} \subset \tau_\Delta\Delta$ by Lemma 12.1(iii). Since $\rho_i < \rho_i^{\mathrm{dom}} < r_i$, Theorem M2.6 gives smooth symplectic embeddings $\psi_i$ of neighbourhoods of $B(\rho_i^{\mathrm{dom}})$ into $(\operatorname{int} V_\Delta(E_0), \Omega_{\mathrm{mix}})$ with pairwise disjoint images in the interior of the finite model, so $\mathcal{K}$ is a compact subset of that interior.

*Affine return.* By Lemma 12.4, $\Phi_2 \circ \chi$ is smooth and injective on $\mathcal{U} = \operatorname{int} \mathcal{M}_R$, with symplectic derivative, hence (inverse function theorem) an embedding into $(\mathbb{C} \times B^\circ_{n-1}(1), \Omega_1)$ preserving $u$. So $\mathcal{K}'$ is compact, $u \leq h_1$ and $\mu < 1$ on $\mathcal{K}'$, and the maximum $\mu_{\mathcal{K}}$ of $\mu$ is attained, so $\mu_{\mathcal{K}} < 1$.

*Slit-shell map.* Here $0 < h_1 < s < 1$, $\mu_{\mathcal{K}} < \hat{\mu} < 1$, $D \le (1-h_1)(1-\hat{\mu})$ and $D \le s - h_1$, so $\delta_{\mathrm{sl}} < (1-h_1)(1-\mu_{\mathcal{K}})$ and $h_1 + \delta_{\mathrm{sl}} < s$. By Proposition 12.8 with $h = h_1$, $M = \mu_{\mathcal{K}}$, the open set $W = V_{\delta_{\mathrm{sl}}} \cap \{x < s\}$ contains $\mathcal{K}'$, and $\mathcal{T}_{\mathcal{K}} = \mathrm{Sl} \circ \Phi_2 \circ \chi$ is a smooth symplectic embedding of the open set $\mathcal{U} \cap (\Phi_2\chi)^{-1}(W)$ into $S_a$. On $\mathcal{K}'$, $\pi|y|^2 = 1 - x > 1 - u - \delta_{\mathrm{sl}} \ge 1 - h_1 - \delta_{\mathrm{sl}}$, and the strip estimate of Proposition 12.8 with $I = h_1$ gives $x \ge \zeta = x_*(h_1) > 0$. Hence $\mathcal{T}_{\mathcal{K}}(\mathcal{K}) \subset \mathcal{C}$. Also $\zeta \le x < h_1 + \delta_{\mathrm{sl}} < s$, so $1 - \zeta > a > \rho_0^+$ and $1 - h_1 - \delta_{\mathrm{sl}} > a$; thus $\mathcal{C} \subset S_a$.

*Assembly.* For $i \ge 1$, $\psi_i^{-1}\big(\mathcal{U} \cap (\Phi_2\chi)^{-1}(W)\big)$ is an open neighbourhood of $B(\rho_i^{\mathrm{dom}})$ in the domain of $\psi_i$, and $f_i = \mathcal{T}_{\mathcal{K}} \circ \psi_i$ is a smooth symplectic embedding of it (Proposition M2.7). The sets $f_i(B(\rho_i^{\mathrm{dom}}))$, $i \ge 1$, are pairwise disjoint because the $\psi_i(B(\rho_i^{\mathrm{dom}}))$ are and $\mathcal{T}_{\mathcal{K}}$ is injective; they lie in $\mathcal{C} \subset \{\pi|y|^2 > a\}$, so are disjoint from $f_0(B(\rho_0^+)) = B(\rho_0^+)$; and $\mathcal{C}, B(\rho_0^+) \subset B(1 - \zeta)$. □

## 13. Dimension six: the polynomial model

We use the cubic and quadric polynomials of [Ope26] and the convention $d^c = i(\bar{\partial} - \partial)/(4\pi)$, so the Hopf primitive at a nonzero vector $Z$ is

$$\lambda_c(Z) = \frac{c}{2\|Z\|^2}\,\mathrm{Im}\langle Z, dZ\rangle = \pi c\, d^c \log \|Z\|^2.$$

### 13.1. **The curve and the parameters.**

**Lemma 13.1** (Affine cubic coordinates)**.** *Let $f(v,w) = vw(v-w) + w - 2v$. The affine curve $C = \{(v,w) \in \mathbb{C}^2 : vw \ne 0,\ f(v,w) = 0\}$ is identified with*

$$\Sigma = \{(t, y_1, y_2) : y_1^2 = t^2 - 1,\ y_2^2 = t^2 - 2\}$$

*by $(v,w) = (y_1 - t, y_2 - t)$ in the chart $u = 1$, with inverse*

$$t = -\frac{v^2+1}{2v}, \qquad y_1 = t + v, \qquad y_2 = t + w.$$

*Its four ends have limits* $[0:1:0], [0:0:1], [0:1:1], [1:0:0]$.

*Proof.* The two equations on $\Sigma$ imply $v, w \ne 0$ and $f(v,w) = 0$; conversely $f = 0$ and $vw \ne 0$ give the displayed inverse by substitution. The four sign choices in the end coordinates of Lemma 10.1, with $m = 2$, give the stated limits. The formulas are holomorphic on their affine domains; smoothness follows from the explicit quadric charts. □

Write $s = 1 - a$, $b = (2-a)/3$ and $V = 1 - a^3 - \sum_i r_i^3 > 0$, with $a \in \mathbb{Q}$, $1/2 < a < 1$ and $r_i < s$. For a nonempty smaller family let $R = \max_i r_i$; the empty family needs no cubic model.

**Lemma 13.2** (Explicit truncation parameters)**.** *Set*

$$\mu = \min(s - R, (2a-1)/6, V/100), \qquad M = \lceil 1/\mu \rceil + 1, \qquad \delta_0 = 1/M,$$

$$h_1 = s - \delta_0, \quad e = s - \delta_0/2, \quad d_1 = s + \delta_0/2, \quad h_2 = b - \delta_0, \quad d_2 = b - \delta_0/2.$$

*Then these parameters satisfy* (68) *and $\int_\Delta H_0 > \sum_i r_i^3/6$. For any integer $Q \ge 0$ one may take*

$$L = (Q+1)\,\mathrm{den}(a)\,\mathrm{den}(e)\,\mathrm{den}(d_2), \quad A_* = Le,\ B_* = Ld_2,\ D = L(2-a),\ S_* = L(1-a).$$

*These are integers, $L > Q$, and $0 < A_* < S_* \le B_*$, $A_* \le B_*$, $3B_* < D$, $3S_* < D$, $2S_* < L$.*

*Proof.* All three entries defining $\mu$ are positive and $0 < \delta_0 < \mu$; also $b - s = (2a-1)/3 > 2\delta_0$, which gives every ordering assertion. Expanding the integrated-profile formula of Lemma 11.7 at $(h_1, h_2) = (s - \delta_0, b - \delta_0)$ gives

$$\int_\Delta H_0 \ge (1 - a^3)/6 - 3\delta_0.$$

Indeed, let $I(h_1,h_2)$ be that formula. Then $\partial_{h_1} I = \frac{9}{2}(b-h_1)^2 - \frac{9}{2}(b-h_2)^2 - (s-h_1)^2$ and $\partial_{h_2} I = 9h_1(b-h_2)$. On the segment from $(s,b)$ to $(s-\delta_0, b-\delta_0)$, $0 \le b-h_2 \le \delta_0 < \frac{1}{12}$, $0 < h_1 \le s < \frac{1}{2}$ and $b < \frac{1}{2}$, so $|\partial_{h_1} I| < 2$ and $|\partial_{h_2} I| < 1$. Hence $I(h_1,h_2) \ge I(s,b) - 3\delta_0$, and $I(s,b) = (1-a^3)/6$ (Lemma 11.7). Since $3\delta_0 < 3V/100 < V/6$, the required integral is positive. The denominator product clears the three rational denominators; the integer inequalities follow by multiplying $0 < e < s < d_2 < b$ by $L$ and using $a > 1/2$. □

13.2. **The fixed coefficient list.**

**Definition 13.3** (Central potential and physical primitive)**.** Use the integers of Lemma 13.2 and set

$$\Lambda = \{k \in \mathbb{Z}^2_{\ge 0} : k_1 \le A_*,\ k_1 + k_2 \le B_*\}, \quad d_k = D - 3|k|, \quad m_k = (S_* - |k|)_+.$$

For $d > m \ge 0$ let

$$q_{d,m} = (1, v^{d-m}, w^{d-m}, v^{d-m}w^m, v^m w^{d-m}), \quad a_k = \|q_{d_k,m_k}\|^2.$$

For $r = (r_1, r_2) \ge 0$ define

$$Z_a(r) = \sum_{k\in\Lambda} a_k r_1^{k_1} r_2^{k_2}, \qquad \mathsf{M}_a(r)_j = r_j \partial_{r_j} \log Z_a(r).$$

For $P \ge 0$ with $P_1 < A_*$ and $P_1 + P_2 < B_*$, let $r_a(P)$ be the unique nonnegative solution of $\mathsf{M}_a(r) = P$, and set

$$\vartheta_k(y,P) = \frac{a_k(y) r_a(P)_1^{k_1} r_a(P)_2^{k_2}}{Z_a(r_a(P))}, \qquad \beta_{d,m} = \frac{1}{L} d^c \log \|q_{d,m}\|^2,$$

$$\Gamma(p) = \sum_{k\in\Lambda} \vartheta_k(y, Lp)\, \beta_{d_k,m_k}, \qquad \Omega_{\mathrm{phys}} = \omega_{\mathbb{C}^2} + d[\Gamma(p(z))], \quad p_j(z) = \pi|z_j|^2.$$

In the last differential $p$ varies with $z$; the primitive is horizontal in the base variable. The radial disk map of Lemma 13.4 relates these physical coordinates to the toric coordinates of the central potential.

**Lemma 13.4** (Radial coordinates)**.** *The inverse $r_a(P)$ exists and is unique on the stated lower trapezoid, and the radial disk map and its inverse are smooth across both axes. The central potential* (69)*, in physical coordinates, has the form $\Omega_{\mathrm{phys}}$ of Definition 13.3. At fixed $p$ its base curvature $d_\Sigma\Gamma(p)$ is positive.*

*Proof.* For positive radii put $r_j = e^{x_j}$. Then $\mathsf{M}_a$ is the gradient of $\log \sum_k a_k e^{\langle k,x\rangle}$; its Hessian, the covariance matrix of $k$, is positive definite because $(0,0), (1,0), (0,1)$ occur. Its gradient maps onto the interior of the trapezoid: strict convexity gives uniqueness, and subtraction of $\langle P, x\rangle$ is coercive when $P$ is interior. On an axis the same argument in one variable applies. Write $\mathsf{M}_a(r)_j = r_j s_j(a,r)$. The numerators of $s_j$ are polynomials with positive coefficients including a positive constant, so $s_j > 0$ on the closed quadrant. The disk map is

$$(z_1, z_2) \longmapsto (\sqrt{s_1(a,|z|^2)}z_1, \sqrt{s_2(a,|z|^2)}z_2).$$

Its inverse is obtained by the unique quadrant root, and its derivative is invertible also on the axes; the parameter implicit-function theorem gives joint smoothness. The bridge from physical to cone coordinates uses the inverse disk map:

$$\Xi(y,z) = \big(y, \mathsf{R}^{-1}_{a(y)}(\sqrt{\pi L}\, z)\big),$$

where $\mathsf{R}_a$ is the displayed radial disk map; for a compact source it is multiplied, away from a neighbourhood of the source, by a plateau inside the two outer moment faces. Differentiating

the logarithmic potential separates the angular primitive $\sum_j p_j d\theta_j$ from the horizontal primitive $\Gamma(p)$, and reduction of the positive central Kähler form gives positivity of the base form. The radial disk pullback identity includes the vanishing angular terms on the axes. □

13.3. **Explicit homogeneous lifts.** Put

$$Q(X) = X_0X_1 - X_2X_3, \qquad \mathsf{q}(u,v,w) = (vw, u^2, uv, uw).$$

For a plane monomial $v^Jw^K$ with $J, K \le n$, define

$$\alpha = (J+K-n)_+, \quad \beta = n+\alpha-J-K, \quad \gamma = J-\alpha, \quad \nu = K-\alpha, \qquad \mathcal{L}_n(v^Jw^K) = X_0^\alpha X_1^\beta X_2^\gamma X_3^\nu,$$

extended linearly to polynomials with those separate degree bounds; all four exponents are nonnegative integers.

**Lemma 13.5** (Polynomial lifts and vanishing orders)**.** *For $k \in \Lambda$ and each entry $q_i$ put*

$$P_{k,i}(X) = Q(X)^{k_1}\mathcal{L}_{L-2k_1}(f^{k_2}q_{d_k,m_k,i})(X).$$

*This is homogeneous of degree $L$ and vanishes to order at least $L - S_*$ at $o = [1:0:0:0]$. If*

$$P_{t,\delta}(X)_{k,i} = \delta^{-k_1}t^{-k_2}P_{k,i}(X), \qquad T_\delta(v,w,z) = (v,w,vw+\delta z),$$

*then $P_{t,\delta}(vw+\delta z, 1, v, w)$ has a polynomial extension in $\delta$ at zero, with value*

$$z^{k_1}(f(v,w)/t)^{k_2}q_{d_k,m_k,i}(v,w).$$

*Proof.* The exponent identities $\alpha+\gamma = J$, $\alpha+\nu = K$, $2\beta+\gamma+\nu = 2n-J-K$ and $\alpha+\beta+\gamma+\nu = n$ give $\mathcal{L}_n(v^Jw^K)(\mathsf{q}(u,v,w)) = u^{2n-J-K}v^Jw^K$. Since $2m_k \le d_k$, each coefficient monomial has separate degrees at most $d_k - m_k$ and total degree at most $d_k$, so $f^{k_2}q_i$ has separate degrees at most $2k_2 + d_k - m_k$ and total degree at most $3k_2 + d_k$. Writing $n = L - 2k_1$, the integer inequalities give

$$2k_2 + d_k - m_k \le n, \quad L - S_* - k_1 \le n, \quad 3k_2 + d_k + (L - S_* - k_1) \le 2n.$$

The last two imply $\beta + \gamma + \nu \ge L - S_* - k_1$ for every monomial in its lift, and the factor $Q^{k_1}$ contributes another $k_1$ to the order at $o$. Homogeneity follows from $2k_1 + n = L$. Finally $Q(vw+\delta z, 1, v, w) = \delta z$, so the negative powers of $\delta$ cancel term by term. At zero use $\mathsf{q}(1,v,w) = (vw, 1, v, w)$. □

13.4. **Area and exhaustion.** In the diagonal-quadric coordinates set $\Sigma_l = \{|t| \le 5 + l\}$, $l \in \mathbb{Z}_{\ge 0}$: compact, nested bordered surfaces exhausting $C$. The explicit handle and its dual form lie in $|t| < 3$, hence in $\operatorname{int}\Sigma_0$. Let $H_l(p) = \int_{\Sigma_l} d_\Sigma\Gamma(p)$.

**Proposition 13.6** (Area profile with a quantitative tail)**.** *On $\Delta$ one has*

$$0 \le H_0(p) - H_l(p) \le E_l \quad (l \ge 32), \qquad H_0(p) = 9(b - |p|) - 2(s - |p|)_+.$$

*Put $R_* = 1/16$ and fix the end cutoff*

$$\chi_{\mathrm{end}}(q) = S\left(\frac{q + R_*}{R_*/2}\right)\left[1 - S\left(\frac{q - R_*/2}{R_*/2}\right)\right].$$

*Let $K = 1 + \max(0, \sup_{q\in\mathbb{R}} |\chi'_{\mathrm{end}}(q)|)$, let $M_*$ bound the finitely many regular end primitives, and let $B_{\log}$ bound the absolute logarithms of the regular coefficient norms, on the four end discs $|\zeta|^2 \le R_*$. These bounds are the maxima plus one. With $q_* = |c|(2B_{\log} + \#\Lambda)$ and $c = 1/(L\pi)$, one may take*

$$E_l = 64KR_*M_*\sqrt{\frac{R_*}{l+1}} + \frac{16q_*KR_*}{-\log(R_*/(l+1))}.$$

*In particular, if* $C_* = 1 + 64KR_*M_* + 16q_*KR_*$ *and* $\eta > 0$*, then*

$$\mathfrak{m}(\eta) = \max\left(32, \left\lceil (2C_*/\eta)^2 \right\rceil, \left\lceil e^{2C_*/\eta} \right\rceil\right)$$

*satisfies* $|H_l - H_0| < \eta$ *for* $l \geq \mathfrak{m}(\eta)$*, uniformly on* $\Delta$*.*

*Proof.* We write $\Omega_p$ for the base form $d_\Sigma\Gamma(p)$ and $\Gamma_p$ for its primitive, a convex combination of the weighted Fubini–Study primitives of the five homogeneous coefficients, with weights the selection probabilities below.

*Step 1: end orders.* At the ends [0:1:0], [0:0:1], [0:1:1], [1:0:0] one has $(v, w) \sim (-2t, -1/t)$, $(-1/(2t), -2t)$, $(-2t, -2t)$, $(-1/(2t), -1/t)$ as $t \to \infty$ (Lemma 13.1). Reading the pole orders of the five entries of $q_{d,m}$ from $(u^d, u^m v^{d-m}, u^m w^{d-m}, v^{d-m} w^m, v^m w^{d-m})$, the maximal order is $o = d - m$ at the marked ends [0:1:0], [0:0:1], $o = d$ at [0:1:1] and $o = 0$ at [1:0:0] (recall $2m \leq d$).

*Step 2: end charts.* The curve has four ends, indexed by the sign choices $\varepsilon \in \{\pm\}^2$ of the two square roots $y_j = \pm t\sqrt{1 - j\,t^{-2}}$. On each end use $\zeta = t^{-1}$. If $|\zeta|^2 \leq R_* = \frac{1}{16}$, then $|j\zeta^2| \leq \frac{1}{8}$ for $j = 1, 2$, so $\mathrm{Re}(1 - j\zeta^2) \geq \frac{7}{8} > 0$; the principal roots $\sqrt{1 - j\zeta^2}$ are smooth and nonvanishing, and the end map $\zeta \mapsto (t, y)$ is a diffeomorphism of the punctured disc $\{0 < |\zeta|^2 \leq R_*\}$ onto a neighbourhood of the end. On this disc every coefficient is $\zeta^{-o}$ times a smooth nonvanishing function of $\zeta$, where $o$ is its end order from Step 1.

*Step 3: decomposition on an end.* On the punctured disc of end $\varepsilon$,

$$\Gamma_p = \beta_\varepsilon - \tfrac{1}{2}\, d_\varepsilon\, d\theta, \qquad d\theta = \text{the angular form of } \zeta,$$

where $\beta_\varepsilon$ is the selection-weighted sum of the regular parts of the coefficient primitives, a smooth one-form with $\|\beta_\varepsilon\| \leq M_*$, and the residue $d_\varepsilon(\zeta) = c\sum_k \pi_k(\zeta)\, o_k(\varepsilon)$ is the average of the weighted end orders under the selection probabilities $\pi_k = \vartheta_k(y, Lp)$ of Definition 13.3, which are positive, sum to one and have mean $\sum_k \pi_k k = Lp$. The decomposition holds because $d^c \log a_k = d^c \log c_k - o_k\, d^c \log|\zeta|^2$ with $c_k = |\zeta|^{2o_k} a_k$ smooth and positive and $d^c \log|\zeta|^2 = d\theta/(2\pi)$; thus $\beta_\varepsilon = L^{-1}\sum_k \pi_k d^c \log c_k$.

*Step 4: the residue.* Put $\ell = -\log|\zeta|^2 > 0$ and $P = Lp$. On end $\varepsilon$ write $a_k = c_k\,|\zeta|^{-2o_k} = c_k e^{\ell o_k}$, with $c_k > 0$ the regular coefficient norm of Step 2, so $|\log c_k| \leq B_{\log}$. The probabilities $\pi_k \propto a_k r^k$, $r = r_a(P)$, are the unique maximiser of

$$\mathcal{J}(\mu) = \sum_k \mu_k \log a_k - \sum_k \mu_k \log \mu_k$$

over probability vectors $\mu$ on $\Lambda$ with $\sum_k \mu_k k = P$: $\mathcal{J}$ is strictly concave, and the Lagrange condition $\log \mu_k = \log a_k + \langle x, k\rangle + \text{const}$ at an interior critical point is the defining form $\pi_k \propto a_k r^k$ with $r_j = e^{x_j}$ (if some $P_j = 0$, the same holds on the face $k_j = 0$).

At the ends [1:0:0] and [0:1:1] the order $o_k$ is 0, resp. $d_k = D - 3|k|$, an affine function of $k$, so $d_\varepsilon = c\sum_k \pi_k o_k$ is $c$ times this affine function at $P$; this is $c_\varepsilon$, with no error. At the two marked ends $o_k = d_k - m_k$ with $m_k = (S_* - |k|)_+$. Let $A(k) = S_* - |k|$ if $|P| < S_*$ and $A = 0$ otherwise; then $A$ is affine, $A(k) \leq m_k$ on $\Lambda$ and $A(P) = (S_* - |P|)_+$. Put $e_k = m_k - A(k) \geq 0$. The set $F = \{k \in \Lambda : e_k = 0\}$ consists of the lattice points of the polygon $\{k \geq 0,\ k_1 \leq A_*,\ |k| \leq S_*\}$, resp. $\{k \geq 0,\ k_1 \leq A_*,\ S_* \leq |k| \leq B_*\}$, whose vertices are integral; since $P_1 < A_*$ and $|P| < B_*$, $P$ lies in this polygon, so some probability vector $\nu$ supported on $F$ has mean $P$. Since $\log a_k = \log c_k + \ell(d_k - A(k)) - \ell e_k$ and $d_k - A(k)$ is affine, $\sum_k \mu_k \ell(d_k - A(k))$ is the same for all admissible $\mu$, and $\mathcal{J}(\pi) \geq \mathcal{J}(\nu)$ gives

$$\ell \sum_k \pi_k e_k \leq \sum_k (\pi_k - \nu_k)\log c_k - \sum_k \pi_k \log \pi_k + \sum_k \nu_k \log \nu_k \leq 2B_{\log} + \log \#\Lambda.$$

As $\sum_k \pi_k e_k = \sum_k \pi_k m_k - (S_* - |P|)_+$, and Jensen's inequality for the convex function $k \mapsto m_k$ gives the lower bound,

$$0 \le \sum_k \pi_k m_k - (S_* - L|p|)_+ \le \frac{2B_{\log} + \#\Lambda}{-\log|\zeta|^2}.$$

With $c_\varepsilon = c\,(d(p) - h(p))$ at the marked ends, $d(p) = D - 3L|p|$, $h(p) = (S_* - L|p|)_+$, this is

$$|d_\varepsilon(\zeta) - c_\varepsilon| \le \frac{q_*}{-\log|\zeta|^2}, \qquad q_* = |c|\,(2B_{\log} + \#\Lambda),$$

uniformly in $p \in \Delta$. The limiting orders are $d-h$, $d-h$, $d$ and 0, so $\pi \sum_\varepsilon c_\varepsilon = L^{-1}(3d(p) - 2h(p)) = H_0(p)$.

*Step 5: the cutoff and Stokes.* Let $\chi_l(\zeta) = \chi_{\text{end}}((l+1)|\zeta|^2)$ on each end disc and $\psi_l = 1 - \chi_l$, extended by 1 off the ends. Since $\chi_{\text{end}} = 1$ on $[-R_*/2, R_*/2]$, $\psi_l$ vanishes where $(l+1)|t|^{-2} < R_*/2 = \frac{1}{32}$, that is for $|t|^2 > 32(l+1)$. For $l \ge 32$ the radius $R_l = 5 + l$ of $\Sigma_l$ satisfies $R_l^2 \ge (l+3)^2 \ge 32(l+1)$, because $(l+3)^2 - 32(l+1) = l^2 - 26l - 23 \ge 0$ for $l \ge 27$. Hence $\operatorname{supp}\psi_l \subset \Sigma_l$. On each end, $d\psi_l$ is supported in the annulus $A_l = \{R_*/(2(l+1)) \le |\zeta|^2 \le R_*/(l+1)\}$, and $|\chi'_{\text{end}}| \le K$. Stokes' theorem gives

$$\int_\Sigma \psi_l\,\Omega_p = -\int_\Sigma d\psi_l \wedge \Gamma_p = \sum_\varepsilon \int_{A_l} \Big( d\chi_l \wedge \beta_\varepsilon - \tfrac{1}{2} d_\varepsilon\, d\chi_l \wedge d\theta \Big).$$

For the first term, $|d\chi_l| \le 2K(l+1)|\zeta|$ and the area of $A_l$ is at most $\pi R_*/(l+1)$, so it contributes at most $16KR_*M_*\sqrt{R_*/(l+1)}$ on each end. For the second, $-\frac{1}{2}\int_{A_l} d\chi_l \wedge d\theta = \pi$, since $\chi_l$ is radial and runs once between its boundary values 1 and 0 across $A_l$, decreasing in the orientation induced from $\Sigma$; so replacing $d_\varepsilon$ by $c_\varepsilon$ produces $\pi c_\varepsilon$ with an error at most $4KR_*\,q_*/(-\log(R_*/(l+1)))$ by Step 4, since $|\zeta|^2 \le R_*/(l+1)$ on $A_l$. Summing over the four ends,

$$\Big| \int_\Sigma \psi_l\,\Omega_p - H_0(p) \Big| \le 64KR_*M_*\sqrt{\frac{R_*}{l+1}} + \frac{16q_*KR_*}{-\log(R_*/(l+1))} = E_l.$$

Since $E_l \to 0$ and $\psi_l \uparrow 1$, monotone convergence gives $\int_\Sigma \Omega_p = H_0(p)$. As $\Omega_p$ is positive, $0 \le \psi_l \le 1$ and $\operatorname{supp}\psi_l \subset \Sigma_l$, we get $\int_\Sigma \psi_l \Omega_p \le H_l(p) \le H_0(p)$, and therefore $0 \le H_0(p) - H_l(p) \le E_l$.

For $l \ge \mathfrak{m}(\eta)$ the first term of $E_l$ is at most $C_*/(4\sqrt{l+1}) < \eta/2$ and the second at most $C_*/\log(16(l+1)) < \eta/2$, which proves the uniform modulus. □

## 14. Dimension six: compact transfers and the large ball

The smaller balls are packed into the polynomial model of Section 13, and their compact images are transferred exactly through the cubic and quadric polynomial families; the large ball is inserted by a bubble at the distinguished point, and one common map goes into the unit ball.

14.1. **Packing over a bordered base.** Let $r_i' \in (\rho_i, r_i)$ and use the parameters of Lemma 13.2. Put $G = H_0 - \sum_i (r_i - |p|)_+$, $\tau = (1 + \max_i r_i/h_1)/2$ and $d_0 = 9(b - h_2)$. The cap supports lie in $\tau\Delta$, and $G \ge d_0$ off $\tau\Delta$. $G$ is concave, being the concave profile $H_0$ minus convex caps. Write $\overline{G} = |\Delta|^{-1}\int_\Delta G$ and set

$$\eta = \frac{1}{32}\min(d_0, \overline{G}) > 0.$$

Set

$$d = \min\big((\overline{G} - \eta)/2, d_0/4\big), \quad \gamma = 11d/16, \quad r_* = 11d/64, \quad l = \max\big(\mathfrak{m}(\eta), \mathfrak{m}(r_*/32) + 2\big).$$

These are positive; the level $l$ reserves two outer collar levels.

**Proposition 14.1** (Inner packing)**.** *The physical form $\Omega_{\mathrm{phys}}$ admits pairwise disjoint symplectic embeddings $\phi_i$ of neighbourhoods of $B(r_i')$, with compact images in $\Sigma_l \times \{z : p(z) \in \Delta\}$ and moments strictly inside the two outer faces, obtained from the finite comparator for $G - \eta$ and the common handle construction.*

*Proof.* Lemma 13.2 gives $\overline{G} > 0$; the positivity bound for $H_0$ gives the other positive entries of $\eta$. Proposition 13.6 gives $H_l > H_0 - \eta$. For the finite comparator use $P = G - \eta$, outer lower bound $d_0/2$, parameter $d$ and gap $\gamma$. The factor $1/32$ ensures $P \geq d_0/2$ off the core and positive mean; also $P < H_l - \sum_i (r_i - |p|)_+$. To represent $-P$ as a maximum of affine functions, take every sum consisting of one negative affine piece of $H_0$, one of the two pieces of each cap, and the constant $\eta$; their maximum is $-H_0 + \sum_i (r_i - |p|)_+ + \eta$. The handle lies in $\operatorname{int}\Sigma_0$. Put $R = 5 + l$. The second entry in the maximum defining $l$ ensures that the curvature mass between $\Sigma_{l-2}$ and $\Sigma_l$ is less than $r_*/32$, uniformly on $\Delta$: positivity and the tail bound give $0 \leq H_l - H_{l-2} \leq H_0 - H_{l-2} < r_*/32$. Use the fixed boundary cutoff and background coefficient

$$\chi_\partial(t) = S\left(\frac{|t|^2 - (R-2)^2}{(R-1)^2 - (R-2)^2}\right), \qquad c_{\mathrm{bg}} = \min\left(\frac{r_*}{128}, \frac{\kappa_{\min}}{2}\right) > 0.$$

The correction atlas is the family of Lemma 10.3 with $m = 2$ and $R = R - 1$; its inner boxes cover $\{|t| \leq R - 1\}$, which contains the support of $1 - \chi_\partial$ and the handle, and its closed outer boxes lie in $\operatorname{int}\Sigma_l$ (Lemma 6.11). Here $\kappa_{\min} > 0$ is the minimum of the ratio $d_\Sigma\Gamma(p)/\sigma$, $\sigma = dd^c \log(1 + |t|^2 + |y_1|^2 + |y_2|^2)$, over the union of these closed boxes times $\Delta$, a compact set on which $d_\Sigma\Gamma(p) > 0$ (Lemma 13.4). Thus the correction region of Definition 6.1(d) is this union, with $\Lambda = \sigma$ and $\kappa_- = \kappa_{\min}$, and it contains the correction core, as required in §8. The reference mass is $\int_{\Sigma_l} \sigma \leq 2^m = 4$ (Lemma 10.5), so the removed form

$$(1 - \chi_\partial)\big(d_\Sigma\Gamma(p) - c_{\mathrm{bg}}\sigma\big)$$

has handle mass $H(p) \geq H_l(p) - r_*/16$: the physical collar costs less than $r_*/32$ and the background costs at most $4c_{\mathrm{bg}} \leq r_*/32$. The compact-coordinate primitive and annular correction are then the fixed formulas of Construction 8.2. In the cylinder use the working half-width $b_H = 1/16$ and circle-clock parameter $1/4$. For $N = k + 1$ the layer step is $D_H = 2b_H/(3N + 1)$, with layer intervals $[-b_H + (3i+1)D_H, -b_H + (3i+2)D_H]$, $0 \leq i < N$. For the enlarged inner capacities $\widetilde{r}_i = (3r_i + r_i')/4$, use the cap smoothing scale $(r_i - \widetilde{r}_i)/2$ and thin-strip parameter $(r_i - \widetilde{r}_i)/4$. All these layer parameters are positive. The comparison inequality and the cap bound give a residual of at least $\gamma - r_*/16 > 0$ after redistribution, so the common layer and Moser formulas apply to $H$ without assuming its concavity and give the stated neighbourhood embeddings. Positivity and the strict outer face margins persist on neighbourhoods of the compact ball images. □

Let $K_1 = \bigcup_i \phi_i(B(r_i'))$; below only this compact set and its successive images need to be controlled.

14.2. **Compact exact transfer.** We build a complete auxiliary flow and then choose its stopping time. All extrema below are of continuous scalar functions on the displayed compact sets.

For a product of the base curve with a vector space, take the finite coordinate boxes of the quadric atlas covering the compact base projection. Write $\rho_i$ for their coordinate plateaux: $\rho_i = 1$ on the inner box and its closed support lies inside the chart. Let $G_i(t, x)$ be the squared determinant of $d\Gamma_t$ in that product chart, evaluated at $x$, and put

$$g(t,x) = \sum_i \rho_i(x_{\mathrm{base}}) G_i(t,x), \qquad \mu = \min\big(1, \min_K g(0,\cdot)\big) > 0,$$

$$R = \max\big(0, \max_K |x_{\mathrm{fibre}}|^2_{\mathrm{coord}}\big).$$

The summands are extended by zero outside their charts, smoothly because the closed support of $\rho_i$ lies strictly inside the chart. Use

$$\chi(t,x) = S(4g(t,x)/\mu - 1)\, S(2-t^2)\, S\big(2(R+1-|x_{\text{fibre}}|^2_{\text{coord}})\big). \tag{75}$$

Its closed support lies where $g \geq \mu/4 > 0$, hence where the form is nondegenerate and at least one of the finite charts applies. It is spatially compact: the base lies in the union of the closed boxes, and $|x_{\text{fibre}}|^2_{\text{coord}} \leq R+1$.

In each chart, pull back the full ambient covector representing $\Gamma_t$, including its vertical components, differentiate it, and solve the finite linear system

$$\iota_{X_t} d\Gamma_t = -\partial_t \Gamma_t.$$

Take the first applicable chart in the fixed finite order and multiply its field by $\chi$. The chart solutions agree on overlaps, so this rule defines a globally smooth, spatially compact field $W_t$, extended by zero away from the cutoff support. Spatial compactness gives its flow $\Psi_s$ on $-2 < s < 2$, determined by the ambient-coordinate ODE and continuity of the lifted curve; the displayed chart inverses prove smoothness of this lift.

**Lemma 14.2** (Exact transfer with a finite stopping rule)**.** *Let $\Gamma_t$ be a smooth one-form family, with $d\Gamma_0$ nondegenerate on a compact set $K$. Let $A_t : U \to N$ be a smooth family, an embedding for $0 < t \leq \delta$, with $A_t^*\Omega_t = d\Gamma_t$ on the open neighbourhood $U$ of $K$. Let $T \subset N$ be open. Suppose the complete field $W$ above comes with finitely many smooth scalar control functions $b_j(t,x)$ such that*

$$b_j(0,x) > 0 \quad (x \in K), \qquad \big[\, b_j(t,x) > 0 \text{ for all } j \,\big] \Longrightarrow \chi(t,x) = 1,\ x \in U,\ A_t(x) \in T.$$

*Define*

$$c_* = \min\big(1, \min_{j, x \in K} b_j(0,x)\big), \tag{76}$$

$$V_* = 1 + \max\left(0, \max_{j,(s,x)\in[-1,1]\times K} \big|\partial_s\big[b_j(s,\Psi_s x)\big]\big|\right), \tag{77}$$

$$\tau_* = \min\left(\delta/2,\ 1/4,\ c_*/(4V_*)\right). \tag{78}$$

*For $0 < t \leq \tau_*$, the map $A_t \circ \Psi_t$ is a symplectic embedding of a neighbourhood of $K$ into $T$. Empty minima and maxima are interpreted by the inserted constants 1 and 0.*

*Proof.* The field is defined before $\tau_*$, so the derivative in (77) is continuous on the compact set $[-1,1] \times K$, and $c_* > 0$, $V_* > 0$. For $|s| \leq 2\tau_*$, the mean-value estimate gives

$$b_j(s,\Psi_s x) \geq b_j(0,x) - V_*|s| \geq c_*/2 > 0 \qquad (x \in K).$$

Thus every control function is positive on the compact image of $[-\tau_*, 2\tau_*] \times K$ under $(s,x) \mapsto (s, \Psi_s x)$, hence on a neighbourhood of it. There $W = X$, so Cartan's formula gives

$$\frac{d}{ds}\Psi_s^* d\Gamma_s = \Psi_s^* d(\partial_s \Gamma_s + \iota_{X_s} d\Gamma_s) = 0.$$

Consequently $(A_t\Psi_t)^*\Omega_t = d\Gamma_0$. The flow map $\Psi_t$ is a diffeomorphism; composing its restriction with the embedding $A_t$ gives the required neighbourhood embedding, and the target control functions put its image in $T$. □

On a vector space the same construction uses a fixed basis, with $\ell = \min_K \det(d\Gamma_0)^2$ and $R = \max_K |x|^2_{\text{coord}}$. Its cutoff is

$$S(4\det(d\Gamma_t)^2/\ell - 2)\, S\big(2(R+1-|x|^2_{\text{coord}})\big)\, S(2-t^2).$$

Its three plateau control functions $4\det(d\Gamma_t)^2/\ell - 3$, $2(R+1-|x|^2_{\mathrm{coord}}) - 1$ and $1-t^2$, together with the source and target control functions, enter (76)–(78). The same rule, applied uniformly, gives a common bubble parameter in Proposition 14.5.

14.3. **The cubic transfer.** Here the smaller family is nonempty; the empty case of Theorem 14.7 is the direct inclusion.

Define the cone polynomial on $\mathbb{C}^2 \times \mathbb{C}^2$ by

$$Z(v,w,z_1,z_2)_{k,i} = z_1^{k_1} z_2^{k_2} q_{d_k,m_k,i}(v,w), \qquad \lambda_{\mathrm{cone}} = Z^*\lambda_{1/(L\pi)}.$$

The $(k,i) = ((0,0),0)$ coordinate equals one, so this primitive is everywhere smooth. For $t > 0$ define

$$\mathcal{N}_t(v,w,z) = (v,w,z,f(v,w)/t), \qquad \lambda_t^{\mathrm{cub}} = \mathcal{N}_t^*\lambda_{\mathrm{cone}}, \quad \Omega_t^{\mathrm{cub}} = d\lambda_t^{\mathrm{cub}}.$$

For a nonempty compact physical source $K_1$, write $\zeta = \sqrt{\pi L}\, z$ and put

$$\delta_b = \frac{1}{2}\min_{(y,z)\in K_1} \min\big(A_* - |\zeta_1|^2,\ B_* - |\zeta_1|^2 - |\zeta_2|^2\big) > 0,$$

$$\chi_b(\zeta) = S_{\delta_b/2,\delta_b}(A_* - |\zeta_1|^2) S_{\delta_b/2,\delta_b}(B_* - |\zeta_1|^2 - |\zeta_2|^2).$$

The bridge is $\Xi(y,z) = (y, \chi_b(\zeta)\mathsf{R}^{-1}_{a(y)}(\zeta))$ on the disk domain, with fibre coordinate zero outside that domain. The plateau is zero near the boundary of the disk domain and one on a neighbourhood of $K_1$, so $\Xi$ is an explicit globally smooth extension of the inverse radial map, with its exact pullback on that neighbourhood.

**Proposition 14.3** (Cubic transfer)**.** *There are an explicit $t > 0$ and a smooth symplectic embedding $g_1$ of a neighbourhood of $K_1$ into $\{(v,w,z) : vw \neq 0\}$, with $g_1^*\Omega_t^{\mathrm{cub}} = \Omega_{\mathrm{phys}}$.*

*Proof.* Use a normal coordinate $\xi$ to the cubic and write $g(z) = |f_v(z)|^2 + |f_w(z)|^2$ for $z = (v,w)$. Let $K_C$ and $H_C$ be the base and disk projections of $\Xi(K_1)$, and set

$$\mathcal{D} = \{(\iota y, \xi) : (y, z_1, \xi) \in K_C \times H_C\}, \qquad m = \frac{1}{2}\min_{y\in K_C} g(\iota y) > 0, \qquad R_1 = 1 + \max_{Y\in\mathcal{D}} \mathrm{nsq}(Y),$$

where $\mathrm{nsq}(v,w,\xi) = |v|^2 + |w|^2 + |\xi|^2$. For the displacement and derivative bounds below use the product maximum norm. Define

$$r_N = \sqrt{R_1} + 1, \qquad R_N = 3r_N^2, \qquad G_N = \max_{\|z\|\le r_N} \|Dg(z)\|,$$

$$h_m(s) = sS_{m/4,m/2}(s) + \frac{m}{2}\big(1 - S_{m/4,m/2}(s)\big), \qquad c_R(s) = 1 - S_{R_N,R_N+1}(s),$$

$$X(v,w,\xi) = c_R(\mathrm{nsq}(v,w,\xi))\left(h_m(g(v,w))^{-1}(\overline{f_v}\xi, \overline{f_w}\xi), 0\right).$$

The denominator is at least $m/4$, and $X$ is spatially compact. Let

$$C_N = \sup_Y \|X(Y)\|, \qquad d_N = \min\left(1, \frac{m}{2(G_N+1)}\right), \qquad \epsilon_N = \min\left(1, \frac{d_N}{C_N+1}\right) > 0.$$

Take $\psi_t$ to be the complete flow of $S^{\mathrm{plateau}}_{-2,-1,1,2}(t)X$. For initial $Y = (v,w,\xi)$ in the open tube

$$\mathcal{W} = \{Y : \mathrm{nsq}(Y) < R_1,\ g(v,w) > m\},$$

the displacement on $|t| \le \epsilon_N$ is less than $d_N \le 1$, which keeps the product maximum norm below $r_N$, the gradient square above $m/2$ and $\mathrm{nsq} < 3r_N^2 = R_N$. The regularized field therefore agrees with the normal lift, and $f(\psi_t(v,w,\xi)_{\mathbb{C}^2}) = f(v,w) + t\xi$ on this tube. The initial data $\mathcal{D}$ lie strictly inside $\mathcal{W}$, since $g \ge 2m$ and $\mathrm{nsq} \le R_1 - 1$; in particular the constraint is $f = t\xi$ when $f(v,w) = 0$. Set $A_t(y,z_1,\xi) = (\psi_t(\iota y,\xi)_{\mathbb{C}^2}, z_1)$. For $t \neq 0$, $\xi = f/t$ is recovered from the image, and reversing $\psi_t$ recovers $y$, so $A_t$ is an embedding on a common neighbourhood. The

map $\mathcal{N}_t \circ A_t$ has the smooth extension $(\psi_t(\iota y, \xi)_{\mathbb{C}^2}, z_1, \xi)$ at $t = 0$. Pull back $\lambda_{\mathrm{cone}}$ along this extension, then along $\Xi$, to obtain $\Gamma_t$; its initial curvature is $\Omega_{\mathrm{phys}}$ by the radial bridge identity. The target $vw \neq 0$ contains $A_0\Xi(K_1)$. For the stopping rule, use the three plateau control functions $4g(t,x)/\mu - 2$, $1 - t^2$ and $2(R + 1 - |x_{\mathrm{fibre}}|^2_{\mathrm{coord}}) - 1$ of (75), whose positivity implies $\chi = 1$, together with

$$A_* - |\zeta_1|^2 - \delta_b, \quad B_* - |\zeta_1|^2 - |\zeta_2|^2 - \delta_b, \quad R_1 - |y|^2,$$
$$|\nabla f(y_{\mathrm{base}})|^2 - m, \quad |(A_s\Xi(x))_v|^2, \quad |(A_s\Xi(x))_w|^2.$$

Here $\zeta$, $\delta_b$, $R_1$, $m$ are as above and $y = (\iota(\Xi(x)_{\mathrm{base}}), \Xi(x)_{\mathrm{normal}})$. These nine smooth control functions are positive at $s = 0$ on $K_1$. Use $\delta = \epsilon_N$ in Lemma 14.2. Extend $\Xi$ to the ambient affine coordinates by the same positive coefficient formulas. If $\widetilde{F}_s$ is the resulting smooth cone-coordinate map, the ambient covector used to compute the field is

$$\alpha_s(p) = \lambda_{\mathrm{cone}}(\widetilde{F}_s(p)) \circ D\widetilde{F}_s(p).$$

Its full Jacobian includes the dependence on both fibre coordinates. Use its complete cutoff flow in (77), and set $t = \tau_*$ from (78). Lemma 14.2 gives the assertion. □

14.4. **The quadric tube.** Let $K_2 = g_1(K_1)$ and use $P_{t,\delta}$ of Lemma 13.5. Define

$$\lambda^{\mathrm{out}}_{t,\delta} = [P_{t,\delta}(z, 1, v, w)]^* \lambda_{1/(L\pi)}, \qquad \Omega^{\mathrm{out}}_{t,\delta} = d\lambda^{\mathrm{out}}_{t,\delta}.$$

Again the zero-weight constant coordinate is one, so the primitive is smooth everywhere in this chart, including where other coefficients vanish.

**Proposition 14.4** (The quadric tube map)**.** *There are an explicit $\delta > 0$ and a symplectic embedding $g_2$ of a neighbourhood of $K_2$ into $\{z \neq 0\}$, with $g_2^*\Omega^{\mathrm{out}}_{t,\delta} = \Omega^{\mathrm{cub}}_t$.*

*Proof.* Put $A_\delta = T_\delta$. Lemma 13.5 supplies a smooth extension $\Gamma_\delta = T_\delta^* \lambda^{\mathrm{out}}_{t,\delta}$ at zero, and its value at zero is $\lambda^{\mathrm{cub}}_t$. For $\delta > 0$, $T_\delta$ is an affine shear with invertible last coordinate, and $T_0(v, w, z)$ has last coordinate $vw$, nonzero on $K_2$. The initial form is nondegenerate there because $g_1$ is a symplectic neighbourhood embedding. Lemma 14.2, with target $\{z \neq 0\}$, gives the assertion. □

Change from the chart $X_1 = 1$ to $X_0 = 1$ by

$$J(v, w, z) = (z^{-1}, v/z, w/z).$$

It is a diffeomorphism on $z \neq 0$. The compact set $K' = Jg_2(K_2)$ avoids the origin. Set $\rho = \min_{x \in K'} \max_j |x_j| > 0$, $e(x) = \sum_{j=0}^{2} |x_j|^2$, and take the disjoint open sets $T_{\mathrm{out}} = \{e(x) > \rho^2/4\}$ and $T_{\mathrm{in}} = \{e(x) < \rho^2/4\}$. If the smaller family is empty, omit these steps.

14.5. **The polynomial bubble.** Let $A < r_0 < a$, and take $L = (Q + 1)\,\mathrm{den}(a)\,\mathrm{den}(e)\,\mathrm{den}(d_2)$ in Lemma 13.2 with $Q = \lfloor 3/(a - r_0) \rfloor + 1$, so that $L > Q$. Set

$$m_0 = \lfloor r_0 L \rfloor, \quad M_1 = m_0 + 1, \quad M_2 = m_0 + 2, \quad M_3 = m_0 + 3, \qquad d(\epsilon) = (1 - \epsilon)/L.$$

Then $r_0 < M_1/L$ and $M_1 < M_2 < L - S_*$, $M_2 < M_3$. Let $V_j$ be the ordered-monomial Veronese map, with coordinates indexed by words of length $j$ and $\|V_j(X)\|^2 = \|X\|^{2j}$. For $X = (X_0, X')$ put $B_\epsilon X = (\epsilon X_0, X')$ and define the degree-$L$ polynomial vector

$$\mathcal{P}_\epsilon(X) = \Big( P_{t,\delta}(X),\ \epsilon^{M_2 - M_1} X_0^{L - M_1} V_{M_1}(B_\epsilon X),\ \epsilon^{M_3} V_L(X) \Big). \tag{79}$$

For $\epsilon \neq 0$ the last block, hence $\mathcal{P}_\epsilon$, has no nontrivial zero. In the distinguished affine chart define

$$\omega_\epsilon = \epsilon\, dd^c \log(1 + |x|^2) + d(\epsilon)\, dd^c \log \|\mathcal{P}_\epsilon(1, x)\|^2. \tag{80}$$

It is Kähler for $0 < \epsilon < 1$.

**Proposition 14.5** (Separated insertion of the large ball)**.** *There is an explicit $\epsilon \in (0, 1/2)$ such that the smaller ball maps deform to symplectic neighbourhood embeddings for $\omega_\epsilon$ with compact images in $T_{\mathrm{out}}$, and $B(r_0)$ has a symplectic neighbourhood embedding for the same form with image in $T_{\mathrm{in}}$.*

*Proof.* On the outer compact set, (80) has a smooth primitive family at $\epsilon = 0$ with initial curvature the outer polynomial form, since $P_{t,\delta}$ is nonzero in the $X_1 = 1$ chart and both added blocks vanish at zero. The uniform compact Moser transfer, with $A_\epsilon$ the identity, uses the two target control functions

$$|z|^2, \qquad 1 + |v|^2 + |w|^2 - (\rho^2/4)|z|^2.$$

They express $z \neq 0$ and $J(v, w, z) \in T_{\mathrm{out}}$ without division. Equations (76)–(78) give a threshold $\tau > 0$ and the smaller maps in $T_{\mathrm{out}}$ for every $0 < \epsilon \leq \tau$.

For the central chart put $x = \epsilon y$ and divide $\mathcal{P}_\epsilon(1, \epsilon y)$ by $\epsilon^{M_2}$. Every entry of $P_{t,\delta}$ vanishes to order at least $L - S_* > M_2$ at $o$, so its divided block extends polynomially and vanishes at zero. The middle block becomes exactly $V_{M_1}(1, y)$, while the final block vanishes because $M_3 > M_2$. So the vector is nonzero at zero for every $y$, and its Hopf primitive extends smoothly in $\epsilon$. The first term of (80), pulled back by $y \mapsto \epsilon y$, also extends smoothly and vanishes at zero. The limiting central form is

$$\frac{M_1}{L}\, dd^c \log(1 + |y|^2).$$

Its affine chart contains a symplectic image of $B(r_0)$, since $r_0 < M_1/L$. Apply the uniform compact transfer to this compact image, with $A_\epsilon(y) = \epsilon y$ and the target control function $\rho^2/4 - e(\epsilon y) > 0$; it gives a threshold $\sigma > 0$ by the same finite stopping rule. Set $\epsilon = \min(\tau, \sigma)/2$, taking both initial time bounds at most $1/2$. The two target sets are disjoint, so the images are disjoint. □

14.6. **A common map to the ball.**

**Lemma 14.6** (Polynomial target form)**.** *Let $P : \mathbb{C}^{n+1} \to \mathbb{C}^q$ be a homogeneous polynomial map of degree $L$ with no nontrivial zero. If $0 \leq d$ and $Ld < 1$, there is an explicit smooth symplectic embedding of*

$$\big(\mathbb{C}^n, (1 - Ld)dd^c \log(1 + |x|^2) + d\, dd^c \log \|P(1, x)\|^2\big) \quad \textit{into } (B^\circ(1), \omega_0).$$

*Proof.* Put $c = 1/\pi$ and, on $\mathbb{C}^{n+1} \setminus 0$, $h_c(X) = c\,\mathrm{Im}\langle X, dX\rangle/(2\|X\|^2) = d^c \log \|X\|^2$; thus $\|X\|^2 h_c = c\lambda_0$, with $\lambda_0$ the standard Liouville form. Fix the plateaux $\chi = S^{\mathrm{plateau}}_{1/4,\,3/8,\,3,\,4}$ and $\psi = S^{\mathrm{plateau}}_{-2,\,-1,\,1,\,2}$, and put

$$\Gamma_s(X) = \|X\|^2\big[(1 - Lsd\,\chi(\|X\|^2))\, h_c(X) + sd\,\chi(\|X\|^2)\, P^* h_c(X)\big], \qquad \Omega_s = d\Gamma_s,$$

extended by $c\lambda_0$ near 0, where $\chi = 0$. Outside the shell $\frac{1}{4} \leq \|X\|^2 \leq 4$, $\Omega_s = c\,\omega_0$.

*Nondegeneracy.* All data are invariant under $X \mapsto e^{i\theta}X$, with generator $iX$. Since $h_c(iX) = c/2$ and $P^* h_c(iX) = h_c(iLP) = cL/2$, we get $\Gamma_s(iX) = c\|X\|^2/2$ for all $s$, so $\iota_{iX}\Omega_s = -d(c\|X\|^2/2)$, i.e. $\Omega_s(iX, V) = -c\,\mathrm{Re}\langle X, V\rangle$. On the complex orthogonal complement $E_X$ of $\mathbb{C}X$, $d\|X\|^2 = 0$, so $\Omega_s|_{E_X} = \|X\|^2\big[(1 - Lsd\chi)\, dh_c + sd\chi\, P^* dh_c\big]|_{E_X}$, which is positive for $s \in [0, 1]$ because $dh_c$ restricts to a positive form on $E_X$, $P^* dh_c \geq 0$ and $1 - Lsd\chi \geq 1 - Ld > 0$. If $\iota_V \Omega_s = 0$ with $V = aX + b\, iX + \xi$, $\xi \in E_X$, then pairing with $iX$ gives $a = 0$, pairing with $E_X$ gives $\xi = 0$, and $\Omega_s(iX, X) = -c\|X\|^2 \neq 0$ gives $b = 0$. So $\Omega_s$ is symplectic on $\mathbb{C}^{n+1}$ for $s \in [0, 1]$; the same argument shows that the complex hyperplane $E = \{X_0 = 0\}$, which contains $\mathbb{C}X$ for $X \in E$, is a symplectic subspace.

*Relative Moser field.* Put $\alpha = \Gamma_1 - \Gamma_0$, supported in the shell, with $\alpha(iX) = 0$. For $b \in \mathbb{C}^n$ let $\mathsf{e}(b) = (0, b)$, and let $\mathsf{N}_t(b, u) = (u, \mathsf{Y}_t(b, u))$ be the unique vector with $X_0$-component $u \in \mathbb{C}$

that is $\Omega_{S(t)}$-orthogonal to $E$ at $\mathsf{e}(b)$; it is real linear in $u$ and smooth in $(t,b)$, because $E^\Omega$ is a real two-dimensional complement of $E$ projecting isomorphically to the $X_0$-coordinate. Put $\mathrm{pr}(X)=X'$ and

$$h_t(X)=-\psi(|X_0|^2)\,\alpha_{\mathsf{e}(\mathrm{pr}\,X)}\big(\mathsf{N}_t(\mathrm{pr}\,X,X_0)\big),\qquad \beta_t=\alpha+dh_t,$$

$\iota_{W_t}\Omega_{S(t)}=-\beta_t$ and $V_t=S'(t)W_t$. Then $h_t$ vanishes on $E$ and is real linear in $X_0$ near $E$, so at $\mathsf{e}(b)$, $dh_t(\mathsf{N}_t(b,u))=-\alpha(\mathsf{N}_t(b,u))$: $\beta_t$ annihilates $E^{\Omega_{S(t)}}$, hence $W_t\in(E^\Omega)^\Omega=E$ along $E$. The function $h_t$ is circle invariant (all data are) and $\beta_t(iX)=\alpha(iX)+dh_t(iX)=0$, so $\Omega_{S(t)}(iX,W_t)=-\beta_t(iX)=0$ gives $\mathrm{Re}\langle X,V_t\rangle=0$. The field $V_t$ is smooth, vanishes for $t\notin[0,1]$, and has compact support in $\{X:\|X\|^2\le 6\}$. Indeed, $\alpha$ is supported in the shell $\frac14\le\|X\|^2\le 4$, whereas the support of $h_t$ is contained in $\{|X_0|^2\le 2,\ \|X'\|^2\le 4\}$, and differentiation does not enlarge support. Thus its flow $\phi_t$ is complete. It preserves every sphere $\|X\|=r$, the hyperplane $E$ and therefore $\{X_0\neq 0\}$, and commutes with the circle action. Since $\frac{d}{dt}\Omega_{S(t)}=S'(t)\,d\alpha=S'(t)\,d\beta_t$, Cartan's formula gives $\frac{d}{dt}\phi_t^*\Omega_{S(t)}=\phi_t^*d(\iota_{V_t}\Omega_{S(t)}+S'(t)\beta_t)=0$, so $\phi_1^*\Omega_1=\Omega_0$.

*Descent.* On the unit sphere $\chi=1$, and $\Omega_s$ restricted to the sphere is the pullback under the Hopf map of $(1-Lsd)\,dd^c\log\|X\|^2+sd\,dd^c\log\|P(X)\|^2$; in the chart $X_0=1$ this is $\omega_s:=(1-Lsd)dd^c\log(1+|x|^2)+sd\,dd^c\log\|P(1,x)\|^2$ (the difference of the potentials is $\log|X_0|^{2(1-Lsd)+2Lsd}=\log|X_0|^2$, which is pluriharmonic). Since $\phi_1$ preserves the unit sphere and $\{X_0\neq0\}$ and commutes with the circle action, it descends to a diffeomorphism $\bar\phi$ of $\mathbb{C}^n$ with $\bar\phi^*\omega_1=\omega_0=dd^c\log(1+|x|^2)$. Put $G=\bar\phi^{-1}$, so $G^*dd^c\log(1+|x|^2)=\omega_1$. The map $x\mapsto x/\sqrt{\pi(1+|x|^2)}$ is a diffeomorphism of $\mathbb{C}^n$ onto $B^\circ(1)$ pulling $\omega_0$ back to $dd^c\log(1+|x|^2)$ (Lemma A.18); its composite with $G$ is the required embedding. □

14.7. **Assembly.**

**Theorem 14.7** (One large ball, $n=3$)**.** *Let $A,\rho_1,\dots,\rho_k>0$ have valid auxiliary data $(a;r)$ with $a\in\mathbb{Q}$, and let $\rho_i<r_i'<r_i$ and $A<r_0<a$. Then there are smooth symplectic embeddings $f_0$ of an open neighbourhood of $B(r_0)$ and $f_i$ of open neighbourhoods of $B(r_i')$, $1\le i\le k$, into $B^\circ(1)$, given by the formulas below, such that the sets $f_0(B(r_0))$ and $f_i(B(r_i'))$ are pairwise disjoint and contained in $B(1-\zeta)$, where*

$$\zeta=\tfrac12\min\Big\{1-\pi|y|^2:\ y\in f_0(B(r_0))\cup\textstyle\bigcup_{i=1}^k f_i(B(r_i'))\Big\}>0.$$

*Proof.* Take $L$ and $S_*$ from Definition 11.6, so that $L>Q=\lfloor 3/(a-r_0)\rfloor+1$, and $m_0=\lfloor r_0L\rfloor$, $M_j=m_0+j$ as in §14.5; then $r_0<M_1/L$, $M_1<M_2<L-S_*$ and $M_2<M_3$. The vanishing order in Proposition 11.8 permits the bubble construction of Proposition 14.5 with the orders $M_1,M_2,M_3$, in which $\epsilon=\min(\tau,\sigma)/2$ for the two explicit thresholds $\tau,\sigma>0$ of that proposition. Its central chart contains a symplectic image of $B(r_0)$ and its outer chart contains the images of Proposition 14.1: starting with the maps $\phi_i$ of that proposition, apply in order $g_1$ (Proposition 14.3), $g_2$ (Proposition 14.4), the outer bubble deformation and $J$. The resulting maps and the large-ball map of Proposition 14.5 have pairwise disjoint compact images for the common degree-$L$ polynomial form (80), with coefficient $(1-\epsilon)/L$. Lemma 14.6 applies because $1-Ld(\epsilon)=\epsilon>0$, and its common embedding into $B^\circ(1)$ preserves all disjointness. The union of the images of $B(r_0)$ and the $B(r_i')$ is compact and lies in $B^\circ(1)$, so the minimum defining $\zeta$ is attained and positive, and $\pi|y|^2\le 1-2\zeta<1-\zeta$ on it. If there are no smaller balls, $f_0$ is the inclusion. □

## 15. Input names and elementary programs

The finite verifier reads the capacity and centre names and the derived radius names, and does integer arithmetic; none of the programs below evaluates the level-L1 construction.

**Definition 15.1** (Names)**.** Put $\varepsilon_p = 2^{-p}$. A rational stream $q$ *names* $x \in \mathbb{R}$ if $|q(p) - x| \le \varepsilon_p$ for every $p \in \mathbb{N}$. Rationals are coded by the natural triples of Definition 17.1, so a name is a function $\mathbb{N} \to \mathbb{N}$. A family $(x_c)_{c\in\mathbb{N}}$ is named by one stream $q(\langle c, p\rangle)$, where $\langle\cdot,\cdot\rangle$ is the pairing bijection $\mathbb{N}^2 \to \mathbb{N}$. A program *computes relative to* given names if it is a partial recursive function relative to those streams as oracles (`Nat.RecursiveIn`); an oracle-free program results when the streams are themselves computable.

**Lemma 15.2** (Strict certification)**.** *If $q, r$ name $x, y$, the rational test $q(p) + \varepsilon_p < r(p) - \varepsilon_p$ implies $x < y$, and if $x < y$ it holds for every $p$ with $4\varepsilon_p < y - x$. Hence, for a family of finite conjunctions of strict inequalities between named reals, given uniformly in the candidate index, a search over pairs (candidate, precision) halts if some conjunction is true, and every pair it accepts has a true conjunction.*

*Proof.* The first assertion follows from the two name bounds. If $4\varepsilon_p < y - x$, then $q(p) + \varepsilon_p \le x + 2\varepsilon_p < y - 2\varepsilon_p \le r(p) - \varepsilon_p$. One precision serves all inequalities of a true conjunction, so the search reaches an accepted pair. □

**Proposition 15.3** (Arithmetic on names)**.** *Sums, negatives, products, absolute values and maxima have uniform total name programs. If an integer $a \ge 0$ with $2^{-a} \le |x|$ is supplied, $1/x$ has one as well: at precision $p$ it inverts the approximation of $x$ at precision $p + 2a + 2$. Each program is total and makes finitely many oracle queries, at indices computed from $p$ and earlier answers.*

*Proof.* Addition reads both inputs at precision $p+1$, and negation changes sign. Absolute value and maximum are 1-Lipschitz and read at precision $p$. For multiplication, put $M_q = \lfloor |q(0)| \rfloor + 2 \ge |x|$, $M_r = \lfloor |r(0)| \rfloor + 2 \ge |y|$ and $K = M_q + M_r + 1$. The program returns $u'v'$ with $u' = q(p+K)$, $v' = r(p+K)$. As $|v'| \le M_r + 1$, $|xy - u'v'| \le |x|\,|y - v'| + |v'|\,|x - u'| \le K 2^{-(p+K)} \le 2^{-p}$. For the reciprocal, the queried $u$ satisfies $|u| \ge 2^{-a-1}$, so $|u^{-1} - x^{-1}| = |u - x|/|ux| \le 2^{2a+1} 2^{-(p+2a+2)} < 2^{-p}$. All rational operations are integer operations on codes. □

**Lemma 15.4** (Quadrature)**.** *Let $f$ be $L$-Lipschitz on $[0, 1]$ and let $N \ge 1$. Then the left Riemann sum $N^{-1} \sum_{j<N} f(j/N)$ differs from $\int_0^1 f$ by at most $L/N$. If every sample is replaced by a rational within $\theta$, the error is at most $L/N + \theta$.*

*Proof.* On $[j/N, (j+1)/N]$ we have $|f - f(j/N)| \le L/N$; summing over $j$ gives $L/N$. The sample errors average to at most $\theta$. □

**Lemma 15.5** (A name of $\pi$)**.** *The identity $\pi = \int_0^1 4/(1 + t^2)\, dt$ gives a computable rational name of $\pi$, and the lower bound $1 \le \pi$ gives a computable name of $1/\pi$.*

*Proof.* The identity holds because $4/(1+t^2) = (4 \arctan t)'$. On $[0, 1]$ the integrand is 8-Lipschitz, since $|4/(1 + x^2) - 4/(1 + y^2)| = 4|x - y|(x + y)/((1 + x^2)(1 + y^2))$. Its samples at rational points are exact rationals, so Lemma 15.4 with $\theta = 0$ and $N \ge 2^{p+3}$ gives error at most $2^{-p}$. Proposition 15.3 with $a = 0$ gives the reciprocal. □

**Lemma 15.6** (Square root)**.** *Let $q$ name $x \ge 0$. For $p \in \mathbb{N}$ put $S = 2^{p+2}$, $u = q(2p + 4)$, $v = \max(u, 0)$, $c = \lceil vS^2 \rceil \in \mathbb{N}$, and let the output be $\lfloor \sqrt{c} \rfloor / S$, where $\lfloor \sqrt{c} \rfloor$ is the integer square root. This is a name of $\sqrt{x}$.*

*Proof.* Since $x \geq 0$, $|v - x| \leq |u - x| \leq S^{-2}$, so $|\sqrt{v} - \sqrt{x}| \leq S^{-1}$. Put $w = \lfloor \sqrt{c} \rfloor / S$. From $vS^2 \leq c < vS^2 + 1$ we get $\sqrt{v} - S^{-1} < w \leq \sqrt{v + S^{-2}} \leq \sqrt{v} + S^{-1}$, hence $|w - \sqrt{x}| \leq 2S^{-1} = 2^{-p-1}$. □

**Corollary 15.7** (Radius names)**.** *Names of positive capacities $c_i$ and $C$ yield, by total programs with no further oracle, names of $r_i = \sqrt{c_i/\pi}$ and $R = \sqrt{C/\pi}$.*

*Proof.* Multiply each capacity name by the name of $1/\pi$ of Lemma 15.5 (Proposition 15.3) and apply Lemma 15.6. □

## 16. Finite scalar compilation

We compile $H$ satisfying $\mathrm{Pack}(n, k, C, c, s)$ into $H_{\mathrm{fin}} = \beta_B \cdot P$, with $B \in \mathbb{N}$, $\beta_B$ a product of flat cutoffs and $P \in \mathbb{Q}[t, z]$, and show that $\varphi^1_{H_{\mathrm{fin}}}$ still carries every closed source ball into the open target ball; this proves Lemma M5.1 and Theorem M5.2. A rounded tensor Bernstein polynomial approximates $H$ in $C^1$ on a box, the box cutoff globalises the estimate, and one Grönwall estimate bounds the displacement, which is paid from the gap to the target boundary.

*Notation.* $|v|$ is the Euclidean norm on $\mathbb{R}^d$, $|v|_\infty = \max_l |v_l|$ (so $|v| \leq \sqrt{d}\,|v|_\infty$), and $\|A\|$ is the operator norm. The real coordinates of $z \in \mathbb{R}^{2n}$ are $\zeta_1, \dots, \zeta_{2n}$. For $G\colon \mathbb{R} \times \mathbb{R}^{2n} \to \mathbb{R}$, we write $X_G = J_0 \nabla_z G$ and $\varphi^t_G$ for its flow from time 0 to $t$ (Paper I, § "Conventions"). Since $J_0$ is orthogonal, $|X_G - X_{G'}| = |\nabla_z G - \nabla_z G'|$.

Fix $n \geq 2$, $k \geq 1$, $C > 0$, $c \in (0, \infty)^k$, $s \in (\mathbb{R}^{2n})^k$ and $H$ satisfying $\mathrm{Pack}(n, k, C, c, s)$ (Definition M1.1). Thus $H$ is $C^\infty$, $H(t, z) = 0$ for $z$ outside a compact $K \subset \mathbb{R}^{2n}$, and $\varphi^1_H(B(s_i, c_i)) \subset B^\circ(C)$. A *compilation datum* for $H$ is a tuple (C1)–(C7) as in §M5.1, with numbers $C_0$, $R_*$, $R_C$, $a_{\mathrm{box}}$, $B$, $L_X$, $C_\beta$, $b_R$, $\eta$, $M_0$, $\delta$, $N_{\mathrm{B}}$, $\tilde{H}_m$.

### 16.1. The box cutoff and global Lipschitz bounds.

**Definition 16.1** (Box cutoff)**.** Let $S$ be the flat step (Definition A.1). For a real $B > 0$, $x \in \mathbb{R}$ and $z \in \mathbb{R}^{2n}$ put

$$\beta_{B,1}(x) = 1 - S\Big(\frac{x^2 - (B/2)^2}{(3B/4)^2 - (B/2)^2}\Big), \qquad \beta_B(z) = \prod_{l=1}^{2n} \beta_{B,1}(\zeta_l).$$

The compilation uses the integer $B$ of (C2).

**Lemma 16.2** (Support in the box)**.** *For every real $B > 0$, the function $\beta_B$ is $C^\infty$ on $\mathbb{R}^{2n}$, $0 \leq \beta_B \leq 1$, $\beta_B = 1$ on $[-B/2, B/2]^{2n}$, $\beta_B = 0$ on the closed set $\{|z|_\infty \geq 3B/4\} = \mathbb{R}^{2n} \setminus (-3B/4, 3B/4)^{2n}$, and $\nabla \beta_B = 0$ on the open set $\{|z|_\infty > 3B/4\}$. For the integer $B$ of (C2), $\beta_B(z) H(t, z) = H(t, z)$ for all $(t, z) \in \mathbb{R} \times \mathbb{R}^{2n}$.*

*Proof.* Use Lemma A.2 and $S = 0$ on $(-\infty, 0]$, $S = 1$ on $[1, \infty)$. The argument of $S$ in $\beta_{B,1}(x)$ is $\leq 0$ for $|x| \leq B/2$ and $\geq 1$ for $|x| \geq 3B/4$, which gives all claims factor by factor. If $\beta_B(z) \neq 1$, then $|z|_\infty > B/2 > a_{\mathrm{box}}$ by (C2), so $H(t, z) = 0$. □

**Lemma 16.3** (Global Lipschitz bounds)**.** *The suprema $\sup_{[0,1] \times \mathbb{R}^{2n}} \|D^2_z H\|$ and $\sup_{\mathbb{R}^{2n}} |\nabla \beta_B|$ are finite, so (C3) can be met. For $L_X, C_\beta$ as in (C3) and every $t \in [0, 1]$, the field $X_H(t, \cdot)$ is $L_X$-Lipschitz on $\mathbb{R}^{2n}$ and $|\nabla \beta_B| \leq C_\beta$.*

*Proof.* Both suprema are finite: $D^2_z H$ vanishes on $\mathbb{R} \times (\mathbb{R}^{2n} \setminus K)$ and is continuous on the compact $[0, 1] \times K$, and $\nabla \beta_B$ is continuous and vanishes outside $[-3B/4, 3B/4]^{2n}$ (Lemma 16.2). Since $\|D_z X_H\| = \|J_0 D^2_z H\| = \|D^2_z H\| < L_X$, the mean value inequality gives the Lipschitz bound. □

16.2. **Tensor Bernstein approximation.**

**Definition 16.4** (Tensor Bernstein operator)**.** For $N \ge 1$ and $0 \le j \le N$ let $b_{N,j}(x) = \binom{N}{j} x^j (1-x)^{N-j}$, and $b_{N,j} = 0$ for $j \notin \{0,\dots,N\}$. For $d \ge 1$, $m \in \{0,\dots,N\}^d$ and $x \in [0,1]^d$ put $W_m(x) = \prod_{\ell=1}^{d} b_{N,m_\ell}(x_\ell)$, and for $g\colon [0,1]^d \to \mathbb{R}$

$$\mathcal{B}_N g(x) = \sum_{m \in \{0,\dots,N\}^d} g(m/N)\, W_m(x).$$

The weights $W_m(x)$ are nonnegative and $\sum_m W_m(x) = \prod_\ell (x_\ell + 1 - x_\ell)^N = 1$.

**Lemma 16.5** (Mean absolute deviation)**.** *For $N \ge 1$ and $x \in [0,1]$,*

$$\sum_{j=0}^{N} b_{N,j}(x) \left| x - \frac{j}{N} \right| \le \sqrt{\frac{x(1-x)}{N}} \le \frac{1}{2\sqrt{N}}, \qquad \sum_{j=0}^{N-1} b_{N-1,j}(x)\, \bigl| j - (N-1)x \bigr| \le \frac{\sqrt{N}}{2}.$$

*Proof.* The $b_{M,j}(x)$ are binomial probabilities with mean $Mx$ and variance $Mx(1-x)$ [Lor86, §1.1], so Cauchy–Schwarz gives $\sum_j b_{M,j}(x)|j - Mx| \le \sqrt{Mx(1-x)}$. For the first bound take $M = N$, divide by $N$ and use $x(1-x) \le 1/4$; for the second take $M = N-1$. □

**Lemma 16.6** (Value error)**.** *Let $g\colon [0,1]^d \to \mathbb{R}$ and $M \ge 0$ satisfy $|g(y) - g(y')| \le M \sum_\ell |y_\ell - y'_\ell|$ for all $y, y' \in [0,1]^d$. Then $|\mathcal{B}_N g(x) - g(x)| \le dM/(2\sqrt{N})$ for all $N \ge 1$ and $x \in [0,1]^d$.*

*Proof.* As the weights sum to 1, $|\mathcal{B}_N g(x) - g(x)| \le M \sum_\ell \sum_m W_m(x) |m_\ell/N - x_\ell|$. Summing out the coordinates other than $\ell$ reduces each inner sum to the first sum of Lemma 16.5. □

**Proposition 16.7** (Tensor Bernstein estimates)**.** *Let $d \ge 1$, $N \ge 1$, and let $g$ be the restriction to $[0,1]^d$ of a $C^2$ function on a neighbourhood of $[0,1]^d$ whose first and second partial derivatives are bounded in absolute value by $M_0$ on $[0,1]^d$. Then for every $x \in [0,1]^d$ and every $j \in \{1,\dots,d\}$:*

*(a)* $|\mathcal{B}_N g(x) - g(x)| \le dM_0/(2\sqrt{N})$;
*(b)* $|\partial_j \mathcal{B}_N g(x) - \partial_j g(x)| \le M_0\bigl[d/(2\sqrt{N}) + 2/N\bigr]$;
*(c) if $(e_m)_{m \in \{0,\dots,N\}^d}$ are reals with $|e_m| \le a$ for all $m$, then $\bigl|\sum_m e_m W_m(x)\bigr| \le a$ and $\bigl|\partial_j \sum_m e_m W_m(x)\bigr| \le 2Na$.*

*Proof.* (a) By the mean value theorem, $|g(y) - g(y')| \le M_0 \sum_\ell |y_\ell - y'_\ell|$; apply Lemma 16.6.

(b) Use $b'_{N,i} = N(b_{N-1,i-1} - b_{N-1,i})$ and summation by parts [Lor86, §1.4]. For arbitrary $h_m$ and the unit vector $e_j$ this gives

(81)
$$\partial_j \sum_m h_m W_m(x) = N \sum_{m\colon m_j \le N-1} \bigl(h_{m+e_j} - h_m\bigr)\, W_m^{(j)}(x), \qquad W_m^{(j)}(x) = b_{N-1,m_j}(x_j) \prod_{\ell \ne j} b_{N,m_\ell}(x_\ell),$$

with weights $W_m^{(j)} \ge 0$ summing to 1. For $h_m = g(m/N)$, $N[g((m+e_j)/N) - g(m/N)] = \int_0^1 \partial_j g((m + s e_j)/N)\, ds$, and $|\partial_j g(y) - \partial_j g(x)| \le M_0 \sum_\ell |y_\ell - x_\ell|$. Hence $|\partial_j \mathcal{B}_N g - \partial_j g|$ is at most $M_0$ times the $W^{(j)}$-average of $\int_0^1 \bigl(\sum_{\ell \ne j} |m_\ell/N - x_\ell| + |(m_j + s)/N - x_j|\bigr) ds$.

Each term with $\ell \ne j$ contributes at most $1/(2\sqrt{N})$ (Lemma 16.5). Writing $\frac{m_j + s}{N} - x_j = \frac{m_j - (N-1)x_j}{N} + \frac{s - x_j}{N}$, the second bound of Lemma 16.5 bounds the last term by $1/(2\sqrt{N}) + 1/N$. This yields $M_0[d/(2\sqrt{N}) + 1/N]$, stronger than the stated bound.

(c) The weights are nonnegative with sum 1. For the derivative, by (81) the derivative is $N$ times a convex combination of the differences $e_{m+e_j} - e_m$, each of size at most $2a$. □

**Definition 16.8** (The finite Hamiltonian)**.** Let a compilation datum (C1)–(C7) be given, put $d = 2n + 1$ and $N = N_{\mathrm{B}}$, and let $\Phi_B \colon \mathbb{R}^d \to \mathbb{R}^d$,

$$\Phi_B(t, z) = \Big(t, \frac{\zeta_1 + B}{2B}, \dots, \frac{\zeta_{2n} + B}{2B}\Big),$$

which maps $[0, 1] \times [-B, B]^{2n}$ onto $[0, 1]^d$, with coordinates $\xi = (\xi_0, \dots, \xi_{2n})$; $\tilde{H} = H \circ \Phi_B^{-1}$ is the function of (C5). With the rationals $\tilde{H}_m$ of (C7) put

$$\hat{P}(\xi) = \sum_{m \in \{0,\dots,N\}^d} \tilde{H}_m \, W_m(\xi), \qquad P = \hat{P} \circ \Phi_B, \qquad H_{\mathrm{fin}}(t, z) = \beta_B(z) \, P(t, z).$$

This $P$ is the polynomial displayed after (C7) in §M5.1. Since $\tilde{H}_m \in \mathbb{Q}$ and $B \in \mathbb{N}$, $P \in \mathbb{Q}[t, z]$, of degree at most $N$ in each variable. Put $a_N = \max_m |\tilde{H}_m - \tilde{H}(m/N)|$; then $a_N < \delta/(2N + 1)$ by (C7).

**Lemma 16.9** (Polynomial error in physical coordinates)**.** *For every compilation datum, every $t \in [0, 1]$, every $z \in [-B, B]^{2n}$ and every $l \in \{1, \dots, 2n\}$,*

$$|P(t, z) - H(t, z)| < \delta, \qquad |\partial_{\zeta_l}(P - H)(t, z)| \le \frac{\delta}{B},$$

*and hence $|\nabla_z P(t, z) - \nabla_z H(t, z)| \le \sqrt{2n}\, \delta/B < \delta$.*

*Proof.* On $[0, 1]^d$ write $\hat{P} - \tilde{H} = (\mathcal{B}_N \tilde{H} - \tilde{H}) + \sum_m e_m W_m$ with $e_m = \tilde{H}_m - \tilde{H}(m/N)$, so $|e_m| \le a_N$. Here $\tilde{H}$ is the restriction of the $C^\infty$ function $H \circ \Phi_B^{-1}$, with $M_0$ as in (C5). By (C6), $dM_0/(2\sqrt{N}) < \delta/2$ and $2M_0/N < \delta/2$. Proposition 16.7 then gives $|\hat{P} - \tilde{H}| < \delta/2 + a_N < \delta$ and $|\partial_{\xi_j}(\hat{P} - \tilde{H})| < \delta + 2Na_N < 2\delta$ for $0 \le j \le 2n$.

Composing with $\Phi_B$ gives the value bound, and $\partial_{\zeta_l}(P - H) = (2B)^{-1}(\partial_{\xi_l}(\hat{P} - \tilde{H})) \circ \Phi_B$ gives $\delta/B$. Hence $|\nabla_z(P - H)| \le \sqrt{2n}\, \delta/B < \delta$, since $B > 2n \ge \sqrt{2n}$ by (C2). □

*Remark* 16.10 (Grid values). Condition (C7) asks only for rationals near the finitely many grid values; these exist by density of $\mathbb{Q}$. Compilation data are used only to prove that the search of Section 18 terminates.

## 16.3. **The global field error.**

**Lemma 16.11** (Product rule estimate)**.** *Let $E$ be a real normed space, $b, p, h, C', \varepsilon \in \mathbb{R}$ and $g_b, g_p, g_h \in E$ with $g_h = bg_h + hg_b$, $0 \le b \le 1$, $\|g_b\| \le C'$, $|p - h| < \varepsilon$ and $\|g_p - g_h\| < \varepsilon$. Then $\|(bg_p + pg_b) - g_h\| < \varepsilon(1 + C')$.*

*Proof.* $(bg_p + pg_b) - g_h = b(g_p - g_h) + (p - h)g_b$ has norm $< \varepsilon + \varepsilon C'$. □

**Lemma 16.12** (Global field error)**.** *For every compilation datum, $H_{\mathrm{fin}}$ is $C^\infty$ on $\mathbb{R} \times \mathbb{R}^{2n}$, $H_{\mathrm{fin}}(t, z) = 0$ for all $t$ whenever $z \notin (-3B/4, 3B/4)^{2n}$, and for all $t \in [0, 1]$ and $z \in \mathbb{R}^{2n}$*

$$|X_{H_{\mathrm{fin}}}(t, z) - X_H(t, z)| = |\nabla_z H_{\mathrm{fin}}(t, z) - \nabla_z H(t, z)| < \delta(1 + C_\beta) < \eta/16.$$

*Proof.* Smoothness and the support claim hold because $H_{\mathrm{fin}}$ is $\beta_B$ times a polynomial (Lemma 16.2). Fix $t \in [0, 1]$.

For $z \in [-B, B]^{2n}$, differentiating $\beta_B H = H$ gives $\nabla_z H_{\mathrm{fin}} - \nabla_z H = \beta_B \nabla_z(P - H) + (P - H)\nabla\beta_B$. Lemma 16.11 with $E = \mathbb{R}^{2n}$, $b = \beta_B$, $p = P$, $h = H$, $g_b = \nabla\beta_B$, $g_p = \nabla_z P$, $g_h = \nabla_z H$, $C' = C_\beta$, $\varepsilon = \delta$, whose hypotheses hold by Lemmas 16.2, 16.3 and 16.9, gives the bound $\delta(1 + C_\beta)$.

For $z \notin (-B, B)^{2n}$, $z$ lies in the open sets $\{|z|_\infty > 3B/4\}$, where $H_{\mathrm{fin}}(t, \cdot) = 0$, and $\{|z|_\infty > a_{\mathrm{box}}\}$, where $H(t, \cdot) = 0$ by (C2), so both gradients vanish. Finally, $\delta(1 + C_\beta) < \eta/16$ by (C6). □

16.4. **The Grönwall estimate.**

**Lemma 16.13** (Grönwall)**.** *Let $E$ be a finite-dimensional real normed space, $L > 0$, $\eta \geq 0$, and let $v, w\colon \mathbb{R}\times E \to E$ be continuous, with $v(t,\cdot)$ $L$-Lipschitz for every $t \in \mathbb{R}$ and $\|w(t,e)-v(t,e)\| \leq \eta$ for all $(t,e)$. If $x, y\colon [0,1] \to E$ are differentiable with $\dot{x}(t) = w(t,x(t))$, $\dot{y}(t) = v(t,y(t))$ for $t \in [0,1]$ (one-sided derivatives at the end points) and $x(0) = y(0)$, then*

$$\|x(1) - y(1)\| \leq \eta\,\frac{e^L - 1}{L}.$$

*If moreover $b > 0$ and $\eta = be^{-L}/2$, then $\eta(e^L-1)/L < b/2$.*

*Proof.* $u = x - y$ is $C^1$ with $u(0) = 0$ and $\|\dot{u}\| \leq \|w(s,x) - v(s,x)\| + \|v(s,x) - v(s,y)\| \leq \eta + L\|u\|$. With $\Omega(t) = \int_0^t (\eta + L\|u\|)\,ds$ we get $\|u\| \leq \Omega$, $\Omega' \leq \eta + L\Omega$, so $(e^{-Lt}\Omega)' \leq \eta e^{-Lt}$ and $\Omega(t) \leq \eta(e^{Lt}-1)/L$; take $t = 1$. For $\eta = be^{-L}/2$, $\eta(e^L-1)/L = b(1-e^{-L})/(2L) < b/2$ as $1 - e^{-L} < L$. □

*Proof of Lemma M5.1.* With $\pi_{[0,1]}(t) = \min(1, \max(0,t))$, put $v(t,e) = X_H(\pi_{[0,1]}(t), e)$, $w(t,e) = X_G(\pi_{[0,1]}(t), e)$. These are continuous, satisfy the hypotheses of Lemma 16.13 for all $t$ by (i), (ii) (with $E = \mathbb{R}^{2n}$, $L = L_X$), and agree with $X_H, X_G$ on $[0,1]$; apply that lemma with $x = \gamma_G$, $y = \gamma_H$. □

**Lemma 16.14** (Flow error of the compiled Hamiltonian)**.** *Let a compilation datum be given and let $G$ be $H$ or $H_{\mathrm{fin}}$.*

- *(a) For every $(t_0, z_0) \in \mathbb{R}\times\mathbb{R}^{2n}$ there is an integral curve $\gamma\colon \mathbb{R} \to \mathbb{R}^{2n}$ of $X_G$ with $\gamma(t_0) = z_0$, and every integral curve of $X_G$ defined on an interval $I \ni t_0$ with value $z_0$ at $t_0$ is the restriction of $\gamma$ to $I$. In particular $\varphi_G^t$ is defined on all of $\mathbb{R}^{2n}$ for every $t$, and $\varphi_G^1$ is injective.*
- *(b) For every $z \in \mathbb{R}^{2n}$,*

$$|\varphi^1_{H_{\mathrm{fin}}}(z) - \varphi^1_H(z)| \leq \eta\,\frac{e^{L_X} - 1}{L_X} < b_R.$$

*Proof.* (a) $X_G = J_0\nabla_z G$ is $C^\infty$ and vanishes outside a $t$-independent compact $K_G$ ($K_H = K$, $K_{H_{\mathrm{fin}}} = [-3B/4, 3B/4]^{2n}$ by Lemma 16.12), so on each $[T_1,T_2]\times\mathbb{R}^{2n}$ it and its spatial derivative are bounded. Picard–Lindelöf gives a unique maximal integral curve through each $(t_0,z_0)$, of which every integral curve through $(t_0,z_0)$ is a restriction; bounded velocity rules out a finite end point, so it is defined on $\mathbb{R}$. If $\varphi^1_G(z) = \varphi^1_G(z')$, the two curves meet at $t = 1$, hence coincide, so $z = z'$.

(b) Define $v, w$ as in the proof of Lemma M5.1 with $G = H_{\mathrm{fin}}$. Then $v(t,\cdot)$ is $L_X$-Lipschitz (Lemma 16.3) and $|w - v| \leq \eta/16 \leq \eta$ (Lemma 16.12); apply Lemma 16.13 to $x = \varphi^t_{H_{\mathrm{fin}}}(z)$, $y = \varphi^t_H(z)$ with $L = L_X$ and $\eta = b_R e^{-L_X}/2$ (C4). □

16.5. **The finite output.**

**Theorem 16.15** (Finite output)**.** *Let $H$ satisfy* $\mathrm{Pack}(n,k,C,c,s)$*. For every compilation datum (C1)–(C7) for $H$, the function $H_{\mathrm{fin}} = \beta_B \cdot P$ of Definition 16.8 is $C^\infty$, $H_{\mathrm{fin}}(t,z) = 0$ for all $t$ whenever $z \notin (-3B/4, 3B/4)^{2n}$, and $|X_{H_{\mathrm{fin}}} - X_H| < \eta$ on $[0,1]\times\mathbb{R}^{2n}$. For every $z \in \bigcup_i B(s_i,c_i)$,*

$$|\varphi^1_{H_{\mathrm{fin}}}(z) - \varphi^1_H(z)| < b_R, \qquad |\varphi^1_{H_{\mathrm{fin}}}(z)| < R_* + b_R < R_C.$$

*In particular $H_{\mathrm{fin}}$ satisfies* $\mathrm{Pack}(n,k,C,c,s)$*, and if the balls $B(s_i,c_i)$ are pairwise disjoint, so are their images under $\varphi^1_{H_{\mathrm{fin}}}$. This is Theorem M5.2.*

*Proof.* Smoothness, support (condition (a) of Definition M1.1) and the field error are Lemma 16.12. For $z \in B(s_i, c_i)$, (C1) gives $|\varphi^1_H(z)| \le R_*$, so by Lemma 16.14(b) and (C4), $|\varphi^1_{H_{\mathrm{fin}}}(z)| < R_* + b_R < R_C$, i.e. $\pi|\varphi^1_{H_{\mathrm{fin}}}(z)|^2 < C$. By Lemma 16.14(a) every integral curve $\gamma\colon [0,1] \to \mathbb{R}^{2n}$ of $X_{H_{\mathrm{fin}}}$ with $\gamma(0) = z$ has $\gamma(1) = \varphi^1_{H_{\mathrm{fin}}}(z) \in B^\circ(C)$, which is condition (b) of Definition M1.1; injectivity of $\varphi^1_{H_{\mathrm{fin}}}$ gives disjointness of the images. □

## 17. Finite trajectory certificates

The verifier uses only the input names, symbolic differentiation of a rational polynomial and exact rational Euler iteration; the geometry enters only through Theorem 17.7. Put $d = 2n$; $\|\cdot\|_\infty$ is the maximum norm and $|\cdot|$ the Euclidean norm, so $\|x\|_\infty \le |x| \le \sqrt{d}\,\|x\|_\infty \le d\|x\|_\infty$.

**Definition 17.1** (Polynomial programs)**.** A polynomial program is a finite expression formed from rational constants, the variables $x_0 = t, x_1 = z_1, \dots, x_d = z_d$, addition and multiplication; a variable $x_j$ with $j > d$ denotes zero. A rational constant is coded by a triple $(a, b, c) \in \mathbb{N}^3$, with value $(a - b)/(c + 1)$. Every rational polynomial in $(t, z)$ is $P_e$ for some program $e$. For an integer $B \ge 1$ the *budget* of a program is the integer

$$h_B(a,b,c) = a + b + 2, \qquad h_B(x_j) = B + 2,$$
$$h_B(e + f) = h_B(e) + h_B(f), \qquad h_B(ef) = 4h_B(e)h_B(f).$$

Write $P_e$ for the polynomial of $e$ and $\Theta = h_B(e)$. Symbolic differentiation $\partial_j e$ is defined by $\partial_j x_i = \delta_{ij}$, $\partial_j(\text{constant}) = 0$ and the sum and product rules.

**Lemma 17.2** (Field bounds)**.** *Let* $\mathrm{clip}(t, z) \in [0,1] \times [-B/2, B/2]^d$ *be the point obtained by clipping* $t$ *to* $[0,1]$ *and each* $z_l$ *to* $[-B/2, B/2]$*, and put*

$$F_e(t, z) = J_0 \nabla_z P_e\big(\mathrm{clip}(t, z)\big).$$

*Then* $\|F_e\|_\infty \le \Theta$, $\|F_e(t,z) - F_e(t,z')\|_\infty \le \Theta\|z - z'\|_\infty$ *and* $\|F_e(t,z) - F_e(t',z)\|_\infty \le \Theta|t - t'|$. *Through every point* $F_e$ *has a unique integral curve on* $[0,1]$*. At rational arguments* $F_e$ *is evaluated exactly by rational arithmetic on the programs* $\partial_j e$*.*

*Proof.* Let $X = [-B, B]^{d+1}$ with the maximum norm. Define integers $\mathrm{bd}(e)$ and $\mathrm{lip}(e)$ recursively: $\mathrm{bd} = a + b + 1$ and $\mathrm{lip} = 0$ for a constant $(a, b, c)$; $\mathrm{bd} = B$, $\mathrm{lip} = 1$ for a variable; both are additive under sums; and for products $\mathrm{bd}(ef) = \mathrm{bd}(e)\,\mathrm{bd}(f)$, $\mathrm{lip}(ef) = \mathrm{bd}(e)\,\mathrm{lip}(f) + \mathrm{bd}(f)\,\mathrm{lip}(e)$. By induction, $|P_e| \le \mathrm{bd}(e)$ on $X$ and $P_e$ is $\mathrm{lip}(e)$-Lipschitz on $X$; for products use $P_eP_f(x) - P_eP_f(y) = P_e(x)(P_f(x) - P_f(y)) + P_f(y)(P_e(x) - P_e(y))$. A second induction shows that $h_B(e)$ bounds the four numbers $\mathrm{bd}(e)$, $\mathrm{lip}(e)$, $\mathrm{bd}(\partial_j e)$ and $\mathrm{lip}(\partial_j e)$ for every $j$. Constants and variables are immediate (note $\mathrm{bd}(\partial_j x_i) \le 2$). Sums are additive. For a product, $\partial_j(ef) = (\partial_j e)f + e\,\partial_j f$, so each of the four numbers for $ef$ is a sum of at most four products of a number for $e$ and a number for $f$, each at most $h_B(e)h_B(f)$.

Symbolic differentiation agrees with the partial derivative by the sum and product rules, so the components of $F_e$ are $\pm P_{\partial_j e} \circ \mathrm{clip}$. Clipping maps into $X$ (as $B \ge 1$) and is 1-Lipschitz in each coordinate. Hence the speed bound follows from $\mathrm{bd}(\partial_j e) \le \Theta$, and both Lipschitz bounds from $\mathrm{lip}(\partial_j e) \le \Theta$. The field $F_e$ is continuous and globally Lipschitz in $z$, uniformly in $t$, so the Picard–Lindelöf theorem gives a unique integral curve on $[0,1]$ through every point. □

**Lemma 17.3** (Euler tubes)**.** *Let* $N \ge 1$, $a \in \mathbb{Q}^d$*, and form the exact rational table*

$$x_0 = a, \qquad x_{j+1} = x_j + N^{-1}F_e(j/N, x_j) \quad (0 \le j < N).$$

*Let* $\gamma$ *be an integral curve of* $F_e$ *on* $[0,1]$ *with* $\|\gamma(0) - a\|_\infty \le \rho$*, and put*

$$E = \left(\rho + \frac{\Theta + \Theta^2}{N}\right) 4^\Theta + \frac{\Theta}{N}. \tag{82}$$

*Then* $\|\gamma(j/N) - x_j\|_\infty \le E$ *for* $0 \le j \le N$, *and for every* $t \in [0,1]$ *some* $j \le N$ *satisfies* $\|\gamma(t) - x_j\|_\infty \le E$.

*Proof.* With $t_j = j/N$ and $e_j = \|\gamma(t_j) - x_j\|_\infty$, the bounds of Lemma 17.2 give $\|F_e(s,\gamma(s)) - F_e(t_j,x_j)\|_\infty \le (\Theta+\Theta^2)(s-t_j) + \Theta e_j$ on $[t_j,t_{j+1}]$, so $e_{j+1} \le (1+\Theta/N)e_j + (\Theta+\Theta^2)/N^2$. As $e_0 \le \rho$ and $(1+\Theta/N)^j \le e^\Theta \le 4^\Theta$, the node bound follows; between nodes $\gamma$ moves by at most $\Theta/N$. □

Let $q_{i,l}(p)$, $q_i^r(p)$, $q^R(p)$ be names, with error at most $2^{-p}$, of the coordinates of $s_i$, of $r_i = \sqrt{c_i/\pi}$ and of $R = \sqrt{C/\pi}$ (radius names from capacity names by Corollary 15.7). Put

$$s_{i,l}^p = q_{i,l}(p), \qquad r_{i,p}^+ = q_i^r(p) + (d+2)2^{-p}, \qquad R_p^- = \max(0, q^R(p) - 2^{-p}). \tag{83}$$

Since $|s_i^p - s_i| \le \sqrt{d}\,2^{-p}$ and $r_i \le q_i^r(p) + 2^{-p}$, $\overline{B}(s_i^p, r_{i,p}^+) \supseteq B(s_i,c_i)$, and $0 \le R_p^- \le R$.

**Definition 17.4** (Finite checks)**.** A candidate is $(B,e,Q,p,m,N)$ with integers $B,Q,N \ge 1$, $p,m \ge 0$ and a polynomial program $e$. It passes if the following finitely many strict rational inequalities hold. First the source-box tests

$$|s_{i,l}^p| + r_{i,p}^+ < Q \qquad (1 \le i \le k,\ 1 \le l \le d).$$

Next, let the grid nodes be the $(2^m+1)^d$ points $-Q + 2Q\nu/2^m$, $\nu \in \{0,\dots,2^m\}^d$; every point of $[-Q,Q]^d$ lies within maximum-norm distance $\rho = 2Q2^{-m}$ of a node. For every node $a$ and every source $i$, either the discard test

$$\sum_l (a_l - s_{i,l}^p)^2 > (r_{i,p}^+ + d\rho)^2 \tag{84}$$

holds, or the Euler table of $a$ (Lemma 17.3, with $E$ computed from this $\rho$) satisfies

$$|x_{j,l}| + E < B/2 \qquad (0 \le j \le N,\ 1 \le l \le d), \tag{85}$$

$$R_p^- - dE > 0, \qquad \sum_l x_{N,l}^2 < (R_p^- - dE)^2. \tag{86}$$

**Theorem 17.5** (Soundness)**.** *If a candidate passes,* $\beta_B P_e$ *satisfies* $\mathrm{Pack}(n,k,C,c,s)$.

*Proof.* By Lemma 16.2, $\beta_B P_e$ is $C^\infty$ and vanishes for $z \notin (-3B/4, 3B/4)^d$ (condition (a) of Definition M1.1); its integral curves exist on $\mathbb{R}$ and are unique as in Lemma 16.14(a).

Fix $y \in B(s_i,c_i)$. Then $|y - s_i^p| \le r_{i,p}^+$, so the source-box tests give $|y_l| < Q$; some node $a$ has $\|a - y\|_\infty \le \rho$, whence $|a - s_i^p| \le r_{i,p}^+ + d\rho$, (84) fails, and (85), (86) hold for $a$. Let $\gamma$ be the integral curve of $F_e$ with $\gamma(0) = y$. By Lemma 17.3 and (85), $|\gamma_l(t)| < B/2$, where clipping is the identity and $\beta_B \equiv 1$ nearby; so $F_e = X_{\beta_B P_e}$ along $\gamma$. By (86), $|\gamma(1)| \le |x_N| + dE < R_p^- \le \sqrt{C/\pi}$, so $\gamma(1) \in B^\circ(C)$; uniqueness gives condition (b). □

**Lemma 17.6** (Retained nodes)**.** *If a node* $a$ *is not discarded for source* $i$, *its maximum-norm distance from* $B(s_i,c_i)$ *is at most*

$$\delta_p = (2d+3)2^{-p} + d\rho.$$

*Proof.* Failure of (84), $|s_i^p - s_i| \le d2^{-p}$ and $r_{i,p}^+ \le r_i + (d+3)2^{-p}$ give $|a - s_i| \le r_i + \delta_p$; radial projection to $\overline{B}(s_i,r_i)$ moves $a$ by at most $\delta_p$ in either norm. □

**Theorem 17.7** (Nonemptiness)**.** *Let* $(n,k,C,c,s)$ *satisfy the hypotheses of Theorem M1, and let names of the coordinates of the* $s_i$ *and of the radii* $r_i$, $R$ *be given. Then some candidate passes the checks of Definition 17.4.*

*Proof. A compiled packing with margins.* $H = H[n,k,C,c,s]$ of Theorem M1 satisfies $\mathrm{Pack}(n,k,C,c,s)$; by compactness $\bigcup_i \phi_H^1(B(s_i,c_i)) \subset B^\circ(C_0)$ for some $C_0 \in (0,C)$; with $C_1 = (C_0+C)/2$, $H$ satisfies $\mathrm{Pack}(n,k,C_1,c,s)$ with (C1) for $C_0$. Let $\mathcal{T} = \{\phi_H^t(y) : t \in [0,1],\ y \in \bigcup_i B(s_i,c_i)\}$ (compact). As each condition is strict, choose a compilation datum (C1)–(C7) for $H$ with target $C_1$, $b_R \le 1$ and $B > 4 + 2\max_{\mathcal{T}} \|\cdot\|_\infty$; let $H_{\mathrm{fin}} = \beta_B P$ and $P_e = P$. By the proof of Lemma 16.13, compiled and exact curves from the same point stay within distance $\eta(e^{L_X}-1)/L_X < b_R \le 1$, so each compiled curve $\gamma_y$ from a source point satisfies $\|\gamma_y(t)\|_\infty < B/2 - 1$, is an integral curve of $F_e$, and by Theorem M5.2 $|\gamma_y(1)| < \sqrt{C_1/\pi} < R$.

*A source neighbourhood.* With $\Theta = h_B(e)$, Lemma 17.2 and Grönwall give $\|\gamma_a(t) - \gamma_y(t)\|_\infty \le e^\Theta \|a-y\|_\infty$, so $\mu = \frac{1}{2}\big(R - \max_y |\gamma_y(1)|\big) > 0$ (maximum over the sources). Choose $\nu > 0$ with $e^\Theta \nu \le \frac{1}{2}$ and $de^\Theta \nu \le \mu$. If $\|a-y\|_\infty \le \nu$ for a source point $y$, then

$$|\gamma_{a,l}(t)| < B/2 - \tfrac{1}{2} \quad (t \in [0,1],\ 1 \le l \le d), \qquad |\gamma_a(1)| \le R - \mu. \tag{87}$$

*The finite parameters.* As $|s_{i,l}^p| \le |s_{i,l}| + 1$ and $r_{i,p}^+ \le r_i + d + 3$, any integer $Q > \max_{i,l}(|s_{i,l}| + r_i) + d + 4$ passes the source-box tests. Since $\delta_p \to 0$ as $p,m \to \infty$ and $E \to 0$ as $m,N \to \infty$, choose $p,m,N$ with $\delta_p \le \nu$, $E < \frac{1}{8}$, $dE < \frac{\mu}{8}$, $2^{1-p} < \frac{\mu}{4}$. By Lemma 17.6, every non-discarded node $a$ satisfies (87). Lemma 17.3 with initial error 0 gives $|x_{j,l}| + E \le |\gamma_{a,l}(j/N)| + 2E < B/2$ and, as $R_p^- \ge R - 2^{1-p}$, $|x_N| + dE \le |\gamma_a(1)| + 2dE < R - \mu + \frac{\mu}{4} < R_p^-$; hence $R_p^- - dE > |x_N| \ge 0$ and (86) holds. So $(B,e,Q,p,m,N)$ passes. □

## 18. The algorithm and its termination

We fix the encoding for the verifier of Section 17 and prove that the machine, with the dimension and the number of balls as runtime inputs, halts.

### 18.1. **Input and output.**

**Definition 18.1** (Input)**.** An input consists of integers $n \ge 2$, $k \ge 1$ and names (Definition 15.1) of $C, c_1, \dots, c_k$ and of the $2nk$ coordinates of $s_1, \dots, s_k$. They are supplied as two oracle streams: the capacity stream, whose value at $\langle i,p\rangle$ is the precision-$p$ approximation of $c_i$ for $1 \le i \le k$ and of $C$ for $i = 0$ and for $i > k$; and the centre stream, whose value at $\langle\langle i,p\rangle, l\rangle$ is the precision-$p$ approximation of the $l$-th coordinate of the $(i+1)$-st centre. A rational input is given by its constant name. The promises, hypotheses of the termination theorem that are not tested, are that $(n,k,C,c)$ is admissible and the closed balls $B(s_i,c_i)$ are pairwise disjoint (Theorem M1).

The output is a natural number $\langle B, \ulcorner e \urcorner\rangle$, where $B \ge 1$ and $e$ is a polynomial program (Definition 17.1). It denotes $H_{\mathrm{fin}} = \beta_B P_e$; expanding the sums and products of $e$ gives the rational coefficients of $P_e$.

An Epoch input is an integer $k \ge 9$ and a name of $\varepsilon \in (0,1)$. It stands for

$$n = 2, \quad R = \left(\frac{k}{1-\varepsilon/2}\right)^{1/4}, \quad C = \pi R^2, \quad c_i = \pi, \quad s_i = (3i-3,0,0,0). \tag{88}$$

### 18.2. **Finite encoding.**

Tuples and lists of naturals are coded by iterated pairing, and rationals by the triple code of Definition 17.1. The rational operations ($+$, $\times$, negation, $|\cdot|$, max, inverse of a nonzero rational, strict comparison by cross multiplication with positive denominators) are integer operations on codes, without reduction to lowest terms. A polynomial program is a list of instructions $\langle \tau, v\rangle$ read right to left with a stack: tag 0 pushes the constant with code $v$, tag 1 pushes $x_v$, tag 2 replaces the top two entries by their sum, and other tags by their product. A missing entry counts as 0, and the value is the final top (0 if the stack is empty). This reading is total, every expression has a code, and the same recursion computes $h_B(e)$ and $\partial_j e$. A vector of dimension $d$ is a list of $d$ rational codes, and any other code reads as 0. Evaluating

$F_e$ at a rational point clips the arguments and interprets $\partial_j e$. An Euler step is one evaluation plus $d$ rational operations, and the table is primitive recursive in $N$. The grid node with index $g < (2^m+1)^d$ has coordinates $-Q + 2Q\nu_l/2^m$, where the $\nu_l$ are the base-$(2^m+1)$ digits of $g$. Table 1 lists the data of the verifier.

The radius names $r_i = \sqrt{c_i/\pi}$ and $R = \sqrt{C/\pi}$ come from the capacity stream by Corollary 15.7. For an Epoch input, the source radius name is the constant 1 and the centres are rational constants. The name of $R$ is obtained from that of $\varepsilon$ by forming $1 - \varepsilon/2$ (Proposition 15.3), inverting it using $1 - \varepsilon/2 > 2^{-1}$, multiplying by $k$, and applying Lemma 15.6 twice. No names of $C$ or $c_i$ are formed.

| Object | Defining rule | Effective data | Reference |
|---|---|---|---|
| Input enclosures | (83) | error $2^{-p}$; outer source radius, inner target radius | Def. 17.4 |
| Polynomial field | $F_e = J_0 \nabla P_e$ at clipped arguments | exact rational interpreter; speed and both Lipschitz bounds $\Theta = h_B(e)$ | Lem. 17.2 |
| Spatial grid | $-Q + 2Q\nu/2^m$ | $(2^m+1)^d$ nodes; covering radius $\rho = 2Q2^{-m}$ | Def. 17.4 |
| Euler table | $x_{j+1} = x_j + F_e(j/N, x_j)/N$ | tube radius $E$ of (82) on $[0,1]$ | Lem. 17.3 |
| Tests | plateau and terminal tests | margins $B/2 - \lvert x_{j,l}\rvert - E$ and $R_p^- - \lvert x_N\rvert - dE$ | Thm. 17.5 |
| Existence | smaller-target packing and Bernstein compilation | neighbourhood radius $\nu$, plateau margin $\frac{1}{2}$, terminal margin $\mu$ | Thm. 17.7 |

TABLE 1. Data of the finite verifier. Every quantity used by a test is rational.

18.3. **The machine.**

**Construction 18.2** (Algorithm)**.** On input $(n,k)$ and the oracle streams, put $d = 2n$. For $\mathrm{st} = 0, 1, 2, \ldots$ decode st as a candidate $(p, B, Q, m, N, e)$ and compute its failure count $\Phi(n,k,\mathrm{st}) \in \mathbb{N}$, a sum of failure indicators: one for the condition $B, Q, N \geq 1$, one for each source-box test, and one for each node-source pair that passes neither the discard test nor both trajectory tests of Definition 17.4, all evaluated with the names at precision $p$. So $\Phi = 0$ exactly when the candidate passes. Stop at the first st with $\Phi(n,k,\mathrm{st}) = 0$ and output $\langle B, \ulcorner e \urcorner \rangle$. The Epoch machine is the same search with $n = 2$ and the names described above; its runtime input is $k$ and its only oracle is the name of $\varepsilon$.

**Lemma 18.3** (Totality of the checker)**.** *For fixed oracle streams, $(n,k,\mathrm{st}) \mapsto \Phi(n,k,\mathrm{st})$ is a total function, recursive in the oracles. Each value uses finitely many oracle queries.*

*Proof.* Decoding is primitive recursive. Radius names are total with finitely many queries per value (Corollary 15.7), and centre approximations are single queries. The checker performs $kd$ source-box tests and, for each of the $k(2^m+1)^d$ node-source pairs, a discard test. If the discard test fails, it also runs an $N$-step Euler table with $N+1$ plateau rows and one terminal test. The quantities $\Theta, \rho, E$ are rational in the candidate, all loop bounds are computed from $(n,k,\mathrm{st})$, and interpretation is total. Hence $\Phi$ arises from the oracles by composition and bounded recursion. □

**Theorem 18.4** (Termination and correctness)**.** *For every input satisfying the promises of Definition 18.1, Construction 18.2 halts, and its output $\langle B, \ulcorner e \urcorner \rangle$ satisfies: $\beta_B P_e$ is $C^\infty$, vanishes for $z \notin (-3B/4, 3B/4)^{2n}$, and satisfies* $\mathrm{Pack}(n,k,C,c,s)$*. The machine is partial recursive relative to the oracle streams, uniformly in $(n,k)$. For an Epoch input it halts with $\beta_B P_e$ satisfying* $\mathrm{Pack}(2,k,\pi R^2,\pi,s)$ *for the data* (88)*, and $k/R^4 = 1 - \varepsilon/2 > 1 - \varepsilon$.*

*Proof.* The machine is an unbounded minimisation of the total oracle-recursive $\Phi$ (Lemma 18.3) followed by a primitive recursive output map, hence partial recursive in the oracles, uniformly in $(n, k)$. By Theorem 17.7 a passing candidate exists, and since decoding inverts encoding its code $\mathrm{st}_0$ satisfies $\Phi(n, k, \mathrm{st}_0) = 0$. So the search halts at some $\mathrm{st} \le \mathrm{st}_0$, and the output is correct by Theorem 17.5.

For an Epoch input, the data (88) are admissible with disjoint closed source balls by the proof of Corollary M5. The machine's names give the centres, the source radius $\sqrt{\pi/\pi} = 1$ and the target radius $R$, so the argument applies with $n = 2$; finally $R^4 = k/(1 - \varepsilon/2)$ and $\varepsilon > 0$. □

*Remark* 18.5 (Epoch form). The same verifier serves the Epoch input and all four cases of the construction (Table M3): the cases enter only the proof that a passing certificate exists.

## Appendix A. Statements quoted from Paper I

This appendix reproduces, with their hypotheses, the statements of Paper I [Son26] used in the proofs of this supplement; each names the statement it repeats, and the proofs are in Paper I. The notation is that of the cited sections of Paper I.

### A.1. **Conventions and analytic lemmas.**

**Definition A.1** (Flat step; Paper I)**.** Let $a_\flat(x) = 0$ for $x \le 0$ and $a_\flat(x) = e^{-1/x}$ for $x > 0$, and

$$S(x) = \frac{a_\flat(x)}{a_\flat(x) + a_\flat(1 - x)} \qquad (x \in \mathbb{R}).$$

**Lemma A.2** (Properties of $S$; Paper I)**.** *$S$ is $C^\infty$ on $\mathbb{R}$, takes values in $[0, 1]$, is nondecreasing on $\mathbb{R}$ and strictly increasing on $[0, 1]$, and $S(1 - x) = 1 - S(x)$. For every $j \ge 1$, $S^{(j)} = 0$ on $(-\infty, 0] \cup [1, \infty)$; in particular $S^{(j)}(0) = S^{(j)}(1) = 0$.*

**Definition A.3** (Radial and interval cutoffs; Paper I)**.** For $0 \le r < r'$ and $u$ in a Euclidean space $\mathbb{R}^d$ put

$$\chi_{r,r'}(u) = 1 - S\Big(\frac{|u|^2 - r^2}{r'^2 - r^2}\Big).$$

For $\ell_0 < \ell_1$ and $0 < w < (\ell_1 - \ell_0)/2$ put $\chi^{\mathrm{I}}_{\ell_0,\ell_1,w}(x) = S\big((x - \ell_0)/w\big)\, S\big((\ell_1 - x)/w\big)$ for $x \in \mathbb{R}$.

**Lemma A.4** (Moser lemma; Paper I)**.** *Let $M$ be a smooth manifold, $U \subset M$ open and $I \subset \mathbb{R}$ an open interval containing $[0, 1]$. Let $\Omega_t$ and $\mathfrak{v}_t$, $t \in I$, be two-forms and one-forms on $U$ depending smoothly on $(t, x) \in I \times U$, such that each $\Omega_t$ is closed and nondegenerate at every point of $U$ and*

$$\partial_t \Omega_t = d\mathfrak{v}_t.$$

*Let $Y_t$ be the vector field on $U$ defined by $\iota_{Y_t}\Omega_t = -\mathfrak{v}_t$, and let $\Psi_t$ be its flow from time 0 ($\Psi_0 = \mathrm{id}$), where defined.*

- *(a) $Y$ is smooth on $I \times U$, and $Y_t(x) = 0$ whenever $\mathfrak{v}_t(x) = 0$ as an element of $T^*_x M$.*
- *(b) If $\Psi_t(x)$ is defined for $t \in [0, 1]$, then $\Psi_t$ is defined and smooth on a neighbourhood of $x$ and $(\Psi_t^*\Omega_t)_x = (\Omega_0)_x$ for $t \in [0, 1]$.*
- *(c) If $Y_t(x) = 0$ for all $t \in [0, 1]$, then $\Psi_t(x) = x$ for $t \in [0, 1]$. In particular $\Psi_t$ fixes pointwise every set at whose points $\mathfrak{v}_t$ is the zero covector for all $t \in [0, 1]$.*
- *(d) (Tangency.) Let $U' \subset U$ be open, $\mu \in C^\infty(U')$ and $V$ a vector field on $U'$ with $\iota_V\Omega_t = -d\mu$ and $\mathfrak{v}_t(V) = 0$ on $U'$ for $t \in [0, 1]$. Then $d\mu(Y_t) = 0$ on $U'$, and $\mu$ is constant along every integral curve of $Y$ on a subinterval of $[0, 1]$ as long as the curve stays in $U'$.*

(e) *(Completeness.) Let $\mathcal{K} \subset U$ be compact, and suppose that every integral curve $\gamma\colon J \to U$ of $Y$, $J \subset [0,1]$ an interval, which meets $\mathcal{K}$ satisfies $\gamma(J) \subset \mathcal{K}$. Then $\Psi_t$ is defined on $\mathcal{K}$ for every $t \in [0,1]$, $\Psi_t(\mathcal{K}) = \mathcal{K}$, $\Psi_t|_{\mathcal{K}}$ is the restriction of a diffeomorphism between open neighbourhoods of $\mathcal{K}$, and $\Psi_t^*\Omega_t = \Omega_0$ at every point of $\mathcal{K}$.*

(f) *(Reparametrisation and concatenation.) If $\varsigma\colon I \to I$ is smooth with $\varsigma(0) = 0$, the pair $(\Omega_{\varsigma(t)}, \varsigma'(t)\mathfrak{v}_{\varsigma(t)})$ satisfies the hypotheses, its field is $\varsigma'(t)Y_{\varsigma(t)}$, and its flow is $\Psi_{\varsigma(t)}$. Consequently, let $(\Omega_s^{(j)}, \mathfrak{v}_s^{(j)})$, $1 \le j \le N$, be pairs on $U$, smooth on $[0,1] \times U$ (up to the boundary), with every $\Omega_s^{(j)}$ closed and nondegenerate on $U$, $\partial_s \Omega_s^{(j)} = d\mathfrak{v}_s^{(j)}$, and $\Omega_1^{(j)} = \Omega_0^{(j+1)}$. With the clocks $\sigma_j$ of Paper I, Lemma (Flat concatenation) put $(\Omega_t, \mathfrak{v}_t) = (\Omega^{(j)}_{\sigma_j(t)}, \sigma_j'(t)\mathfrak{v}^{(j)}_{\sigma_j(t)})$ on $[t_{j-1}, t_j]$, $(\Omega_t, \mathfrak{v}_t) = (\Omega_0^{(1)}, 0)$ for $t \le 0$ and $(\Omega_t, \mathfrak{v}_t) = (\Omega_1^{(N)}, 0)$ for $t \ge 1$. Then $(\Omega_t, \mathfrak{v}_t)_{t\in\mathbb{R}}$ satisfies the hypotheses with $I = \mathbb{R}$, and its time-one map is $\Psi_1 = \Psi_1^{(N)} \circ \cdots \circ \Psi_1^{(1)}$ wherever the right side is defined.*

(g) *(Equivariance.) If a diffeomorphism $g$ of $U$ satisfies $g^*\Omega_t = \Omega_t$ and $g^*\mathfrak{v}_t = \mathfrak{v}_t$ for all $t$, then $g_*Y_t = Y_t$ and $g \circ \Psi_t = \Psi_t \circ g$ wherever defined.*

**Lemma A.5** (The affine Fubini–Study chart; Paper I)**.** *On $\mathbb{C}^N$, put $D = 1 + |z|^2$ and*

$$\lambda_{\mathrm{FS}} = \frac{1}{2\pi D}\sum_{j=1}^{N}(x_j\,dy_j - y_j\,dx_j), \qquad \omega_{\mathrm{FS}} = d\lambda_{\mathrm{FS}}.$$

*Then:*

(a) *$\lambda_{\mathrm{FS}} = d^c \log D$ is smooth. Where every coordinate is nonzero it equals $\sum_j p_j^{\mathrm{FS}}\,d\theta_j$, with $p_j^{\mathrm{FS}} = |z_j|^2/D$ and $\theta_j = \arg z_j/(2\pi)$; its differential is $\sum_j dp_j^{\mathrm{FS}} \wedge d\theta_j$ there.*

(b) *The map $\mathrm{ch}(z) = z/\sqrt{\pi D}$ is a smooth bijection of $\mathbb{C}^N$ onto $B^\circ(1)$ with $\mathrm{ch}^*\omega_0 = \omega_{\mathrm{FS}}$, with smooth inverse $w \mapsto \sqrt{\pi}\,w/\sqrt{1-\pi|w|^2}$.*

(c) *$\mathrm{ch}^*\lambda_{\mathrm{std}} = \lambda_{\mathrm{FS}}$, where $\lambda_{\mathrm{std}} = \frac{1}{2}\sum_j(x_j\,dy_j - y_j\,dx_j)$, and $\pi|\mathrm{ch}(z)|^2 = 1 - D^{-1} < 1$.*

A.2. **The coordinate rearrangement.**

**Lemma A.6** (Increasing rearrangement on an interval; Paper I)**.** *Let $L \ge 0$, let $f\colon [0,L] \to \mathbb{R}$ be continuous and convex, and for $0 \le x \le L$ put*

$$f^\sharp(x) = \min_{0\le a\le L-x} \max\bigl(f(a), f(a+x)\bigr).$$

(a) *For $b \in \mathbb{R}$ and $0 \le x \le L$: $f^\sharp(x) \le b$ if and only if $\{f \le b\} \ne \emptyset$ and $\mathcal{L}_f(b) \ge x$. Hence $f^\sharp(x) = \min\{b \ge \min f : \mathcal{L}_f(b) \ge x\}$, the increasing rearrangement of $f$.*

(b) *$\mathcal{L}_{f^\sharp} = \mathcal{L}_f$; in particular $\int_0^L f^\sharp = \int_0^L f$.*

(c) *$f^\sharp$ is nondecreasing, $f^\sharp(0) = \min f$ and $f^\sharp(L) = \max f$.*

A.3. **The planar machine.**

**Definition A.7** (Isotopy data; Paper I)**.** Let $(U, V)$ be an admissible potential pair over $\mathcal{Y}$ with disc area $h$, and consider positive numbers

$$p_{\mathrm{on}},\ p_{\mathrm{zero}},\ p^\dagger,\ p^*_{\mathrm{lo}},\ p^*_{\mathrm{hi}},\ E_c,\ \Delta P,\ \Delta\theta,\ R_q,\ \delta_x,\ \delta_\theta,\ \delta_s,\ \bar{E},\ \bar{X},\ \bar{Y},\ \lambda_0,\ P_{\min},\ P_{\max},\ B_{\mathrm{lin}},\ B^*,\ c_{\mathrm{sh}},$$

where the radial range bounds satisfy

$$P_{\min} \le \tfrac{1}{2}\min(p^*_{\mathrm{lo}}, \Delta P), \qquad P_{\max} \ge \max(\tfrac{3}{2}p^*_{\mathrm{hi}}, h).$$

From these numbers we form the *derived constants*

$$M_C = \frac{1}{2\sqrt{\pi P_{\min}}} + 2\sqrt{\pi P_{\max}},$$

$$R^\dagger = \sqrt{p^\dagger/\pi}, \quad R_h = \sqrt{h/\pi}, \quad r_b^+ = \sqrt{p^*_{\rm hi}/\pi},$$

$$R_{\rm supp} = \max\big(R^\dagger,\ r_b^+ + c_{\rm sh}B_{\rm lin},\ r_b^+ + c_{\rm sh}B^*,\ R_q\big), \quad P_{\rm supp} = \pi R_{\rm supp}^2, \quad p_{\rm bd} = \tfrac{1}{2}(h + P_{\rm supp}).$$

The numbers are *isotopy data* for the pair if, for every $y \in \mathcal{Y}$:

(D1) $p_{\rm on} < p_{\rm zero} < p^\dagger$ and $p^*_{\rm lo} \le p^*(y) \le p^*_{\rm hi} < h$;

(D2) (endpoint margins) for all $u, v$ with $u^2 < E_c$, $v^2 < E_c$:

$$\Delta P \le p^*(y) + X_y(u) \le h - \Delta P, \qquad p^*(y) + X_y(u) \le \pi R_q^2 < h, \qquad |Y_y(v)| \le \tfrac{1}{2} - \Delta\theta;$$

(D3) (interpolation error) $|a_0(y)u - X_y(u)| \le \delta_x$ and $|b_0(y)v - Y_y(v)| \le \delta_\theta$ for $u^2 < E_c$, $v^2 < E_c$;

(D4) (late range) $\delta_s \le \frac{1}{4}$, $\delta_s\delta_x \le \frac{1}{2}\Delta P$, $\delta_s\delta_\theta \le \frac{1}{2}\Delta\theta$ and either $(1-\delta_s)^2\mathcal{A}_y(E_c) > p^\dagger$, or the uniform-energy alternative $\mathcal{A}_{y,s}(E_c) > p^\dagger$ for all $s \in [0,1]$; the slice construction uses the latter with $E_c = \bar{E}$;

(D5) (uniform energy) $\mathcal{A}_{y,s}(\bar{E}) \ge p^\dagger$ for all $s \in [0,1]$, and $\max(a_0(y)|u|, |X_y(u)|) \le \bar{X}$, $\max(b_0(y)|v|, |Y_y(v)|) \le \bar{Y}$ for $u^2 \le \bar{E}$, $v^2 \le \bar{E}$;

(D6) (early range) $\lambda_0 \le \frac{1}{2}$, $\lambda_0\bar{X} \le \frac{1}{2}p^*_{\rm lo}$ and $\lambda_0\bar{Y} \le \frac{1}{4}$;

(D7) (ranges of the path)

$$R^\dagger \max\big(1, |(\mathbf{A}_y)_{11}|, |(\mathbf{A}_y)_{22}|\big) \le B_{\rm lin}, \qquad B^* \ge \min\Big(M_C\sqrt{\bar{X}^2 + \bar{Y}^2},\ \frac{\bar{X}}{\sqrt{\pi p^*_{\rm lo}}} + 2\pi r_b^+\bar{Y}\Big),$$

$$c_{\rm sh} \le \min\Big(\frac{1}{4},\ \frac{R_h - r_b^+}{8\max(R^\dagger, B_{\rm lin}, B^*)}\Big).$$

The schedule $\lambda(s)$ of Paper I, Definition (Density family, rescaled charts and schedule)(c) is formed with these $\lambda_0$ and $\delta_s$.

**Lemma A.8** (Existence and order of the data; Paper I)**.** *(a) The equal-ball potential family has the explicit isotopy data given in Paper I, § "The data of the whole-disc isotopy".*

*(b) For any isotopy data: $0 < \delta_s \le \frac{1}{4}$, $p^\dagger < \mathcal{A}_y(E_c) < h$, $E_c \le E_{\rm rect}(y)$ and $p^\dagger < I_{\rm rect}(y)$ for every $y$, and*

$$R_{\rm supp} < R_h, \qquad i.e. \qquad p_{\rm on} < p^\dagger \le P_{\rm supp} < p_{\rm bd} < h.$$

For the convex pair of this paper, Definition 3.3 gives isotopy data (Lemma 3.4).

**Lemma A.9** (Morse inverse charts; Paper I)**.** *Let $(U, V)$ be an admissible potential pair over $\mathcal{Y}$.*

*(i) The minimum point $y \mapsto p^*(y)$ and $y \mapsto U_{\min}(y)$ are $C^\infty$ on $\mathcal{Y}$.*

*(ii) On the open set $\mathcal{D}_U = \{(y,x) : y \in \mathcal{Y},\ p^*(y) + x \in I_U\}$ put*

$$\Sigma_U(y,x) = \operatorname{sign}(x)\sqrt{U_y(p^*(y)+x) - U_{\min}(y)}.$$

*This is the signed square root, with its removable singularity at $x = 0$; and $\Sigma_U$ is $C^\infty$ with $\partial_x\Sigma_U > 0$. For each $y$, $\Sigma_U(y,\cdot)$ is an increasing diffeomorphism of $I_U - p^*(y)$ onto $\mathbb{R}$; its inverse $X_y$ is $C^\infty$ jointly in $(y,u) \in \mathcal{Y} \times \mathbb{R}$, and*

$$U_y\big(p^*(y) + X_y(u)\big) - U_{\min}(y) = u^2,$$

$$\operatorname{sign} X_y(u) = \operatorname{sign} u, \qquad X_y'(0) = a_0(y) := \sqrt{2/U_y''(p^*(y))}.$$

*(iii)* *In the same way* $\Sigma_V(y,\theta)=\operatorname{sign}(\theta)\sqrt{V_y(\theta)}$ *is an increasing diffeomorphism of* $I_V$ *onto* $\mathbb{R}$ *for each* $y$*, and its inverse* $Y_y$ *is* $C^\infty$ *in* $(y,v)$*, with* $V_y(Y_y(v))=v^2$*,* $\operatorname{sign} Y_y(v)=\operatorname{sign} v$ *and* $Y_y'(0)=b_0(y):=\sqrt{2/V_y''(0)}$.

*(iv)* *The Morse chart* $F_y(u,v)=(X_y(u),Y_y(v))$ *is a* $C^\infty$ *diffeomorphism of* $\mathbb{R}^2$ *onto the Morse rectangle* $\mathcal{R}_y=(I_U-p^*(y))\times I_V$*, jointly smooth in* $(y,w)$*, with*

$$F_y^*(dx\wedge d\theta)=J_y\,du\wedge dv,\qquad J_y(u,v)=X_y'(u)\,Y_y'(v)>0,\qquad J_y(0)=j_0(y):=a_0(y)b_0(y),$$

*and* $U_y(p^*(y)+x)-U_{\min}(y)+V_y(\theta)=u^2+v^2$ *for* $(x,\theta)=F_y(u,v)$*. Consequently, for* $E\ge 0$*,*

$$\int_{|w|^2\le E} J_y(w)\,dw=\operatorname{area}\big\{(p,\theta)\in I_U\times I_V:\ U_y(p)-U_{\min}(y)+V_y(\theta)\le E\big\},$$

*and these areas increase to* $|I_U|\,|I_V|\in(0,\infty]$ *as* $E\to\infty$.

*(v)* *If the pair is exactly quadratic near the minimum, then* $X_y(u)=a_0(y)u$ *for* $|u|\le\varepsilon_U/a_0(y)$ *and* $Y_y(v)=b_0(y)v$ *for* $|v|\le\varepsilon_V/b_0(y)$*; hence* $J_y=j_0(y)$ *on the disc* $|w|\le\rho_T(y)$*,* $\rho_T(y)=\min(\varepsilon_U/a_0(y),\varepsilon_V/b_0(y))$.

**Lemma A.10** (The density family and the uniform energy bound; Paper I)**.** *Let* $(U,V)$ *be an admissible potential pair over* $\mathcal{Y}$*,* $y\in\mathcal{Y}$*,* $s\in[0,1]$*, and abbreviate* $a_0=a_0(y)$*,* $b_0=b_0(y)$*,* $j_0=a_0b_0$.

*(a)* $X_{y,s},Y_{y,s}$ *are* $C^\infty$ *and strictly increasing, and*

$$J_{y,s}=(1-s)^2j_0+s(1-s)\big(a_0Y_y'+b_0X_y'\big)+s^2J_y\ \ge\ (1-s)^2j_0+s^2J_y>0,\qquad J_{y,s}(0)=j_0.$$

*(b)* *For* $E\ge 0$*,*

$$\mathcal{A}_{y,s}(E)\ \ge\ \pi j_0(1-s)^2E+s^2\mathcal{A}_y(E)\ \ge\ \mathcal{A}_y^-(E):=\frac{\pi j_0E\,\mathcal{A}_y(E)}{\pi j_0E+\mathcal{A}_y(E)}\quad(E>0).$$

*In particular* $\mathcal{A}_{y,s}(\infty)=\infty$ *for* $s<1$*, and* $\mathcal{A}_{y,1}(\infty)=\mathcal{A}_y(\infty)$.

*(c)* *(Uniform energy bound.)* $\mathcal{A}_y^-$ *is continuous and strictly increasing. If* $\bar E>0$ *and* $\mathcal{A}_{y,s}(\bar E)\ge p^\dagger$ *for every* $s\in[0,1]$ *(in both families this holds for the explicit* $\bar E$ *of Paper I, Lemma (Bounds for the instance)(d), respectively Definition 3.3), then* $E_{y,s}(I)\le\bar E$ *for* $0\le I\le p^\dagger$ *and* $s\in[0,1]$.

*(d)* $f_{y,s}$ *is a symplectic embedding on the source domain* $\mathcal{D}_y^{\mathrm{src}}$ *of Theorem A.12; the map* $(y,s,z)\mapsto f_{y,s}(z)$ *is smooth for* $s\in[0,1]$ *and* $z\in\mathcal{D}_y^{\mathrm{src}}$*. In the equal-ball case it is a diffeomorphism onto* $\mathbb{R}^2$*;* $f_{y,s}(0)=0$ *and* $Df_{y,s}(0)=\mathbf{L}_y$ *for every* $s$*;* $f_{y,0}=\mathbf{L}_y$ *and* $f_{y,1}=f_y$*. If* $(u,v)=T_{y,s}^{-1}D_{y,s}(z)$*, then* $f_{y,s}(z)=(X_{y,s}(u),Y_{y,s}(v))$ *and* $u^2+v^2=E_{y,s}(\pi|z|^2)$.

*(e)* *If the pair is exactly quadratic near the minimum, then* $J_{y,s}=j_0$ *on* $|w|\le\rho_T(y)$ *and* $f_{y,s}(z)=\mathbf{L}_yz$ *for* $|z|\le\sqrt{j_0}\,\rho_T(y)$*, for every* $s\in[0,1]$.

*The polar chart* (Paper I, § "Area-preserving disc maps from Morse charts"). For $y\in\mathcal{Y}$ let

$$C_y(x,\theta)=-\sqrt{\frac{p^*(y)+x}{\pi}}\,e^{2\pi i\theta},\qquad(x,\theta)\in\mathcal{C}_y=\{p^*(y)+x>0,\ |\theta|<\tfrac12\}.\tag{89}$$

It is injective and symplectic, $C_y^*(dp\wedge d\vartheta)=dx\wedge d\theta$, with $\pi|C_y(x,\theta)|^2=p^*(y)+x$; its image is $\mathbb{C}$ minus the closed positive real axis, and $C_y(0)=(-r_b(y),0)$ with $r_b(y)=\sqrt{p^*(y)/\pi}$. A computation gives $DC_y(0)=\operatorname{diag}\big(-1/(2\pi r_b(y)),\,-2\pi r_b(y)\big)$.

**Definition A.11** (The inner map; Paper I)**.** Let $(U,V)$ be an admissible potential pair over $\mathcal{Y}$ and $y\in\mathcal{Y}$. Apply the formulas of Paper I, Definition (Rotation and area normalisation) to the density $J_y=X_y'Y_y'$ of Lemma A.9 (parameters $\sigma=y\in\mathcal{P}=\mathcal{Y}$), and write $T_y$, $D_y$, $\mathcal{A}_y$, $E_y$ for

its normalisations. In the slice case the claims about these maps are made on the disc $D(p^\dagger)$ of Definition 3.3 and are proved with the constant-tail extension of Paper I, § "Area-preserving disc maps from Morse charts". Put

$$f_y = F_y \circ T_y^{-1} \circ D_y \colon D(\mathcal{A}_y(\infty)) \to \mathcal{R}_y.$$

Let $E_{\mathrm{rect}}(y) \in (0,\infty]$ be the infimum of $U_y(p) - U_{\min}(y) + V_y(\theta)$ over $(I_U \times I_V) \setminus \big((0,h) \times (-\frac{1}{2},\frac{1}{2})\big)$ (and $+\infty$ if this set is empty), let $I_{\mathrm{rect}}(y) = \mathcal{A}_y(E_{\mathrm{rect}}(y))$ (with $\mathcal{A}_y(+\infty) = \mathcal{A}_y(\infty)$), and define the *inner map*

$$q_{\mathrm{in},y} = C_y \circ f_y \colon D\big(I_{\mathrm{rect}}(y)\big) \to \mathbb{C}.$$

We also put $c_x(y) = \sqrt{a_0(y)/b_0(y)}$ and $\mathbf{L}_y = \mathrm{diag}(c_x(y), c_x(y)^{-1})$.

With $\mathbf{A}_y = DC_y(0)\,\mathbf{L}_y$ (Theorem A.12),

$$\mathbf{A}_y = \mathrm{diag}\Big(-\frac{c_x(y)}{2\pi r_b(y)}, -\frac{2\pi r_b(y)}{c_x(y)}\Big), \qquad \det \mathbf{A}_y = 1. \tag{90}$$

**Theorem A.12** (The inner map; Paper I)**.** *Let $(U,V)$ be an admissible potential pair. Put $\mathcal{D}_y^{\mathrm{src}} = \mathbb{C}$ for the equal-ball family and $\mathcal{D}_y^{\mathrm{src}} = D(p^\dagger)$, with $p^\dagger$ of Definition 3.3, for the slice family.*

*(a) For each $y$, $f_y$ is a symplectic embedding of $\mathcal{D}_y^{\mathrm{src}}$ into the Morse rectangle $\mathcal{R}_y$, and $(y,z) \mapsto f_y(z)$ is smooth on the corresponding source set. In the equal-ball case it is a diffeomorphism onto $\mathcal{R}_y = \mathbb{R}^2$. Moreover $f_y(0) = 0$ and $Df_y(0) = \mathbf{L}_y$, a diagonal matrix of determinant 1.*

*(b) (Energy identity.) If $f_y(z) = (x,\theta)$, then*

$$U_y\big(p^*(y) + x\big) + V_y(\theta) = U_{\min}(y) + E_y(\pi|z|^2),$$

*so $f_y$ maps each circle $\{\pi|z|^2 = I\}$ onto the level curve of $U_y(p^* + \cdot) - U_{\min} + V_y$ which encloses area $I$.*

*(c) The inner map $q_{\mathrm{in},y}$ is a symplectic embedding of $D(I_{\mathrm{rect}}(y)) \cap \mathcal{D}_y^{\mathrm{src}}$ into $D(h) \setminus [0,\infty)$, jointly smooth in $(y,z)$, with $q_{\mathrm{in},y}(0) = b_y$, $Dq_{\mathrm{in},y}(0) = \mathbf{A}_y := DC_y(0)\,\mathbf{L}_y$, and, for $f_y(z) = (x,\theta)$,*

$$\pi|q_{\mathrm{in},y}(z)|^2 = p^*(y) + x, \qquad U_y\big(\pi|q_{\mathrm{in},y}(z)|^2\big) + V_y(\theta) = U_{\min}(y) + E_y(\pi|z|^2).$$

*In particular $U_y(\pi|q_{\mathrm{in},y}(z)|^2) \le U_{\min}(y) + E_y(\pi|z|^2)$, with equality on the centre line $\theta = 0$.*

*(d) If the pair is exactly quadratic near the minimum, then $f_y(z) = \mathbf{L}_y z$ for $|z| \le \sqrt{j_0(y)}\,\rho_T(y)$.*

**Theorem A.13** (Whole-disc Hamiltonian isotopy with prescribed endpoint on the core; Paper I)**.** *Let $(U,V)$ be an admissible potential pair over $\mathcal{Y}$ with disc area $h$, and let isotopy data be given (Definition A.7). Let $\hat{q}$ and $Q$ be obtained from the five-segment path of Paper I, Construction (The five-segment path) by Paper I, Lemma (Normalisation of the whole path), with the levels $p_{\mathrm{on}} < p_{\mathrm{zero}}$ of the data, and Paper I, Construction (The Hamiltonian). Then $P_{\mathrm{supp}} < h$, and for every $y \in \mathcal{Y}$:*

*(a) each $\hat{q}_{y,t}$ is a symplectic diffeomorphism of $\mathbb{C}$ which is the identity on $\{\pi|z|^2 > P_{\mathrm{supp}}\}$ and maps $\bar{D}(P)$ onto itself for every $P \ge P_{\mathrm{supp}}$, in particular $\bar{D}(h)$ and $D(h)$;*

*(b) $\hat{q}_{y,t} = \mathrm{id}$ for $t \le 0$ and $\hat{q}_{y,t} = \hat{q}_{y,1}$ for $t \ge 1$;*

*(c) (same endpoint on the core) $\hat{q}_{y,1}(z) = q_{\mathrm{in},y}(z)$ for $\pi|z|^2 \le p_{\mathrm{on}}$;*

*(d) $Q_{y,t}$ vanishes with all derivatives on $\{\pi|x|^2 > P_{\mathrm{supp}}\}$ and for $t \notin (0,1)$, and generates $\hat{q}_{y,t}$: $\partial_t \hat{q}_{y,t} = \xi_{y,t} \circ \hat{q}_{y,t}$ with $\iota_{\xi_{y,t}} \omega_{\mathbb{C}} = -dQ_{y,t}$;*

*(e) (core formula)* $\hat q_{y,t}(z) = \mathfrak{e}_{y,t}(z/\sqrt{\mu_t})$ *for* $\pi|z|^2 \le \mu_t p_{\mathrm{on}}$.

**Theorem A.14** (Parametrised version; Paper I)**.** *In the situation of Theorem A.13:*

*(a) the maps* $(y,t,z)\mapsto \hat q_{y,t}(z)$, $(y,t,x)\mapsto \hat q_{y,t}^{-1}(x)$ *and* $(y,t,x)\mapsto Q_{y,t}(x)$ *are* $C^\infty$ *on* $\mathcal{Y}\times\mathbb{R}\times\mathbb{C}$, *and* $(y,s,z)\mapsto q^{\mathrm{in}}_{y,s}(z)$ *is smooth on* $\mathcal{Y}\times[0,1]\times D(p^\dagger)$; *the numbers* $p_{\mathrm{on}}$, $P_{\mathrm{supp}}$, $c_{\mathrm{sh}}$, *and hence the collar* $\{\pi|z|^2 > P_{\mathrm{supp}}\}$, *the core* $\bar D(p_{\mathrm{on}})$ *and the common core* $\bar D(c_{\mathrm{sh}}^2 p_{\mathrm{on}})$, *do not depend on* $y$;

*(b) if* $\mathcal{Y}_0\subset\mathcal{Y}$ *is compact, then the derivatives of order* $\ge 1$ *of* $\hat q$ *and* $\hat q^{-1}$, *and all derivatives (including order* $0$*) of* $\hat q - \mathrm{id}$, $\hat q^{-1}-\mathrm{id}$ *and* $Q$, *are bounded on* $\mathcal{Y}_0\times\mathbb{R}\times\mathbb{C}$.

## A.4. The shear and the disc-to-strip map.

**Lemma A.15** (Multi-angle shear; Paper I)**.** *Let* $A$ *be a smooth function on an open set* $W\subset\mathbb{R}^m$ *and* $\mathrm{Sh}_A(x,T,w) = (x,T,\mathrm{Sh}_{A,T}(w))$, *where*

$$\mathrm{Sh}_{A,T}(w)_j = \exp\big(-2\pi i\, T\,\partial_j A(p(w))\big)\, w_j \qquad (1\le j\le m),$$

*on* $\{(x,T,w) : p(w)\in W\}\subset\mathbb{R}^2\times\mathbb{C}^m$. *Then* $\mathrm{Sh}_A$ *is a diffeomorphism of this set onto itself, preserves* $p$, $x$ *and* $T$, *and*

$$\mathrm{Sh}_A^*\big(\omega_V + dx\wedge dT + dA(p)\wedge dT\big) = \omega_V + dx\wedge dT.$$

*The strip map* (Paper I, § "The disc-to-strip map and the layer maps"). For $\delta>0$ put $b_{\mathrm{st}} = \delta/\pi$ and, for $x>0$, $0<T<1$,

$$R_{\mathrm{st}}(x) = x + \frac{b_{\mathrm{st}}^2}{4x} - b_{\mathrm{st}}, \qquad V_{\mathrm{st}}(T) = b_{\mathrm{st}}\Big[\frac{1}{2\sqrt{T(1-T)}} - 1\Big],$$

Morse functions with minimum 0 at $x = b_{\mathrm{st}}/2$ and $T=\frac12$. Let $F_{\mathrm{st}}(\xi) = (X_{\mathrm{st}}(\xi_1), Y_{\mathrm{st}}(\xi_2))$, where $X_{\mathrm{st}}(\xi_1) = \frac12[\xi_1^2 + b_{\mathrm{st}} + \xi_1\sqrt{\xi_1^2+2b_{\mathrm{st}}}]$ and $Y_{\mathrm{st}}(\xi_2) = \frac12 + \xi_2\sqrt{\xi_2^2+2b_{\mathrm{st}}}/(2(\xi_2^2+b_{\mathrm{st}}))$ are the signed square-root inverses ($R_{\mathrm{st}}(X_{\mathrm{st}}(\xi_1)) = \xi_1^2$, $V_{\mathrm{st}}(Y_{\mathrm{st}}(\xi_2)) = \xi_2^2$), with Jacobian $\mathcal{J}_{\mathrm{st}}$. Let $\mathsf{T}_{\mathrm{st}}$, $\mathsf{D}_{\mathrm{st}}$ be the area normalisations of Paper I, Definition (Rotation and area normalisation) for the density $\mathcal{J}_{\mathrm{st}}$, which preserve the circles about 0, and let $\mathcal{E}_{\mathrm{st}}$ be the inverse of the function $\mathcal{E}\mapsto (dx\,dT$-area of $\{R_{\mathrm{st}}(x)+V_{\mathrm{st}}(T)\le\mathcal{E}\})$. The strip map is $\Psi_\delta = F_{\mathrm{st}}\circ\mathsf{T}_{\mathrm{st}}^{-1}\circ\mathsf{D}_{\mathrm{st}}\colon\mathbb{C}\to(0,\infty)\times(0,1)$.

**Proposition A.16** (Disc-to-strip map; Paper I)**.** *Let* $\delta>0$ *and* $b_{\mathrm{st}} = \delta/\pi$. *Then:*

*(i)* $\Psi_\delta$ *is a smooth diffeomorphism of* $\mathbb{C}$ *onto* $(0,\infty)\times(0,1)$ *with* $\Psi_\delta^*(dx\wedge dT) = \omega_{\mathbb{C}}$;

*(ii)* $R_{\mathrm{st}}(x) + V_{\mathrm{st}}(T) = \mathcal{E}_{\mathrm{st}}(\pi|z|^2)$ *for* $(x,T) = \Psi_\delta(z)$;

*(iii) for every* $z\in\mathbb{C}$, *with* $(x,T) = \Psi_\delta(z)$,

$$0 < x < \pi|z|^2 + \frac{\delta}{2}, \qquad 0<T<1; \tag{91}$$

*(iv) for* $I_{\max}\ge 0$, $\mathcal{E}_{\max} = I_{\max} + b_{\mathrm{st}}(\frac{\pi}{2}-1)$ *and* $\pi|z|^2\le I_{\max}$,

$$x \ge \tfrac12\Big[\mathcal{E}_{\max} + b_{\mathrm{st}} - \sqrt{\mathcal{E}_{\max}(\mathcal{E}_{\max}+2b_{\mathrm{st}})}\Big] > 0, \qquad \big|T-\tfrac12\big| \le \tfrac12\sqrt{1-(1+\mathcal{E}_{\max}/b_{\mathrm{st}})^{-2}} < \tfrac12.$$

**Definition A.17** (Enlarged capacities; Paper I)**.** For $1\le i\le k$ put

$$\delta_i = \frac{a_i-\rho_i}{2}, \qquad \bar\rho_i = a_i - \frac34\delta_i, \qquad \hat\rho_i = a_i - \frac{\delta_i}{2}.$$

Here $\bar\rho_i$ is the closed enlarged source capacity and $\hat\rho_i$ the open domain capacity, and $\rho_i < \frac{\rho_i+a_i}{2} = a_i - \delta_i < \bar\rho_i < \hat\rho_i < a_i$. Put $\varepsilon_i = (a_i-\hat\rho_i)/4 = \delta_i/8$; the $i$-th ball is placed by $\Psi_{\varepsilon_i}$.

## A.5. **The ball chart.**

**Lemma A.18** (The ball chart on representatives; Paper I)**.** *On* $S^5_* = \{Z \in \mathbb{C}^3 : |Z| = 1,\ Z_0 \neq 0\}$ *put* $b_{\mathrm{ball}}(Z) = \frac{\bar{Z}_0}{|Z_0|}\frac{(Z_1,Z_2)}{\sqrt{\pi}}$*. This map is gauge invariant,* $\pi|b_{\mathrm{ball}}(Z)|^2 = 1 - |Z_0|^2 < 1$*, and in the affine coordinates* $z = (Z_1/Z_0, Z_2/Z_0)$ *it is the chart* $\mathrm{ch}(z) = z/\sqrt{\pi(1+|z|^2)}$ *of Lemma A.5(b). Moreover* $b^*_{\mathrm{ball}}\omega_0 = \frac{1}{\pi}\sum_{j=0}^{2} dx_j \wedge dy_j\big|_{S^5_*} = \omega_{\mathrm{FS}}|_{S^5_*}$*, where* $Z_j = x_j + iy_j$*. Hence* $(b_{\mathrm{ball}} \circ \varphi)^*\omega_0 = \varphi^*\omega_{\mathrm{FS}}$ *for every smooth map* $\varphi$ *of a manifold into* $S^5_*$*.*

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